

Particulate Matter Monitoring & Health Effects

August 2026 | Entry window 1–31 August 2026, 31 daily issues | 556 records pooled

556

RECORDS POOLED
(31 DAILY ISSUES)

140

SENSING /
INSTRUMENTATION
(25%)

148

EFFECT ESTIMATES
FROM 63 SOURCES

48

RECORDS FROM
AMT + ACP

12

ULTRAFINE /
NANOSCALE (2%)

Signal of the month

Low-cost PM sensor calibration stopped being a collection of one-site case studies in August. Eight independent groups published work whose common subject is not *whether* a correction works but *how far it travels* — across cities, across seasons, across distance from the nearest reference.

The clearest statement is Ginsburg et al. (*Aerosol Sci Technol*), who tested whether the US EPA PurpleAir correction transfers worldwide and found it does not without regional recalibration. Zhivkov et al. (*Atmosphere*) put a number on the spatial half of the same problem: using 24 months from Sofia with four co-located and eighteen remote sensors, calibration uncertainty grows **3.84–5.62% per kilometre** from the nearest reference station, so co-location sites 6 km apart hold the network midpoint inside a 30% uncertainty increase — a design rule, derived empirically, that a network planner can actually use. Wang et al. (*Environ Sci Technol*) extended correction to a nationwide dense network without local references at all; Balasubramanian et al. (*Environ Pollut*) built the US-wide PurpleAir model; Blaga et al. (*PLOS Clim*) took a city-scale network to regulatory-compliant output; Chung et al. (*Sensors*) characterised PurpleAir *longevity* and uncertainty across 2016–2025 deployments, which is the failure mode nobody calibrates for; Barkjohn et al. (*Atmosphere*) shipped ASNAT, an R-Shiny QA/QC and reference-comparison tool, i.e. the tooling layer arriving after the methods; and Gäbel & Hertig (*AMT*) tied recalibration practice to the current sensor-technology literature. Two constraints published alongside them matter more than any of the corrections. Chyzhykov et al. (*J Hazard Mater*) quantified **particle-bound water as a bias on gravimetric PM mass** — that is the reference being wrong, not the sensor — and Benjamin et al. (*Sensors*) showed environmental extrapolation failing outside the training envelope for both CO and PM_{2.5} nodes. **The month's verdict: transferability, drift and reference validity are now the three named open problems, and the field has stopped pretending a single regression solves any of them.**

What changed in August

- **The pipeline gap July's rollup named was closed, and it is the largest single change in this corpus.** July's monthly stated that there was "no date-windowed channel into *Atmos. Meas. Tech.* or *Atmos. Chem. Phys.*" and proposed a Crossref by-journal feed keyed on created. That feed was built and widened through the month. August's result: **Atmos Chem Phys is the single largest venue in the corpus at 37 records, with Atmos Meas Tech at 11** — 48 together, ahead of *Environ Sci Technol* (25) and *Environ Pollut* (22). The pure atmospheric-measurement literature that was structurally invisible in July is now the backbone of the sensing cluster.
- **Sensing is now the largest subtopic, at 140/556 (25%).** July recorded 23. Most of that difference is the AMT/ACP channel plus a full 31-issue cadence rather than five issues, so it is **not** evidence that the field grew — it is evidence of how badly the July corpus under-sampled instrumentation. Exposure assessment & modelling is second at 129 (23%). Together the two non-clinical clusters hold 48% of the month.
- **Ultrafine particles are still the metric nobody measures.** Twelve records (2%) use an ultrafine or nanoscale metric, *down* from July's 5% share. Yet Zhong et al. (*Environ Int*) showed photochemical ultrafine particles posing an inflammatory risk *at low PM_{2.5} mass*; Rashid et al. (*Toxics*) published a scoping review of UFP mortality evidence and exposure assessment; Blanco et al. (*JESEE*) showed mobile-monitoring design choices propagating into UFP exposure models; and Zhang et al. (*Environ Sci Technol*) built a wearable that measures particle deposition in the human respiratory tract directly. The prediction made in July's rollup — that a field calibration of a low-cost particle-number or black-carbon sensor against a CPC or aethalometer reference would appear within two monthly issues — **did not come true in August**. The nearest approaches are Kirchhoff et al.'s CPC working-fluid evaluation (*AMT*) and Shlipak et al.'s source-apportioned black carbon in Addis Ababa, neither of which is a low-cost-instrument validation. That prediction now carries into September, and one more miss should be read as a hardware-cost bottleneck rather than a research-attention one.
- **Effect estimates: 148 from 63 sources, and 110 of them exclude the null in the harmful direction.** Only six were protective, and every one of those has a benign explanation (an ozone term, a wildfire-season influenza-like-illness contrast, thyroxine and menopause-timing shifts). Thirty-two cross the null. The largest ratios are all extreme-contrast designs — Xu et al. report OR ≈ 4.3 –4.4 for cerebral hypoperfusion

against $\text{PM}_{2.5}$ and black carbon, and Li et al. OR 5.08 (3.45–7.47) for paediatric allergic-rhinitis exacerbation comparing $\text{PM}_{2.5} > 75$ against $< 35 \mu\text{g m}^{-3}$ — and are not comparable to per-10 $\mu\text{g m}^{-3}$ estimates.

- ▶ **Compound and multi-stressor designs consolidated.** Mandalapu et al. (medRxiv) mapped joint heat and $\text{PM}_{2.5}$ exposure across US metropolitan areas against historical HOLC redlining grades (**OR 3.97, 2.79–5.66** for hotspot residence in the worst-graded areas); Yu et al. (*Environ Sci Technol*) gave a probabilistic treatment of $\text{PM}_{2.5}$ –ozone co-occurring extremes under wildfire smoke; Wang L et al. closed the month with a lipid-fraction $\times \text{PM}_{2.5}$ interaction on stroke. Single-pollutant, single-endpoint time-series work continues to read as the low-information tail.
- ▶ **Sub-Saharan Africa quadrupled, from 4 records in July to 16, and it still is not enough.** Shlipak et al. (Addis Ababa source apportionment), Mushore et al. (road-edge effect across three campaigns), Rehbein et al. (low-cost network source triangulation in Ile-Ife) and Makacha et al. (personal $\text{PM}_{2.5}$ exposure determinants) are all methodologically serious. Against 121 records from China, the imbalance remains the corpus's most persistent structural feature.
- ▶ **A dedup defect was found by this rollup and has been gated.** See the provenance box: twelve records were shipped twice.

Corpus analytics — August 2026

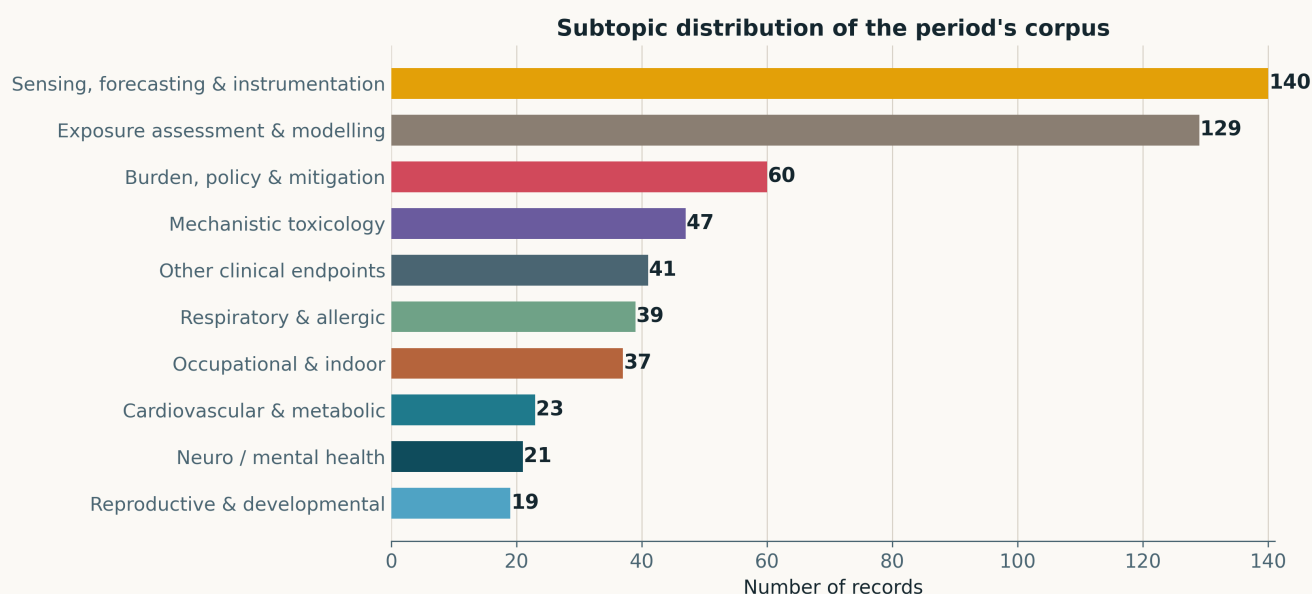


Fig. Subtopic distribution across all 556 pooled records. All ten subtopics are populated, as they were in every week of the month. *Sensing, forecasting & instrumentation* (140) and *Exposure assessment & modelling* (129) together take 48% of the corpus — a share that reflects the new AMT/ACP harvesting channel as much as the underlying literature. *Cardiovascular & metabolic* (23) and *Reproductive & developmental* (19) are the thinnest clusters. Assignment is single-label by dominant contribution.

What the daily cadence missed, by subtopic (2026-08-01 to 2026-08-31)

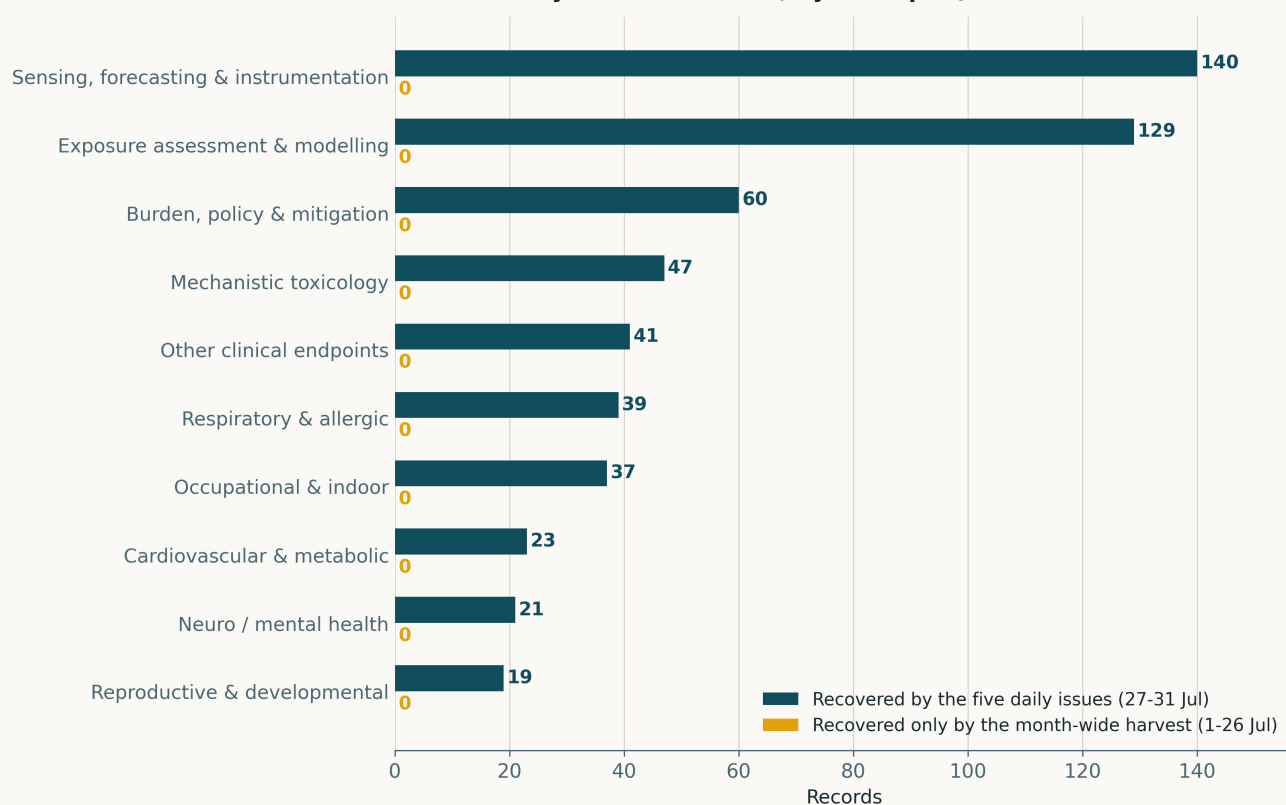


Fig. Records per issue date by subtopic across the 31 August issues. Unlike July's equivalent panel this is a genuine daily series — every bar is one entry-date window, with no backfill batch. Day-to-day variation is large (8 records on 29 August, 36 on 27 August) and is driven by deposit timing at a handful of high-volume publishers rather than by anything in the literature. The ISO-week totals are steadier: 56, 127, 132, 111, 114 and 16 for the part-week ending 31 August. Read week totals, not day totals.

Study architecture

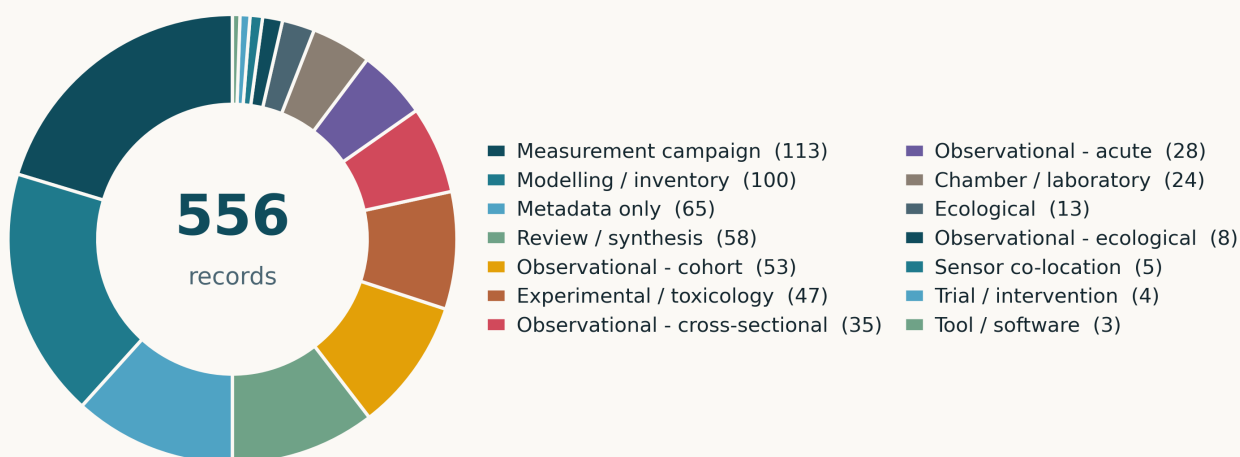


Fig. Study architecture over the month. Measurement campaigns lead at 113, ahead of modelling/inventory (100), reviews (58) and observational cohorts (53). The “Metadata only” group at 65 records (12%) is the number to hold onto: those are records whose abstract was not deposited anywhere reachable at issue time. They are counted in every total in this issue but carry no assessed content. **Trials or interventions: 4 records in 556.** That ratio has not improved since July and is the standing weakness of the evidence base.

Health endpoint distribution — reported as text, not as a figure

The endpoint figure is **omitted from this issue on purpose**. The endpoint field is free text and has never been canonicalised — an open defect since 15 August — so the 556 pooled records carry **144 distinct endpoint strings**, and the pooled chart degenerates into roughly ninety unreadable rows. The distribution itself is short enough to state: **210 records (38%) carry no human health endpoint at all**, split across “None (monitoring)” 75, “No health endpoint” 71, “None (exposure)” 33 and “None (instrument)” 31 — four spellings of one category, which is the defect in miniature. A further **31 records have an endpoint that was not extracted**. Of the remainder the named categories are **respiratory 36, cardiovascular 22, other clinical 14, reproductive 13, neuro/cognitive 11, in-vitro toxicological 11, mortality 10, aerosol chemistry 9, oncologic 9**, with the balance spread thinly across more than a hundred one- and two-record strings. **Fix required before the September rollup**: a bounded `ENDPOINT_CANON` map in `plots.py`, in the same pattern as `design_group` and `geo_group`, both of which already work.

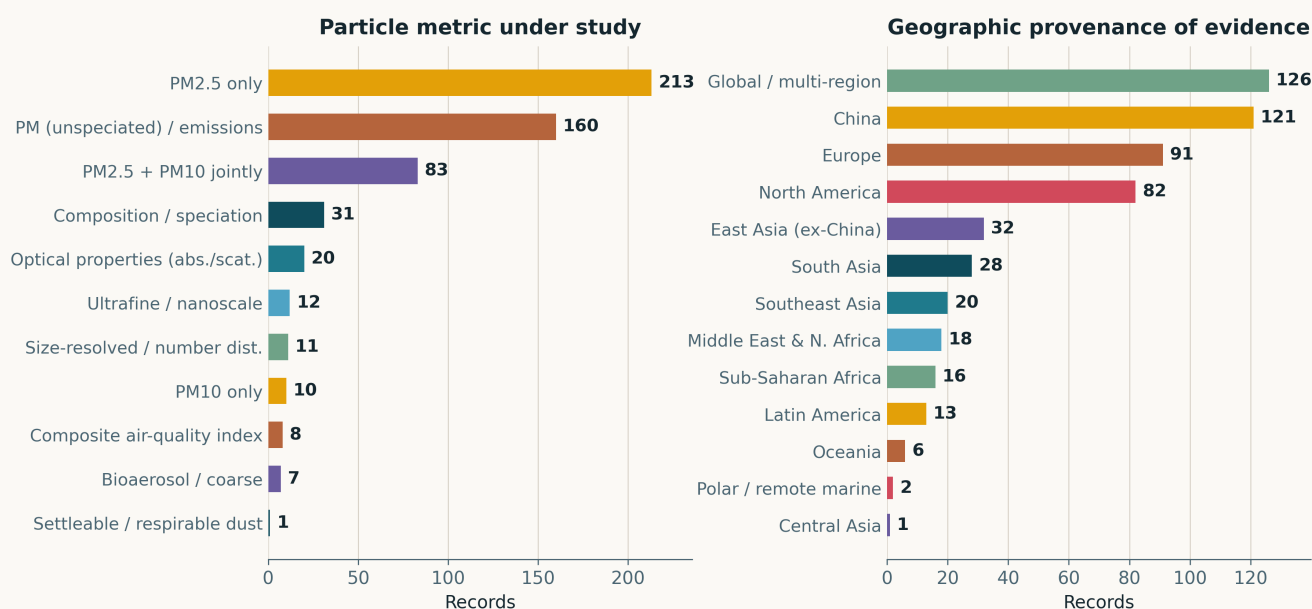


Fig. Left — particle metric. PM_{2.5} alone in 213 records, unspecified PM or emissions in 160, PM_{2.5} with PM₁₀ in 83. **Ultrafine or nanoscale metrics appear in 12 records (2%) and size-resolved number distributions in 11**, against 31 for composition/speciation and 20 for optical properties. Right — geography. *Global / multi-region* 126, *China* 121 (22%), *Europe* 91, *North America* 82, *East Asia excluding China* 32, *South Asia* 28, *Southeast Asia* 20, *Middle East & N. Africa* 18, *sub-Saharan Africa* 16, *Latin America* 13, *Oceania* 6, *polar/remote marine* 2, *Central Asia* 1.

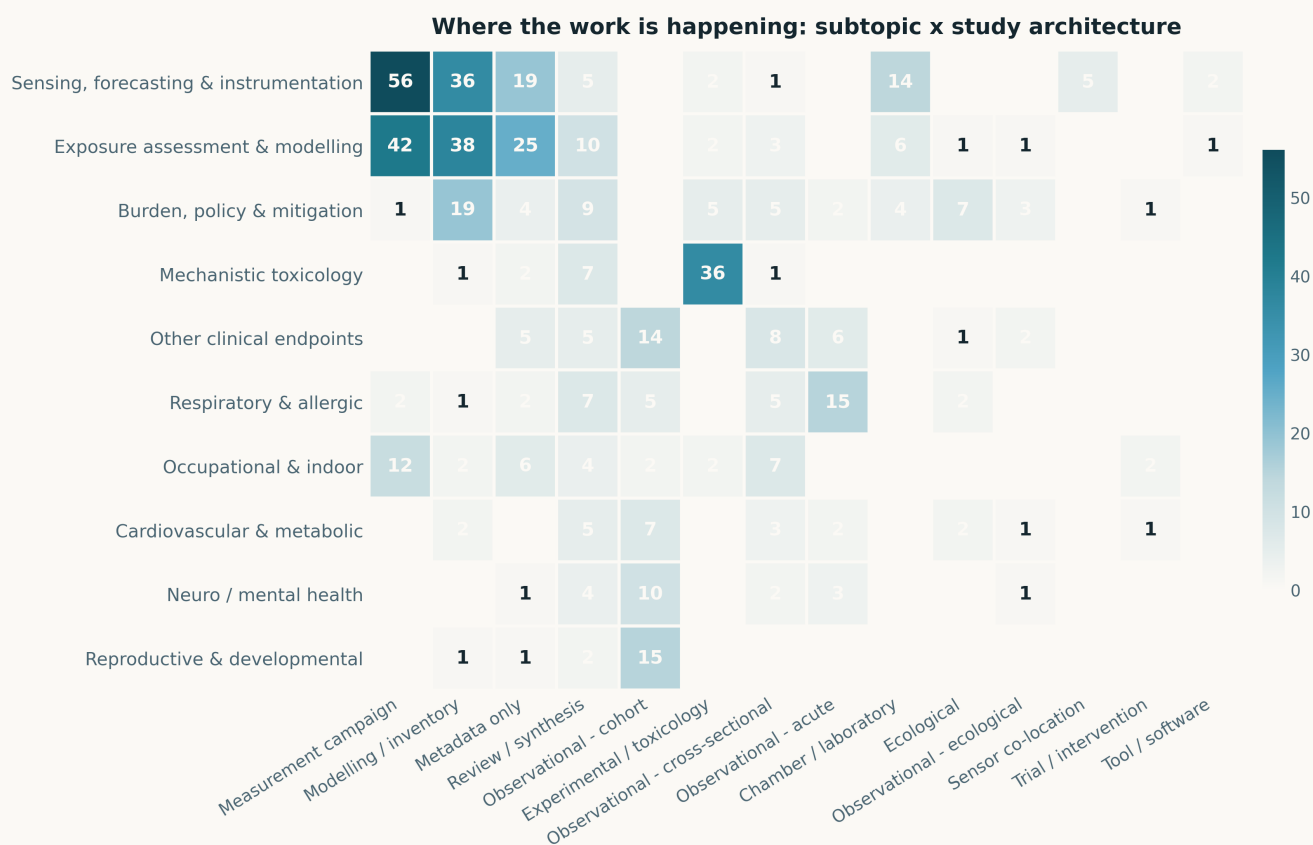


Fig. Subtopic × study-architecture cross-tabulation over the month. The dense cells are expected: sensing × measurement campaign, exposure × modelling/inventory. The informative absences are the same two the daily issues keep reporting — the trial column is nearly empty across every health subtopic, and the sensing and exposure rows contribute almost nothing to the effect-estimate pool.

Quantitative associations reported in the period's corpus (1 of 5)

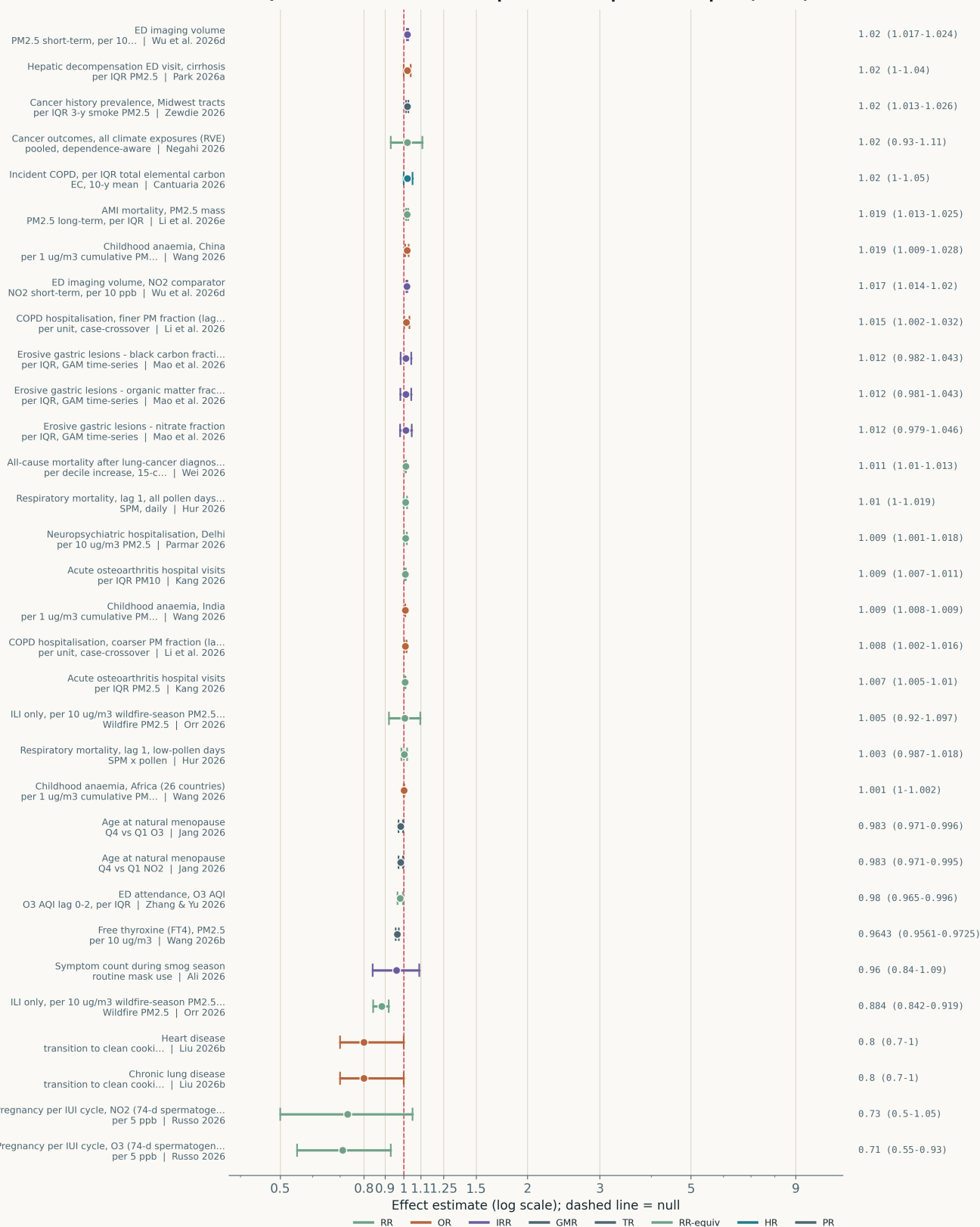


Fig. Quantitative effect estimates extracted across August, panel 1 of 5. **This is not a meta-analysis and no pooled summary is drawn.** Exposure contrasts are not commensurable — per-IQR, per-SD, per- $10 \mu\text{g m}^{-3}$, threshold, quartile and mixture-index contrasts all appear — and the outcome scale mixes RR, OR, aOR, HR, IRR, GMR, PR and time ratios across unrelated diseases. **Heterogeneity is therefore not estimable and none is reported.** Of the 148 estimates, 110 exclude the null in the harmful direction, 6 in the protective direction and 32 cross it.

Quantitative associations reported in the period's corpus (2 of 5)

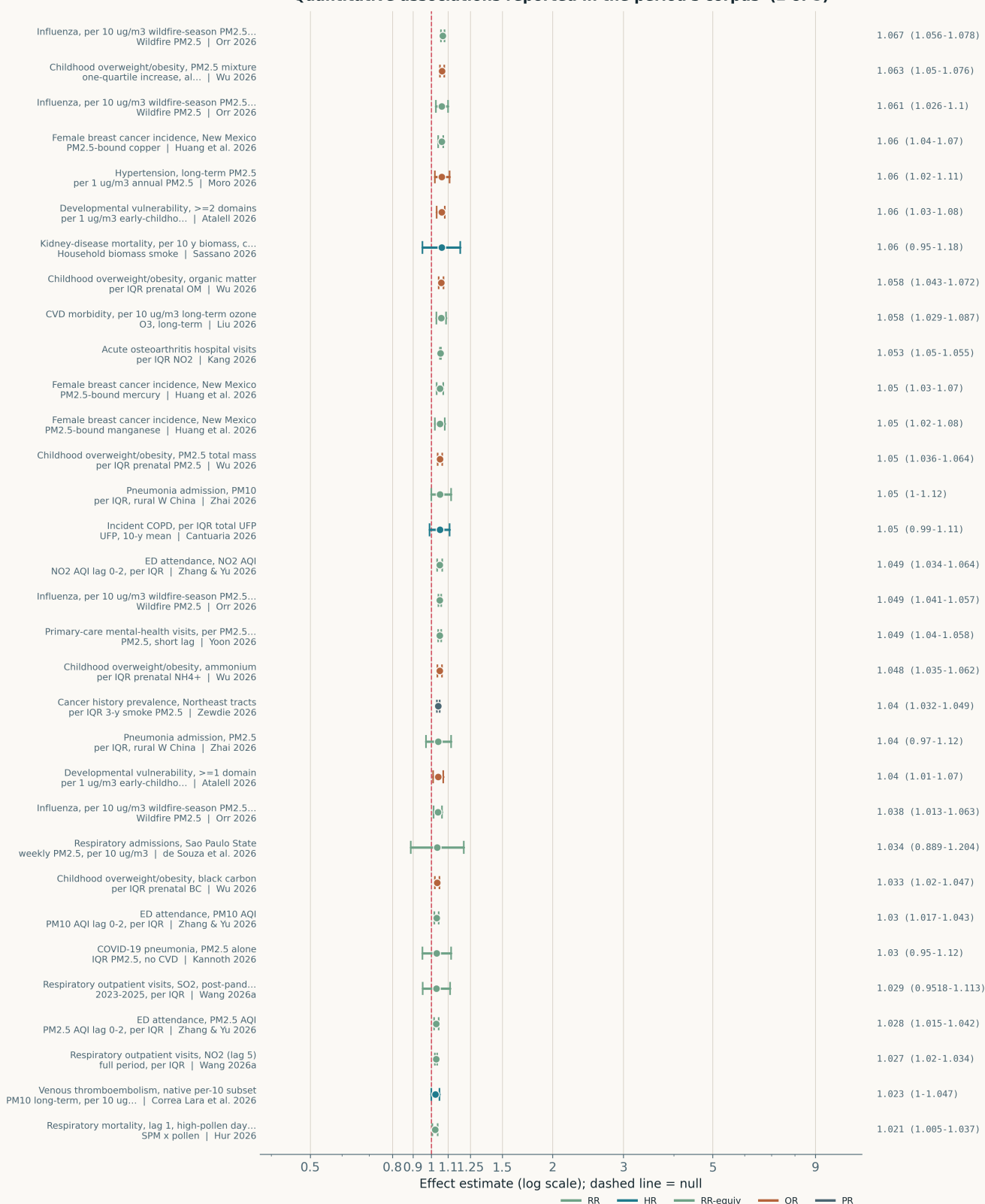


Fig. Effect estimates, panel 2 of 5. Same caveats as panel 1.

Quantitative associations reported in the period's corpus (3 of 5)

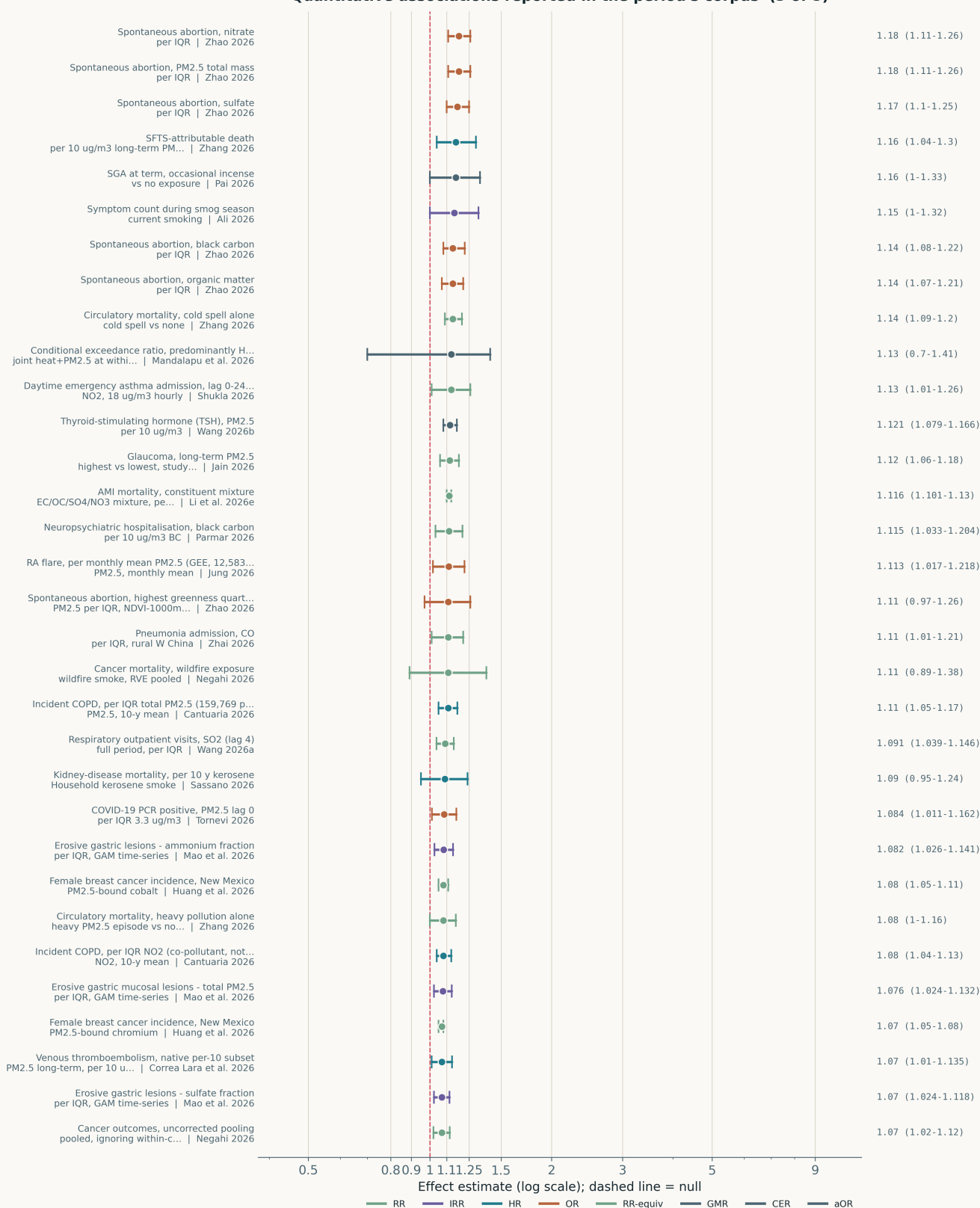


Fig. Effect estimates, panel 3 of 5. Same caveats as panel 1.

Quantitative associations reported in the period's corpus (4 of 5)

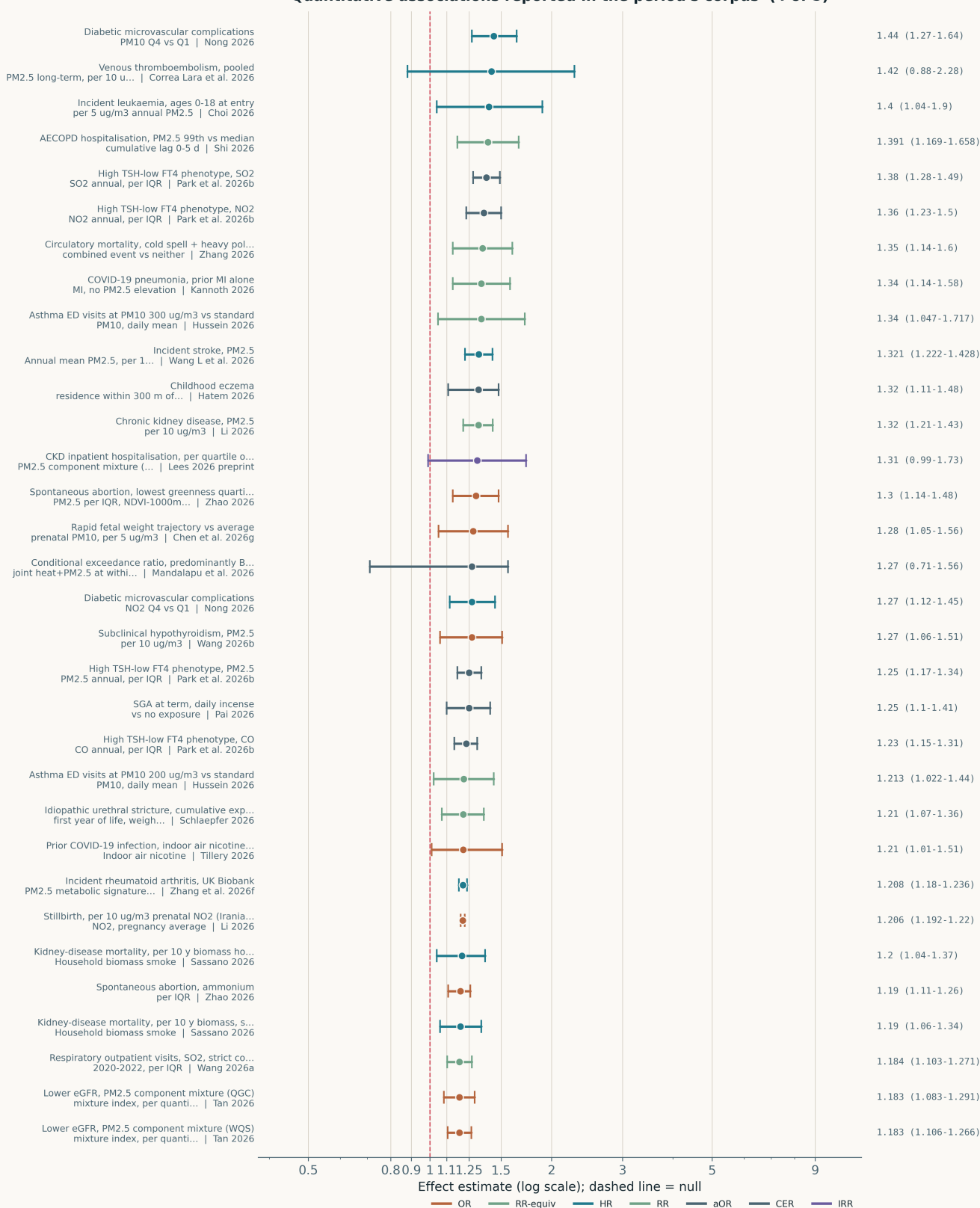


Fig. Effect estimates, panel 4 of 5. Same caveats as panel 1.

Quantitative associations reported in the period's corpus (5 of 5)

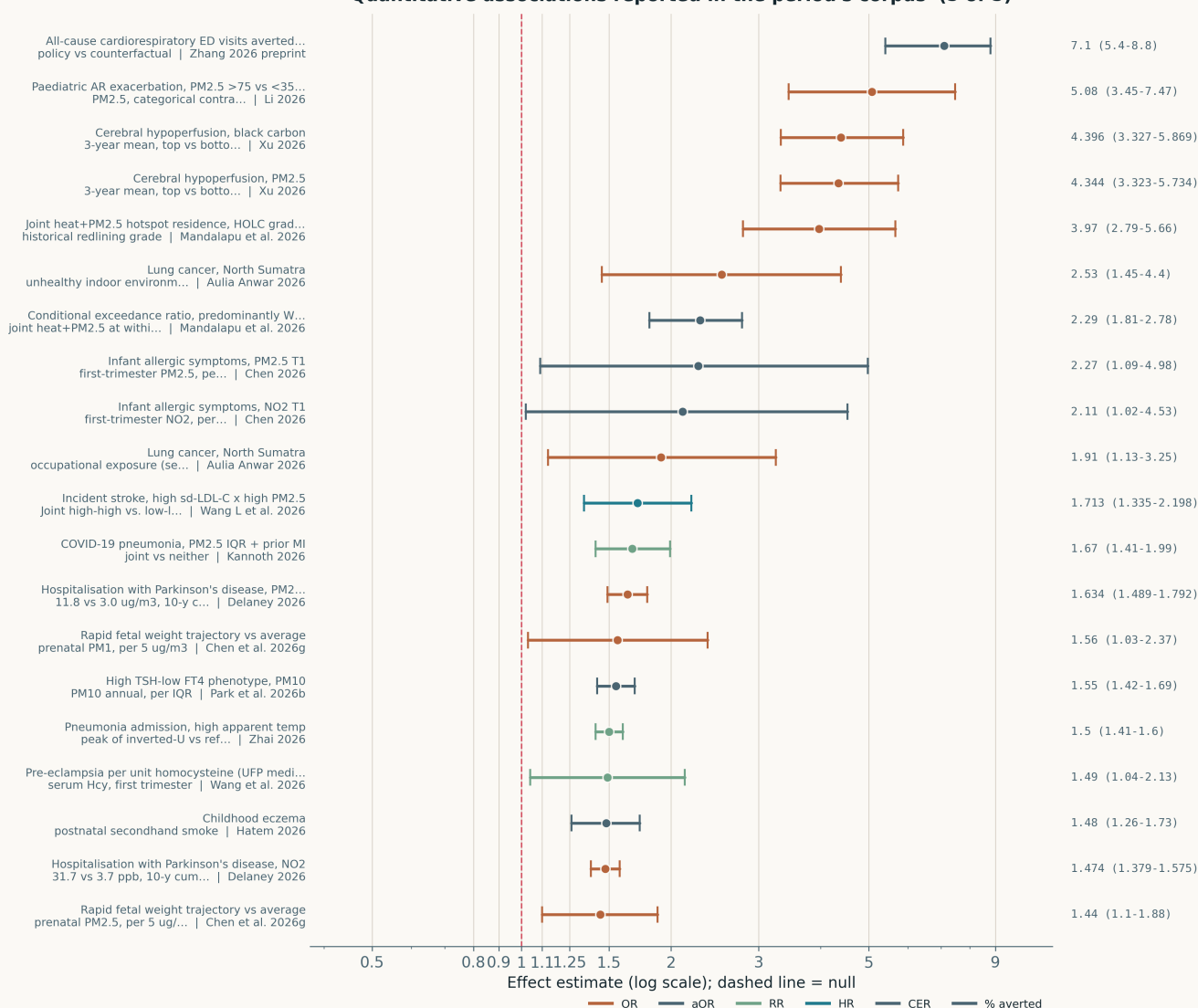


Fig. Effect estimates, panel 5 of 5. Same caveats as panel 1. Read the panels as a catalogue of what was reported, not as a ranking: the visually largest estimates are uniformly extreme-contrast designs (threshold comparisons, redlining-grade contrasts, mixture indices) and the tightest intervals belong to the largest cohorts.

Life-course windows addressed this period (records may span several stages)

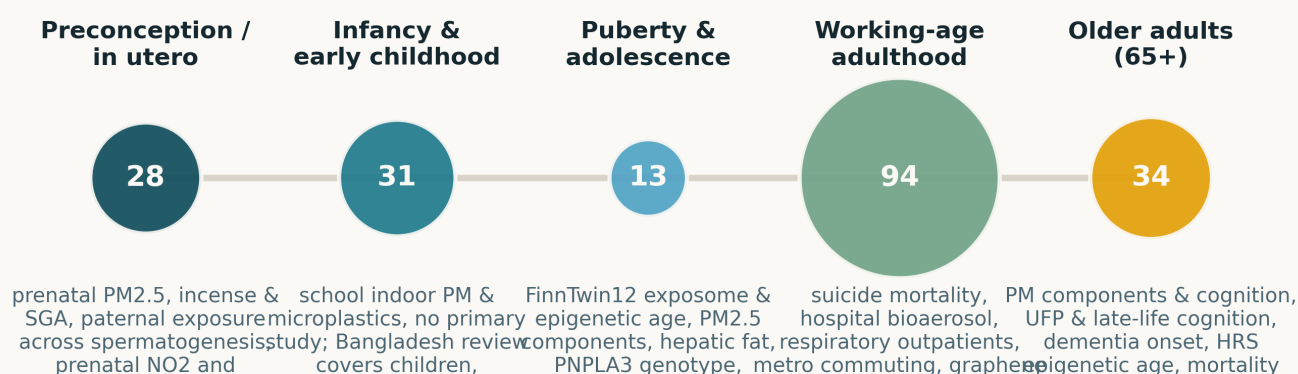


Fig. Life-course windows addressed across the month. Counts are non-exclusive and cover only records with an identifiable human life-course window; *in vitro*, ex-vivo, animal and pure-instrumentation records are excluded, which is why the totals are far below 556. Adolescence remains the thinnest window, as it was in July and in almost every individual daily issue.

Emerging, fading, and recurring

Emerging. (i) **Calibration transferability as a named problem** — see the signal box; the shift is from “here is a correction for our site” to “here is how far a correction travels, and here is the uncertainty as a function of distance, season and instrument age”. (ii) **The reference itself under scrutiny** — Chyzykhov et al. on particle-bound water biasing gravimetric mass, Anon et al. on a gravimetric-method assessment campaign for PM₁₀ and PM_{2.5}, and Alexa et al. on separating gaseous nitric acid from particulate nitrate. When the calibration target is itself biased, every downstream R^2 inherits it. (iii) **Toxicity- and source-weighted exposure** — Chen et al. on source-specific oxidative potential of PM_{2.5} in Xi’an, Kang et al. on vehicle-origin condensable PM, and Song et al.’s closing perspective on shifting control from mass to toxicity. (iv) **Citizen science formalised** — Marcon et al. (*Curr Environ Health Rep*) reviewed methodological approaches and effectiveness, and Cho et al. ran a community-scientist-led mobile campaign; this had been anecdotal in July.

Fading. With 31 genuine daily windows this month, a decline claim is finally defensible in principle — but one month against one partly-backfilled month is still not a trend, so nothing is asserted here as fading. The weakest-signal observation, and it is only an observation: **single-city, single-pollutant time-series studies continued to appear and continued to add nothing**, generally reporting PM as the weakest term in their own multi-pollutant models. That was July’s observation too, unchanged.

Recurring groups and venues. **No author-cluster analysis is possible in this corpus and none is offered.** Records carry a first-author surname but no affiliation field, so apparent repeats (Li, Zhang, Wang, Chen, Liu) are surname frequency in a China-heavy corpus, not the same investigators. This has been an open defect since 15 August and remains one. Venue concentration is the analysis that *is* supportable: *Atmos Chem Phys* 37, *Atmos Environ* 34, *Environ Sci Technol* 25, *Environ Pollut* 22, *Environ Monit Assess* 17, *Atmos Pollut Res* 14, *Atmos Meas Tech* 11, *Atmosphere* 11, *J Hazard Mater* 10, *Environ Int* 8. Preprints contributed 31 records — Research Square 21 under *three* spellings, medRxiv 6 under two, arXiv 4 — which is a venue-normalisation defect, logged below. Citation counts on records days to weeks old are uninformative and were not used to rank anything here; “most-cited work of the period” is therefore not reported, because any such ranking would be an artefact of indexing lag.

Trial registry watch

Seven registry sweeps ran across August. The accumulated watch list now holds **24 trials**, of which **21** were registered, updated or first seen inside August. Two new interventional registrations appeared in the month:

NCT07111208 (MARKOPOLO/METSGREEN, $n=180$, recruiting), added 4 August, and **NCT07797413** — the **Savannah Environmental Nutrition Study**, registered 31 August by Georgia Southern University. SENS is a community-based antioxidant- and nutrient-rich food intervention in residents near the Port of Savannah, $n=20$, not yet recruiting, starting 1 September 2026. `analyze_endpoints` was run: the *primary* endpoint is **participant retention** ($\geq 80\%$ target through week 12) — this is a feasibility study, and no efficacy claim should be attached to it. The secondary endpoints are the interesting part: change from baseline in **fractional exhaled nitric oxide** (NIOX Mini, ppb) and **hydrogen peroxide in exhaled breath condensate** (RTube collection, Amplex Red fluorescence, μM), plus blood pressure, resting heart rate and HbA1c. **No PM exposure measurement is registered at all** — proximity to the port is the entire exposure contrast. That is the recurring weakness of this registry segment: the intervention is characterised in detail and the exposure is not characterised at all.

The rest of the August watch list is dominated by **portable air cleaner and filtration trials** — PATHFINDER in heart failure ($n=400$), a multi-centre clean-air RCT in COPD ($n=770$), NCT05874479 on blood pressure in New York ($n=440$), HEPA in rural Alaska ($n=15$), ERV plus air cleaner for mould and asthma in low-income housing ($n=80$), and cost-effective filtration with PM monitoring in low-income housing ($n=55$) — plus three MEDiterranean climate-health resilience arms ($n=102$ each, outdoor workers, hypertensive 65+, and pregnant women) and INTERLUNG in Bhaktapur ($n=800$). One completed cluster-randomised comparison of HEPA against ionisation/plasma air cleaners ($n=180$) is the most directly instrument-relevant result to watch for publication. **The interventional evidence base for personal and household PM reduction is thickening in the registry but has not yet reached the literature: four trial-class records in 556 published papers this month.**

Open gaps and read on direction

Gaps in the literature

(i) **Ultrafine measurement still does not exist at network scale.** Twelve ultrafine records in 556, no low-cost particle-number instrument with a published field calibration against a CPC, and a scoping review (Rashid et al.) that documents the epidemiological interest without closing the measurement gap. July's falsifiable prediction on this point failed; it is re-issued for September. (ii) **The calibration reference is itself unvalidated.** Particle-bound water biases gravimetric mass (Chyzhykov et al.); a full assessment campaign of the gravimetric method was needed and published (Anon et al.). Every sensor R^2 in this corpus is conditional on a reference nobody has re-audited. (iii) **Four trial-class records in 556.** Every causal claim in the month rests on observational designs with modelled exposure. (iv) **Geographic imbalance persists.** 121 China against 16 sub-Saharan Africa and 13 Latin America — better than July's 44 against 4, still not proportionate to burden, and Song et al.'s toxicity-weighting perspective puts the highest toxicity-adjusted exposure precisely in the regions this corpus covers least. (v) **Drift and ageing are measured but not modelled.** Chung et al. characterise PurpleAir longevity; Kumpika et al. correct sensor ageing; no record in the month puts an ageing term inside an operational network's correction pipeline.

Gaps and defects this pipeline creates

(i) **DUPLICATES SHIPPED — found by this rollout.** Pooling August returned **556 records over 548 unique DOIs**. A full-store audit found **12 records published twice**: seven on 2 August (four first shipped 1 August, three first shipped 29 July), two on 3 August, one on 9 August (first shipped 7 August), and the two 10.3779/j.issn.1009-3419 records already disclosed in the W35 weekly. In every case the DOI was *already correctly recorded in state/seen.json* under the first issue date — the index was being written and not read, and the duplicate shipped under a reworded title that defeated the title-signature leg. **Fixed forward on 1 September: `check_dois.py`**, which already gates every `-post` build, now hard-fails on DUPLICATE-DOI by scanning the shipped corpus itself rather than `seen.json`, so it cannot be defeated by a write-side bug either. Shipped issues have *not* been retro-edited — rewriting published issues is worse than disclosing the error — so **every count in this issue is over 556 records containing 8 within-August duplicates, and subtopic counts are inflated accordingly by roughly 1.4%**. (ii) **65 records (12%) are classification-only.** No abstract was deposited anywhere reachable at issue time. They are counted but carry no assessed content, and the absence of criticism on those records is not approval. (iii) **No first-author affiliation field**, so no author- or institution-cluster analysis is possible. Open since 15 August. (iv) **Venue names are not normalised** — Research Square appears as “Res Sq (preprint)” (9), “Research Square (preprint)” (9) and “Research Square” (3), and medRxiv as two spellings (5 and 1). Any venue table in this series under-reports split spellings; the top-ten list above is correct only for venues with a single spelling. (v) **OpenAlex has been dead all month** (HTTP 403 paid-plan block). Its coverage is routed through Consensus, which has no entry-date filter, so that leg contributes dated backfill rather than windowed harvest.

Read on direction. July's read was that the field is moving from mass to composition and number while the instrumentation base stands still. August refines it: the instrumentation base is moving, but along the *transferability* axis rather than the *new-quantity* axis. Eight calibration papers, a QA/QC tool, a longevity study and a distance-uncertainty law all improve how well we can trust a $\text{PM}_{2.5}$ mass network — and not one of them measures anything a $\text{PM}_{2.5}$ mass network does not already measure. Meanwhile Zhong et al. demonstrated inflammatory risk from photochemical ultrafine particles *at low mass*, and Song et al. argued that toxicity, not mass, should be the control variable. **The two halves of the field are optimising different objective functions, and the gap widened in August rather than narrowing.** Falsifiable predictions for September, both re-issued and sharpened: (1) a field calibration of a low-cost particle-number or black-carbon sensor against a CPC or aethalometer reference appears — this failed in August and a second miss should be read as a hardware-cost bottleneck, not a research-attention one; and (2) at least one paper explicitly incorporates a *sensor-ageing* term into an operational network correction, rather than characterising ageing and correcting it offline. If (2) also fails, the practical conclusion for a distributed-sensing programme is that drift management remains an unclaimed contribution.

Paper-level digest — the month by subtopic

Of the 556 pooled records, **548 carry a full assessed summary** lifted verbatim from the daily issue that first published it, and **8 are title-only**. A title-only entry is **not an assessment**: it has not been checked against the paper, no limitation has been identified, and the absence of criticism is not approval. Read those entries as a pointer to a DOI and nothing more. Note also that the 548 figure counts *records*, not distinct works — see defect (i) above.

Sensing, forecasting & instrumentation

140 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (138 OF 140 RECORDS IN THIS SUBTOPIC)

Acton-Bond et al. 2026 Atmos Environ

Rolling positive matrix factorisation for aerosol optical apportionment

METADATA ONLY — no abstract in Crossref, Europe PMC or Scholar Gateway. Listed, not summarised. Carried because *rolling* PMF applied to optical rather than chemical data is a specific and consequential claim: conventional PMF source apportionment requires filter sampling and offline speciation, whereas an optical implementation with a rolling window could in principle run on an aethalometer or a multi-wavelength absorption photometer at deployment cadence. That would move source attribution from a laboratory capability to a network capability, which is the distinction that decides whether a distributed sensing programme can report anything beyond mass. Whether the paper supports that reading is unknown until an abstract appears.

doi: 10.1016/j.atmosenv.2026.122295

Adams et al. 2026 Environ Sci Atmos

Sampling sub-pixel scale variability in total column NO₂ within a coastal urban environment

Long-term Pandora spectrometer records of column-integrated NO₂ at Boston University and Harvard University, used to characterise variability within a single satellite pixel. Directly relevant to any satellite–surface fusion product: sub-pixel heterogeneity is an irreducible error term in the validation of column retrievals, and two sites a few kilometres apart in the same coastal urban area are the cleanest way to bound it. Two instruments in one metropolitan area is a small basis for a general claim about pixel-scale variability, and NO₂ column heterogeneity is not a direct proxy for PM heterogeneity.

doi: 10.1039/d6ea00033a

Adeniji et al. 2026 PLOS Glob Public Health

Twelve years of hourly criteria-pollutant and meteorological coupling in Abeokuta, Nigeria

Hourly PM_{2.5}, PM₁₀, NO₂, SO₂, CO and O₃ over 2012–2023 with six meteorological covariates. **PM_{2.5} and PM₁₀ dominate and persistently exceed WHO guidelines**, peaking in the December–March Harmattan; regression against temperature gives R² 0.48–0.52 with negative coefficients on rainfall and humidity, consistent with thermal stagnation and wet scavenging. The aggregate AQI sits at 39–51 (“moderate”) while the health-risk-adjusted index exceeds 90 seasonally — **a 2-fold divergence between the two indices at identical concentrations**, which is the paper’s real contribution and an argument for HAQI-style reporting on any network serving a PM-dominated airshed. Single station for a metropolitan area, and the provenance and calibration history of a 12-year record in a data-sparse setting are not documented.

doi: 10.1371/journal.pgph.0006683

Akazezuwa et al. 2026 Environ Monit Assess

SFFO-RL intelligent environmental monitoring: smart sensing networks with artificial intelligence

Proposes an “Intelligent Environmental Monitoring Framework” combining adaptive filtering, Isolation Forest outlier removal, min–max normalisation, multi-order neighbour feature fusion and a reinforcement-learning optimiser, reporting **97.1 % accuracy, 97.8 % F1, 97.6 % recall and 98.21 % precision**. Those figures cannot be interpreted. The abstract names no pollutant, no dataset, no sensor, no units, no sample size and no held-out split; “accuracy” is quoted for what is elsewhere described as predicting continuous pollutant concentrations, which is a category error unless the task was silently discretised. A four-metric cluster in the 97–98 % band on environmental time series with autocorrelation is the signature of evaluating on temporally adjacent data. Carried as tier C because the framework class (edge-side denoising plus learned correction) is squarely on this watch’s axis and the record should be visible in the register; the reported numbers should not be cited.

doi: 10.1007/s10661-026-15724-0

Albashir 2026 Res Sq (preprint)

A physics-informed digital twin framework for health vigilance and indoor air quality decision support

Preprint proposing a physics-informed digital twin for indoor air quality in healthcare-oriented environments, arguing that monitoring-only systems do not support proactive decisions. **Not peer reviewed, single author, no deployment, no validation against measured concentrations reported in the abstract**. Carried only because the architectural claim — coupling a physical model of a room to a sensor stream so the system can extrapolate beyond sensor locations — is the same idea as physics-informed calibration on the ambient side, and the indoor literature is where it is being tried first. Treat as a pointer to a design pattern, not as evidence. Relevance to distributed sensing: physics-informed correction is the leading candidate for generalising calibration away from co-location sites, and this is a low-cost read on how that framing is developing indoors.

doi: 10.21203/rs.3.rs-10034990/v1

Alexa et al. 2026 Talanta

Simultaneous continuous determination of gaseous HNO₃ and particulate nitrate by chemiluminescence

Gaseous HNO₃ is sampled by wet diffusion denuder and particulate NO₃[−] in parallel by continuous aerosol sampler, both analysed online by continuous flow analysers with chemiluminescence detection — no preconcentration step, signal recorded at 1 s intervals. Detection limits (3S/N) are **5.1 nM and 12.8 nM in solution, corresponding to 0.13 μg/m³ (49.7 ppt) for HNO₃ and 0.10 μg/m³ for NO₃[−]**. Applied to urban air in Brno across winter and summer campaigns 2022–2023 and compared against filter sampling with NaF-coated dry denuders. The gas–particle partitioning of nitrate is temperature- and humidity-driven on timescales of minutes, which filter-based methods average away entirely and optical low-cost sensors cannot resolve at all; a method at this time resolution is the reference any nitrate-aware correction scheme would need. The abstract reports no field intercomparison statistics against the reference method, only that the comparison was made.

doi: 10.1016/j.talanta.2026.130457

Alharbi et al. 2026 Atmosphere*Morphology, surface chemistry and mineralogy of coastal PM_{2.5}, Jeddah*

Three coastal sites over summer 2017 characterised by SEM, EDS mapping, XPS and XRD, linking bulk concentration (**22.2, 18.9 and 14.2 $\mu\text{g}/\text{m}^3$** north to south) to particle morphology, surface oxidation state and crystalline phase. The authors are unusually candid about their own artefact: **borosilicate glass-fibre substrates dominate the SEM field**, so aerosol is resolved only as discrete sub-micrometre particles and agglomerates decorating the fibres, and the morphological interpretation is framed accordingly. That admission is worth more than the result — filter-substrate choice constrains what electron microscopy of a PM sample can establish, and most papers in this genre do not say so.

doi: 10.3390/atmos17090830

Anon et al. 2026 Atmosphere*Metrological consistency of the EN 12341 gravimetric reference method*

National interlaboratory comparison run by ISPRA at Castel Romano with **18 Italian regional environmental agencies** sampling side by side, motivated by the absence of Certified Reference Materials for PM and by known sampler-design discrepancies. Systematic between-laboratory deviations were driven mainly by **relative humidity and precipitation** acting on PM thermodynamic and mass stability, with a minor filter-matrix effect. Because gravimetry is the only traceability source for automated measurement systems, the bias propagates into every AMS calibrated against it — and Directive (EU) 2024/2881 tightens AMS residual uncertainty budgets sharply from 2030, at concentrations that are falling. No numeric bias magnitudes appear in the abstract, which is the main gap; the full text is needed to size the effect. Crossref lists no first author.

doi: 10.3390/atmos17090828

Aparicio et al. 2026 Sensors (Basel)*Sensor system design and deployment for environmental and visitor monitoring in cultural heritage sites*

Design, build and field validation of low-power low-cost nodes across five pilot sites in four European countries under the EU ARGUS project, in two configurations: static nodes at fixed positions and dynamic nodes carried by a quadruped robot or drone. Payload spans thermal-humidity probes, volumetric water content, wind, pollution monitors and two generations of visitor-counting devices, with Baltanás used as the iterative prototyping site before wider deployment. Carried for the mobile-platform architecture — a sensor-on-robot deployment pattern that is directly transferable to indoor and semi-enclosed PM mapping — rather than for any air-quality result. **The abstract reports no PM performance figures, no reference co-location and no calibration procedure**, which is the whole question for a low-cost node, so nothing here constrains data quality. Tier C on that basis.

doi: 10.3390/s26154941

Astaneh et al. 2026 Atmos Environ*Source-tagged PM tracers in random-forest bias correction of ozone design-value predictions*

METADATA ONLY. Using source-tagged particulate tracers as features in an ML bias-correction layer over a photochemical model is a transferable construction even though the target here is ozone, not PM.

doi: 10.1016/j.j.atmosenv.2026.122288

Ayele et al. 2026 Environ Sci Atmos*Fine-scale black carbon and TSP composition at a heavy-traffic intersection, Addis Ababa*

METADATA ONLY — the deposited abstract is truncated to its opening sentences and reports no quantities. What is verifiable from it: the study addresses spatiotemporal variability of equivalent black carbon and the molecular composition of organics in total suspended particulate at a roadside intersection, and states that Ethiopian urban aerosol remains poorly characterised. Today's only sub-Saharan record.

doi: 10.1039/d6ea00100a

Babu et al. 2026 Environ Monit Assess*Capturing the invisible: hyperlocal variability of size-resolved particulate matter in a smart city*

Mobile monitoring campaign in Madurai, Tamil Nadu, during the northeast-monsoon retreat: a vehicle-mounted size-resolved optical instrument sampling every 6 s across $\sim 2.4 \text{ km}^2$ aggregated to a **50 m $\times$ 50 m** grid. Local enhancement above route background was largest for PM₁₀ at **188.91 $\pm$ 149.07 $\mu\text{g}/\text{m}^3$** , attributed to road dust and construction; evening PM₁ (**42.10 $\mu\text{g}/\text{m}^3$**) exceeded morning levels; the size distribution was **bimodal**, with coarse (16.77–32.37 μm) and submicron (0.275–0.53 μm) peaks. Getis-Ord Gi* identified persistent hotspots on traffic corridors and cold spots in restricted-access zones; SEM-EDX on 22 elements linked size to composition. The limitation that threatens the inference is that the reported dispersion (SD 79 % of the mean) is not decomposed: on a moving platform, true spatial variance, uncalibrated optical response, inlet losses and 6 s instrument response time are confounded, and no co-location against a reference monitor is described. Directly relevant to distributed sensing: this is the spatial resolution a mobile network buys, reported without the calibration chain that would make it transferable.

doi: 10.1007/s10661-026-15746-8

Bajaj & Settle 2026 Environ Monit Assess*Rapid assessment of air pollution data quality using a composite index*

A Data Quality Index aggregating five 0–100 dimensions — accuracy (30%), calibration (20%), completeness (20%), representational consistency (15%), timeliness (15%) — with weights the authors state should vary by stakeholder. Demonstrated on hourly OpenAQ PM records for Los Angeles, 1–5 July 2025: a reference-grade monitor scored ≈ 99 and a low-cost sensor ≈ 58 , the latter penalised for calibration bias, low completeness, extreme outliers during the 4 July episode, and zero timeliness once reporting ceased on 5 July. The value is in keeping the components legible rather than collapsing to a single opaque score. The demonstration is $n = 2$ over five days and includes a fireworks episode — an outlier regime chosen, apparently, for illustration — so the numbers are a worked example, not a benchmark.

doi: 10.1007/s10661-026-15780-6

Balasubramanian et al. 2026 Environ Pollut*Calibration model for PurpleAir-monitored PM_{2.5} across the United States*

Random-forest correction for PurpleAir PM_{2.5} trained on 2019–2021 and evaluated on a held-out 2022, using only sensor-reported inputs — PM₁, PM_{2.5}, PM₁₀, temperature, pressure, relative humidity — with no external meteorology. Pearson r against co-located EPA monitors is **0.90–0.96** across buffers from 500 m to 5 km; multiple calibrated products are released at different buffer radii. Deliberately interpretable rather than a deep network, and explicitly aimed at regions without regulatory infrastructure. *The limitation is the metric: r measures covariation, not agreement, and is blind to the multiplicative bias hygroscopic growth actually produces.* Without RMSE, slope or bias the accuracy claim is unverified. Still the most directly transferable record of the day for a distributed PM network — a national, sensor-only, temporally validated correction is the exact artefact a deployment needs.

doi: 10.1016/j.envpol.2026.128935

Band & Mohammadi 2026 (preprint) arXiv*Shift-aware dual-encoder transfer learning for short-horizon PM_{2.5} forecasting*

A source encoder pretrained on hourly PM_{2.5} from 10 US monitoring locations is fused with a target-specific encoder and adapted to **77 Taiwanese stations** over two years under a chronological train/validation/test split — the correct protocol, and one many comparable papers do not use. Against four baselines the frozen-source dual encoder gives **MSE 21.8960, MAE 3.1597, $R^2 = 0.8725$** (a 7.1% and 4.1% MSE reduction over two transfer-learning variants); allowing the source encoder to adapt improves this to **MSE 21.6575, MAE 3.1383, $R^2 = 0.8739$** . Ablation shows the Taiwan-specific branch matters most, and SHAP attributes predictions mainly to recent PM_{2.5} and meteorology. The framing — explicit handling of source/target distribution shift to avoid negative transfer — is the right one for anyone standing up a network in a region with no long record. Two reservations: the absolute gain over the better baselines is a few percent of MSE and no significance or repeated-seed variance is reported, and both domains are reference-grade regulatory data, so transfer onto low-cost nodes (a far larger shift) is untested. Preprint, not peer reviewed.

doi: 10.48550/arXiv.2608.14456

Barkjohn et al. 2026 Atmosphere*An R-Shiny application for sensor-versus-reference QA/QC*

Tool paper describing an application that lets air-quality agencies (i) assess sensor data quality against nearby regulatory reference monitors, (ii) apply baseline QA/QC and (iii) characterise local conditions, with reproducibility as an explicit design goal. Demonstrated on two cases: PM_{2.5} sensor performance under Midwestern wildfire smoke, and a year of O₃ sensor performance. The authors are explicit that advanced correction remains out of scope. **Limitation: no new method and no quantitative evaluation of the tool itself** — the case studies illustrate rather than validate. Carried because availability, not novelty, is what determines whether a small group's sensor data ever gets screened at all; this lowers that barrier and standardises the first pass.

doi: 10.3390/atmos16111270

Benjamin et al. 2026 Sensors (Basel)*Limitations of environmental extrapolation in low-cost carbon monoxide and PM_{2.5} sensors*

Multi-season co-location of a Plantower PMS5003 and an Alphasense CO-B4 against Federal Equivalence Method monitors, with calibration models deliberately trained on restricted ranges of relative humidity, temperature and pollutant concentration and then scored outside those ranges. **Within-envelope bias was minimal; crossing any envelope wall introduced bias; crossing the upper concentration wall specifically increased random error.** The design is the contribution — most co-location papers report a single in-sample R^2 and say nothing about the envelope. Limitations: one site, one sensor model per pollutant, and no aerosol-composition contrast, so the concentration effect cannot be separated from a possible composition shift at high concentration.

doi: 10.3390/s26154988

Blaga et al. 2026 [backfill] PLOS Clim*City-scale calibration of a 44-node PM_{2.5} network for regulatory-compliant assessment*

44 Clarity Node-S sensors across Bucharest against National Environmental Protection Agency reference stations; nine co-located spatial clusters, one used to fit and eight held out. Manufacturer pre-calibration **systematically underestimated** PM_{2.5}, worst in summer when secondary aerosol formation dominates. A two-step seasonal model on temperature and relative humidity alone moved Pearson r from **0.06 to 0.65** (dry season) and from **−0.28 to 0.89** (humid season). Applied network-wide, calibrated sensors average **>60 days/yr** above the EU 25 $\mu\text{g m}^{-3}$ daily limit against 35 permitted; pre-calibration, only 8 of 44 (18%) exceeded. Weakness: calibration degrades in hot dry low-concentration conditions and inter-sensor variability in calibration performance remains large, so a single network-wide model is not defensible. The 1:8 train/validate split is the design detail worth copying.

doi: 10.1371/journal.pclm.0000780

Bobbia et al. 2026 Data in Brief*Multi-source dataset of NO₂ and PM from mobile and fixed sensors in Rouen (France)*

BACKFILL (PubMed entry 2026-07-09; surfaced by the Amass connector, not by the data window). Data descriptor for the TIGA “Smart Mobility for All” deployment in Rouen: **10 mobile AtmoTrack** units on buses at 10 s, **18 fixed AirSenseEUR** units on traffic lights at 1 min, **6 reference stations** at 15 min, plus 2 AtmoTrack units co-located beside a reference station, over **15 January–30 November 2025**. Mobile records carry bus speed, GPS position and onboard RH/T/pressure; fixed records add O_x, NO and CO. The stated purpose is explicitly **multi-hop calibration** — transferring a reference-derived correction through fixed sensors to mobile units that never see a reference site — and heterogeneous data fusion. This is the single most directly usable record of the issue for distributed low-cost PM sensing, because it is the rare case where the co-location topology needed to test calibration transfer is published rather than described. Limitation: it is a data descriptor, so it contains no calibration result, no error metric and no validation; the value is entirely in the availability of the raw structure.

doi: 10.1016/j.dib.2026.113025

Büttner et al. 2026 Atmos Meas Tech*PIA v3.0.0: automated quality assurance for PINE ice-nucleating-particle data*

Software description for the analysis chain behind the Portable Ice Nucleation Experiment. Two components documented in detail: automated data quality control, and the algorithm separating aerosols and droplets from ice crystals by optical size. The second half gives quality-assurance recommendations for PINE operation. Not a PM_{2.5} paper, but the most directly transferable artefact in the issue — a published, versioned QA/QC specification for a continuously operating particle instrument is precisely what distributed sensing networks lack.

doi: 10.5194/amt-19-5007-2026

Burke et al. 2026 Atmos Environ*Polar-grid nonparametric trajectory analysis recovers near-field source structure*

Nonparametric trajectory analysis combines continuous concentrations with local-scale back trajectories from wind speed and direction to map source-area impact. The regular Cartesian grid plus kernel smoothing used previously suppresses spatial variability exactly where a near-source monitor needs it; re-gridding onto polar coordinates gives finer resolution close to the site. Tested on one year of continuous data from a near-road network site in St. Louis. Areas nearest the roadway contributed most (1.4 $\mu\text{g/m}^3$ BC, 39 ppb NO_x), and **polar NTA raised maximum estimated contributions by 22% (BC) and 39% (NO_x)**. Background-subtracted PM_{2.5} tracked the traffic-related species but also revealed nearby point-source contributions. **Limitation: one site, one year, no independent validation** of the recovered source field — the improvement is demonstrated as internal consistency, not against known emissions. Immediately relevant to siting and to interpreting single-node data in a sparse network.

doi: 10.1016/j.atmosenv.2026.121951

Cao et al. 2026 Atmos Environ

Characterising on-road NO₂ and particulate matter across urban road environments using city-scale mobile monitoring

Abstract not deposited at Crossref on the entry date. The most directly relevant record of the day for mobile and distributed sensing and the one whose loss to the abstract famine costs most. Queued for the Saturday sweep.

doi: 10.1016/j.atmosenv.2026.122302

Chen et al. 2026e Atmos Environ

Aerosol and VOC pollution characteristics along the lower Yangtze, from shipborne observation

Metadata-only deposit. Shipborne mobile measurement of aerosol and VOCs along an inland waterway corridor — a platform class this watch sees rarely and one with a specific methodological problem: the measurement frame moves through the plume it is sampling, so concentrations are Lagrangian-contaminated in a way that fixed-site data are not, and self-sampling of the vessel's own exhaust is a standing artefact that requires wind-relative rejection to control. Whether the authors address either is not determinable from the title. Carried at tier C; the record is retained rather than rejected because inland-shipping emissions are a genuinely under-monitored PM source and the corridor is one of the most heavily trafficked in the world.

doi: 10.1016/j.atmosenv.2026.122331

Chen et al. 2026k Environ Sci Technol

Lake-breeze effects on urban air quality resolved by a city-wide low-cost sensor network

One year (2022) of the Microsoft Eclipse LCS network plus meteorological stations over Chicago, composited into **42 lake-breeze events** and 51 weak events confined within a few kilometres of the shore. Weekday events reduce O₃ by up to ~6 ppb along the northeast shoreline — attributed to NO titration in a shallow boundary layer, inflow of cleaner lake air, and suppressed photochemistry at lower temperature — while raising PM_{2.5} by up to ~3 µg m⁻³ via accumulation of local emissions and semivolatile partitioning. The limitation that matters: event-composited means over an LCS network inherit that network's calibration bias, and no reference-monitor validation of the PM_{2.5} increment is reported, so the 3 µg m⁻³ figure is a network-relative anomaly rather than a traceable concentration. Directly relevant precedent for using a dense LCS deployment as the primary instrument in a mesoscale study.

doi: 10.1021/acs.est.6c09290

Cho et al. 2026 Sci Total Environ

Greenspace, land use and PM_{2.5} across pandemic phases: a community scientist-led mobile sensing campaign

Community scientists ran low-cost mobile PM_{2.5} monitoring across Norwood, Ohio, for two summer periods — the initial pandemic phase (2020) and the reopening phase (2021) — with a meteorology-adjusted difference-in-differences framework for the year contrast and street-level linear land-use regression for spatial structure. **Higher 2021 concentrations were not fully explained by meteorology.** In the 2021 afternoon model, PM_{2.5} was elevated within 50 m of office and manufacturing land use and inversely associated with NDVI within 100 m. **Cross-validated R² 0.17–0.25 (2020) rising to 0.55–0.61 (2021).** The paper attributes the gap to source recovery, but a low-contrast summer is an equally good explanation and is not tested. Health-survey scores by ward are reported alongside but are not linked to measured exposure at the individual level. The transferable part is the QC and sampling protocol for neighbourhoods with no regulatory coverage, which is stated explicitly as the deliverable.

doi: 10.1016/j.scitotenv.2026.182124

Chung et al. 2026 Sensors

PurpleAir deployment trends, longevity and uncertainty across the United States

Every publicly shared outdoor PurpleAir unit, 2016–2025, with deployment duration taken as the span between first and last public observation, summarised nationally, by Census region and by state. **Most sensors persist beyond three years;** density and longevity are highest in the western US. Descriptive PM_{2.5} summaries for California, Minnesota, Pennsylvania and Texas by deployment duration and urban–rural class. The limitation is that it characterises *availability*, not accuracy — persistence of a data stream is not persistence of calibration, and the paper explicitly does not address drift. That is exactly the join left open against Kumpika et al. below.

doi: 10.3390/s26061789

Chyzykhov et al. 2026 J Hazard Mater

Particle-bound water and the composition dependence of gravimetric PM mass

Karl Fischer coulometric titration with stepwise heating on 35 single-component chemicals and CRMs at two filter loadings (0.6 and 2.0 mg, summer- and winter-like), plus a compositionally realistic artificial PM mixture built both by sequential addition and in bulk. Deliquescent inorganic salts retain the most water; metals, metal oxides, PAHs and graphitic carbons sit near background. Water-to-solute molar ratio falls as absolute loading rises — compacted powder is less vapour-accessible — and temperature-resolved release clusters into three material groups differing mainly in strongly bound (250 °C) water. Bulk artificial-PM release profiles matched ambient PM₁ filters. **Limitation: powders are not ambient particles;** mixing state, morphology and coating history are not reproduced, so the transfer to real filters is argued from profile similarity, not demonstrated. For distributed sensing this is the uncomfortable upstream result — the mass reference itself carries a composition-dependent water term, so an empirical RH correction fitted against it is partly correcting the reference.

doi: 10.1016/j.jhazmat.2026.143142

Clark & Bashar 2026 Risk Anal

How IoT enables the Protective Action Decision Model during a wildfire event

Survey of 1,200 Oregon residents (Jun–Aug 2021) about their responses to the 2020 fire season, framed against the Protective Action Decision Model and the availability of hyper-local real-time PM_{2.5} from IoT devices. The finding that matters for network designers is negative: residents were **overwhelmed by the data and reported inadequate guidance** on how to act on it. Retrospective self-report a year after the event, and the survey does not establish who actually had sensor access — so this constrains interface and messaging design rather than measuring sensor efficacy. Still the rare paper in this watch that treats the last mile of a dense network as the problem.

doi: 10.1111/risa.70319

Colarusso et al. 2026 Internet of Things

Mobile urban sensing: spatio-temporal observability analysis and optimisation

Fleet-aware framework for hybrid mobile–fixed sensing on an Edge–Fog–Cloud architecture, evaluated on real data from three sensing vehicles covering 11.37 km² in Benevento, Italy. Introduces spatial and temporal observability indices and a value-of-fleet metric, then uses spatio-temporal interpolation plus what-if analysis to quantify how fleet size, route redundancy, sampling density and data loss propagate into reconstructed-map accuracy; a greedy co-design procedure places fixed nodes against the residual. Limitation: a single city, a single small fleet, and interpolation-based ground truth — the observability indices are internally consistent but not externally validated. The framing is the transferable part: opportunistic mobile coverage becomes a measurable quantity rather than a qualitative claim.

doi: 10.1016/j.iot.2026.101907

Dal Porto et al. 2026 Aerosol Sci Technol

Humidified CAPS-PM_{SSA} with a revised truncation-correction methodology

Metadata only. Taylor & Francis withheld the abstract on this run and Semantic Scholar was rate-limited throughout, so nothing below the title, authors and journal is verified and no result is asserted. On the registered title alone the paper describes a humidified variant of the cavity-attenuated phase-shift single-scattering-albedo monitor together with an improved method for the scattering truncation correction — the two quantities that dominate uncertainty when an SSA measurement is used as a reference for hygroscopic growth. Relevant enough to the humidity-correction problem to carry rather than defer; flagged for abstract retrieval on the next run.

doi: 10.1080/02786826.2026.2711663

DeMarsh et al. 2026 (backfill) Atmosphere

Community PurpleAir network versus regulatory monitors for PM_{2.5} in Fresno County

The SJV-CAIR network at UC Merced, built with community-based organisations, compared against regulatory monitoring across Fresno County for June–July 2023. Community monitors returned **concentration estimates comparable to regulatory instruments** where the two co-locate, but identified **four weeks in which some locations episodically exceeded the NAAQS that the regulatory network did not flag**, and interpolation maps built with the denser network resolved spatial structure the regulatory-only map did not. Two summer months in one county is a short and seasonally atypical window for the San Joaquin Valley, and no formal co-location correction is reported. Carried as backfill — Crossref created 11 February 2026, outside this entry window — and must be excluded from weekly trend lines.

doi: 10.3390/atmos17020187

Diana et al. 2026 Talanta

Validated multi-technique workflow for inorganic characterisation of brake-wear PM₁₀

Thirty-two PM₁₀ samples from WLTP-based brake-dynamometer tests on four tribological couples — low-steel (LS) and non-asbestos organic (NAO) pads against grey cast-iron (GCI) or coated discs. A combined acid digestion (HNO₃/H₂O₂ plus HNO₃/HF/H₃BO₃) was validated against NIST SRM 1648a and quantified by ICP-OES and ICP-MS, giving **27 elements with recoveries above 75% for most analytes**; FESEM-EDS confirmed the trends at particle scale and PCA/HCA separated the tribological families cleanly. **LS-GCI produced the highest PM₁₀ emission factors and NAO-GCI the lowest**; Fe dominates LS-derived material while Ba, Ti, K and Zr mark NAO. An Mn-based enrichment factor improved discrimination of rotor- versus pad-derived contributions. The value here is a defensible marker set for non-exhaust source apportionment, which matters more each year as tailpipe PM falls. Caveats are the usual dynamometer ones: WLTP braking is not real-world braking duty, only four couples were tested, and no on-road or roadside validation is attempted.

doi: 10.1016/j.talanta.2026.130448

Eger et al. 2026 Aerosol Air Qual Res

Compact low- and mid-cost sensors for inland-ship exhaust emissions

One year of onshore co-location on the Rhine against reference instruments, screening sensors on noise floor, correlation and response time — criteria chosen because ship plumes are transients, not the slowly-varying urban background most validation studies target. Qualifying devices: **MA350 black carbon** ($R^2 = 0.95$, slope 0.77), **Partector 2 Pro ultrafine particle number** ($R^2 = 0.85$), **K30 FR CO₂** ($R^2 = 0.98$), **Alphasense NO** ($R^2 = 0.99$) and **NO₂** ($R^2 = 0.81$). The limitation the authors state plainly is the useful part: sensor noise and the temperature dependence of the electrochemical cells — which they describe as difficult to correct — cut the number of analysable ship peaks. No PM mass sensor cleared the bar, which is itself a result.

doi: 10.1007/s44408-026-00163-w

Fahim & Karim 2026 Res Sq (preprint)

Cost-aware benchmark of time-series foundation models for PM_{2.5} forecasting

Twenty-nine cities, two domains (hourly PM_{2.5}, 2 m temperature), pre-specified protocols with measured energy and a cost-adjusted decision rule, comparing zero-shot foundation models, transfer learning and tuned specialists. **On PM_{2.5} the zero-shot model and the specialist are statistically indistinguishable (MASE 0.662 vs 0.692)**; transfer learning never significantly beats zero-shot at any budget; frequent retraining costs the specialist roughly ten times the measured energy. The methodological contribution is the demonstration that the specialist's temperature advantage is a *perfect-foresight artefact* of feeding it covariates unknowable at forecast time. Unreviewed, and 29 cities is a small sample for a claim about generalisation — but the artefact critique applies to published benchmarks too.

doi: 10.21203/rs.3.rs-10515545/v1

Fawaz et al. 2026 Sensors (Basel)

Surface cleaning of SAW-based microparticle sensors in a cascade impactor via SAW-induced droplet displacement

Rayleigh waves generated on a lithium-niobate chip actuate a water droplet that sweeps deposited particles out of the sensing zone of surface-acoustic-wave microparticle sensors integrated into a cascade impactor. Optimal droplet size and required RF power were determined; the droplet displaced both NaCl-solution residue and SiC particles without degrading sensor performance. Surface saturation is the binding constraint on long unattended deployment of mass-sensitive acoustic sensors, so a self-regenerating cycle changes the maintenance model rather than the sensitivity. Laboratory demonstration only — no ambient aerosol, no duty-cycle endurance data, and no accounting for what the droplet does to semi-volatile mass.

doi: 10.3390/s26154666

Fei et al. 2026 Front Immunol

SHAP-interpretable machine learning integrating exposures and multi-omics in COPD-lung cancer comorbidity

A sequential cross-scale framework: GBD epidemiology 1990–2021 across 204 countries, exposome-wide random-forest classification with SHAP interpretation, immune characterisation via CIBERSORT and single-cell RNA sequencing, candidate-gene prioritisation by SMR and LASSO, then wet-lab validation including a macrophage-epithelial Transwell co-culture. **SHAP identified smoking, particulate matter pollution and residential radon as the major shared risk factors**; TREM1 and ODF2L emerged as shared hub genes, downregulated in both diseases, and macrophage-specific silencing of either attenuated macrophage-driven A549 migration and proliferation. The exposure side is GBD-level and coarse — this is not an exposure-assessment paper — but it is a clean worked example of putting an exposome variable through an interpretable-ML pipeline and ending at a testable target.

doi: 10.3389/fimmu.2026.1875146

Filioglou et al. 2026 Atmos Chem Phys*Particle linear depolarisation ratio at 1565 nm from Halo Doppler lidar*

Assesses whether a wind-profiling Halo Photonics StreamLine Doppler lidar can type aerosol at 1565 nm, a wavelength where depolarisation is rarely characterised, across three tropospheric case studies. **Extremely fresh smoke gave 0.017 ± 0.004** , aged long-range-transported smoke marginally higher, and **volcanic ash layers 0.45 ± 0.01** — a separation wide enough to type layers unambiguously. Smoke mass concentration retrieved from the 1565 nm signal agreed with concurrent in-situ observations. The value here is that an instrument already deployed for wind now also delivers aerosol typing and a mass estimate on the same beam; three case studies is thin for a conversion factor, and the aged-smoke result is qualitative.

doi: 10.5194/acp-26-11525-2026

Gao et al. 2026 Atmos Meas Tech*Feasibility of an air sensor array for real-time detection and characterisation of VOCs*

Commercial VOC air sensors are broadband — each returns one scalar reflecting an ill-defined sensitivity envelope, which is why they have seen little research use. This paper tests whether an *array* spanning three fundamentally different sensor types can recover speciation from the joint response. The measurand is VOCs, not PM, and it is carried here for the method: the same argument applies directly to optical PM sensors whose single scalar folds together size distribution, refractive index and hygroscopic growth. Feasibility study — laboratory characterisation, and the practical question of whether the array's discrimination survives ambient interferent mixtures is not settled.

doi: 10.5194/amt-19-5337-2026

Giannetti et al. 2026 Environ Pollut*Ants as bioindicators of urban traffic pollution*

Thirty traffic-graded sites plus four controls in Parma; magnetic susceptibility and SEM-EDS on *Crematogaster scutellaris* exoskeleton and on co-located soil at 5 cm. **Ant magnetic susceptibility tracked traffic intensity at $\rho = 0.825$ ($p < 0.001$)**; **soil did not** ($\rho = 0.167$, $p = 0.346$). On a log scale the relation is linear ($R = 0.66$, $p < 0.001$) with **1.41-fold per additional 1,000 vehicles/day**, ~700-fold across the gradient. SEM-EDS found Fe-oxide particles doped with Zn, Cu and Ti concentrated on the thorax, morphologically consistent with traffic-derived ultrafine PM. The weaknesses are that no slope CI is given, magnetic susceptibility is a proxy for the ferrous fraction only (it says nothing about organic or secondary mass), and nest-density confounding with traffic corridors is not addressed. For distributed sensing the interest is orthogonal: a passive, unpowered integrator with month-scale averaging at effectively zero unit cost, usable as a siting prior for where to put real instruments rather than as a replacement for them.

doi: 10.1016/j.envpol.2026.129010

Gilardoni et al. 2026 Atmos Chem Phys*First continuous aerosol measurements at Testa Grigia at 3480 m a.s.l.: aerosol populations and dust transport dynamics in the southern European Alps*

First continuous **number size distribution and absorption coefficient** record at 1 h resolution from the Testa Grigia Observatory (3480 m a.s.l., Italian Alps), spanning **September 2021–May 2023**. Three aerosol population types are separated by transport pathway: a coarse population indicating long-range Saharan transport, a fine population tied to Po Valley and alpine-valley boundary-layer dynamics and broader continental flow, and low-number periods representing free-tropospheric background. The record is a baseline dataset rather than a hypothesis test, and the limitation is the one every single-site campaign carries — population attribution rests on back-trajectory and covariance reasoning, not on independent source measurement. Relevant to distributed sensing for a narrow but useful reason: a high-altitude site with characterised free-tropospheric periods gives an absolute-zero reference for optical instruments that low-altitude urban co-location cannot provide, and the absorption-coefficient record is one of the few instrument-side quantities published at deployment cadence.

doi: 10.5194/acp-26-11135-2026

Ginsburg et al. 2026 Aerosol Sci Technol*Global and regional calibration of low-cost PM_{2.5} sensors (PurpleAir)*

1,013 PurpleAir units co-located within 500 m of reference monitors on six continents, Jan 2023–Mar 2025, assessed at 1-h and 24-h averaging by R^2 , RMSE, nRMSE, precision (CV) and AQI category correctness. Raw readings overestimate PM_{2.5} in most regions and miss EPA performance targets; US-derived corrections show limited transferability outside North America; a data-driven global quadratic helps in Australia/NZ and hurts elsewhere; **regional equations meet EPA 24-h thresholds in multiple regions and lift 24-h AQI correctness to $\geq 77\%$** . Limitation: co-location proximity (≤ 500 m) is a proxy for comparability, so some regional “bias” is spatial-gradient error, and continents are represented very unevenly. Still the strongest available empirical case that LCS network calibration must be local.

doi: 10.1080/02786826.2026.2676293

Gómez et al. 2026 Sensors*Machine- and deep-learning calibration of low-cost PM₁₀/PM_{2.5} nodes*

LCS nodes co-located with a T640X reference at Universidad Distrital, Bogotá, with RH, temperature and absorption flow from an Automet station. Pipeline: filtering, segmentation, **FastDTW** alignment, then six calibration models (two statistical, three ML, one DL). Random Forest and XGBoost performed best; the authors argue that the FastDTW preprocessing step and the inclusion of absorption flow — not the model choice — carry the accuracy gain. Single-site co-location with no transfer test to an uncalibrated node, so generalisation is unestablished.

doi: 10.3390/s26030796

Guo et al. 2026 ACS ES&T Air*Mechanistic RH correction by separating aerosol liquid water from dry mass*

Two-stage framework for a nephelometer plus optical particle counter: (i) Mie-theory optical calibration accounting for size distribution, refractive index and hygroscopic growth; (ii) quantification of aerosol liquid water mass from EPA Chemical Speciation Network composition to recover dry-equivalent PM_{2.5} at high RH. Corrected values agree closely with EPA reference measurements, and the recovered ALW is itself useful for aqueous-phase aerosol chemistry. Practical guidance is given for sites without co-located CSN speciation. **Limitation: the method is only as good as the speciation surrogate** — outside the CSN footprint the ALW term becomes an assumption, and the paper does not report how much accuracy is lost when the surrogate is wrong. This is the right-shaped answer to the humidity problem for anyone unwilling to refit an empirical curve at every new site.

doi: 10.1021/acsestair.5c00225

Gäbel & Hertig 2026 Atmos Meas Tech*Recalibration of low-cost O₃ and PM_{2.5} sensors: linking practices to recent air sensor test protocols*

Yearlong urban-background co-location at Augsburg of Alphasense OX-B431 and Sensirion SPS30 against reference instruments, spanning −10 to 36 °C, 19–96 % RH and PM_{2.5} to 153 µg m^{−3}, with MLR, ridge, random forest and XGBoost calibration functions evaluated against the current US EPA and CEN air-sensor test protocols. A five-month baseline training gives PM_{2.5} R^2 0.93–0.97 in training and **0.84–0.93 in the following test period**, RMSE 2.04–3.02 µg m^{−3} — the decay is the finding. Testing pairwise recalibration cycles, **monthly in-season recalibration** was required to keep uncertainty inside the data-quality objective for indicative measurements. Limitation: one site, one year, two sensor models, so the cadence is an existence result rather than a general rule. **Out-of-window backfill** (created 19 Feb 2026); *AMT* is not indexed by PubMed. Exclude from weekly trend lines.

doi: 10.5194/amt-19-1293-2026

Atmos Environ 2026b METADATA ONLY Atmos Environ*Stable carbon isotopic composition of PM₁ emissions from biomass and coal combustion under monitored combustion conditions*

METADATA ONLY — no abstract retrievable at build time. Carried because $\delta^{13}\text{C}$ signatures for PM₁ measured under *monitored* combustion conditions are the reference values that source-apportionment studies need and usually borrow from decades-old literature. Biomass and coal are the two sources whose separation matters most in residential-heating regions, and combustion-condition control addresses the standing objection that isotopic signatures drift with burn phase. If the reported ranges are tight this is a reference dataset; if they overlap it undermines $\delta^{13}\text{C}$ apportionment. Unresolvable without the text.

doi: 10.1016/j.atmosenv.2026.122296

Aerosol Air Qual Res 2026 METADATA ONLY Aerosol Air Qual Res*Identifying winter PM_{2.5} sources in an urban environment using receptor modeling and nonparametric trajectory analysis*

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar. Carried because the pairing of receptor modelling with *nonparametric* trajectory analysis is methodologically specific: it drops the functional-form assumption that conventional trajectory statistics impose on the source–receptor relationship, which is the assumption most likely to fail in complex winter meteorology with shallow boundary layers. Site, sample size and factor count are all unstated. Listed for the register, not weighted.

doi: 10.1007/s44408-026-00157-8

Huang et al. 2026 Atmos Pollut Res*Metal ions and pH as controls on the optical properties of atmospheric brown carbon*

Crossref deposit of 7 August, **metadata only — no abstract retrievable**, so no summary is offered. Logged because brown-carbon absorption is the aerosol optical property that most directly confounds the fixed refractive-index assumptions built into low-cost optical particle counters.

doi: 10.1016/j.apr.2026.103167

Jafari et al. 2026 IEEE Open J Intell Transp Syst*Joint optimization of bus-based mobile sensing and reference sensor deployment for urban air quality monitoring*

Mathematical-programming formulation selecting simultaneously **which buses to instrument with low-cost sensors and where to place reference stations**, maximising spatiotemporal network coverage under a budget constraint while requiring that every mobile sensor pass a reference site frequently enough to be recalibrated. Demonstrated on a Rotterdam case study covering residential and industrial areas. The contribution is structural rather than empirical: recalibration cadence enters the siting objective as a constraint, so coverage and data quality are traded off in one problem instead of sequentially. Limitation: coverage is geometric — optimised against route and population geometry, with no validation against a measured concentration field, and no treatment of the sampling bias that fixed bus routes impose. This is the record held on 4 August and recovered on 5 August via Semantic Scholar.

doi: 10.1109/ojits.2026.3706855

Jin et al. 2026 Atmos Chem Phys*Emissions, chemistry and health impact of aged wildfire smoke over Missoula, Montana*

Hourly surface observations of PM_{2.5}, CO, NO_x, O₃ and **75 speciated VOCs** through a strong 2020 smoke episode, used as a direct test of how wildfire emissions, ageing chemistry and health effects are represented in GEOS-Chem. Smoke three or more days old, transported from California and the Pacific Northwest, raised CO, PM_{2.5} and total measured VOCs by **factors of 2–8**, with hourly maxima of 800 ppb CO and **120 µg/m³ PM_{2.5}**. The value here is the speciated VOC suite alongside mass: it is the composition record against which a mass-only low-cost network's health interpretation has to be judged, and it is one city and one episode, so the transport chemistry is not generalisable without replication at other receptor sites.

doi: 10.5194/acp-26-11047-2026

Kafi et al. 2026 Sci Total Environ*Evaluating the suitability of Sentinel-5P data for air quality monitoring in Tehran*

Six years (2019–2024) of TROPOMI Level-2 CO, NO₂, SO₂, O₃ and UV Aerosol Index against 22 ground stations in a megacity with complex topography, assessed by Pearson and Spearman correlation and Kling–Gupta Efficiency, plus a spatiotemporal trajectory comparison. **Only NO₂ showed strong agreement; CO, SO₂ and O₃ were weak**, and the trajectory analysis diverged as well. The authors are explicit that UVAI represents column aerosol loading rather than surface particulate concentration and warn against reading it as PM — a caveat routinely ignored in the satellite-PM literature and worth citing for that reason alone. The diagnosis (coarse pixel size against a topographically constrained urban basin) is plausible but not separated from the competing explanation of station siting, and no vertical-profile or boundary-layer-height information is brought in to test it.

doi: 10.1016/j.scitotenv.2026.182069

Kalkavouras et al. 2026 Atmos Pollut Res*Ten-year trends in ultrafine particle number concentration and size distribution, Athens*

METADATA ONLY — no abstract was deposited, so no trend, magnitude or instrument detail is asserted here. Flagged because a decade-long urban particle-number and size-distribution record is exactly the reference series that a low-cost optical network cannot reproduce and must be evaluated against; worth retrieving in full.

doi: 10.1016/j.apr.2026.103181

Kang et al. 2026b J Hazard Mater*Emission characteristics and health impacts of vehicle-originated condensable PM*

Condensable PM from **eight vehicle types** characterised with a purpose-built real-time system (PNU-CPM) plus offline physicochemical analysis, with emission factors derived from the real-time record. Untreated **diesel levels 4 and 5 emitted the most CPM and total PM**; CPM mass fractions were **63.2 % (diesel L4), 8.2 % (GDI) and 58.8 % (LPG 2)**. Organic CPM was ~53.5 % alkanes and ~5.1 % esters; dominant water-soluble ions were SO_4^{2-} and Na^+ . MPPD dosimetry gives higher deposited mass per body weight in younger age groups. Directly relevant to optical sensing: CPM is ultrafine, semi-volatile and forms after the tailpipe, so it is missed by hot-filter source testing and will not follow the scattering or hygroscopic-growth assumptions that low-cost sensor corrections inherit from combustion soot. No on-road fleet measurement — these are dynamometer emission factors.

doi: 10.1016/j.jhazmat.2026.143367

Kirchhoff et al. 2026 Atmos Meas Tech*Evaluation of DMSO as working fluid in condensation particle counters*

Laboratory and field comparison of a DMSO-operated CPC against a butanol counterpart across pressure, temperature and aerosol type, with a modified automatic refill system run for six months. Growth depends strongly on saturator temperature and saturator–condenser ΔT , confirmed by downstream optical particle counting and saturation/droplet-size simulation. **Mean cutoff 5.8 ± 0.9 nm (DMSO) vs. 5.6 ± 0.5 nm (butanol)**, with substantially lower fluid consumption; DMSO–H₂O mixtures extend the operating range. The wider cutoff spread on the DMSO unit is the honest cost, and matters if you are differencing two size channels. Non-flammability is what makes airborne and unattended remote deployment tractable.

doi: 10.5194/amt-19-5243-2026

Kumpika et al. 2026 Environ Pollut Bioavailab*Correction of humidity and sensor ageing effects in low-cost PM_{2.5} light-scattering sensors*

Field campaign in Chiang Mai with PMS7003 units across a wide relative-humidity range, validated against a government BAM and cross-checked with dry-air measurement through a diffusion dryer. Polynomial corrections up to third order were fitted; **agreement with BAM reached $R^2 = 0.9737$** after correction, with the largest gains at elevated RH. The second result is the more interesting one: **restricting operation to 20 min/day measurably reduced long-term drift** relative to continuous running, implicating cumulative optical-chamber exposure rather than elapsed calendar time as the ageing mechanism. Single-site, single sensor model, single season, so the polynomial coefficients are not transferable — but duty-cycling is a design lever, not a calibration constant, and it applies to any continuously powered node.

doi: 10.1080/26395940.2026.2616098

Kumpika et al. 2026 Environ Pollut Bioavailab*Correction of humidity and sensor ageing effects in low-cost PM_{2.5} light-scattering sensors*

Field campaign in Chiang Mai with PMS7003 units across a wide relative-humidity range, validated against a government BAM and cross-checked with dry-air measurement through a diffusion dryer. Polynomial corrections up to third order were fitted; **agreement with BAM reached $R^2 = 0.9737$** after correction, with the largest gains at elevated RH. The second result is the more interesting one: **restricting operation to 20 min/day measurably reduced long-term drift** relative to continuous running, implicating cumulative optical-chamber exposure rather than elapsed calendar time as the ageing mechanism. Single-site, single sensor model, single season, so the polynomial coefficients are not transferable — but duty-cycling is a design lever, not a calibration constant, and it applies to any continuously powered node.

doi: 10.1080/26395940.2026.2616098

Kumpika et al. 2026 Environ Pollut Bioavailab*Correction of humidity and sensor ageing effects in low-cost PM_{2.5} light-scattering sensors*

Field campaign in Chiang Mai with PMS7003 units across a wide relative-humidity range, validated against a government BAM and cross-checked with dry-air measurement through a diffusion dryer. Polynomial corrections up to third order were fitted; **agreement with BAM reached $R^2 = 0.9737$** after correction, with the largest gains at elevated RH. The second result is the more interesting one: **restricting operation to 20 min/day measurably reduced long-term drift** relative to continuous running, implicating cumulative optical-chamber exposure rather than elapsed calendar time as the ageing mechanism. Single-site, single sensor model, single season, so the polynomial coefficients are not transferable — but duty-cycling is a design lever, not a calibration constant, and it applies to any continuously powered node.

doi: 10.1080/26395940.2026.2616098

Laffolley et al. 2026 ACS Omega*A 3D-printed two-stage virtual impactor (μ SPLIT) for radioactive aerosol size classification and direct analysis*

Stereolithography-printed virtual impactor separating aerosol into **>10 , $1\text{--}10$ and <1 μm** aerodynamic classes onto integrated filters for direct post-collection α counting, designed for Fukushima Daiichi decommissioning where the instrument must be cheap enough to incinerate rather than decontaminate. Flow path optimised by CFD with Lagrangian particle tracking; predicted cutoffs **9.0 and 1.3 μm** . The paper is unusually candid about its own failure modes: simulations identify small-particle contamination in the second-stage minor flow and large-particle contamination in the major flow, and dimensional metrology on the printed parts found channel contraction and corner rounding — i.e. the as-built geometry is not the as-designed geometry, which for an impactor moves the cutoff directly. Experimental validation is a proof of concept only: incense smoke, and Rn-progeny-bearing NaCl at 1.70 μm GMAD putting peak α activity on the middle filter. No standardised-aerosol calibration and no quantitative wall-loss budget, both of which the authors name as required next steps. Relevant well beyond radiological work — additive manufacture of size-selective inlets is the cheapest available route to giving a low-cost node a real aerodynamic size cut instead of an optical inference.

doi: 10.1021/acsomega.6c04521

Le et al. 2026 Atmos Meas Tech*Operational performance of the Vaisala CL61 ceilometer for atmospheric profiling*

Three years of operational data from multiple CL61 units — a depolarisation-capable elastic backscatter lidar — assessed for inter-unit consistency, drift and bias. Most early production units showed **progressive laser-power decline with rising background noise**, and manual calibration against atmospheric targets at Lindenberg showed that **once laser power falls below 40% the internal calibration factor stops compensating**. Instrumental background and bias vary with temperature; a correction and uncertainty-estimation method is given. Under low aerosol load the bias correction shifts volume linear depolarisation ratio by up to 0.005, while the volume-to-particle depolarisation difference reaches 0.1 — so molecular correction, not instrumental bias, dominates quantitative aerosol interpretation at 910.55 nm. One unit's signal loss was traced to window fogging from insufficient firmware-driven heating. Instructive as a template: silent, firmware-mediated degradation is exactly the failure mode a distributed network cannot detect without co-location.

doi: 10.5194/amt-19-4923-2026

Lefevre et al. 2026 Data Brief*Urban microclimate dataset from low-cost sensor networks, Reunion Island*

A data descriptor for **25 self-built LoRa weather stations** deployed on residential buildings across two contrasting neighbourhoods (Saint-Denis, Le Port), October 2024 to June 2025, at **10-minute cadence** for air temperature, relative humidity, wind speed, global horizontal solar radiation and grey-globe temperature. Station metadata, QC procedures and maintenance logs are published alongside, and incomplete stations are identifiable. **There is no PM channel** — this is a microclimate network, not an air-quality one — so its value here is as a deployment reference: dense neighbourhood-scale LoRa siting, self-built hardware, published QC, and an explicitly stated reuse path for model calibration (ENVI-met, STEVE). Tropical hot-humid deployments are exactly the regime in which PM sensor RH correction fails worst, and this is a rare open record of what such a network's data actually look like.

doi: 10.1016/j.dib.2026.113166

Legaria-Santiago et al. 2026 Atmosphere*Leveraging remote traffic data for local air pollutant estimation across London monitoring sites*

Four interpretable tree ensembles (Random Forest, Extra Trees, LightGBM, XGBoost) across six predictor scenarios, from remote traffic + meteorology + temporal variables alone up to including one and four neighbouring stations, benchmarked against ridge regression, spatial interpolation and a cross-site validation. Without neighbouring stations, NO₂ RMSE fell from **9.73–11.66 to 8.72–11.52 $\mu\text{g}/\text{m}^3$** when traffic information was added, and SHAP shows traffic variables contributing at a level comparable to neighbouring-station measurements in traffic-dominated settings. The practical claim — remote traffic counts can partially substitute for network density — is demonstrated for NO₂; the PM channels are reported but not where the gain is.

doi: 10.3390/atmos17080806

Lei et al. 2026 Atmos Chem Phys*Eight-year speciated PM₁ record in Dublin: heating emissions versus transboundary transport*

Continuous chemically speciated PM₁, 2016–2023, resolving two distinct pollution regimes: short intense events (hours, PM₁ >100 $\mu\text{g}/\text{m}^3$) driven by local solid-fuel home heating, and moderate multi-day events (<60 $\mu\text{g}/\text{m}^3$) from continental transport. Annual PM₁ fell from 6.5 to <5.0 $\mu\text{g}/\text{m}^3$, led by nitrate and ammonium (−0.11 and −0.09 $\mu\text{g m}^{-3} \text{yr}^{-1}$). **Transported oxygenated organic aerosol rose (+0.34 $\mu\text{g m}^{-3} \text{yr}^{-1}$)** while locally formed OOA barely fell, so the composition shifted even as mass declined — and 2023 showed a heating-driven rebound. A demonstration that a mass-trend headline can be exactly wrong about what a network is now measuring.

doi: 10.5194/acp-26-10927-2026

Li et al. 2026 Atmos Chem Phys*Summertime ice-nucleating particle concentrations over the Eurasian Arctic seas*

Shipborne immersion-mode INP measurements from the Arctic Century Expedition across the Barents, Kara and Laptev Seas and adjacent high-Arctic islands, August–September 2021. N_{INP} was lower than at continental high-latitude sites, most clearly below −15 °C, implying sparse mineral-dust INP; concentrations peaked in ice-free water, marginal ice zones and near land, and were very low inside the ice pack, with the maximum north of Novaya Zemlya under back-trajectories from the western Siberian coast. The authors conclude sources are local to regional with little long-range transport signature. Limitation: one summer and one cruise track, so the seasonal and interannual envelope is unconstrained. Peripheral to PM exposure, but relevant to interpreting particle counts in clean marine air, where the concentration floor is set by local sources rather than by transport.

doi: 10.5194/acp-26-10835-2026

Li et al. 2026 Photoacoustics*Spatiotemporally synchronised dual-wavelength photoacoustic-scattering measurement of aerosol optical properties*

A sphere-tube coupled resonator that removes the transport, response-time and microenvironment mismatch inherent to split absorption/scattering instruments. Two integrating spheres act simultaneously as scattered-light collection cavities and as acoustic buffers forming a low-frequency **differential Helmholtz** resonator, while the central connecting tube supports a high-frequency **second-order longitudinal** mode; 450 and 532 nm lasers are modulated at **2572 and 169 Hz** and demodulated by an FPGA digital lock-in, so both wavelengths and both quantities are recovered from one sampling path. Absorption channels calibrated against NO₂, scattering against N₂/CO₂ Rayleigh scattering; at 1 s integration and 3 σ the 532 nm limits of detection are **1.2 Mm^{−1}** (absorption) and **1.32 Mm^{−1}** (scattering). The validation is the weak part: water mist, controlled combustion and a field check establish channel isolation and feasibility but not accuracy against a reference absorption method, and no Allan-variance drift figure is given, so the 1 s LOD does not tell you the useful averaging time. For distributed sensing the transferable idea is architectural rather than economic — this is a laboratory instrument, but co-locating absorption and scattering in a shared volume is what would let a node report single-scattering albedo instead of a mass proxy.

doi: 10.1016/j.pacs.2026.100864

Li et al. 2026 Environ Sci Technol*Crowdsourced, community-engaged VOC monitoring and source apportionment near a landfill*

Community citizen-science campaign near the Bristol, Virginia solid-waste landfill using a modified ultrasonic personal air sampler under EPA Method TO-17. BTEX remained below EPA reference concentrations throughout but was significantly elevated relative to comparison locations. Gas-phase, not particulate — carried here for the deployment methodology rather than the analyte. The transferable content is the study-design integration of resident-reported odour events into sampling triggers, which is the same problem a community PM network faces when episodic sources drive the exposure and a fixed duty cycle misses them.

doi: 10.1021/acs.est.6c04657

Li et al. 2026 Front Pediatr*Interpretable ML for 90-day paediatric allergic-rhinitis exacerbation from clinical and ambient data*

Development cohort $n = 2,000$ linking NHANES (2005–06, 2011–18) to EPA AQ5 and NOAA data; external cohort $n = 50$ from Wuhan. L2-regularised logistic regression was prespecified and beat random forest, XGBoost, LightGBM and SVM under nested 5-fold CV. Internal AUC 0.81 (0.74–0.88), external 0.71 (0.50–0.92). Baseline symptom score, same-day PM_{2.5} and total IgE dominate SHAP; **PM_{2.5} >75 vs <35 $\mu\text{g}/\text{m}^3$ gives OR 5.08 (3.45–7.47)**. The external cohort is 50 children on a different continent — the authors say plainly this is preliminary transportability, not external validity, and they are right.

doi: 10.3389/fped.2026.1865399

Li et al. 2026c Plants (Basel)*Early warning of cucumber angular leaf spot from airborne pathogen aerosols estimated with PM sensors*

Two-stage greenhouse surveillance in which **low-cost PM sensors act as continuous physical proxies** for suspended pathogen aerosol, and qPCR air sampling is triggered only when an estimated aerosol risk crosses a threshold. Evaluated on **732 plot-days** across spatially independent trials: an Extra Trees model predicted pathogen-aerosol accumulation at $R^2 = 0.884$, and an XGBoost model integrating aerosol load with microclimate forecast the daily disease-index change rate at $R^2 = 0.874$. SHAP identified aerosol concentration and vapour pressure deficit as the primary drivers. The limitation is stated by the authors and is the right one: PM sensors have no biological specificity, and the design compensates by scheduling a specific assay rather than by claiming the sensor sees pathogens. Cross-greenhouse rather than cross-site validation, and no reference-instrument comparison for the PM channel itself.

doi: 10.3390/plants15162510

Liu T et al. 2026 Environ Sci Technol*Aerosol solute strength drives TMI-catalysed sulfate aerosol formation*

Aerosol flow-reactor experiments on atmospherically relevant aqueous aerosols, decoupling solute strength, interfacial chemistry and Fe(III)–Mn(II) interaction. Sulfate formation from Mn(II) chemistry runs **>2 orders of magnitude faster** than models assuming dilute cloudwater; substantial insoluble Fe(III) suppresses the rate below an ionic-strength threshold of **16 mol kg⁻¹**. Implemented in an atmospheric model, this raises sulfate abundance during polluted haze episodes by **>50%**. The flow reactor cannot reproduce real aerosol organic coatings or mixing state, so the transfer to ambient particles is an extrapolation — but this is a rare case of a laboratory rate measurement closing a known model–measurement gap rather than opening a new one.

doi: 10.1021/acs.est.6c09521

Liu et al. 2026 Atmos Chem Phys*From κ to χ : particle-resolved evaluation of hygroscopicity-based mixing-state retrieval*

PartMC-MOSAIC urban-plume ensembles spanning emissions, ageing and meteorology were used to compare the particle-resolved reference mixing-state index against the Yuan–Zhao retrieval from HTDMA κ distributions, per diameter over 50–250 nm. The retrieval performs well for many populations but **systematically overestimates χ when externally mixed intermediate-hygroscopicity components break its binary more-/less-hygroscopic assumption**. The useful output is the size-resolved error distribution, which the authors then apply as uncertainty bounds to long-term HTDMA records at urban, continental and coastal sites. Model-derived error bars on an observational retrieval — the validation is only as good as PartMC's own composition realism, which the paper does not independently test.

doi: 10.5194/acp-26-10965-2026

Liu et al. 2026c Atmos Meas Tech*Polarization calibration of spaceborne lidar using dense cirrus-scattered solar background*

Solar background scattered by optically thick ice cloud provides a semi-continuous daytime polarization calibration that complements onboard pseudo-depolarizers. At 1064 nm (CATS) molecular scattering is negligible; at shorter wavelengths it polarizes the background and biases the calibration. The paper adds a vector-radiative-transfer molecular-scattering correction and shows the daytime bias at CALIOP 532 nm is **~1%**, removed by the correction to agree with onboard calibration; EarthCARE ATLID at 355 nm is treated by model calculation only. Carried at tier C: depolarization ratio drives satellite aerosol typing, which in turn feeds AOD-to-PM_{2.5} conversion, but the link to surface concentration is two inferential steps away.

doi: 10.5194/amt-19-5325-2026

Liu et al. 2026d Sensors (Basel)*LRM-YOLO: lightweight YOLOv10n model for forest fire smoke detection in UAV images*

YOLOv10n with a RepViTBlock feature-extraction module, a lightweight efficient convolutional detection head, and MPDIoU bounding-box regression loss. Against baseline: **–36.7% parameters, –42.3% GFLOPs, +1.3% mAP50, +3.7% mAP50-95**, recall +3.2%, precision +3.6%, on a purpose-built UAV-perspective dataset plus Wildfire Smoke V1. Smoke detection in imagery is a plume-presence proxy, not a concentration measurement, and the self-constructed dataset means the headline gains are partly a property of the benchmark. Relevant only as an early-warning trigger upstream of a real exposure measurement.

doi: 10.3390/s26154887

Magdaong et al. 2026 ISPRS Ann Photogramm Remote Sens Spat Inf Sci*Criteria and an algorithm for siting a low-cost IoT air-quality network near roads*

GIS multi-criteria analysis combining street aspect ratio, traffic emissions, global horizontal irradiance and road proximity into a 0–6 suitability surface for Quezon City, then a heuristic selecting nodes by joint suitability and population raster under a minimum-spacing constraint. A 40-sensor configuration covers **1.27–1.35 million residents, 23.0–24.4% of the city**, stably across weighting schemes while keeping spatial dispersion balanced. Limitation: coverage is defined geometrically against a population raster with no validation against measured concentration fields, so “covered” means “near a node”, not “well estimated”. Still the clearest statement this watch has seen that siting is an optimisation problem with a stated objective, and the criteria transfer directly to any near-road deployment in a low- or middle-income city.

doi: 10.5194/isprs-annals-x-5-w4-2025-315-2026

Marcon et al. 2026 Curr Environ Health Rep*Citizen science in air pollution monitoring: methodological approaches, effectiveness and integration with health research*

Critical review of participatory air-quality monitoring — low-cost sensors, wearables, community co-creation — and its uptake into environmental health research. The review identifies AI-assisted calibration and distributed sensor networks as the methodological frontier and argues that hyperlocal citizen data materially improves spatial resolution and risk communication. Its useful contribution is the ordering of the barriers: **representativeness and policy uptake, not sensor accuracy**, are named as the binding constraints. No quantitative synthesis, no pooled performance figures and no inclusion criteria are reported in the abstract, so it is a framing document rather than an evidence base. For a distributed low-cost network this is the argument that siting design deserves as much engineering attention as the calibration pipeline.

doi: 10.1007/s40572-026-00559-1

Masoom et al. 2026 Atmos Chem Phys*Long-range transport of Canadian wildfire smoke to Europe in 2023: aerosol properties and spectral features*

Combined column remote sensing and in situ measurement of short- and long-range plumes across North America and Europe through the April–October 2023 fire season. The late-season episode is the payload: **concave AOD spectral curvature with the maximum above the 340 nm minimum wavelength, giving negative Ångström exponents below 500 nm**; accumulation-mode volume median diameters near 800 nm; and non-monotonic single-scattering albedo, rising through the UV–visible then flattening or falling at longer wavelengths, consistently across most stations. Limitation: a rare ~one-week event characterised observationally, with no closure between the retrieved optics and a measured composition, so the cause of the curvature is inferred rather than demonstrated. For distributed optical sensing the implication is that smoke-episode response cannot be extrapolated from background calibration.

doi: 10.5194/acp-26-10801-2026

Mohapatra et al. 2026 IEEE Access*IoT-machine-learning proof-of-concept framework for real-time urban particulate matter prediction*

PMS7003 on an Arduino module at campus scale, with outlier treatment, normalisation and temporal indexing feeding random-forest and polynomial regression, plus k -means clustering to identify persistent high-exposure zones. Predictive performance **exceeds 85% for $PM_{1.0}$ and $PM_{2.5}$ but degrades sharply at 5 and 10 μm** , which the authors correctly attribute to discretised, low-variability sensor output rather than to the model — a useful negative result about the coarse channels of optical counters generally. No reference co-location, single site, and the models predict the sensor's own future readings rather than true concentration, so "performance" here is self-consistency. Included for the size-channel finding only.

doi: 10.1109/access.2026.3651996

Moteki et al. 2026 J Aerosol Sci*Refractive index of fractal-like aggregates from complex forward-scattering amplitudes*

Metadata-only deposit — title and DOI without an abstract. The method named in the title is the single-particle complex forward-scattering amplitude $S(0^\circ)$ approach, previously used to constrain the black-carbon refractive index at $\lambda = 0.633 \mu m$; the new element is **mass classification of the aggregates** before the optical measurement, which removes the size-distribution term that dominates the uncertainty in bulk retrievals. If it delivers, this is the most directly useful sensing record in the window: every optical PM instrument, from a nephelometer to a \$20 laser scatter sensor, converts scattered intensity to mass through an assumed refractive index, and for soot-dominated aerosol that assumption is the largest single error term. Carried at tier C on the title alone and placed at the top of the abstract retry list.

doi: 10.1016/j.jaerosci.2026.106884

Neuberger et al. 2026 Atmos Chem Phys*Importance of hydrated aerosol particles for aerosol-fog relationships in the Italian Po Valley*

FAIRARI 2021/2022 aerosol-fog observations at San Pietro Capofiume, analysed with κ -Köhler theory and the MIMICA large-eddy simulation. Median hygroscopicity separated by population: **fog residuals $\kappa = 0.45$, interstitial particles 0.40, out-of-fog aerosol 0.34**, the gradient reflecting inorganic enrichment in activated droplets. Excluding hydrated but non-activated particles from the analysis **inflated effective diameter by 81 % (11.6 $\rightarrow$ 21.0 μm) and depressed droplet number concentration by 87 % (97.4 $\rightarrow$ 12.4 cm^{-3})**; hydrated particles supplied **21 % of liquid water content** and drove **36 % of sub-kilometre visibility events without any activation**. The LES leg shows fog prediction hinging on the largest dry particles. The limitation that binds is site specificity — one heavily polluted valley over two winters, with a κ distribution shaped by Po Valley nitrate that will not transfer to a marine or biomass-burning regime. For distributed low-cost sensing the transferable point is that the hydrated/activated distinction is a *composition* problem, so an RH correction fitted at one site during one season inherits that site's κ and no more.

doi: 10.5194/acp-26-11991-2026

Ng et al. 2026 Atmos Chem Phys*Impacts of droughts on biomass burning emissions, air quality and public health in the Amazon*

Satellite fire activity plus GFEDv4s emissions driving GEOS-Chem High Performance, 2010–2015, with paired fire-on/fire-off simulations. In non-drought years biomass burning contributes **34.0% (32.0–35.5%) of dry-season CO and $PM_{2.5}$** ; in drought years that rises to **55.4% (44.1–76.7%)**. Fire-induced $PM_{2.5}$ and ozone raise premature mortality (GEMM) by 6.0% and 18.6% in non-drought years, 8.9% and 24.4% in drought years — *at stable deforestation rates*, which is the point: the climate-driven amplification is separable from the land-use driver. For network design, the drought-year contribution range is wide enough that a fixed seasonal baseline will misestimate the fire fraction by a factor of two.

doi: 10.5194/acp-26-11835-2026

Nimmala et al. 2026 Environ Monit Assess*Drone and ground sampling of transgenic switchgrass pollen with Lagrangian dispersal modelling*

Release-recapture with 200 GE switchgrass plants expressing orange fluorescent protein under a constitutive (PvUBI1) and a pollen-specific (Zm13) promoter, sampled to **1 km** by fixed and drone-borne collectors, with Lagrangian stochastic simulation driven by high-resolution wind. Emission rate estimated by combining simulated and measured concentrations; a strong diurnal trend correlates positively with wind speed, temperature and vapour pressure deficit and negatively with RH. Incorporating changing wind direction improved estimates under low-wind meandering — the transferable result for anyone inverting a dispersion model against sparse fixed sensors. Held back from the 2 August issue on its per-record [EDAT]; it is not a gap.

doi: 10.1007/s10661-026-15632-3

Nurseitov et al. 2026 Information*Design, deployment and field evaluation of a low-cost IoT monitoring network for urban particulate matter*

Six custom nodes (Winsen ZPHS01B on Raspberry Pi Zero 2 W, ~US\$100/node) deployed across Almaty over a February–April campaign, **>70,000 measurements**, with three engineering contributions: an active airflow-stabilisation enclosure decoupling sampling from wind, context-aware adaptive edge filtering cutting transmitted volume ~40%, and a segmented telemetry pipeline. Cross-validation was against a consumer monitor with documented traceability rather than a reference analyser ($R^2 = 0.89$ – 0.95), and the authors state explicitly that the network provides *internally consistent* rather than reference-equivalent concentrations — unusually honest framing. **Daily mean $PM_{2.5}$ exceeded the WHO 24 h guideline on 84% of days** (February 54.4 vs March 21.9 $\mu g/m^3$); a $PM_{2.5}/PM_{10}$ ratio near 0.96 plus higher weekend levels points to residential coal heating. That ratio is high enough to warrant suspicion of the coarse channel rather than confidence in the source inference.

doi: 10.3390/info17070642

Orr et al. 2026 (JMIR) JMIR Pediatr Parent*In-home digital air quality monitors in families of children with asthma: feasibility and acceptability*

Pilot mixed-methods study, **N = 20** families, combining interviews and surveys to assess whether a consumer in-home air quality monitor can be implemented in paediatric asthma care. Use was reported as both feasible and acceptable, with **affordability identified as the principal barrier to clinical adoption**. The report is very short and carries no exposure data, no adherence metric and no asthma outcome, so it establishes only that the deployment step is not the bottleneck. That is still the piece of the low-cost-sensing translation chain that is least often evidenced — most LCS literature stops at accuracy and never asks whether a clinical population will host the device. Needs a powered trial with objective device-up-time data before it can bear weight.

doi: 10.2196/92070

Oybek et al. 2026 Am J Appl Sci Technol

Chamber bounds on the temperature and humidity penalty, by sensor chemistry

Controlled-chamber factorial over six temperature (15–35 °C) and relative humidity (30–90%) combinations for an optical PM_{2.5} counter, a metal-oxide NO₂ sensor and an electrochemical CO₂ sensor. PM_{2.5} error rises from **1.2 µg/m³ at 30% RH to ~9 µg/m³ at 85% RH**; a simple linear correction recovers up to 74% of it. **Limitations are substantial**: one unit per chemistry, no replication, an unstated test aerosol, a venue with no visible peer-review record, and a linear correction applied to a process (hygroscopic growth) that is not linear. Carried as a magnitude anchor only — the number is consistent with better studies and useful for sizing an error budget, but nothing here should be cited as a correction method.

doi: 10.37547/ajast/volume06issue05-25

Pérez-Pastor et al. 2026 Environ Sci Pollut Res Int

Robust quantification of anhydrosugars and reference material evaluation

Accelerated solvent extraction (dichloromethane:methanol), BSTFA derivatisation, GC/MS for levoglucosan, galactosan and mannosan. Linearity $R > 0.9987$, repeatability <6%, LOD <0.02 mg L⁻¹; relative error 1–7.4% against SRM 2787 (fine PM). **SRM 1649b (urban dust) gave extraction-condition-dependent levoglucosan values**, which the authors attribute to matrix ageing — a direct warning against using an old urban-dust SRM to validate a biomass-burning tracer method. Chromatographic calibration dominates the uncertainty budget. Expanded uncertainties meet forthcoming EU air quality monitoring requirements.

doi: 10.1007/s11356-026-38136-6

Perišić et al. 2026 Sci Total Environ

Explainable AI resolves ten recurrent environmental settings behind urban PM₁₀ variability

Four years (2020–2023) of routine regulatory data from a *single* station — PM₁₀, PM_{2.5}, NO₂, SO₂, O₃ and meteorology — fed to ensemble machine-learning models with metaheuristic hyperparameter optimisation, reaching $R^2 = 0.913$, then clustered on SHAP-derived predictor-impact profiles rather than on concentrations. **Ten recurrent settings emerge**. The strongest positive contributions are cold-season accumulation regimes: E0 is frequent and predominantly nocturnal at a mean impact of **50.9 µg m⁻³**; E7 is rarer but reaches **82.9 µg m⁻³** and persists day and night. Warm, well-mixed settings run **26–29 µg m⁻³ below** the model baseline. The substantive claim — that equal concentrations can encode different pollutant-meteorology configurations — is well supported and directly useful where speciation is unaffordable. The limitation is structural and the authors say so: one station, one climate, no external validation, and SHAP attributions are model-conditional, so the “settings” are properties of a fitted ensemble before they are properties of the atmosphere. For distributed low-cost networks the appeal is that the entire input set is what a modest node plus a met sensor already produces.

doi: 10.1016/j.scitotenv.2026.182195

Qian et al. 2026 Atmos Meas Tech

Enhancing accuracy of indoor air-quality sensors via automated machine-learning calibration

Multi-stage AutoML calibration chaining field sensors to intermediate drift-correction reference sensors to a reference-grade instrument, with **separate models fitted below and above a concentration break** for clean air versus pollution events. Evaluated in a controlled indoor chamber with two sensor models against diverse real indoor sources under uncontrolled ambient conditions: $R^2 > 0.90$, **NRMSE and sMAPE roughly halved** against uncalibrated output, bias effectively removed. The split-range design is the transferable idea — a single global calibration is being asked to serve two distinct optical regimes and it fails at both ends. Chamber conditions make the result an upper bound on field performance, and no ageing term is included, which the Kumpika result suggests is the dominant long-run error.

doi: 10.5194/amt-19-603-2026

Qian et al. 2026 Atmos Meas Tech

Enhancing accuracy of indoor air-quality sensors via automated machine-learning calibration

Multi-stage AutoML calibration chaining field sensors to intermediate drift-correction reference sensors to a reference-grade instrument, with **separate models fitted below and above a concentration break** for clean air versus pollution events. Evaluated in a controlled indoor chamber with two sensor models against diverse real indoor sources under uncontrolled ambient conditions: $R^2 > 0.90$, **NRMSE and sMAPE roughly halved** against uncalibrated output, bias effectively removed. The split-range design is the transferable idea — a single global calibration is being asked to serve two distinct optical regimes and it fails at both ends. Chamber conditions make the result an upper bound on field performance, and no ageing term is included, which the Kumpika result suggests is the dominant long-run error.

doi: 10.5194/amt-19-603-2026

Rapp et al. 2026 Atmos Meas Tech

Curve fitting algorithm for multimodal particle size distributions: a theoretical basis

Open-source lognormal decomposition with no predefined mode templates and no mode-count constraint. Each mode's N_i , σ_g and $\bar{D}_{pg}$ are fitted by Levenberg–Marquardt and iteratively subtracted until convergence. Evaluated against a ten-year ambient record at Storm Peak Laboratory and against laboratory PSDs of size-selected ammonium sulfate coated in oxidised biogenic SOA. The result that matters: **a tri-modal assumption is inadequate for 55.7% of new-particle-formation data**, four or more modes being required — so the default configuration of most PSD fitting software is wrong in the majority of the regime it is applied to. Agreement with existing models at a tri-modal setting is strong with substantially lower computation. Supplemental routines format output from common mobility particle size spectrometers, which lowers the adoption cost considerably.

doi: 10.5194/amt-19-5373-2026

Rehbein et al. 2026 Clean Air J

Source identification and triangulation from a low-cost sensor network in Ile-Ife

Reanalysis of a published Nigerian LCS dataset using k -means over particle-number concentration profiles, resolving **six clusters** that separate day from night and local from regional influence. Stagnant conditions near industrial activity gave PM roughly 10× the daily mean (PM_{2.5} 30.6±40.7, PM₁₀ 83.4±259.2 µg m⁻³); cleanest conditions were daytime south-westerlies (PM_{2.5} 11.5±4.7), still above WHO annual guidance. Standard deviations exceeding the means signal a heavily skewed distribution that k -means handles poorly — the triangulation is indicative, not quantitative apportionment.

doi: 10.17159/caj/2026/36/1.24186

Reis 2026 Sensors*Low-power embedded sensor node for environmental monitoring with on-board machine-learning inference*

Architecture and system-level optimisation of a multi-parameter node coupling a low-power microcontroller to a neural inference accelerator, with adaptive duty-cycling, hierarchical power domains and quantised fixed-point models doing three-class normal/anomalous/critical classification at the edge. The reported figures — **94% inference accuracy, 0.87 ms latency, ≈2.9 mWh average power, ≈88% reduction in LoRaWAN transmissions** — come from MATLAB/Simulink simulation, not deployment, and there is no co-location against a reference instrument anywhere in the study. Treat the energy budget as a design target and the accuracy figure as unvalidated. The duty-cycling and hierarchical-power argument is the reusable part, and it is also the part that does not depend on the classifier working.

doi: 10.3390/s26020703

Reis 2026 Sensors*Low-power embedded sensor node for environmental monitoring with on-board machine-learning inference*

Architecture and system-level optimisation of a multi-parameter node coupling a low-power microcontroller to a neural inference accelerator, with adaptive duty-cycling, hierarchical power domains and quantised fixed-point models doing three-class normal/anomalous/critical classification at the edge. The reported figures — **94% inference accuracy, 0.87 ms latency, ≈2.9 mWh average power, ≈88% reduction in LoRaWAN transmissions** — come from MATLAB/Simulink simulation, not deployment, and there is no co-location against a reference instrument anywhere in the study. Treat the energy budget as a design target and the accuracy figure as unvalidated. The duty-cycling and hierarchical-power argument is the reusable part, and it is also the part that does not depend on the classifier working.

doi: 10.3390/s26020703

Roska et al. 2026 Environ Sci Technol*Beyond targeted calibrations: comprehensive quantification of oxygenated organic compounds in urban atmospheres*

A collision-induced-dissociation voltage-scanning method to derive sensitivities for an ammonium-adduct CIMS, so OVOCs without an authentic standard can still be quantified. Validated in the laboratory, then applied to the **2023 AEROMMA airborne campaign and the 2024 SAPHIR-CHANEL chamber campaign**: coverage rose to **98% of total signal in chamber and 78% in the field**, and adding the newly quantified species raised estimated **OH reactivity by ~5% and SOA formation potential by ~62%**. Coverage degrades with molecular size and oxidation state, and chamber-derived sensitivities are being transferred to ambient air on the strength of trend agreement rather than field standards — the paper says so. The 62% figure is the one that matters downstream: SOA yields used in chemical-transport models that feed exposure surfaces are calibrated against measurements that were missing most of this signal.

doi: 10.1021/acs.est.6c04149

Rubio et al. 2026 IEEE Consum Electron Mag*LoRa-connected micromobility pollution monitor for the Internet of Vehicles*

On-board PM_x/CO₂/meteorology unit for personal mobility vehicles, evaluated across Cartagena's three-gateway LoRa network. The engineering contribution is compact low-power design plus LoRa in a genuinely mobile setting, where duty-cycle limits and Doppler normally break LPWAN links. No reference co-location and no uncertainty quantification are reported, so the pollution data quality is unestablished — the communications and positioning results are the defensible part.

doi: 10.1109/mce.2026.3684542

Saad et al. 2026 Environ Monit Assess*Short-term air pollutant monitoring at the Alexandria port–urban exposure zone*

Two monitoring campaigns, May–October 2022, across 11 in-port and 13 surrounding urban sites, using a portable multi-gas monitor and particulate sensors, with hourly meteorology from Ras El-Tin (campaign-mean wind 3.0–7.2 m s⁻¹, northerly to north-westerly). Fuel and bulk-fertiliser berths recorded significantly higher CO and NH₃ than urban sites; SO₂ and traffic pollutants outside the port dominate the urban background. **Particulate levels were low across the whole study area** — the authors say so, and the paper is correspondingly framed as screening-level spatial evidence, not exposure assessment. No sensor model, no reference co-location and no uncertainty budget are reported, which caps what the PM numbers can support. Carried mainly because African monitoring records are rare in this corpus.

doi: 10.1007/s10661-026-15828-7

Samburova et al. 2026 Atmos Pollut Res*Effects of western US wildfire smoke on urban air quality in northern Nevada*

METADATA ONLY — no abstract available in Crossref, Europe PMC or Semantic Scholar. Long-range transported wildfire smoke over an urban receptor is the same problem class as Jin et al. (ACP, 7 August) on aged smoke in Missoula; worth a re-check once the publisher deposits the abstract.

doi: 10.1016/j.apr.2026.103168

Sanders et al. 2026 Aerosol Sci Technol*Nonionic surfactant composition and hygroscopic growth of supermicron NaCl particles*

Crossref deposit of 7 August, **metadata only — no abstract deposited anywhere in the retry chain**, so no summary is offered. Recorded here because surfactant control of hygroscopic growth is the physical mechanism underlying humidity correction in light-scattering PM sensors, and the supermicron size range is where nephelometric response is most sensitive to growth factor.

doi: 10.1080/02786826.2026.2703042

Saravagna Sai Lakshmi et al. 2026 J Vis Exp*CORTA-Net: hybrid deep learning with attention-based interpretability for PM_{2.5} forecasting*

CorrXGBoost-Rank feature screening, transfer-learning LSTM encoder, multi-head attention for feature- and timestep-level attribution, over CPCB/DPCC station data, IMD meteorology and MODIS fire counts for Delhi 2012–2024. Reported to reduce prediction error against random forest, XGBoost, LSTM and attention-LSTM baselines — but **no RMSE, MAE or R² appears in the abstract for either the model or the baselines**, so the size of the improvement is unstated and unverifiable here. A methods-video venue rather than a modelling journal, and the contribution is an assembly of existing components. Included for the fire-count integration, which is the one input choice that is specific to the Delhi problem rather than generic.

doi: 10.3791/71004

Senarathna et al. 2026 [backfill] Environ Monit Assess

Seasonal and spatial non-transferability of LCS calibration across two tropical climatic zones

TSI BlueSky sensors in a harmonised network across Colombo (wet-zone) and Kandy, compared under linear regression, multiple linear regression and an echo-state network. Harmonised sensors agreed within **10%** of one another. Performance improves monotonically with averaging window: at 24 h, wet season $R^2 = 0.60$, MAE $0.93 \mu\text{g m}^{-3}$, MAPE 7.81%; dry season $R^2 = 0.85$, MAE 1.98, MAPE 9.17%. The two transfer results are the point — applying a wet-season model to dry-season data yields **26.57% MAPE**, and a Colombo-fitted model degrades in Kandy, while a Kandy-specific model with temperature and RH reaches $R^2 = 0.84$ – 0.92 . Limitation: no independent reference co-location is reported for the second zone at the same duration, so the spatial-transfer claim rests partly on inter-sensor agreement rather than reference truth.

doi: [10.1007/s10661-026-15623-4](https://doi.org/10.1007/s10661-026-15623-4)

Sengupta et al. 2026 [backfill] arXiv (preprint)

LSTM calibration of low-cost PM and NO₂ sensors against a regulatory equivalence test

Sequence model trained on co-located OxAria reference data in Oxford for PM_{2.5}, PM₁₀ and NO₂, benchmarked against a Random Forest baseline. The framing argument is that RF treats each hourly observation independently and therefore cannot represent sensor lag or delayed environmental response; the LSTM with time-lagged features, harmonic time encodings and interaction terms beats it on R^2 across train, validation and test. The result that carries weight is the external check: validation against Equivalence Spreadsheet Tool 3.1 gives expanded uncertainties of **9.1% (PM_{2.5})**, **12.42% (PM₁₀)** and **22.11% (NO₂)** — i.e. PM meets, and NO₂ misses, the indicative-monitoring bar. Unrefereed, single network, and no drift analysis over a multi-year horizon, which is where sequence models are most likely to fail.

doi: [10.48550/arXiv.2604.21527](https://doi.org/10.48550/arXiv.2604.21527)

Shirkhani et al. 2026 J Aerosol Sci

Design and development of a high-throughput multipattern aerosol deposition device (MARTHA) for respiratory toxicity and health-risk assessment

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar at the time of this issue (S2 returns 404 on the DOI). Listed rather than summarised. It is flagged as the day's highest-priority re-check for a specific reason: the title states an aerosol-delivery instrument built for respiratory toxicity assessment, which is the exact instrument × health-endpoint coupling that the 2–8 August weekly identified as structurally absent across 146 records. Nothing beyond the title and the author list (Shirkhani, Salana, Amanatidis, Washeem, Eiguren-Fernandez, Spielman, Nannu Shankar, Lewis) supports that reading yet.

doi: [10.1016/j.jaerosci.2026.106882](https://doi.org/10.1016/j.jaerosci.2026.106882)

Shlipak et al. 2026 ACS EST Air

Spatial and temporal trends of ambient PM_{2.5} and source-apportioned black carbon in Addis Ababa

Three continuous years (Apr 2022–Mar 2025) from the MAIA surface network: 10 PurpleAir units across the city calibrated against collocated Beta Attenuation Monitors at two contrasting U.S. Embassy sites (Central, Jacros), plus AethLabs MA350 microaethalometers at those two sites. Network mean PM_{2.5} **30 $\mu\text{g}/\text{m}^3$** ; BC 3.7 and 7.7 $\mu\text{g}/\text{m}^3$; the WHO 24 h guideline of 15 $\mu\text{g}/\text{m}^3$ was exceeded on **99% of days**. Diurnal peaks at morning traffic and under nighttime inversion, strongest in the wet season; highest concentrations west of the city centre. Aethalometer-model apportionment attributes **94% of BC to fossil combustion**, 6% to biomass — the opposite of the usual prior for the region. Directional and back-trajectory analysis raises long-range shipping transport from the Gulf of Aden as a regional contributor. Limitation: only two sites carry a reference anchor and BC, so the calibration transfer across the other eight nodes is assumed rather than demonstrated. The single most directly transferable deployment design in this window.

doi: [10.1021/acsestair.6c00140](https://doi.org/10.1021/acsestair.6c00140)

Silva et al. 2026 J Air Waste Manag Assoc

Low-cost sensors for indoor air-quality monitoring: systematic review of accuracy, applications and limitations

PRISMA review, **24 studies retained from 2,736 screened**, covering optical particle counters and NDIR CO₂ against reference-grade instruments in real settings. Strong correlations are achievable under controlled or semi-controlled conditions, but performance degrades with humidity, temperature and source type, and the authors flag methodological heterogeneity, near-absent inter-device validation, regional bias and **un-addressed sensor ageing and cross-sensitivity** as systematic gaps. The operational conclusion — that most devices do not meet regulatory compliance requirements and standardised validation protocols do not exist — is well supported and worth citing directly. As a review it inherits the heterogeneity it documents, and a 24-study base is thin for the breadth of the claims.

doi: [10.1080/10962247.2026.2624795](https://doi.org/10.1080/10962247.2026.2624795)

Silva et al. 2026 J Air Waste Manag Assoc

Low-cost sensors for indoor air-quality monitoring: systematic review of accuracy, applications and limitations

PRISMA review, **24 studies retained from 2,736 screened**, covering optical particle counters and NDIR CO₂ against reference-grade instruments in real settings. Strong correlations are achievable under controlled or semi-controlled conditions, but performance degrades with humidity, temperature and source type, and the authors flag methodological heterogeneity, near-absent inter-device validation, regional bias and **un-addressed sensor ageing and cross-sensitivity** as systematic gaps. The operational conclusion — that most devices do not meet regulatory compliance requirements and standardised validation protocols do not exist — is well supported and worth citing directly. As a review it inherits the heterogeneity it documents, and a 24-study base is thin for the breadth of the claims.

doi: [10.1080/10962247.2026.2624795](https://doi.org/10.1080/10962247.2026.2624795)

Siu et al. 2026 Environ Res Health

Disparities in fine particulate matter (PM_{2.5}) monitoring capacity in eastern Canada

Locally adaptive kernel density estimation of total PM_{2.5} monitor density — regulatory plus low-cost — across Ontario, Quebec and Atlantic Canada, 2020–2024, with each local bandwidth set from the surrounding variability of surface PM_{2.5} and its sensitivity to resolving an exceedance. Mixed-effects logistic regression gives **25% higher odds of monitoring deprivation for residential instability** and **54% higher odds for situationally vulnerable populations**, after adjusting for surface PM_{2.5} and emission-facility density; suburban Montreal and Toronto, the Windsor and Niagara border regions, southeast Kitchener and remote northwest Ontario are named. Limitation: monitor density is a proxy for information yield, and low-cost and regulatory devices are pooled without weighting by data quality, so a dense but uncalibrated cluster counts as capacity. The adaptive-bandwidth construction is directly reusable for designing where to add nodes rather than merely auditing where they are.

doi: [10.1088/2752-5309/ae87c0](https://doi.org/10.1088/2752-5309/ae87c0)

Somnugpong 2026 Environ Monit Assess*Similarity-weighted seasonal-anchor Stineman interpolation for gappy air-quality series*

SW-SAST keeps a local Stineman anchor and selectively borrows cross-day information when nearby daily analogues are compatible with the local estimate. Evaluated on UCI Air Quality and Beijing PM_{2.5} under MCAR, MAR and MNAR at 10–50% missingness, 30 repeated runs per condition, against a univariate LSTM baseline and with contiguous gap-length experiments. **It improves on plain Stineman for UCI, chiefly under MCAR and MAR, and fails to beat it on Beijing.** The contribution is explicitly a boundary condition on when seasonal borrowing helps, not a replacement interpolator — which is the right claim and an unusually disciplined one. Directly applicable to reconstructing low-cost-sensor series where dropouts are neither random nor uniform.

doi: 10.1007/s10661-026-15770-8

Suryawanshi et al. 2026 Int J Environ Res*Spatiotemporal variation and hotspot identification of PM_{2.5} from low-cost sensors*

Abstract not indexed at the time of harvest; carried as a metadata-only record so the register is complete and the next rollup does not re-surface it as new. No claims are made about its content. Full text required.

doi: 10.1007/s41742-026-01169-3

Tian et al. 2026 Atmos Meas Tech*A heterogeneity index for when lidar-retrieved CCN vertical profiles stop being trustworthy*

ARM ground-based Raman lidar plus surface CCN measurements retrieve vertically resolved CCN number concentration on the assumption of vertically homogeneous composition. A *k*-nearest-neighbours classifier on lidar ratio and linear depolarisation ratio infers dominant aerosol type by altitude, and a derived heterogeneity index quantifies vertical complexity. Validated against HI-SCALE aircraft in situ data: **retrieval closure degrades monotonically as the index rises**, so the metric functions as a self-diagnostic flag on the retrieval. Single campaign, so the index thresholds are not yet portable.

doi: 10.5194/amt-19-5051-2026

Tsai et al. 2026 IEEE Sens J*Vision-based vertical PM_{2.5} retrieval from a 5G-connected UAV*

Dark-channel-prior atmospheric transmittance extracted from UAV imagery, fused with meteorology into a random-forest regression with 7-D polynomial feature engineering hosted on a 5G-core virtual host; Gemini used for semantic interpretation of the vertical profile. Reported **R² = 0.954, RMSE 1.27 µg m⁻³**. Treat with caution: no held-out spatial or temporal validation is described, the DCP proxy is illumination-dependent, and the generative-AI layer contributes interpretation, not measurement. Carried for the vertical-profiling concept, not the reported accuracy.

doi: 10.1109/jsen.2026.3695609

Vivanco et al. 2026 Environ Pollut*Responsiveness of an air quality model to real-world emission changes*

Metadata-only deposit — title and DOI without an abstract, so the design, model, domain and emission perturbations are all unknown at the time of writing. The question is a live one (whether CTMs used for planning actually reproduce the concentration response to observed emission changes rather than only to scenario deltas) and the record is carried at tier C so it is not lost. Flagged for abstract retry.

doi: 10.1016/j.envpol.2026.129031

Wang et al. 2026 Environ Monit Assess*Traffic-induced environmental health assessment at intersections using roadside LiDAR*

Roadside LiDAR extracts high-resolution vehicle trajectories at one urban and one suburban intersection, from which traffic emissions, noise and ground vibration are estimated and combined into a fuzzy comprehensive assessment score, with weights set by quadratic programming under ranking constraints, variable-weight grey entropy, and game theory to reconcile the two. The urban intersection scored **69.45 by day, falling to 57.12 at night (~18%)**, while the suburban site was stable at 72.80/74.54; population-dimension scores were consistently below environmental-dimension scores. Limitation: PM is *inferred* from trajectories through an emission model and never measured, so this is a modelled exposure surrogate validated against nothing — the elaborate weighting machinery sits on top of an unvalidated emissions estimate. Included because trajectory-derived emission proxies are an increasingly common input to near-road sensor fusion, and their weak point should be explicit.

doi: 10.1007/s10661-026-15750-y

Wang et al. 2026 Atmos Environ*Measuring heavy metals in atmospheric particulate matter: operational definitions, method comparability and environmental interpretation*

METADATA ONLY — no abstract in Crossref, Europe PMC or Scholar Gateway. Listed, not summarised. Carried because the title states the problem this watch treats as central, transposed to a different measurand: for heavy metals in PM, the reported concentration depends on the digestion protocol, the filter substrate, the size cut and whether total or bioaccessible fraction is meant — so “comparability” is a property of the method chain, not of the air. The same argument applies to low-cost optical mass measurement, where the reported µg/m³ is a function of assumed density, refractive index and hygroscopic growth. A systematic treatment of operational definitions on the metals side would be worth reading against the sensor-calibration literature.

doi: 10.1016/j.atmosenv.2026.122291

Wang et al. 2026 Environ Sci Technol*Remote calibration of a nationwide dense low-cost sensor network for hyperlocal air surveillance*

Removes physical collocation from the LCS calibration loop. A spatiotemporal pairing algorithm selects representative training data linking uncollocated sensors to reference stations, with metropolis-specific monthly model refresh; validated on **>33 million hourly observations** across ten US metros. Demonstrated downstream on high-granularity land-use regression and wildfire plume tracking. Two things to hold against it. First, the abstract reports agreement qualitatively and gives *no* R², RMSE or bias, so it cannot be placed against the colocated calibration literature without the full text. Second, and more usefully, the paper quantifies its own denominator problem: LCS density differs significantly by race/ethnicity and income, with African American communities least covered. Remote calibration widens the usable network; it does not move a sensor into a neighbourhood that has none.

doi: 10.1021/acs.est.6c03270

Wang et al. 2026d Atmos Chem Phys*Perchloric acid drives new particle formation enhanced by dimethylamine, ammonia and sulfuric acid*

Quantum-chemical calculations coupled to the Atmospheric Clusters Dynamic Code compare the nucleation enhancement of DMA, NH₃ and H₂SO₄ for perchloric-acid-based new particle formation, motivated by recent Arctic observations of atmospheric HClO₄. Both DMA and NH₃ bind PA strongly through hydrogen bonding and proton transfer, and **PA nucleates more readily with DMA than sulfuric acid does**; the PA–DMA cluster formation rate exceeds PA–NH₃ even at overwhelming NH₃ excess, with 1:1 PA:DMA clusters controlling growth. Purely computational — no chamber or field verification, and the claimed NH₃ excess of “10–100 orders of magnitude” is a transcription problem in the abstract rather than a physical statement. Relevant as a candidate nucleation channel in clean marine and polar boundary layers where sulfuric acid is scarce, which is where number-concentration instruments see unexplained events.

doi: 10.5194/acp-26-12171-2026

Wei et al. 2026 iScience*Stacked Transformer-XGBoost model with SHAP interpretability for AQI prediction in Linfen*

Transformer temporal encoder stacked with XGBoost through temporally consistent out-of-fold stacking, on 2013–2024 air quality and meteorological series from a heavily industrialised Chinese city. Reported $R^2 = 0.955$, **RMSE 9.05**, **MAE 2.91**; SHAP attributes the variance to PM_{2.5}, NO₂, lagged AQI and seasonal terms. The performance claim is against conventional ML and DL baselines on the same series, with no external site held out, so it is an in-sample-city result. The out-of-fold stacking discipline is the part worth borrowing — naive stacking on autocorrelated air quality series leaks badly.

doi: 10.1016/j.isci.2026.116396

Werderman et al. 2026 Atmos Pollut Res*Laboratory evaluation of Sensirion SEN66, Piera IPS-7100, PurpleAir PIXEL and OPC-N3 low-cost aerosol sensors across aerosol types and concentration ranges*

Metadata only. Crossref by-ISSN sweep; title, authorship (Werderman, Richards, Berkuta, Sousan) and DOI verified, no abstract deposited and no numbers extractable. It is nevertheless the single most on-topic record of the window: a chamber characterisation of *four* current-generation units across aerosol type and concentration range, three of which — SEN66, IPS-7100, PurpleAir PIXEL — have no prior evaluation in this corpus, against the OPC-N3 as the established reference point. The same last author’s 2021 four-sensor evaluation established that environmental and occupational settings need separate calibration factors, so the plausible contribution here is the aerosol-type dependence of that factor for the newer optics. **Highest-priority full-text retrieval on the next run**; nothing is asserted about its findings until then.

doi: 10.1016/j.apr.2026.103185

Wongsasuluk et al. 2026 Environ Sci Pollut Res Int*PM_{2.5} exposure and respiratory health risk for vulnerable populations in an urban underpass: a predictive MLR approach, Chiang Mai*

Five DustBoy low-cost stations along the Don Kaew road tunnel, August 2022 to July 2023, yielding **385,622 records** across PM_{2.5}, PM₁₀, CO, CO₂, NO₂, SO₂, O₃, TVOC, CH₂O and meteorology. A stepwise multiple linear regression stratified by weekday/weekend and day/night gives a high reported goodness of fit, and a comparative risk assessment converts predicted concentrations into inhaled dose and relative risk against Thai and WHO exposure factors. The inference-threatening gap is that **no co-location against a reference monitor is reported**: a semi-enclosed tunnel is a high-humidity, high-loading environment, precisely where nephelometric low-cost sensors over-read most, and every downstream risk number inherits that bias uncorrected.

doi: 10.1007/s11356-026-38171-3

Wu & Qu 2026 Environ Monit Assess*Interpretable short-term PM_{2.5} forecasting using meteorology–pollution coupling across Beijing stations*

420,768 hourly observations from 12 Beijing stations, 2013–2017, under a strict chronological train–validation–test split with Diebold–Mariano testing, feature-group ablation and station-holdout transfer. At the 6 h horizon CatBoost gave **RMSE 51.24 µg/m³**, **MAE 30.87**, $R^2 = 0.617$, a 6.73% RMSE reduction over persistence; station-holdout LightGBM reached 53.11 µg/m³, 5.65% below persistence. Dominant predictors were recent PM history, wind speed and direction, diurnal cyclicity and the temperature–dew-point gap. **The honest part is the framing**: a 6.7% gain over persistence is a modest operational return, performance degrades during high-pollution episodes, and the authors state that the 2013–2017 regime would require retraining for contemporary deployment.

doi: 10.1007/s10661-026-15799-9

Xiong et al. 2026 Atmos Chem Phys*In situ real-time determination of SO₂ photochemical oxidation in nanoscale sea salt aerosols based on dark-field microscopy*

Planar-waveguide dark-field microscopy tracks hygroscopic growth factors of single NaCl nanoparticles under UV irradiation and SO₂, with the Zdanovskii–Stokes–Robinson rule converting growth to composition and hence to reaction rate. Reported first-order rate constant **0.6523 h^{−1} for 100 nm NaCl**, with a **non-monotonic size dependence** — rising from 50 to 200 nm, falling above 200 nm — attributed to competition between surface curvature and specific surface area, and cross-checked against transition-state theory and a double-film mass-transfer treatment. Single salt system under controlled UV; the extrapolation to ambient mixed-composition marine aerosol is not demonstrated. Methodologically this is the interesting part: an optical single-particle technique that yields kinetics without collection, at sizes below what most PM instrumentation resolves at all.

doi: 10.5194/acp-26-12183-2026

Yan 2026 Sensor Review*What 90+ ML calibration studies actually agree on*

Structured review of more than 90 peer-reviewed studies (2013–early 2024) across optical, electrochemical, metal-oxide and NDIR sensors and across PM, O₃ and NO₂. Consensus findings: ML calibration cuts error by >50% and lifts R^2 above 0.8–0.9; **optical PM sensors post-calibration cluster at R^2 0.8–0.95 with slopes near unity under long-term co-location**, while electrochemical NO₂/O₃ sit at 0.6–0.9 with much larger site-to-site spread. Neural networks and gradient boosting beat simple regression when data are plentiful; regression is more stable when they are not. Persistent problems named: drift, absent standard protocols, poor cross-site generalisation, and regulator resistance to black-box models. **Limitation: the search ends in early 2024** and the review is narrative in its weighting, with no quantitative pooling. Useful as a citation anchor for the claim that transferability, not fit, is the open problem.

doi: 10.1108/sr-10-2025-0761

Yang et al. 2026 Atmos Chem Phys

Diurnal sea breeze worsens coastal air quality and complicates monitoring of remote North Atlantic air

Ten years of observations from Penlee Point Atmospheric Observatory on the northeast Atlantic coast, from which **428 diurnal sea-breeze events** were identified, most frequent in spring and summer under low wind and strong irradiance. Daytime surface O_3 , NO_x , CH_4 , CO_2 , SO_2 , total aerosol number, $PM_{2.5}$ and PM_{10} were all elevated during events, with mean daytime O_x near **45 ppb** and increased regulatory exceedance for O_3 . Accumulated ship-emitted SO_2 appears to drive new particle formation and a pronounced diurnal cycle in aerosol number. The monitoring payload: **excluding sea-breeze events reduces the diurnal amplitude and alters the seasonal cycle of O_3 and CH_4** under southwesterly flow — a coastal “background” site is conditionally not background. Limitation: one site, so the event classification and its consequences are not shown to transfer. Number-without-mass is precisely the regime a mass-based low-cost network cannot see.

doi: 10.5194/acp-26-10909-2026

Yang et al. 2026b Atmos Chem Phys

Dust transport versus local emissions in shaping urban ice-nucleating particles

Month-long winter campaign in Taiyuan, a heavily industrialised Chinese city: immersion-mode INP concentration and ice-nucleation active-site density alongside size distributions and chemical composition. N_{INP} $0.05\text{--}13.37\text{ L}^{-1}$ at and above -15°C . During a desert-dust event **7.47 L^{-1} (95% CI 6.64–8.41) versus 0.61 L^{-1} otherwise**, n_s 1.57×10^7 versus $9.52\times 10^5\text{ m}^{-2}$. During pollution periods N_{INP} correlated only weakly with sulfate, nitrate and OC ($|r| < 0.3$), and none of five PMF-resolved $PM_{2.5}$ source factors — industrial, dust, secondary, coal and traffic, fireworks — covaried significantly with it despite dominating the mass. One city, one winter, one dust event carrying the contrast. Still the cleanest recent demonstration that a $PM_{2.5}$ mass record does not index a co-located particle property.

doi: 10.5194/acp-26-11771-2026

Yaqoob et al. 2026 IEEE Access

Calibration that survives a change of city and a change of reference instrument

PurpleAir and Alphasense OPC-N3 in Antwerp, Oslo and Zagreb, hourly $PM_{2.5}$ and PM_{10} , calibrated with Random Forest, a compact feedforward network, and HybriCal-AQ (a hybrid NN taking a standardised out-of-fold RF hint). Models are evaluated separately against TEOM, FIDAS and generic reference grades to mirror real reference availability, with a train-first split/impute/encode pipeline to avoid leakage. HybriCal-AQ wins on $PM_{2.5}$ (Oslo-FIDAS/PurpleAir RMSE $1.79 \rightarrow 1.18$, R^2 $0.94 \rightarrow 0.97$) while **plain RF usually remains better for PM_{10}** (Antwerp 4.76 vs 6.58). A sparse-reference test withholding 25% of reference observations probes robustness. **Limitation: three cities is not a transferability test** — every model is still trained and evaluated within-city, so this bounds instrument sensitivity, not spatial generalisation. The explicit reference-instrument stratification is the transferable methodological idea.

doi: 10.1109/access.2026.3656120

Yaqoob et al. 2026 Sci Rep

Semiparametric hybrid air quality forecasting using Prophet ensemble and feature selection

Prophet models the trend and seasonality of daily $PM_{2.5}$; the residuals are then modelled by bagging, boosting, stacking and voting regressors, with Lasso, SHAP and recursive feature elimination applied to control the high-dimensional lag/interaction feature space. Tested on Seoul daily data and a high-dimensional simulated dataset against XGBoost, SVR and LSTM baselines; Prophet + boosting with Lasso and SHAP gave the lowest RMSE. The residual-modelling decomposition is a sensible way to keep an interpretable trend component, but the abstract reports no RMSE values, no forecast horizon and no comparison against a persistence baseline, which is the benchmark that actually discriminates in daily PM forecasting.

doi: 10.1038/s41598-026-64590-w

Yee et al. 2026 Atmos Meas Tech

UCB-GLOBES: open mass-spectral library for identified and unidentified organic aerosol

~27,000 GC/EI-MS spectra of trimethylsilyl derivatives of semi-volatile organic aerosol from US and Central Amazon ambient samples plus laboratory SOA, catalogued *without regard to known chemical identity* and annotated with sample provenance and predicted properties. Reanalysis of prior ambient data gave an 11% net increase in MS-based source categorisation for the southeastern US and lifted Central Amazon matches from 136 to 375 of 1,513 analytes. **~70% of UCB-GLOBES spectra match nothing** in the library or in NIST — the honest headline is the size of the unidentified fraction, not the size of the library.

doi: 10.5194/amt-19-5027-2026

Yohanna & Lim 2026 Chemosphere

Regional and seasonal $PM_{2.5}$ and ozone variation across Malaysia, XGBoost + SHAP

Daily DOE network observations 2018–2023 for five Malaysian regions, augmented with ERA5 boundary-layer height, solar radiation and cloud cover; XGBoost against a multiple linear regression baseline, interpreted with SHAP. $PM_{2.5}$ $R^2=0.576$, RMSE $4.87\text{ }\mu\text{g m}^{-3}$; O_3 $R^2=0.543$, RMSE 3.4 ppb — rising to $R^2=0.85$ only when persistence is included, which is the number to be sceptical of, since a persistence term will carry most of the skill in any daily-mean series. Kruskal-Wallis confirms regional ($\chi^2=3239.5$) and seasonal ($\chi^2=477.8$) structure. SHAP attributes PM variability to combustion tracers and meteorology, ozone chiefly to humidity and temperature. Useful as a tropical-monsoon reference for anyone tuning a regional model; not a calibration result.

doi: 10.1016/j.chemosphere.2026.145074

Yu et al. 2026b Indoor Air

Machine learning-based prediction of occupant symptom risk using indoor and outdoor environmental quality data in university dormitories

Seven monitoring periods across two Milan student residences, May 2024 to June 2025, pairing continuous indoor and outdoor environmental quality data with occupant symptom surveys. From **74 occupant samples** and 131 candidate predictors (reduced per symptom to 3–40 by permutation importance), seven binary symptom classifiers were fitted under leave-one-period-out cross-validation. Performance is weak and honestly reported: **mean AUC 0.595, range 0.398–0.705**, four of seven above 0.6; the higher combined accuracy figures appear only after dropping the two least specific symptoms. With $n = 74$ against 131 candidate predictors the feature-selection step is almost certainly overfitting even under LOPO-CV. The transferable finding is the feature ranking: **outdoor statistics and distributional shape features beat mean indoor concentrations**, which argues directly against summarising a sensor time series by its mean.

doi: 10.1155/ina/8230467

Yukhymchuk et al. 2026 Aerosol Air Qual Res

Low-cost particulate matter sensor performance under polluted conditions in Northeast China

Crossref deposit of 7 August with **no abstract in Crossref, Europe PMC or Semantic Scholar**. Carried on title, journal and DOI only; **no summary is offered because none can be written without inventing content**. On title alone this is the single most directly relevant record of the window — LCS performance characterisation at the high concentrations where light-scattering response is least well constrained — and it should be revisited as soon as an abstract appears.

doi: 10.1007/s44408-026-00159-6

Zeng et al. 2026 Atmos Environ*Aerosol vertical distributions determine the seasonal relationships between column AOD and surface PM_{2.5} over Europe*

Metadata only. No abstract deposited. The title states a result that matters directly to satellite-based PM_{2.5} estimation — the AOD-to-surface conversion being governed by vertical structure and therefore seasonally non-stationary — so this is the highest-priority record of the six for abstract retry or full-text retrieval on the next run.

doi: 10.1016/j.atmosenv.2026.122305

Zhang et al. 2026 Atmos Chem Phys*High frequency of urban new particle formation on the Tibetan Plateau*

In-situ particle number size distribution measurements at Tibet University, Lhasa (3,650 m), across spring and autumn campaigns, with days classified NPF / Undefined / Non-Event from PNSD evolution plus size-segregated counts and growth rates. NPF event frequency was **86.2% in autumn and 62.1% in spring** — extremely high by urban standards — and estimated CCN was $(2.3 \pm 1.5) \times 10^3 \text{ cm}^{-3}$ on NPF days versus $(1.3 \pm 0.7) \times 10^3$ on non-event days. Autumn events were associated with greater sulfur-precursor availability; no dominant control was identifiable in spring, which the authors state plainly. Relevant here as a reminder that number-based instrumentation resolves a regime mass-based monitoring cannot see at all; the high-altitude, low-condensation-sink setting limits transfer to lowland urban networks.

doi: 10.5194/acp-26-11473-2026

Zhang et al. 2026 Environ Sci Technol*Wearable device for direct assessment of particle deposition in the human respiratory system*

Face-mask-format sampler collecting inhaled and exhaled aerosol simultaneously, so deposition is measured by difference rather than inferred from a lung-deposition model. Bench validation confirmed exhaled-particle capture with minimal ambient interference; deployed on **12 healthy volunteers** in subway environments. **~66% of inhaled PM deposited**, highest for PM₁; Fe, Cu and Pb dominated the subway aerosol and deposition varied with lung function and exercise habit. Twelve subjects in one microenvironment is a feasibility study, not a dosimetry dataset, and the size-resolved difference method is vulnerable to hygroscopic growth of exhaled particles at 100% RH — which the abstract does not address. The device concept is nonetheless the most directly relevant instrumentation record of the issue: it measures the delivered dose that every ambient network is a proxy for.

doi: 10.1021/acs.est.6c05543

Zhang et al. 2026b Atmos Meas Tech*Impact of aerosol-type assumption on Landsat 8 atmospheric correction over land*

600 Landsat 8 scenes over 100 AERONET sites, comparing the Urban Clean aerosol type hardwired into the LaSRC surface-reflectance algorithm against MOD04- and MOD09-based dynamic types. **MOD04 gave the best AOD accuracy: $R^2 = 0.7236$, RMSE 0.0437, bias 0.0052.** Relevant here not for surface reflectance but because the same fixed-aerosol-type assumption propagates into satellite AOD products used as predictors in satellite-surface PM_{2.5} fusion models — an R^2 of 0.72 against AERONET is the ceiling any such predictor inherits. AERONET sites are systematically non-representative of high-PM urban environments, so this is likely optimistic.

doi: 10.5194/amt-19-5281-2026

Zhang et al. 2026c Atmos Environ*Characterisation of air pollution transport using absorbing aerosol index from multiple payloads*

Abstract not deposited at Crossref on the entry date. Multi-payload AAI synergy is a transport-attribution method rather than a surface-concentration one; logged for completeness. Queued for the Saturday sweep.

doi: 10.1016/j.atmosenv.2026.122298

Zhang et al. 2026c Atmos Chem Phys*Buffering of atmospheric nanoparticle growth by temperature-dependent shifts in molecular composition, volatility and diffusivity*

A kinetic multiphase treatment applied to previously unreconciled field and chamber nanoparticle growth-rate data. The standing puzzle is that observed growth rates are smaller than theory predicts and *less* sensitive to condensable vapour concentration than theory predicts. The resolution offered is two antagonistic temperature dependencies: **slow surface-to-bulk transport limits vapour uptake by semi-solid, low-diffusivity particles, while Clausius–Clapeyron shifts in the volatility distribution enhance growth at low temperature and low concentration.** Netted out, they buffer the vapour-concentration dependence, which is why measured growth rates converge on a few nm/h across widely varying conditions. No new measurement is reported — this is a reinterpretation of existing datasets, and its persuasiveness rests on the volatility-distribution parameterisation rather than on independent validation. The relevance here is upstream of sensing: the 50–100 nm photochemical mode that Zhong et al. flagged yesterday as invisible to mass-based assessment is produced by exactly this growth process, and a mechanism that makes its rate insensitive to precursor loading also makes it insensitive to the emission controls that reduce PM_{2.5} mass.

doi: 10.5194/acp-26-12037-2026

Zhang et al. 2026h Atmos Chem Phys*Contrasting nighttime heterogeneous and daytime photochemical aging drive the optical evolution of black carbon*

Integrated SP2 and single-particle aerosol mass spectrometry over an urban Shenzhen campaign, mean refractory BC **$1.2 \mu\text{g m}^{-3}$** , core mass median diameter 155–170 nm. The coating-to-core mass ratio rises continuously through the diurnal cycle by two mechanistically distinct routes — nighttime heterogeneous coating growth with enhanced nitrate signal, and daytime photochemical aging with rapid nitrate accumulation followed by sulfate and oxidised organics. Both converge on similar mass absorption cross section enhancement at 532 nm: **$\sim 0.8 \text{ m}^2 \text{ g}^{-1}$ overnight, $\sim 1.0 \text{ m}^2 \text{ g}^{-1}$ by day.** No uncertainty is given on the MAC enhancements and it is a single city over one campaign, so the magnitudes should not be transported. Directly relevant to distributed sensing: any optical channel with a fixed absorption or scattering assumption re-expresses this aging cycle as concentration drift with a diurnal period.

doi: 10.5194/acp-26-12197-2026

Zhao et al. 2026 Biomed Environ Sci*WHA air-LSTM: composite morbidity-driven pollution index for next-day outpatient forecasting*

A five-level composite index is built *per city* from city-specific exposure–response relationships across >223.7 million hospital visits, then fed with temperature, humidity and visit history into an LSTM predicting next-day respiratory outpatient load. Beijing $R^2 = 0.963$, RMSE 53.5; removing the index degrades RMSE by 44.1%; performance holds on held-out cities. The index is fitted on morbidity and then used to predict morbidity, so the incremental-value claim is partly circular — the external validation is what carries it. Attractive template for turning a dense sensor network into an operational early-warning product.

doi: 10.3967/bes.2026.062

Zhao et al. 2026 J Aerosol Sci*Monte-Carlo inversion of aerosol complex refractive index from multi-wavelength lidar*

Metadata only. Elsevier returned no abstract through Crossref and ScienceDirect did not serve the record; no finding is asserted. The registered title describes an improved Monte-Carlo sampling algorithm that inverts the aerosol complex refractive index from multi-wavelength lidar signals combined with an independently measured particle size distribution. Carried at tier C because refractive index is the parameter that converts an optical measurement into a mass estimate, and every low-cost nephelometric sensor assumes one implicitly; whether this inversion is well-posed cannot be judged from the title.

doi: 10.1016/j.jaerosci.2026.106883

Zhivkov et al. 2026 Atmosphere*How fast does a calibration decay with distance from its reference station?*

Twenty-four months of hourly data (Aug 2023–Jul 2025) from Sofia: five official reference stations and 22 AirThings low-cost units, four co-located. Random Forest calibration achieved R^2 0.53–0.75 across $PM_{2.5}$, PM_{10} , NO_2 and O_3 , improving on multiple linear regression by 40% (O_3) to 408% ($PM_{2.5}$). Leave-one-station-out spatial cross-validation yields pollutant-specific uncertainty growth rates α of **3.84–5.62% per km** (significant for PM_{10} and O_3 only, $p < 0.05$), applied to 18 non-co-located sensors to generate 1.2 M calibrated hourly values with 95% prediction intervals. **Limitation:** α is estimated from five stations in one city, so its significance for $PM_{2.5}$ is not established and the linear-in-distance form is an assumption. Still the most directly usable network design result in the window: 6 km anchor spacing keeps the midpoint uncertainty increase below 30%.

doi: 10.3390/atmos17040335

Atmos Environ 2026a METADATA ONLY Atmos Environ*Reliability-constrained identification of primary and elevated aerosol structures from a single tilted Mie lidar under dust-influenced conditions*

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar. Listed, not summarised. Carried because the title states a specific and testable capability: retrieving both the primary and the elevated aerosol layer from a *single tilted* Mie lidar, with an explicit reliability constraint, under dust loading that normally defeats single-wavelength inversion. Single-instrument retrieval of layer structure is the difference between a research lidar and one a network can afford, and dust is the condition under which the retrieval usually fails. Whether the paper delivers that is unknown until an abstract appears.

doi: 10.1016/j.atmosenv.2026.122297

Zundel et al. 2026 (preprint) medRxiv*Wearable personal $PM_{2.5}$ monitoring during wildfire smoke episodes in Metro Detroit*

AirBeam3 wearables on four participants (one adolescent, three adults) during southeast Michigan smoke episodes, with neighbourhood outdoor $PM_{2.5}$ taken as the mean of the three nearest PurpleAir units and smoke days identified from state advisories. **Personal 28.2 ± 20.0 vs $5.1 \pm 4.7 \mu g/m^3$** on smoke vs non-smoke days; neighbourhood **87.6 ± 80.2 vs 12.3 ± 7.0** . Within participants, each $10 \mu g/m^3$ of neighbourhood $PM_{2.5}$ mapped to only **$2.3 \mu g/m^3$** personal — personal exposure is both attenuated and poorly tracked by the outdoor proxy, yet still put >25% of participant-days over the EPA 24 h standard. $n = 4$ and no formal uncertainty on the slope: this is a feasibility demonstration, and the attenuation factor should not be reused as a calibration constant. Preprint, not peer reviewed.

doi: 10.64898/2026.07.30.26359253

d'Hondt et al. 2026 Research Square (preprint)*Neural downscaling of air-quality simulations requires structural correction before spatial refinement*

Deep-learning downscaling of CAMS-class chemical transport model output, which is capped near 10 km for continental forecasts. The argument is that existing approaches either treat the coarse field as a smoothed copy of the fine field or learn a direct coarse-to-fine map, and both conflate two different things: cross-resolution differences in *model behaviour*, and genuinely missing spatial detail. Separating a structural correction step from the super-resolution step is what the paper proposes and tests. Directly transferable to LCS-plus-CTM fusion, where the network supplies the detail the model lacks and the model supplies the physics the network lacks — but a bias-corrected coarse field is the precondition. Preprint, not peer reviewed; carried on relevance, and no numeric claim from it is used in this issue's analytics.

doi: 10.21203/rs.3.rs-10627504/v1

NOT CARRIED BY A DAILY ISSUE — EXTRACTIVE PRECIS OR METADATA ONLY, NOT AN ASSESSED SUMMARY (2 RECORDS)

Park et al. 2026 Atmos Environ*Diagnosing explainability of machine learning: transformer-based $PM_{2.5}$ estimation in Seoul*

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1016/j.atmosenv.2026.122264

Zheng et al. 2026 Atmos Environ*Augmented transformer framework for $PM_{2.5}$ chemical composition via one-step LDI single-particle aerosol mass spectrometry*

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1016/j.atmosenv.2026.122260

Exposure assessment & modelling

129 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (128 OF 129 RECORDS IN THIS SUBTOPIC)

Abadi et al. 2026 Atmos Res*Future projections of surface dust concentrations in major dust source regions of the Mesopotamian plains*

Metadata only. Crossref sweep, no abstract deposited. Projected mineral-dust concentrations for a source region that exports coarse PM across the Gulf and into South Asia; no scenario, horizon or model family is extractable from the deposited metadata.

doi: 10.1016/j.atmosres.2026.109282

Adeniji et al. 2026 PLOS Glob Public Health*Twelve years of hourly criteria-pollutant and meteorological coupling in Abeokuta, Nigeria*

Hourly $PM_{2.5}$, PM_{10} , NO_2 , SO_2 , CO and O_3 over 2012–2023 with six meteorological covariates. **$PM_{2.5}$ and PM_{10} dominate and persistently exceed WHO guidelines**, peaking in the December–March Harmattan; regression against temperature gives R^2 0.48–0.52 with negative coefficients on rainfall and humidity, consistent with thermal stagnation and wet scavenging. The aggregate AQI sits at 39–51 (“moderate”) while the health-risk-adjusted index exceeds 90 seasonally — **a 2-fold divergence between the two indices at identical concentrations**, which is the paper's real contribution and an argument for HAQI-style reporting on any network serving a PM-dominated airshed. Single station for a metropolitan area, and the provenance and calibration history of a 12-year record in a data-sparse setting are not documented.

doi: 10.1371/journal.pgph.0006683

Agpaoa et al. 2026 Environ Monit Assess*Occurrence, temporal variation and meteorological drivers of airborne fungal spores in Laoag City, northern Philippines*

First aeromycological record for a monsoon-influenced coastal site in northern Luzon: 216 sampling days, March to November 2022, cumulative index **190,096 spores m⁻³**, dominated by *Cladosporium* and *Fusarium*-like types. Wet-season sampling carries most of the cumulative load but the highest daily concentrations fall in the dry season, and diurnal peaks sit at 00:00–06:00. Positive associations with relative humidity and rainfall, negative with temperature, and no significant wind association. Single site, single year, morphology-based identification only, and no PM co-measurement — so the bioaerosol fraction cannot be placed within a mass-based PM budget. Included because overnight spore peaks are a documented confounder for optical particle counters in humid climates.

doi: [10.1007/s10661-026-15814-z](https://doi.org/10.1007/s10661-026-15814-z)

Alaimo et al. 2026 Atmosphere*Elemental composition and Pb isotopic signatures in pine needles from the industrialized Milazzo area*

Forty *Pinus pinea* needle samples by ICP-OES/ICP-MS across 36 elements, 25 of them also for ²⁰⁶Pb/²⁰⁷Pb and ²⁰⁸Pb/²⁰⁶Pb. Enrichment factors above 10 for Cd, Cu, Mo, Sb, V and Zn against EF 0.5–2 for Al, Fe and Ti cleanly separate anthropogenic from geogenic contributions. Pb ratios of 1.153–1.192 and 2.063–2.108 indicate mixing of industrial emissions with local geological background and, notably, *limited* residual leaded-gasoline signal. A good template for isotope-assisted source apportionment where no monitoring network exists.

doi: [10.3390/atmos17080805](https://doi.org/10.3390/atmos17080805)

Albugami 2026 Atmosphere*Dust event characteristics, transport pathways and source regions over Riyadh*

34 dust events over 2024–2025 identified from AERONET by a dual optical criterion (AOD₅₀₀ ≥ 0.50 and 440–870 nm Ångström exponent ≤ 0.50), cross-checked against NOAA ISD present-weather and visibility and traced with HYSPLIT back-trajectories, PSCF and CWT. **82% fell in March–May**; mean Ångström exponent 0.28, mean fine-mode fraction 0.26. Surface present-weather codes confirmed 26 of 34; the other 8 (24%) were classified as elevated dust invisible at the surface station — a clean demonstration that column and surface instruments disagree by design, not by error. Source–receptor analysis points to the Mesopotamian Basin, with the northern Rub’ al-Khali secondary. Single-author, single-station, two years.

doi: [10.3390/atmos17080802](https://doi.org/10.3390/atmos17080802)

Alshehri & El-Hamid 2026 (preprint) Research Square*Sentinel-5P and MODIS land-surface coupling of pollutant fields over Riyadh, 2018–2025*

Eight years of TROPOMI NO₂, SO₂, CO, O₃ and Absorbing Aerosol Index combined with MODIS land-surface temperature and NDVI and Sentinel-2 land cover in Google Earth Engine, analysed by correlation, PCA and multiple linear regression. NO₂ and CO concentrate over urban and industrial areas, O₃ peaks downwind, LST falls from 43.10 °C (2021) to 37.07 °C (2025) while NDVI rises after 2022. **The regression is weak: R² ranges from 0.124 (O₃) to 0.231 (SO₂)**, and PCA explains 55.8% of variance; the authors correctly conclude that land-surface descriptors explain only a minor share and that emissions and transport dominate. Carried as a negative methodological result for an arid-region setting where ground networks are sparse — and as a caution, because **no surface PM product appears anywhere in the study** despite the framing as air pollution monitoring; AAI is a column absorption index, not a PM concentration. Preprint, not peer reviewed, and the LST trend is reported without any test for its significance against interannual variability.

doi: [10.21203/rs.3.rs-10538140/v1](https://doi.org/10.21203/rs.3.rs-10538140/v1)

Anon et al. 2026b Environ Int*Condensable particulate matter and severe winter PM_{2.5} in remote Canadian mountain communities*

Metadata-only deposit — title and DOI without an abstract. The topic is directly relevant and under-covered: condensable PM forms downstream of the stack as vapour cools, is invisible to filter-based source testing conducted at temperature, and in subzero valley-inversion conditions can dominate ambient PM_{2.5} in communities with no industrial source of any size. If the finding is what the title implies, it also bears on optical sensing — semi-volatile condensable material has different scattering and hygroscopic behaviour from the combustion soot most correction factors assume. Carried at tier C and flagged as the top abstract retry.

doi: [10.1016/j.envint.2026.110472](https://doi.org/10.1016/j.envint.2026.110472)

Anon et al. 2026c Atmos Pollut Res*Source-specific PM_{2.5} oxidative potential in a national shipbuilding complex*

Metadata-only deposit. The title asserts that metal processing and combustion dominate source-apportioned oxidative potential, which if supported is another instance of the recurring result that OP-weighted and mass-weighted source contributions rank differently. No apportionment method, OP assay, site count or period is available. Tier C, second on the abstract retry list.

doi: [10.1016/j.atm.2026.103189](https://doi.org/10.1016/j.atm.2026.103189)

Ashta et al. 2026 Atmos Chem Phys*Wet and dry atmospheric deposition of microplastics across a Swiss site gradient*

Four-weekly wet and dry deposition sampling over one year (May 2024–May 2025) at urban (Zurich), suburban (Dübendorf), rural (Magadino, Payerne) and mountainous (Chaumont) sites, with focal-plane-array μ -FTIR over the 20–215 μ m range and an explicit measurement-uncertainty treatment. Urban deposition **881 MP m⁻² d⁻¹ (95% CI 562–1199)** and **53 μ g m⁻² d⁻¹ (17–107)**; the other four sites are similar to each other at 249–331 MP m⁻² d⁻¹ (140–478). Scaled to land use, 219 t or 3.8×10^{14} particles per year below 2000 m nationally, excluding tyre wear. The 20 μ m lower cut is the binding limitation: the inhalable fraction sits mostly below it.

doi: [10.5194/acp-26-11803-2026](https://doi.org/10.5194/acp-26-11803-2026)

Awan et al. 2026 Aerosol Air Qual Res*BRIQ-PM_{2.5}: quantile-mapping attribution of industrial excess PM_{2.5} across Pakistan*

Proposes estimating industrial impact as the residual between an observed satellite PM_{2.5} distribution and the same distribution quantile-mapped onto a province-specific “low-industry” baseline city, then interpolating the residual by variogram modelling and universal kriging across four provinces. Reported validation is distributional — histogram and ECDF alignment, improved RMSE after correction. **The attribution is definitional, not estimated**: whatever separates a target city from its chosen baseline is labelled industrial, so meteorology, dust, biomass burning and population density all load onto the same term, and no ground-monitor comparison or sensitivity analysis over baseline choice is reported. Useful as a hotspot screen; not an emissions attribution.

doi: [10.1007/s44408-026-00160-z](https://doi.org/10.1007/s44408-026-00160-z)

Baldo et al. 2026 Atmos Chem Phys

Chemical and mineralogical composition of Southern African mineral dust

First comprehensive characterisation of dust from a major and under-sampled source region. Laboratory-generated aerosol from soils at natural and emerging anthropogenic source areas, characterised by Si/Al, (Ca+Mg)/Al and K/Al ratios plus clay content. Those ratios separate arid western coastal sources from humid eastern inland sources and encode source-area weathering regime. The K-feldspar finding matters beyond composition — K-feldspar is the dominant mineral ice-nucleating agent, which links this record directly to the PINE/PIA instrumentation above. Laboratory-generated, so size distribution and mineralogy need not match what is actually emitted and transported.

doi: 10.5194/acp-26-10727-2026

Bhardwaj & Subramanian 2026 arXiv (preprint)

Identifiability-aware source apportionment for sparse city-scale sensor networks

Reframes inventory-constrained apportionment as a wind-conditioned lagged inverse problem: time-varying source activity is represented on a low-dimensional nonnegative temporal basis, each source-basis coefficient produces a sensor-time fingerprint, and a separate low-dimensional background space is projected out. **Exact identifiability requires full column rank of the projected lagged response matrix**; noise-robust attribution is governed by its singular values, coefficient visibility, background absorption, pairwise coherence and ray distance. The practical payoff is diagnostic — you can test before deployment whether a proposed sensor geometry can separate two inventory groups at all. Preprint, not peer reviewed.

doi: 10.48550/arXiv.2608.00050

Blanco et al. 2026 J Expo Sci Environ Epidemiol

Influence of on-road mobile monitoring design on ultrafine particle exposure models and cognitive health inferences

A reference roadside UFP model built from repeated measurements at 309 locations was used with the Adult Changes in Thought cohort ($n = 5,283$) to test late-life cognitive function, then **600 km of on-road measurements were subsampled** to emulate alternative campaign designs varying visit frequency, spatial balancing and sampling time, with universal kriging plus partial least squares. The reference model gave a null: **CASI-IRT +0.007 (95% CI −0.013 to 0.027) per 1,900 pt/cm³**. Alternative designs reproduced the null but attenuated it; route-based sampling and very short 4-visit campaigns produced the most variable estimates, while temporal and plume adjustments changed little. Testing design sensitivity against a null limits what can be concluded — attenuation toward zero is unobservable when the reference is zero — and the authors say so. The design guidance, that visit count and out-of-hours coverage matter more than post-hoc adjustment, is what carries.

doi: 10.1038/s41370-026-00845-y

Blanco-Alegre et al. 2026 Atmos Chem Phys

Emission factors and particle morphology from prescribed fires in Mediterranean shrubland

Field campaign on *Calluna vulgaris* and *Genista hispanica* burns in north-west Spain combining gravimetry, thermal-optical transmittance, ion chromatography, FTIR and SEM. *Calluna* burned flaming (MCE 90.6%), *Genista* smouldering (MCE 70.8%); gas-phase concentrations were higher for *Calluna* but emission factors for incomplete-combustion products were higher for *Genista*. **PM_{2.5} EFs 53.5 g kg^{−1} (*Calluna*) and 44.7 g kg^{−1} (*Genista*)**, with OC and EC at 28.1% and 32.9% of PM_{2.5} mass and water-soluble inorganic ions at 6.9% and 4.5%. SEM showed abundant tar-ball-like submicron particles and thickly coated aggregates. Two vegetation types at one site — the EFs are not a regional average — but combustion-regime-resolved and directly usable.

doi: 10.5194/acp-26-11695-2026

Blanco-Villafuerte et al. 2026 (backfill) Curr Environ Health Rep

Hybrid low-cost-sensor, satellite and chemical-transport modelling of ambient PM_{2.5} in Peru

Review of national PM_{2.5} exposure-model construction in LMICs, anchored on the NIH-funded Peru GeoHealth Hub's **176-monitor PurpleAir network** — 62.5% urban, 37.5% rural, across all 24 regions, established with national environmental agencies. The hybrid model fuses LCS, satellite retrievals, chemical transport output and machine learning; initial application to Lima reached **R² = 0.88** against regulatory monitors for 2010–2023, with extension to daily national 5 km² fields for 2024–2026 in progress. R² against regulatory monitors in a single city is a weak validation for a national product, and the review is written by the network's own investigators. The deployment-ratio and institutional-integration detail is the transferable content. Crossref created 5 January 2026 — backfill, exclude from trend lines.

doi: 10.1007/s40572-025-00508-4

Boogaard et al. 2026 Curr Environ Health Rep

Advancing long-term outdoor air pollution exposure assessment for health studies

Overview of the completed HEI multinational initiative on novel long-term exposure assessment across North America and Europe. Exposure predictions from models that account for daily mobility patterns and outdoor–indoor infiltration correlated **>0.7** with conventional residential estimates, and the health effect estimates they produced were similar — differing in magnitude but *not direction*. The paper is honest that this is a comparison among models, not a validation against measurement: representative validation data for long-term exposure simply does not exist, which is the initiative's own headline gap. Read as the strongest current argument for spending resources on validation campaigns rather than additional model structure, and as a caution against over-reading small differences between LUR variants in cohort re-analyses.

doi: 10.1007/s40572-026-00560-8

Buhendwa et al. 2026 Environ Pollut

Modelling the coupled PM_{2.5}–ozone system in China: mechanisms, model biases, perspectives

Review of the compound-pollution era in which PM_{2.5} and O₃ share a precursor pool (VOCs, NO_x, SO₂, NH₃) and single-pollutant control can worsen the co-pollutant — the “pollution seesaw”. The useful content is the bias inventory: chemical transport models **underestimate secondary organic aerosol by about 40%**, with substantial spread across models, seasons and regions; daytime HONO sources and OH reactivity are missing or understated; ammonia and episodic agricultural-burning inventories are uncertain; nocturnal boundary-layer mixing is poorly represented. These errors propagate into data assimilation and into machine-learning models trained on CTM output. Limitation: a narrative synthesis with no quantitative pooling, so the 40% figure is a reported central tendency, not a meta-analytic estimate. Directly relevant if CTM fields are used as priors or features in a sensor-fusion product.

doi: 10.1016/j.envpol.2026.128887

Cai et al. 2026 (ACP) Atmos Chem Phys

Molecular characteristics and formation pathways of particulate organosulfur compounds in Shanghai

Paired urban and suburban Shanghai sampling with high-resolution mass spectrometry, resolving **1,964, 1,914 and 2,689 organosulfur molecular formulas** across the sampled conditions and separating spatial, seasonal and day–night variation in formation pathway. Organosulfates are ubiquitous but their production regime is not: the comparison across contrasting sites is what isolates the pathways rather than the inventory itself. The limitation that matters for downstream use is that molecular formula assignment from exact mass does not fix structure, so pathway attribution is inferential. Relevant to sensing as a concrete demonstration that a large, chemically distinct fraction of urban PM_{2.5} mass is invisible to any mass-only instrument.

doi: 10.5194/acp-26-11067-2026

Cai et al. 2026 Atmos Environ

Effect of ammonia on reactive oxygen species generation in secondary organic aerosols

Metadata only. Crossref sweep, no abstract deposited, and no study location stated. Ammonia-modulated oxidative potential of secondary organic aerosol sits precisely between the previous issue's Nanjing ammonia-thermodynamics result and today's oxidative-priming mechanism papers, so it is the second-highest-priority retry after Zeng et al.

doi: 10.1016/j.atmosenv.2026.122307

Chai et al. 2026 Environ Pollut

Role of ammonia, aerosol pH and liquid water content in secondary inorganic aerosol formation in suburban Nanjing

Hourly water-soluble ions and precursor gases, January–March 2021, coupled to ISORROPIA-II and WRF-Chem. SIA is **49.55% of PM_{2.5} mass** (mean 34.4 $\mu\text{g}/\text{m}^3$), nitrate-dominated. Continuous ammonia excess fully neutralises the anion pool; rising aerosol liquid water enhances aqueous oxidation and drives nitrate and ammonium into the particle phase, while pH governs both semi-volatile partitioning and the sulfate pathway (transition-metal-ion catalysis dominant). Thermodynamically the system is **total-nitrate-limited**: PM_{2.5} responds minimally to moderate NH₃ cuts and needs >60% reduction to move, whereas cutting HNO₃ alone matches joint HNO₃+NH₃ control. The policy inversion — prioritise NO_x, treat NH₃ as a local adjunct — is the transferable result.

doi: 10.1016/j.envpol.2026.128998

Chebaicheb et al. 2026 Environ Int

A phenomenology of submicron organic aerosol components in urban France

The first nationwide harmonised organic-aerosol source apportionment for France: time-resolved Aerosol Chemical Speciation Monitor measurements at 12 (sub)urban sites over one to seven years each, with a single rolling PMF protocol applied under common constraints and acceptance criteria across all datasets. Primary factors — hydrocarbon-like, biomass-burning and cooking-like OA — account for **≈33% of OA mass**; oxygenated OA dominates at every site and splits into less- and more-oxidised components everywhere except Poitiers and Strasbourg. Site-specific factors resolved a mixed shipping/industrial component at Marseille Longchamp and an amine-related component at Strasbourg and Creil. Wintertime LO-OOA correlates with BBOA, implicating residential combustion in secondary as well as primary mass; summer LO-OOA tracks biogenic precursors. **CHIMERE shows systematic component-level bias against the observations.** The harmonisation is the contribution — rolling PMF applied inconsistently is why the existing European OA literature does not pool — and the limitation is that (sub)urban sites only cannot constrain the rural background these factors are transported through.

doi: 10.1016/j.envint.2026.110453

Chen et al. 2026 Environ Sci Technol

Inventory uncertainty and transboundary transport of biomass-burning black carbon in southern China

Year-round radiocarbon measurement of biomass-burning black carbon at two regional and one background site in southern China, compared against tailored atmospheric-transport simulations. The model reproduces seasonality but the normalised mean bias spans **−68% to +28%** — a factor-of-two-plus spread in a quantity inventories report without error bars, attributed largely to unmonitored biomass burning in mainland China. Residential burning across East Asia dominates; Southeast Asian transport matters in summer and, via free-tropospheric pathways, **persists in winter** when continental air masses are assumed to exclude it. *Limitation*: three sites cannot constrain a regional inventory, and ¹⁴C apportionment inherits uncertainty from the assumed fossil/non-fossil end-members. Relevance: this is the cleanest current demonstration that source-diagnostic *measurement*, not finer model resolution, is what bounds inventory error — the argument for putting source-resolving capability into distributed networks.

doi: 10.1021/acs.est.6c00449

Chen et al. 2026f Toxics

Source-specific oxidative potential of PM_{2.5}: water-soluble metals via DTT and interpretable ML

Winter and summer ambient PM_{2.5} in Xi'an assayed for DTT oxidative potential, with XGBoost–SHAP attribution over composition and receptor-model source contributions. **Volume-normalised OP (DTT_v) was markedly higher in winter**, and although water-soluble metals are a small fraction of PM_{2.5} mass, **potassium and manganese dominated the OP attribution** — a biomass-burning tracer and a redox-active transition metal. Source apportionment separates the two seasons cleanly: dust drives winter OP through mass loading, traffic drives summer OP through intrinsic per-mass toxicity. The abstract reports no numeric DTT_v values, confidence intervals or model performance, which is a real gap for a paper whose central claim is quantitative attribution; SHAP importance is also not a causal decomposition and correlated constituents will trade rank. The qualitative conclusion — OP tracks composition and source, not mass — is the second independent instance of it this week.

doi: 10.3390/toxics14080646

Chi et al. 2026 Atmos Chem Phys

Heterogeneous nitrosation of amines by N₂O₄: a missing source of particulate nitrosamines

Born–Oppenheimer molecular dynamics plus metadynamics on methylamine and dimethylamine reacting with N₂O₄ at the air–water interface. Two competing routes: a *barrierless* direct N-nitrosation yielding nitrosamine cations and NO₃[−], and an amine-mediated N₂O₄ hydrolysis generating interfacial HONO in **2–16 ps**, which then nitrosates over a 7.65 kcal mol^{−1} barrier at 300 K. Purely computational, two amines, idealised interface — there is no measurement here and no yield that a chemical transport model can use directly. Its value is as a mechanism for particulate carcinogen formation that current CTMs omit entirely.

doi: 10.5194/acp-26-12261-2026

Chillrud et al. 2026 Environ Epidemiol

pcpr: principal component pursuit for exposure pattern recognition and outlier detection

R package implementing PCP with three environmental-health extensions: a non-negativity constraint on the low-rank matrix, missingness handling, and a specialised penalty for below-LOD observations. Demonstrated on 26 PM_{2.5} constituents measured every few days in Queens, NYC, 2015–2021: four patterns recovered (secondary/tailpipe, traffic, crustal dust, salt) plus **211 outlying events** including July Fourth firework spikes. PCP's sparse component is an outlier detector, not a source; calling the 211 events “exposure events” presumes an interpretation the decomposition does not supply. Still the most immediately usable artefact in today's corpus for anyone holding speciated network data.

doi: 10.1097/EE9.0000000000000515

Chuang et al. 2026 Atmos Res

Simulating PM_{2.5} in southern Taiwan under high-pressure return flow (Kao-Ping Experiment)

METADATA ONLY — a KPEx modelling study of a synoptic regime that traps PM_{2.5} over the Kao-Ping region. No abstract in the deposit and no indexed full text on 20 August. No quantity inferred.

doi: 10.1016/j.atmosres.2026.109286

Cui et al. 2026 Environ Pollut

Daily 1 km satellite-based PM_{2.5} exposure estimates in Bangladesh, 2020–2025

Two-stage machine learning over one of the most monitoring-sparse high-burden regions in the world. Stage one reconstructs MAIAC aerosol optical depth, of which **more than 60% of potential retrievals are missing**, with LightGBM; stage two feeds the gap-filled AOD plus meteorological, composition and geospatial predictors into an attention-based TabNet to estimate ground PM_{2.5}. Independent test performance $R^2 = 0.95$, **RMSE 17.41 $\mu\text{g}/\text{m}^3$, MAE 9.38**, against a national annual mean of **68–75 $\mu\text{g}/\text{m}^3$** and winter peaks above 130. The limitation is structural: with almost no ground reference network, the held-out set is small and drawn from the same sparse monitors used in training, so $R^2 = 0.95$ describes interpolation skill more than transferability, and the DHS linkage is division-level and ecological.

doi: 10.1016/j.envpol.2026.129051

Defrance 2026 (preprint) Research Square

Beyond the fire perimeter: the smoke footprint that constrains outdoor activity dwarfs the burned area of the July 2026 Gironde–Landes wildfire episode

ERA5-Land wet-bulb globe temperature combined with CAMS analysis PM_{2.5} at hourly resolution over 25–31 July 2026, against 40 background monitoring stations. The two fires burned 340 km² and forced ~220,000 evacuations; the footprint above the European AQI “Poor” threshold (hourly PM_{2.5} > 50 $\mu\text{g}/\text{m}^3$, where reduced outdoor activity is advised) reached **149 times the burned area** and about **370 km downwind**, with 139 $\times$ on the EU 2030 daily limit of 25 $\mu\text{g}/\text{m}^3$ and at least 121 $\times$ against the larger operational fire perimeter. Plume co-location rose from 73% hourly to 94% daily. At the stations, a same-day heat–smoke relay affected 28% of affected station-days at 25 $\mu\text{g}/\text{m}^3$ and 19% at 50, against 0–2% in the co-located grid cells — **the 10 km analysis smooths away the peaks that drive the relay**, and reproduces the episode at daily $r = 0.83$ while underestimating peaks by a median factor of ~2.2. Preprint, single author, single episode; the self-directed error analysis is unusually thorough for the format.

doi: 10.21203/rs.3.rs-10757961/v1

Deng et al. 2026 Atmos Chem Phys

Impact of black carbon on daytime valley and slope winds in idealised simulations

High-resolution idealised WRF–Chem experiments, run with and without radiatively interactive black carbon, on thermally driven valley circulation. Absorption warms the upper boundary layer and cools lower levels, increasing stability, reducing surface heating, and leaving up-slope winds **weaker and confined to a shallower layer**. The feedback is self-reinforcing: the circulation that ventilates a valley is suppressed by the aerosol it would otherwise disperse. *Limitation*: two idealised simulations with prescribed BC, no observational constraint, no real terrain — the sign of the effect is credible, the magnitude is not transferable. Relevance to monitoring: in mountain basins the relationship between emissions and measured concentration is not linear, and siting a network to sample a circulation that the pollution itself modifies requires the vertical dimension, not a denser surface grid.

doi: 10.5194/acp-26-11171-2026

Dias et al. 2026 Bull Environ Contam Toxicol

Micro-scale spatial patterns of PM_{2.5} across urban green and non-green areas in an arid city

Six mobile monitoring campaigns along a fixed 4.3 km transect with **75 georeferenced sampling points** in San Juan, Argentina, winter 2026, classified green versus non-green by surrounding vegetation cover. Generalised linear mixed models adjusting for traffic intensity, distance to nearest road and repeated measurement give **~31% lower adjusted PM_{2.5} in green spaces**. The adjustment set is the weak point — traffic intensity and distance-to-road are the obvious confounders but green spaces in an arid city also differ in surface dust availability and irrigation, which is an unadjusted pathway that would produce the same sign. A single winter and a single transect. Useful mainly as a demonstration that mobile monitoring resolves within-block gradients that a regulatory network cannot, in a South American arid setting where almost no such data exist.

doi: 10.1007/s00128-026-04318-8

Ding et al. 2026 Geo-spat Inf Sci

Urban–rural balanced estimation framework for PM_{2.5} exposure across China

Targets a structural bias rather than a model-fit problem: because ground stations cluster in cities, AOD-to-PM_{2.5} retrievals systematically overestimate rural concentrations. Integrates AOD, reanalysis fields and ground measurements under data-assimilation principles, adaptively constraining non-urban estimates with low-weighted reanalysis samples, then ensemble-averages across dynamically weighted background/observation settings. China 2015–2021: **clean-site RMSE 18.18 $\rightarrow$ 8.31 $\mu\text{g}/\text{m}^3$, national mean PM_{2.5} down ~55%**, with corresponding correction of PM-attributable mortality in low-density areas. The 55% figure is the paper’s own counterfactual, validated at clean sites that are still station locations — true rural accuracy remains unvalidated by construction. If even partly right, it implies an inherited upward exposure bias across the satellite-based exposure–response literature.

doi: 10.1080/10095020.2026.2624335

Do et al. 2026 Environ Sci Technol

Field-constrained diurnal viscosity of urban PM_{2.5} and implications for N₂O₅ uptake

Poke-and-flow measurement with fluid-dynamics inversion on PM_{2.5} collected in Ansan, South Korea (summer 2024): **6.2 $\times$ 10⁴–1.3 $\times$ 10⁷ Pa s at 20–40% RH**, exceeding 10⁸ Pa s below ~10% RH. Pooled with Seoul and Beijing data into a Vogel–Tammann–Fulcher temperature–RH parametrisation and pushed through an ML surrogate over hourly meteorology for 14 megacities. Predicted viscosity is systematically lower at night everywhere (RH plasticisation beating nocturnal cooling), giving nighttime N₂O₅ uptake coefficients of **0.01–0.07** against the conventional liquid assumption of 0.1. Weakness stated by the authors: the parametrisation rests on 24 h composition, so no time-resolved chemistry enters the diurnal prediction it is being used to make.

doi: 10.1021/acs.est.6c04811

Dovrou et al. 2026 Environ Sci Atmos

Wintertime wildfires reshape urban atmospheric chemistry: the January 2025 Los Angeles fires

METADATA ONLY — the deposited abstract is truncated before any result. Verifiable from it: the study concerns how dominant chemical regimes and meteorology govern wildfire air-quality impacts in urban coastal regions, using the January 2025 Los Angeles fires. Flagged for retrieval given how much of this watch’s recent health signal has come from source-resolved wildfire PM_{2.5}.

doi: 10.1039/d6ea00096g

Duan et al. 2026 Atmos Chem Phys

Regime-dependent aerosol pH in ammonia-rich urban Beijing from explainable machine learning

Field observations plus thermodynamic modelling, interrogated with interpretable ML to apportion pH variability. Temperature dominates, at roughly -0.6 pH units per 10°C . Excess ammonia, nitrate-to-sulfate ratio and PM_{10} mass are positively associated with pH, with strongest sensitivity at low values and saturation at high ones. Relative humidity is the interesting one: its sign *reverses* by regime — higher RH increases acidity below $\sim 15^\circ\text{C}$, at $\text{N/S} > \sim 1.25$, or at $\text{PM}_{10} > \sim 50 \mu\text{g}/\text{m}^3$, and decreases it under warm, sulfate-dominant or low-mass conditions. Feature attribution is association, not causation, and pH is itself a model output rather than a measurement. The RH regime reversal is a useful caution for anyone applying a single humidity correction across seasons.

doi: 10.5194/acp-26-11683-2026

Environ Sci Atmos 2026a Environ Sci Atmos

Multi-scale variability and lagged meteorological controls of $\text{PM}_{2.5}$ in Delhi

Wavelet coherence and causality analysis of Delhi $\text{PM}_{2.5}$ against meteorological drivers, decomposing the record by timescale rather than fitting a single regression. **The abstract is truncated at source** — it stops mid-sentence after stating the aim, and no coherence bands, lag lengths, data period or station count are recoverable. What can be said is that the method is the right one for the problem: Delhi's $\text{PM}_{2.5}$ is driven by processes on separable scales (diurnal boundary-layer collapse, synoptic ventilation, seasonal crop-residue burning) that a time-domain model conflates, and lagged causality is what forecasting a distributed network needs. Carried for the register with its numbers unstated. Re-check on the weekly abstract sweep.

doi: 10.1039/d6ea00071a

Fan et al. 2026 Atmos Environ

Santa Ana winds control wildfire smoke transport and related health impacts in California during January 2025

METADATA ONLY — no abstract in Crossref, Europe PMC or Scholar Gateway. Listed, not summarised. Carried because it is the first record in this watch to attach a health-impact estimate to the January 2025 Los Angeles fires, and because the physical premise is specific and testable: offshore Santa Ana flow reverses the usual coastal transport direction, so smoke-attributable exposure is not predicted by proximity to the burn perimeter. Satellite work on the same event (Li et al., *AGU Advances*, Feb 2026) reports spread rates to 3.7 km h^{-1} , residential-area fire intensities of $72\text{--}297 \text{ W m}^{-2}$ and a total-PM emission coefficient of $10.2\text{--}21.1 \text{ g MJ}^{-1}$, with 24 h mean $\text{PM}_{2.5}$ within 7.5 km reaching $300 \mu\text{g}/\text{m}^3$ — context, not this paper's result. The health-impact methodology is unknown until an abstract appears, and for a wildland–urban–interface fire burning structures the applicability of vegetation–fire concentration–response functions is exactly the assumption that needs checking.

doi: 10.1016/j.atmosenv.2026.122292

Fan et al. 2026 iScience

Multiple amides synergistically enhance sulfuric acid–ammonia nucleation

Quantum-chemical cluster thermodynamics plus kinetic simulation of new particle formation in binary, ternary and quaternary systems containing sulfuric acid (SA), ammonia, methylacetamide (MAA) and methylpropionamide (MPA). $(\text{SA})_1 \cdot (\text{MPA})_1$ and $(\text{SA})_1 \cdot (\text{MAA})_1$ are more stable than $(\text{SA})_1 \cdot (\text{NH}_3)_1$ and even than $(\text{SA})_1 \cdot (\text{DMA})_1$, and the two amides together enhance nucleation by an order of magnitude more than either alone, most strongly at low SA concentration and low temperature. A displacement hierarchy is identified: MPA replaces NH_3 , MAA replaces MPA. Carried because secondary particle number — not mass — is what an optical low-cost sensor is blindest to, and because amide sources (industrial, agricultural, carbon-capture solvent degradation) are rising and largely unmonitored. It is a purely computational result with no chamber or ambient confirmation, the amide concentrations assumed are not tied to measured ambient values, and cluster-level thermodynamics has a poor track record of quantitative transfer to observed formation rates.

doi: 10.1016/j.isci.2026.117013

Florêncio et al. 2026 Atmos Environ

Comprehensive analysis of PM_{10} in the Metropolitan Area of São Paulo, Part I: chemical characterisation and sources

Abstract not deposited at Crossref on the entry date. Title indicates chemical speciation and source apportionment for a megacity with almost no component-resolved PM literature in this watch to date. Queued for the Saturday sweep.

doi: 10.1016/j.atmosenv.2026.122286

Gonçalves et al. 2026 Atmos Pollut Res

Extreme Saharan dust outbreak: air quality crisis in NW Spain (March 2022)

Metadata only. Surfaced through the Crossref journal sweep with no abstract deposited. Title, journal, DOI and authorship are verified; design, sample size and quantitative outcome are not extracted and are not inferred. Flagged for abstract retry on the next run.

doi: 10.1016/j.apr.2026.103183

Gong et al. 2026 Res Sq (preprint)

Regional analysis of urban physical factors influencing air pollution: evidence from Seoul's living areas

Constructs a 100 m grid over the Seoul metropolitan area, kriges a continuous surface from **193 monitoring stations** for SO_2 , CO , NO_2 , O_3 , PM_{10} and $\text{PM}_{2.5}$ combined into a Comprehensive Air-quality Index, reduces 15 urban-form variables to four principal components (urban enclosure, low-density buildings, high-density buildings, development intensity) and fits geographically weighted regression. CAI rises from **88–92** in northeastern Seoul to **96–104** in the southwest. The headline model comparison — adjusted R^2 **0.9928** for GWR against **0.0182** for OLS — should be read as a warning rather than a result. A 54-fold jump in explained variance on a kriged surface means the GWR is largely reproducing the interpolation, since kriged values at 100 m between 193 stations are smooth by construction and locally varying coefficients will fit that smoothness almost exactly. No spatial cross-validation or effective-degrees-of-freedom correction is reported. Tier C. Carried because the urban-form-to-exposure framing is the right one for siting a sensor network, and because it is a clean illustration of how interpolated exposure surfaces manufacture model fit.

doi: 10.21203/rs.3.rs-10443285/v1

Gould & Jagarnath 2026 Research Square (preprint)

Satellite monitoring for air quality and public health research in Africa: a scoping review

JB method, PRISMA-ScR reporting, five databases plus grey literature to May 2026, for studies using satellite-derived air quality data against human health outcomes in Africa. **Twenty-six records met inclusion** — for an entire continent. Most post-2015, most ecological time-series or cross-sectional, concentrated in West Africa, with $\text{PM}_{2.5}$ the dominant exposure. This is the first record in this watch to put a number on the African evidence gap that its own geography figure has shown as an empty bar every day since 27 July. Preprint, not peer reviewed; a scoping review counts literature and does not weigh it.

doi: 10.21203/rs.3.rs-10623736/v1

Guo et al. 2026 Atmos Environ*Meteorology-driven long-range transport to an ultra-low emission region during the monsoon*

Metadata only. Elsevier withheld the abstract; Semantic Scholar returns 404 for the DOI. The registered title indicates a Hainan case study of monsoon-driven long-range transport into a low-background region. Nothing beyond the title is asserted. Transport-into-clean-background studies are the useful test case for source attribution because the local emission term is small, which is why this is carried rather than rejected — but it carries no extractable content until the abstract is released.

doi: 10.1016/j.atmosenv.2026.122293

Guo et al. 2026b Atmos Chem Phys*Kinetic isotope fractionation framework for sulfate source apportionment*

Seasonal PM_{2.5} observations in Nanjing combined with Bayesian isotope mixing and process-specific Rayleigh fractionation. Conventional isotope models assume complete SO₂ oxidation and therefore discard the kinetic isotope effects generated during incomplete processing; correcting for them shows transition-metal-ion-catalysed and NO₂-mediated pathways dominate secondary sulfate, and moves apportionment by **−10% coal combustion**, **+8% traffic** during summer photochemical episodes. Single-city, single-year, and the Rayleigh treatment assumes a closed reservoir, which is a strong assumption for an urban boundary layer. The direction of the correction nevertheless agrees with Liu T et al. in the same issue, from wholly independent evidence.

doi: 10.5194/acp-26-12231-2026

Atmos Environ 2026c METADATA ONLY Atmos Environ*Source-specific black carbon responses to short-term interventions and cross-year emission changes in coastal Qingdao*

METADATA ONLY — no abstract retrievable. Carried because source-specific BC response to *short-term* interventions is a natural-experiment design, and coastal sites separate shipping from land-based combustion in a way inland sites cannot. Read alongside Chen et al. in this issue, which finds inventory-based BC attribution biased by **−68% to +28%**: an intervention study measures the response directly and sidesteps that. Sample period, apportionment method and intervention definition are all unstated.

doi: 10.1016/j.atmosenv.2026.122299

Helbich et al. 2026 Environ Int*Socioeconomic patterning of the urban external exposome depends on how the exposome is defined*

4,640 adults across Bulgaria's five largest cities, with residential NO₂, PM, traffic noise, green space, blue-space distance, impervious surface, walkability and night-time light. City-specific latent profile analysis was compared against composite indices, then both were regressed on individual- and area-level SES. High-risk profiles bundle low green space, high air pollution, noise and imperviousness *with* higher walkability. **Higher area-level SES was positively associated with high-risk profiles in most cities** — the inverse of the standard environmental-justice expectation — and composite indices only partially reproduced the latent-profile findings. The methodological warning is the finding: operationalisation choice changes the sign of the inequality conclusion.

doi: 10.1016/j.envint.2026.110419

Hernández Pardo et al. 2026 Atmos Chem Phys*Downward transport of tropical upper-tropospheric aerosols*

Multi-year global EMAC simulations releasing 19 idealised tracers in the tropical upper troposphere under resolved advection, parameterised convection, turbulent mixing and deposition. Age of air at 500 hPa is ≈ 45 d for tropics-wide sources but > 250 d for regional continental ones — transport diagnostics are dominated by source extent, not by transport physics. Advection dominates; injection height matters more than particle size in the 20–100 nm range. Tracers are idealised, carry no mass or composition, and there is no surface validation, so this constrains *where* upper-tropospheric particles arrive, not *how much* they contribute to surface PM. Carried at tier C for that reason.

doi: 10.5194/acp-26-12275-2026

Hernandez et al. 2026 Atmos Environ*Urbanization and humidity impact atmospheric deposition of anthropogenic microparticles (including microplastics) in a humid subtropical city*

METADATA ONLY — no abstract in Crossref, Europe PMC or Scholar Gateway. Listed, not summarised. Carried at tier C because atmospheric microplastic is a particle class that optical monitors detect but do not identify: fibres in particular violate the spherical-equivalent-diameter assumption underlying every optical particle counter, so a deposition record that quantifies their abundance as a function of urbanisation and humidity bears on the size and shape distribution a low-cost sensor is actually sampling. The humidity dependence is the part worth reading if an abstract appears, since RH is simultaneously the dominant artefact in optical PM measurement and, here, an exposure variable.

doi: 10.1016/j.atmosenv.2026.122294

Hopke et al. 2026 Environ Monit Assess*A PM_{2.5} magnitude-rank index for summarizing annual concentration-frequency structure*

Introduces the PM_{2.5} Magnitude–Rank Index — the largest integer k such that at least k days in a year have daily mean PM_{2.5} $\geq k$ $\mu\text{g}/\text{m}^3$ — as a threshold-free joint summary of magnitude and frequency. Applied to **918 qualifying MSA-years across 96 US metropolitan areas, 2013–2022**, PMRI ranges 6–38 (median 16) and correlates with the annual mean at Spearman $\rho = 0.83$, the annual 98th percentile at 0.95, and days ≥ 15 $\mu\text{g}/\text{m}^3$ at 0.96. That last figure is the honest problem with the metric: at $\rho = 0.96$ against an existing exceedance count it adds very little information, so its case rests on being policy-threshold-free and intuitively readable rather than on discriminating power. No health-outcome validation is attempted.

doi: 10.1007/s10661-026-15858-1

Hu et al. 2026 Geo-spat Inf Sci*Fusion of ground monitoring, satellite retrieval and CTM for seamless hourly 1-km PM_{2.5}*

Fills satellite AOD gaps with continuous WRF-Chem AOD, injects physics-informed features (PM_{2.5} temporal components, AOD vertical structure) from the CTM, then predicts with a geospatially aware LightGBM that exploits monitoring-site spatial structure. Site-based cross-validation R^2 0.87, RMSE 13.42 $\mu\text{g}/\text{m}^3$, MAE 8.68 $\mu\text{g}/\text{m}^3$, with day-and-night coverage demonstrated on one pollution episode. Site-based CV overstates performance away from stations — the standard weakness — and an RMSE of 13.4 is large relative to the WHO annual guideline, so this is useful for episode mapping and weak for long-term exposure assignment. The physics-feature injection is the transferable step.

doi: 10.1080/10095020.2026.2680855

Ianiri et al. 2026 Environ Pollut

Street- versus roof-level deposition fluxes of POPs and metals, normalised to sedimentable PM

Thirteen monthly deposimeter samples at two heights on one Rome building, Aug 2023 – Aug 2024, with PCDD/Fs, PCBs and PAHs by GC-HRMS and metals by ICP-MS, reported both as fluxes and normalised to collected sedimentable PM. **Annual SPM flux 42 vs 26 mg m⁻² d⁻¹ street versus roof**, but a Wilcoxon signed-rank test found **no significant street–roof difference in SPM-normalised concentrations** ($p > 0.05$): height changes deposited quantity, not deposited composition. Seasonality was significant for PCDD/Fs, PCBs, Cr, As, Ni and Sb (winter peaks) and for PAHs (summer/autumn). Single building, single city, $n = 13$ per height — a design that supports the composition claim far better than the flux ratio.

doi: 10.1016/j.envpol.2026.128965

Ibrahim 2026 Research Square (preprint)

Air pollution status and temporal patterns in Bukit Rambai and Muar, 2013–2017

Five years of Malaysian Department of Environment station data (API, O₃, CO, SO₂, PM₁₀, NO₂) at two southern Peninsular stations, analysed in openair with calendar plots. Peak API 477 (Bukit Rambai) and **762 (Muar)** in June 2013 during a transboundary haze episode. Descriptive, two stations, no exposure or health linkage, and not peer reviewed. Carried only because episodic Southeast Asian haze is otherwise unrepresented in this watch and the 762 value is a useful anchor for dynamic-range requirements on low-cost optical sensors, which saturate well below it.

doi: 10.21203/rs.3.rs-10666765/v1

Isworo & Oetari 2026 Environ Sci Pollut Res Int

Risk-integrated industrial air quality: Gaussian dispersion linked to a health-weighted pressure index

Gaussian-plume dispersion for a Central Java textile facility combined with a hazard-quotient-normalised Atmospheric Environmental Pressure Index, using an emission inventory, ERA5 2015–2024 meteorology and six receptor measurements. The headline is a compliance paradox worth stating: **modelled concentrations meet Indonesia's national standards while PM_{2.5} exceeds the WHO 2021 guideline**, and PM_{2.5} and PM₁₀ dominate the index. AEPI values (0.06 baseline, 0.08 predicted) sit in the authors' own "very low" band, which makes the index insensitive at exactly the concentrations where the health argument bites — a scaling problem the paper does not confront. One facility, six receptors, no validation of the plume model against the measurements. Tier C: the framework is transferable, the result is not evidence.

doi: 10.1007/s11356-026-38159-z

Jin et al. 2026 Atmos Pollut Res

Hourly 2 km Himawari-8 aerosol fine-mode fraction retrieval improves PM_{2.5} mapping

Metadata only. Elsevier withheld the abstract and the record is too recent for Semantic Scholar (404) or Scholar Gateway, whose corpus currently ends in May 2026. The registered title asserts that adding a fine-mode-fraction retrieval to Himawari-8 geostationary observations improves surface PM_{2.5} estimation over the Beijing–Tianjin–Hebei region at hourly, 2 km resolution. **No performance figure, validation design or comparison baseline is asserted here, because none was available.** Flagged for retrieval on a later run. Geostationary fine-mode retrieval is the most direct satellite analogue of the composition-resolution problem raised elsewhere in this issue, so the numbers are worth chasing.

doi: 10.1016/j.apr.2026.103180

Joshi et al. 2026 (preprint) Research Square

Environmental drivers of foliar particulate accumulation, leaf structure and biochemistry in urban trees in Delhi

Ten native species, four contrasting Delhi sites, summer and winter 2025, with branch-level triplicates. Total leaf-deposited PM rose from **120.13 ± 30.99 to 423.69 ± 30.99 μg cm⁻²** between seasons, fine PM from 6.82 ± 1.32 to 25.75 ± 1.32. *Plumeria alba* retained most (414.92), *Azadirachta indica* least (140.82). Higher loads tracked ascorbic acid and foliar pH positively and carotenoids negatively, and were *not* explained by specific leaf area, leaf dry matter content or thickness; APTI showed no seasonal change. The useful negative result is that capture capacity and physiological tolerance are separate axes, so single-metric species screening is inadequate.

doi: 10.21203/rs.3.rs-10524655/v1

Kalumbi et al. 2026 J Expo Sci Environ Epidemiol

Age-specific time-spent patterns among residents living near gold mine tailings storage facilities in the West Rand, South Africa

Cross-sectional structured-questionnaire survey of **330 households in Kagiso**, West Rand, resolving indoor and outdoor time across five age categories with Kruskal–Wallis tests and Bonferroni-adjusted Dunn's comparisons. Indoor time runs **66.6–84.8% of a weekday and 76.0–82.7% of a weekend day** (18.28±3.02 h weekdays, 19.00±1.99 h weekends); young children and older adults are indoors most, and weekend indoor time rises significantly for children, adolescents and adults but not for the two extreme age groups. The headline for exposure modelling is that indoor time is **nearly 10 percentage points below USEPA Exposure Factors Handbook defaults**. Self-reported recall with no diary or wearable validation, and a single district, so the magnitude is soft — the direction of the bias in imported default time budgets is the usable result.

doi: 10.1038/s41370-026-00972-6

Kan et al. 2026 Atmosphere

Short-term observations of airborne microplastics in Phnom Penh: concentrations, aerodynamic size distribution and polymer composition

First AMP observations for Cambodia, by μFTIR-ATR imaging, with meteorology, gaseous co-pollutants, water-soluble ionic tracers and HYSPLIT back-trajectories. Four independent 72-h periods: **0.97 ± 0.30 MP m⁻³ in TSP** (range 0.55–1.27) and **0.33 ± 0.10 MP m⁻³ in PM_{2.5}** (0.23–0.49); 134 particles identified in TSP, 46 of them inside the PM_{2.5} fraction. PE, PP and PET dominate, 52% fragments, 82% below 30 μm Feret diameter. The authors are explicit that $n = 4$ precludes source attribution — read the polymer split as a baseline, not an apportionment.

doi: 10.3390/atmos17080804

Kang et al. 2026 Atmos Environ

Characteristics and dry deposition fluxes of particulate parent, alkyl, and nitro-polycyclic aromatic hydrocarbons in the Bohai Sea Strait, China

Metadata only. Crossref sweep, no abstract deposited. Nitro-PAHs are among the more directly mutagenic particle-bound species and are under-measured relative to the parent compounds, and a marine-strait dry-deposition budget is a receptor-side measurement rather than an emission inventory — but no flux, no concentration and no sampling period are extractable from the deposited record. Flagged for retry, low priority relative to the wind-correction paper.

doi: 10.1016/j.atmosenv.2026.122322

Kim et al. 2026 Environ Pollut

Wildfire plumes can lower downwind sulfate by scavenging OH

CMAQ driven by the high-resolution FINN fire inventory, applied to the July 2014 Siberian wildfire episode. Long-range transport generally raised particulate matter across Northeast Asia, but during **26–29 July an anomalous sulfate decrease of up to $-1.56 \mu\text{g m}^{-3}$** appeared over Northeast China, the Shandong Peninsula and the Yellow Sea. The mechanism is competitive: fire-emitted CO and VOCs scavenge hydroxyl radicals, suppressing SO₂ oxidation. Source apportionment attributes 92.30% of the suppressed sulfate to inland anthropogenic SO₂ and 7.70% to oceanic sources. **Limitations: a single episode, model-only, with the mechanism inferred from the chemistry solver rather than measured OH**; FINN emission uncertainty propagates directly into the sign of the result. Relevant because it breaks the assumption that fire contributions to PM are additive — a source-apportionment scheme that assumes additivity will misattribute here.

doi: 10.1016/j.envpol.2026.128872

Kim et al. 2026 Atmos Environ

Application of machine learning-based wind corrections to coupled WRF–CMAQ simulations: a comparison of dynamic assimilation and static replacement

Metadata only. Crossref by-ISSN sweep; title, authorship and DOI verified, no abstract deposited. The comparison it names — injecting an ML-corrected wind field into CMAQ dynamically during the run versus statically replacing the WRF field beforehand — is a live methodological question for anyone producing exposure surfaces, because wind error is the dominant transport error in urban CTM output and the two strategies have very different mass-conservation properties. Nothing is asserted about which won. Scholar Gateway cannot reach it (corpus ends May 2026) and Consensus has not indexed it. Second-priority abstract retry behind Werderman et al.

doi: 10.1016/j.atmosenv.2026.122324

Kommula et al. 2026 Environ Sci Technol

Comparable organic aerosol mass enhancement during photochemical and dark oxidation of biomass burning emissions

Chamber aging of emissions from savannah grass, savannah wood and boreal forest surface fuel under varied combustion conditions, comparing daytime photochemical with nighttime dark oxidation. OA is 82–99% of PM₁ in the fresh emissions. Both aging routes yield **comparable net OA mass**, which PMF traces to more efficient primary-OA loss during daytime offsetting more efficient gas-phase oxidation. The resulting mass enhancement of **0.75–1.3×** matches field observations and closes a long-standing lab–field discrepancy. Applies to low-NO_x remote conditions; the authors say so, and that is where the proposed model simplification is and is not safe.

doi: 10.1021/acs.est.5c15841

Lazo et al. 2026 J Environ Qual

Biomonitoring trace metal deposition with Ligustrum lucidum: Albanian fire incidents

Passive biomonitoring across 14 urban, industrial and rural sites in Elbasan, leaves collected two weeks after an industrial-waste fire and a landfill fire. Al, V, Fe, Cu, Cd, Pb and Zn showed low-to-moderate variability; Cr, Co, Ni, As and Sb were highly variable with industrial-site outliers, and cluster analysis separated the two groups. Method demonstration rather than exposure assessment — there is no control period and no airborne concentration — but it is the kind of post-incident evidence that is otherwise unobtainable.

doi: 10.1002/jeq2.70243

Leibensperger et al. 2026 Atmos Chem Phys

Sea surface warming suppresses primary sea spray aerosol number production

Controlled wave-breaking experiments in the Scripps Ocean-Atmosphere Research Simulator, which reproduces wind, waves and sea surface temperature together. Raising SST from 2 to 23 °C **suppressed subsurface bubble concentration by 1.5×**, **sea spray aerosol number by 3×** and **SSA number flux by 3×**, with a non-monotonic SST dependence in the supermicron fraction. This is a single simulator with one seawater matrix and no biological aging, so absolute fluxes are not transferable — the value is the sign and rough magnitude of the SST dependence, which the field has not agreed on. Relevance here is indirect but real: marine background number concentration is the baseline against which coastal low-cost network anomalies are read.

doi: 10.5194/acp-26-12211-2026

Li Q et al. 2026 Environ Sci Technol

Health risk of fine PM across China with a pollutant-specific index, 2013–2023

A 1 km daily PM_{2.5} surface for 2013–2023 built by machine learning (cross-validated R^2 **0.88 spatially, 0.82 temporally**), fused with meta-analysed short-term effect estimates into an AQHI_{PM_{2.5}}. National mean fell **55.06 → 26.50 $\mu\text{g m}^{-3}$** (2013→2022) then rebounded to 27.46 in 2023. Per 10 $\mu\text{g m}^{-3}$ over 3 d: **+0.42% all-cause, +0.61% cardiovascular, +0.84% respiratory mortality**. Northern and southwestern regions retain >80 medium/high-risk days a year; eliminating them would avert 6,985 (6,419–7,588) deaths, 7.42% of the attributable burden. The weak point is the meta-analysed exposure — response step, which imports between-study heterogeneity into a spatially resolved index and then reports it at 1 km as if the precision were the model's. For sensing work: the AQHI construction is directly portable to a dense network, but only if the network can supply the exposure surface it assumes.

doi: 10.1021/acs.est.6c00144

Li et al. 2026 Environ Monit Assess

Multi-media, multi-pathway PAH exposure and carcinogenic risk under three scenarios

PAH concentrations in air, water, soil, dust and food compiled from Chinese studies published 2010–2024; average daily doses computed for diet, inhalation, ingestion and dermal contact; risk expressed as benzo[a]pyrene equivalents and incremental lifetime cancer risk, with Monte Carlo and sensitivity analysis. Industrial areas dominate air and water, urban areas dust and soil. **Dietary intake dominates both total exposure and carcinogenic risk**, and the industrial–urban–rural ranking is stable across dietary percentile scenarios even though absolute ILCR is not. Compiled from heterogeneous published measurements spanning 14 years — exposure misclassification is structural, not incidental.

doi: 10.1007/s10661-026-15753-9

Li et al. 2026a Environ Sci Technol

Upgrading diesel vehicle after-treatment technology significantly changed the composition of reactive nitrogen in exhaust

Chassis-dynamometer and controlled testing of light-duty diesel vehicles across China IV, V and VI standards. Nitrogen speciation shifts fundamentally: **NO exceeds 80% of reactive nitrogen under China IV**, while NH₃ and N₂O become prominent under China VI. Mechanisms are attributed component-wise — diesel oxidation catalysts raise the NO₂/NO_x ratio, SCR produces NH₃ slip, ammonia-slip catalysts generate N₂O. SHAP analysis of tree models identifies emission standard, exhaust temperature and vehicle speed as dominant drivers. *Limitation*: no fleet sample size or vehicle count is given, and dynamometer cycles under-represent cold starts and real-world transients where SCR is least effective. Relevance: NH₃ is a secondary-PM precursor and N₂O a greenhouse gas, so successive NO_x controls have moved the problem rather than closed it. A traffic-adjacent network measuring NO_x alone now under-reports the emission it is meant to characterise.

doi: 10.1021/acs.est.6c00994

Li et al. 2026d Environ Pollut

Surface PM transport under a closed earth-air model

One year of continuous in-situ monitoring in the semi-enclosed basement of the Mogao Grottoes conservation department, testing whether **earth-air vertical movement** driven by atmospheric-pressure fluctuation controls surface PM independently of wind and human activity. Structural equation modelling gives **CO₂ as a tracer for EVM-driven PM dynamics**, with path coefficients of **0.452 (PM_{2.5})**, **0.470 (PM₁)** and **0.386 (PM₁₀₀)**; ²²²Rn regulates PM positively but far more weakly, and relative humidity is strongly inhibitory. PM_{2.5} and PM₁ are strongly affected by EVM, PM₁₀₀ only weakly. Single site, single enclosure, and SEM path coefficients on autocorrelated one-year time series overstate precision badly. The mechanism claim is interesting for indoor and heritage monitoring; the effect sizes should not be transported.

doi: 10.1016/j.envpol.2026.128983

Lin et al. 2026 Atmos Chem Phys

Elevated aerosol layers within the daytime mixed layer over ecologically sensitive areas of northwest China: diurnal variation, formation mechanisms and regional effects

Identifies an elevated aerosol layer embedded *within* the daytime mixed layer over the Hexi Corridor using intensive vertical-profile campaigns in **2010 and 2012** at Dunhuang and Minqin. Excluding dust events, elevated layers occurred at **0.6–2 km** on **82 %** of days at Dunhuang and **44 %** at Minqin, with the paper analysing formation mechanism, air-quality consequence and radiative effect. The relevance to surface monitoring is direct and under-appreciated: a concentration maximum aloft but inside the mixing layer breaks the well-mixed assumption that lets a surface monitor stand for a column, and it is precisely the configuration in which satellite AOD and ground PM_{2.5} decouple — which is the failure mode of every AOD-to-surface-PM scaling relationship. Limitations: two campaigns fourteen years old, two sites, and an arid region whose boundary-layer behaviour does not generalise to the humid urban settings where most low-cost networks are deployed.

doi: 10.5194/acp-26-11111-2026

Lin et al. 2026 Environ Sci Technol

Three-millennia alpine-lake records of divergent atmospheric Cd, Pb, Sb and Hg pollution in southwestern China

Radiometrically dated sediment cores from three alpine lakes at ~4000 m in Yunnan reconstruct 3000 years of atmospheric Sb, Pb, Cd and Hg deposition. Onset is metallurgy-specific and spatially staggered: Sb and Pb from the mid-1400s in northeastern Yunnan, ~200 years later in the west with Cu/Ag smelting; Cd only after the early 1700s with Zn and Cu production. Preindustrial peaks fall between the early 1800s and 1900 CE. **Since the 1950s Cd has reached unprecedented levels while Sb and Pb remain below their preindustrial maxima**, and Hg rises monotonically from the mid-1400s. *Limitation*: lake sediment integrates catchment erosion as well as atmospheric deposition, and the separation depends on normalisation assumptions not visible in the abstract. Relevance: a 3000-year baseline is the only way to know what “background” means for a metal, and metal-resolved PM is where low-cost sensing has no capability at all.

doi: 10.1021/acs.est.5c17960

Liu et al. 2026 Atmos Res

Identification of dominant weather patterns and circulation regulation mechanisms of winter PM_{2.5} pollution in Iran

Metadata only. Crossref sweep, no abstract deposited. Synoptic classification of winter PM_{2.5} episodes; one of very few Iranian PM records the watch has surfaced, which is why it ships as a stub rather than being dropped.

doi: 10.1016/j.atmosres.2026.109287

Lu et al. 2026 Atmos Environ

Moving-vehicle wakes and noise barriers in traffic emission dispersion

Metadata-only record. Surfaced by the Crossref by-journal leg as a new *Atmospheric Environment* deposit dated 2 August 2026; Elsevier deposits no abstract to Crossref and the article is not yet in PubMed or Europe PMC, so **no summary is offered and none should be inferred from the title**. Listed so that the window’s coverage is complete and so the record can be picked up and summarised when an abstract becomes available. Authors: Lu, Li, Wang, Wu, Liu, Yang, Zhao, Zheng. The topic — vehicle-induced turbulence interacting with barrier geometry — is squarely relevant to near-road sensor siting, which is why it is carried rather than dropped.

doi: 10.1016/j.atmosenv.2026.122256

Madalipay et al. 2026 Atmos Pollut Res

Establishing PM_{2.5} conversion factors for South Korea at provincial level

METADATA ONLY. Provincial PM_{2.5} conversion factors bear directly on how a sparse regulatory network is extrapolated to unmonitored areas, which is the same inference a low-cost network is deployed to replace. Priority re-check alongside the Madrid classroom study.

doi: 10.1016/j.apr.2026.103171

Mahia et al. 2026 J Environ Manage

Street geometry, air quality and real-time physiological response in elderly pedestrians

Fourteen elderly participants walked five routes in Perugia during spring while wearable sensors logged physiology and portable stations logged microclimate and air quality. Paths with higher height-to-width ratios (0.92–1.27) and elevated NO₂ and PM₁₀ (to 21 ppb and 18.77 µg/m³) significantly degraded air-quality perception ($\beta = -0.273$, $p = 0.028$ for NO₂; $\beta = -0.312$, $p = 0.013$ for PM₁₀), and both pollutants tracked skin-conductance change and self-reported emotional state; higher Green Visual Index went with more relaxed responses. With $n = 14$ over one season the coefficients carry little weight, and perception and physiology are both plausibly driven by street geometry directly rather than by the pollutant. Read it for the measurement stack — paired personal air-quality and physiological logging at walking cadence — not the estimates.

doi: 10.1016/j.jenvman.2026.130681

Mainka & Mainka 2026 Preprints.org

Urban cycling versus PM_{2.5} exposure: does the benefit outweigh the risk?

Scenario analysis of **162 real commuting trips by two cyclists** in Gliwice across heating and non-heating seasons, with personal PM_{2.5} measured by low-cost sensor and inhaled dose reconstructed from concentration, trip duration and cycling-specific ventilation. Median trip 33.0 min/day. The median physical-activity relative risk was **0.836** (a 16.4 % all-cause mortality reduction) against a median PM_{2.5} relative risk of **1.00182** (0.182 % increase), for a **net RR of 0.840 (IQR 0.815–0.879)**; **160 of 162 trips retained net benefit** under sensitivity analysis. Two hard limits: $n = 2$ **commuters**, so the exposure distribution is a convenience sample of two routes, and the risk functions are transported from published cohorts rather than estimated. Unreviewed preprint. The useful part is the method — trip-resolved dose from a low-cost sensor is the right exposure object for this question, and almost no one computes it.

doi: 10.20944/preprints202608.1778.v1

Makacha et al. 2026 Front Glob Womens Health

Spatio-temporal and structural determinants of personal PM_{2.5} exposure in three sub-Saharan African geographies

Five days per season of minute-resolved personal PM_{2.5} monitoring on 343 parous women in The Gambia, Kenya and Mozambique, 3,190 person-days, aligned to microenvironment, fuel use, occupation, wealth and meteorology, with facility-based ambient monitors for comparison. Daily medians were modest — 10.4 (7.6–14.7), 8.1 (6.8–9.8) and 10.9 (9.3–13.5) $\mu\text{g}/\text{m}^3$ — but episodic spikes dominated, and WHO 24-h exceedance was strongly seasonal (The Gambia **55.9% dry vs 13.6% wet**). **Ambient–personal concordance was 0.01–0.56 with slopes <1 throughout**. Exposure was higher away from home with a 12:00–16:00 peak, so clean-fuel transition alone will not close the gap. The 5-day-per-season design is thin for characterising episodic exposure and the authors do not report how much of the between-person variance it captures.

doi: 10.3389/fgwh.2026.1843221

Mandalapu et al. 2026 medRxiv (preprint)

Joint heat and PM_{2.5} co-exposure across 48 US metropolitan areas

42,304 census tracts, 48 large CBSAs, ~174.6 million residents, summers 2015–2020. The framework combines a composite co-exposure index, a joint exceedance indicator, a conditional exceedance ratio against within-group independence, and an upper-tail dependence parameter estimated both non-parametrically (Capéraà–Fougères–Genest) and with a Gumbel copula, with BCa intervals from a 5,000-replicate block bootstrap. **13.21 % of residents of predominantly Black tracts versus 3.33 % of predominantly White tracts live in joint hotspots; HOLC grade D vs grade A gives OR 3.97 (2.79–5.66)** after adjustment for contemporary racial composition, poverty, renter occupancy and pre-1960 housing. Magnitudes attenuate under near-surface air temperature rather than surface temperature. Unreviewed; the tail-dependence machinery is the transferable contribution.

doi: 10.64898/2026.08.20.26360970

McCerery et al. 2026 Atmos Environ

Arctic transport of North American and Canadian wildfire aerosols

METADATA ONLY. The only record in today's corpus with a polar or remote-marine receptor; long-range transport to a low-background site is where instrument detection limits, not calibration slope, become the binding constraint.

doi: 10.1016/j.atmosenv.2026.122289

Meena et al. 2026 Environ Pollut

Transplanted lichen trace-metal accumulation along a Jaipur urban gradient

Phaeophyscia hispidula thalli collected from a wildlife-sanctuary reference site and transplanted to ten urban and one reference location in Jaipur for 90 days over the March–June 2024 pre-monsoon, with Zn, Pb, Cd, Cr, Cu and Ni by ICP-MS and chlorophyll and carotenoid concentrations by spectrophotometry. Urban sites accumulated more metal, and **accumulation tracked chlorophyll loss while carotenoids rose** at several exposed sites, consistent with a photoprotective adjustment. PCA and hierarchical clustering resolved a spatial gradient. **No co-located instrumental PM measurement means the biomonitor is never calibrated against a concentration**, so this yields a relative ranking of sites, not an exposure estimate. Useful as an ultra-low-cost spatial screening layer where no network exists; not a substitute for one.

doi: 10.1016/j.envpol.2026.128919

Mohammed 2026 Build Environ

Spatiotemporal dynamics of atmospheric vulnerability in hot arid cities: introducing the Urban Microclimate Air Stress Index (UMASI)

Metadata only. Crossref sweep, no abstract deposited, no study city named in the metadata. A composite index combining microclimate and air-quality stress is exactly the kind of construct that needs its weighting scheme inspected before it is used, so nothing is characterised here. Flagged for retry; low priority relative to Werderman et al.

doi: 10.1016/j.bulldenv.2026.115166

Moryani et al. 2026 J Appl Toxicol

PM_{2.5}-bound Cu, Cd, Ni, Pb and Zn across 20 schools in two contrasting South China cities

Paired summer and winter sampling at 20 schools in Guangzhou and Maoming, ICP-MS for five metals, with geoaccumulation index, enrichment factor and PCA source apportionment, then inhalation risk assessment for children and adults. Cu and Cd carried the highest enrichment (EF 27.72 and 27.32 in Guangzhou), winter exceeded summer, and PCA pointed to traffic, industrial combustion and urban emissions. **All hazard quotients and carcinogenic risks fell below accepted thresholds**, with Pb (4.85×10^{-3}) and Cu (3.27×10^{-3}) leading non-carcinogenic risk and Cd (7.30×10^{-6}) carcinogenic. Fieldwork dates from 2018, so it characterises an emissions regime seven years old; the null risk result is the honest headline and is under-stated in the abstract.

doi: 10.1002/jat.70398

Moryani et al. 2026 Toxics

Bioaccessibility and speciation of heavy metals in PM_{2.5} road dust

48 grid-based samples across four functional zones (residential, school, hospital, office) in Karachi (>27 million) and Shikarpur (~0.3 million), with the PM_{2.5} fraction of road dust extracted in artificial lysosomal fluid (**pH 4.5, 37 °C, 24 h**) and speciation modelled in PHREEQC and Visual MINTEQ. **Cr, As and Pb showed moderate inhalation bioaccessibility**; contamination and ecological-risk indices were computed alongside. The design addresses a real gap — total metal content is a poor predictor of the dose that reaches lung tissue — but ALF extraction is a simulation with no in-vivo anchor, and the abstract reports no bioaccessible fractions, no concentrations and no zone contrasts, so nothing quantitative can be lifted. Road dust is also resuspended, not directly emitted, which the pollution indices do not distinguish.

doi: 10.3390/toxics14080704

Mueangsai et al. 2026 Res Sq (preprint)

PM_{2.5} health risk quotient near stone crushing plants, Chonburi

Community cross-sectional study, 368 residents by stratified random sampling, with PM_{2.5} from two fixed stations near a crushing plant averaged and pushed through the USEPA health risk assessment to a hazard quotient. The design's weakness is structural and fatal to individual-level inference: **two fixed monitors supply a single exposure value for every participant**, so all between-person exposure contrast is assumed away and the HQ varies only through the questionnaire covariates. Carried at tier C. The setting — crushing plants generate coarse, mineral-dominated PM that optical sensors size poorly — is the reason it is here at all.

doi: 10.21203/rs.3.rs-10160540/v1

Mushore et al. 2026 Atmos Environ X*Daily outdoor PM_{2.5} and the road-edge effect across The Gambia, Kenya and Mozambique*

Multi-temporal personal exposure measurements (2022–23) interpolated to continuous surfaces by inverse-distance weighting, validated leave-one-out at $R^2 > 0.55$, MAE $< 4 \mu\text{g}/\text{m}^3$, RMSE $< 10 \mu\text{g}/\text{m}^3$; multi-distance buffers at 10 m intervals to 1000 m from major roads. Most sites exceed WHO guidance, The Gambia highest across all land-cover classes. **The road-edge gradient reverses sign between sites** — PM_{2.5} falls with distance in Rabai and Mozambique and *rises* elsewhere, implicating wetland, fire and industrial sources that a traffic-proximity exposure proxy would misattribute entirely. IDW over sparse personal measurements is a weak spatial model; the buffer result is the transferable part.

doi: 10.1016/j.aeaoa.2026.100463

Myat et al. 2026 Environ Pollut*Ground, satellite and model assessment of PM_{2.5} and attributable mortality in Myanmar*

Multi-source assessment for a least-developed country with minimal regulatory monitoring: two Yangon ground stations 2019–2024, TROPOMI HCHO and NO₂ nationally, and a generalised additive model for meteorological dependence. Mean PM_{2.5} was **24.4 $\mu\text{g}/\text{m}^3$ (SD 18.6)**, rising to **39.4 $\mu\text{g}/\text{m}^3$** in the November–February winter, with sulfate the leading measured component year-round and nitrate and Ca²⁺ seasonally. Compliance with the 5 $\mu\text{g}/\text{m}^3$ WHO guideline would have averted **2,106 (95% CI 345–3,818) premature deaths annually** among adults ≥ 30 in Yangon over 2020–2021. TROPOMI HCHO/NO₂ ratios indicate NO_x-sensitive ozone chemistry nationally. **Two stations cannot represent a city of five million**, and the health impact assessment inherits every assumption of the imported concentration–response function.

doi: 10.1016/j.envpol.2026.128918

Naghizade et al. 2026 Atmos Environ*Multi-class organic marker profiling of urban PM₁₀ in Strasbourg*

METADATA ONLY — seasonal distribution of exhaust, non-exhaust and biomass-burning tracers in northeastern France. No abstract in the Crossref deposit; Semantic Scholar and Europe PMC both returned nothing on 20 August. No quantity is inferred here. Flagged for the next issue that can reach the full record.

doi: 10.1016/j.atmosenv.2026.122306

Nguyen et al. 2026 Eng Res Express*Satellite-driven PM_{2.5} prediction: machine learning and explainable AI*

Systematic review of 2015–2025 work integrating satellite products (LEO and geostationary AOD, top-of-atmosphere reflectance, Ångström exponent) with ML/DL. Finds Random Forest still a robust benchmark, hybrid and transformer architectures better at spatiotemporal dependence, and AOD gap-filling plus normalisation the dominant determinants of model stability; positions XAI as the route to physical plausibility. No pooled quantitative comparison and no protocol registration, so the ranking claims are narrative. Adds little beyond Hu and Ding above; retained as a citation anchor for the physics-guided-learning argument.

doi: 10.1088/2631-8695/ae7028

Parajuli et al. 2026 Atmos Environ*Which soil NO_x parameterisation you pick changes modelled ozone and PM*

As fossil-fuel NO_x declines, ozone becomes more sensitive to non-fossil nitrogen, yet most air-quality models still use simple soil-NO parameterisations. This study compares Yienger & Levy and BDSNP against the mechanistic FEST-C* fertiliser emissions tool, propagating each through CMAQ over the United States and evaluating against rural NO₂ monitors and satellite NO₂ columns. FEST-C* agricultural soil NO/HONO was **double YL and two-thirds of BDSNP**, improved agreement at some rural sites, and had intermediate correlation but the lowest bias over agricultural areas. **Limitation: the PM response is reported only qualitatively** — the paper is NO_x- and ozone-centred, so a PM-focused reader gets the direction and not the magnitude. Carried because it identifies a structural uncertainty in a model layer that PM exposure surfaces routinely inherit without examination.

doi: 10.1016/j.atmosenv.2026.122225

Park et al. 2026 Environ Pollut*City-scale non-exhaust PM₁₀ from emissions to exposure with a coupled chemistry–CFD framework*

Two non-exhaust emission-factor frameworks (EMEP/EEA Guidebook and KIMM) implemented in the Fine-scale Air Quality Simulation Unit, a 3-D coupled CFD–chemistry model, over a densely built Seoul district, combined with the 2022 Seoul Time-Activity Profile. **Total non-exhaust PM₁₀ was similar under both frameworks; source partitioning was not** — brake wear dominant under EMEP/EEA, road wear under KIMM. Non-exhaust reached **up to 70% of total PM₁₀** in near-road and dense cells; outdoor activity-weighted contributions of 9–10% in commercial and traffic zones exceeded concentration-based estimates because peak traffic coincides with peak outdoor presence. No measurement validation of the partitioning is presented, which is the obvious next step and the reason a receptor-model or tracer campaign would be complementary.

doi: 10.1016/j.envpol.2026.128941

Pavlovic et al. 2026 J Air Waste Manag Assoc*Aligning satellite Earth-observation capability, use case and capacity for air-quality health*

Twenty-three-author synthesis from the GEO Health Community of Practice Air Quality and Respiratory Health Work Group. Argues satellite EO translates into public health benefit only where three things align: technical capability of the retrieval, a defined use case, and the human, computational, operational and governance capacity to sustain a solution. Surveys applications from long-term trend tracking and epidemiological exposure assignment to short-term emissions detection and early warning, then names gaps in each dimension. A consensus document rather than new evidence — useful as a citation anchor for why a given satellite product is or is not fit for a stated purpose.

doi: 10.1080/10962247.2026.2698602

Phalen et al. 2026 Inhal Toxicol*Inhaled aerosol dosimetry update*

Synthesis of the Fourth International Aerosol Dosimetry Conference (Irvine, 2024) covering computational deposition models, in vitro and in vivo exposure systems, and deposition variability. Reported advances: integration of model types to span all major respiratory-tract regions; improved in vitro particle delivery and dose measurement; the upper airways recognised as a significant entry route. Two findings matter for ambient PM work — **dose variability emerged as the critical cross-cutting issue**, and **deposited ultrafine particles may acquire surface coatings after deposition and translocate into tissue, including fetal tissue**. Limitation: a conference report, not a systematic review — coverage reflects who presented, there are no effect sizes and no pooling, and the fetal-translocation claim is stated without the primary evidence. Useful mainly as a statement of where dosimetric uncertainty sits, which is the term that converts a measured ambient concentration into an internal dose.

doi: 10.1080/08958378.2026.2691193

Pinakana et al. 2026 NPJ Clean Air*Influence of transboundary wildfire smoke on fine particulate matter concentrations at South Texas border schools*

Low-cost sensors deployed at schools in Roma and Rio Grande City, Starr County, on the US–Mexico border for **146 days, November 2023 to April 2024**, corrected by collocation-based calibration against a research-grade instrument and read alongside meteorology, back trajectories and satellite smoke and fire products. Daily mean PM_{2.5} was significantly higher on smoke-impacted days ($p < 0.001$), with **site-level increases of 8.61% to 54.9%** — a sixfold spread across sites within one small county, which is itself the finding: transboundary smoke impact is not spatially uniform at the scale a single regulatory monitor would represent. Trajectories attribute the source to Mexican wildfire emissions. Limitation: one season, no indoor measurement despite the school framing, and attribution rests on trajectory plus satellite coincidence rather than on composition. A clean template for low-cost sensor deployment in an under-monitored community.

doi: 10.1038/s44407-026-00096-z

Qing et al. 2026 Atmos Pollut Res*Seasonal and urban-suburban characteristics, emission and spatial sources of PM_{2.5}-bound PAHs and n-alkanes in Wuhu City*

Metadata only. Crossref sweep, no abstract deposited. A source-apportionment study of the particle-bound organic fraction, which is the composition class the previous issue's mortality result pointed at — flagged for retry rather than characterised here.

doi: 10.1016/j.j.apr.2026.103186

Rahmani et al. 2026 Rev Environ Health*Airborne microplastics indoors and outdoors: pooled concentrations and the instrument problem*

PRISMA systematic review to October 2023: 27 eligible studies across 16 countries, 21 pooled under a random-effects model. Indoor concentrations exceed outdoor — moderately polluted indoor environments 1,380.28 vs. 102.45 MP m⁻³ outdoors — with indoor healthcare settings and outdoor urban areas the most contaminated sub-environments. The authors present the pooled figures as a descriptive summary rather than an estimate, correctly: heterogeneity is **I² > 99%** and the dominant moderator is **detection instrument**, with FTIR reporting consistently higher counts than microscopy or Raman spectroscopy; sampling duration also matters. Sensitivity analysis excludes single-outlier dominance. For instrumentation work this is a clean external demonstration that method-driven variance can exceed environmental variance, which is exactly the risk in heterogeneous LCS fleets.

doi: 10.1515/reveh-2025-0059

Rai et al. 2026 Environ Monit Assess*Land surface temperature, land cover and aerosol optical depth over Ahmedabad, 1993–2023*

Multi-sensor remote sensing: Landsat LST and urban thermal field variance index for 1993, 2002, 2014 and 2023, with MODIS AOD and LST from 2002–2023 for the diurnal response. **Urban extent grew by 171.15 km² over 30 years** with sharp losses of fallow and vegetated land, and the thermal environment polarised — fallow land carries the worst thermal comfort, water and vegetation the best, and NDVI/NDBI track the expected cooling and warming signs. **The AOD–LST correlation is dual-signed**, which the paper reports but does not mechanistically decompose into aerosol direct radiative effect versus co-varying surface change; that confounding is the main threat to the inference. Read as a heat-planning input rather than a PM exposure product — AOD here is a columnar quantity never converted to surface concentration.

doi: 10.1007/s10661-026-15748-6

Raptaki et al. 2026 Atmos Environ*Atmospheric BTEX in Heraklion, Crete: levels, sources and contribution to secondary aerosol formation*

METADATA ONLY. Enters on the secondary-organic-aerosol precursor link rather than on any direct PM measurement; tier C.

doi: 10.1016/j.atmosenv.2026.122283

Rashid et al. 2026 Toxics*Ultrafine particles and mortality: scoping review of evidence and exposure-assessment limits*

PRISMA-ScR scoping review of PubMed and Scopus: **21 of 704 screened articles**, 2007–2025, split **14 short-term** (time-series or case-crossover against central-site particle number counters) and **7 long-term** (land-use regression, chemical transport or machine-learning surfaces). **10 of 14 short-term and 6 of 7 long-term studies reported positive associations**, strongest for respiratory mortality. The finding that matters for this watch is the second-order one: **the strongest associations came from the studies with higher-resolution or source-specific exposure assessment**, a pattern consistent with reduced Berkson-type misclassification rather than with true effect heterogeneity. A scoping review does not pool, so there is no summary estimate and no formal heterogeneity statistic; the exposure-method gradient is observed, not tested. Still the clearest statement to date of why UFP regulation stalls on measurement.

doi: 10.3390/toxics14080664

Ray et al. 2026 Atmos Environ*Evaluating CCN activity through size-resolved hygroscopicity: synergistic observations and simulations in a tropical montane region*

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar. Listed, not summarised. Size-resolved hygroscopicity (κ) is the physical quantity underneath every relative-humidity correction applied to an optical PM sensor, and tropical montane conditions are the high-RH regime where those corrections fail worst, so this is a priority re-check even though the paper's own framing is cloud-condensation nuclei rather than air quality.

doi: 10.1016/j.atmosenv.2026.122284

Ren et al. 2026 iScience*Vertical contrast in PM_{2.5} composition and secondary formation, Shiyuan, central China*

One year across three urban sites and one mountain site. Annual mountain PM_{2.5} is **70% of urban levels**, but sulfate–nitrate–ammonium is a **larger fraction** there (49.3%). SNA is vertically near-homogeneous in spring, summer and autumn (coefficient of divergence <0.2) and stratifies only in winter, when most components are urban-enriched. Higher nitrogen oxidation ratio (0.27) and SOC/OC (0.65) at altitude indicate a more aged aerosol. Limitation: a single mountain site cannot separate altitude from the site's own upwind exposure, and no vertical profiling instrument is used — the "vertical distribution" claim rests on a two-point contrast. Relevant to siting logic for distributed networks: composition, not mass, is what varies with height here, and a mass-only node would report the mountain site as simply cleaner.

doi: 10.1016/j.isci.2026.117090

Reyna-Bello et al. 2026 Environ Monit Assess

Seasonal variability of PM_{2.5}-bound metals and meteorology in an urban-industrial area of northern Mexico

MiniVol low-volume sampling across four seasons in Saltillo during 2023, with ten metals (Al, Cd, Cu, Fe, K, Mn, Na, Ni, Pb, Zn) by flame atomic absorption and PCA against temperature, humidity, wind, precipitation and pressure. Mass is strongly cold-season weighted — **152.9±23.6 µg/m³ in autumn and 115.3±37.2 in winter against 40.7±6.8 in spring** — consistent with suppressed dispersion, while several metals invert that pattern and peak in spring and summer. The reported metal concentrations are implausibly high for PM_{2.5}-bound species (Cd at 3.13 µg/m³ is orders of magnitude above typical urban values), which points to a units or digestion-blank problem; treat the metal magnitudes as unverified and the seasonal contrast as the usable result.

doi: 10.1007/s10661-026-15856-3

Roky et al. 2026 Environ Pollut

Distance-resolved lead dispersion and crop uptake from open battery-plate burning

Controlled field burns of lead-acid battery plates in Kushtia, Bangladesh (three plates per event, 149.7 ± 4.5 g total mass loss), monitored on a triangular sampling grid to 45 m with simultaneous PM_{2.5}/PM₁₀, passive Pb deposition on filter papers, and Pb accumulation in red amaranth. All three response variables showed strong distance-dependent gradients with significant distance, week and interaction effects ($p < 0.0001$); **PM_{2.5} tracked both deposition and crop Pb more strongly than PM₁₀**, and background plots stayed below detection, supporting atmospheric deposition as the dominant pathway. Crop-response model $R^2 = 0.948$. **Limitation: the threshold distance is not a single number** — the Codex leafy-vegetable benchmark is fresh-weight and the plant data dry-weight, so it had to be evaluated across a plausible moisture range, and 45 m does not bound the plume. A clean demonstration that a short-range PM_{2.5} gradient is a usable proxy for a metal deposition field.

doi: 10.1016/j.envpol.2026.128853

Saarikoski et al. 2026 Atmos Chem Phys

Role of hydrocarbons and oxygenated organic species in traffic-derived aerosol

Four Aerodyne AMS campaigns (2018–2024) at a Helsinki traffic site with PMF source apportionment. Beyond hydrocarbon-like OA peaking at the morning rush, a second traffic-related factor (TrOA) peaks in the morning but stays elevated longer, with mass spectra between HOA and biomass-burning OA and distinct C₂H₄O₂⁺ (m/z 60), C₂H₅O₂⁺ (61) and C₃H₅O₂⁺ (73) fractions. **Using HOA alone to estimate traffic primary OA underestimates it by ~50%**. The authors are explicit that TrOA's origin — atmospheric processing versus direct emission from modern vehicles — is unresolved, which is exactly the ambiguity that makes the 50% figure a bound rather than a correction factor. Directly relevant to any speciated or source-resolved sensing product.

doi: 10.5194/acp-26-11817-2026

Saeedi et al. 2026 Environ Sci Technol

Feature selection and cross-validation strategy in land-use regression for mobile-monitored ultrafine particles

Toronto mobile UFP campaign used to compare LUR models trained under random, spatial, temporal and spatiotemporal cross-validation, with and without forward feature selection, each evaluated on a hold-out test set and against independent stationary backyard measurements. **Average percentage error fell from ~217% under random CV to ~79% under spatiotemporal CV with FFS**. Random-CV models overfitted, were sensitive to outliers, produced spatial artefacts and exaggerated variable effects, and failed on independent samples — while reporting good internal validation statistics. Spatial folds were designed at scales reflecting UFP autocorrelation, which is the methodological core. The blunt implication: a large share of published LUR performance numbers are measuring data reproduction, and any exposure surface fed into an epidemiological model should carry a spatiotemporally validated error, not a random-CV one.

doi: 10.1021/acs.est.5c12601

Stevens et al. 2026 Atmos Environ

Identifying unique biomass burning marker fingerprints for Australian hardwoods, softwoods and grasses

Abstract not deposited at Crossref on the entry date. Fuel-specific tracer fingerprints are the missing input for wildfire source attribution outside the Northern Hemisphere; pairs with Pérez-Pastor et al. below on the metrology side. Queued for the Saturday sweep.

doi: 10.1016/j.atmosenv.2026.122303

Sui et al. 2026 J Environ Manage

Contrasting annual variation of organic and elemental carbon in the Pearl River Delta

Three years of continuous OC/EC observation (2019–2021) spanning the COVID-19 lockdown, with PMF source apportionment. OC declined steadily, driven by suppressed secondary organic carbon formation in the PRD's NO_x-rich regime — a relationship the authors report as *reversing* during lockdown when NO_x fell sharply, which is the interesting result and the one that identifies the regime. EC was flat or slightly rising, so fossil combustion from industry and mobile sources has not responded to the control programme. **Fossil-related emissions are ~70% of carbonaceous aerosol** and that share did not fall over the period. A single-site three-year record cannot separate emission change from meteorological variability, and no such adjustment is described; the lockdown period is doing most of the causal work and is a one-off natural experiment with a confounded meteorology.

doi: 10.1016/j.jenvman.2026.130708

Temu et al. 2026 Environ Sci Pollut Res

A six-domain framework for lichen biomonitoring including symbiont and microbiome response

Narrative synthesis proposing that lichen biomonitoring be extended from the classical triad of pollutant accumulation, physiological injury and community composition to six domains, adding functional traits, symbiotic interaction and lichen-associated microbiome dynamics as **earlier-responding indicators** than bulk accumulation. The argument that microbiome shifts precede visible physiological injury is plausible and is the part worth testing, but the review supplies **no quantitative comparison of detection limits or response latency across the six domains**, so the central claim of earlier warning is asserted rather than demonstrated. The authors name geographic bias, species-specific response and absent methodological standardisation as open problems — the same standardisation gap that limits low-cost sensor networks, and for structurally similar reasons.

doi: 10.1007/s11356-026-38123-x

Walker et al. 2026 medRxiv (preprint)

Directly measured noise, PM_{2.5} and night light across Rhode Island

176 sites, 704 site-sessions over roughly a year, spanning the Providence–Central Falls–Pawtucket urban core and four rural municipalities, with A- and C-weighted noise, PM_{2.5}, night-time illuminance and relative humidity carried through an identical model as a negative control. A-weighted noise rose with disadvantage on six of eight area-level markers (**1.3–1.8 dBA per SD; 6.6 dBA between least and most racially diverse neighbourhoods**) and C-weighted levels followed the same gradient, exceeding their A-weighted counterparts at block-group scale for renter occupancy and vehicle absence. **Short-term PM_{2.5} was roughly an order of magnitude weaker**; humidity showed nothing. The negative-control design is what makes the PM null interpretable rather than merely underpowered. Short-term measurement cannot speak to annual-average disparity, and the authors do not claim it does.

doi: 10.64898/2026.08.20.26360933

Wandji Nyamsi et al. 2026 Atmos Chem Phys*Broadband aerosol optical depth reconstructed from sunshine-duration records since 1900*

Uses sunshine duration as a proxy for broadband AOD together with cloud-cover observations from **2,700 sites**, producing the first historical BAOD estimates for several regions back to the late 19th century. The motivation is the persistent spread in modelled anthropogenic aerosol radiative forcing. The proxy is indirect and confounded by cloud regime and 120 years of instrument-network change; the authors' own framing — a constraint on century-scale trend, not a concentration product — is the right reading.

doi: 10.5194/acp-26-10767-2026

Wang Q et al. 2026 Atmos Chem Phys*Quantitative analysis of aerosol acidity and its driving factors in the Guanzhong area*

One year of water-soluble inorganic ions in PM_{2.5} on the semi-arid Guanzhong Plain, with thermodynamic pH reconstruction and sensitivity-based driver attribution. Annual mean pH **3.8±1.0** (winter > spring > summer > autumn), shifting from 2–5 on clean days toward 3–6 on polluted days. Temperature contributes 22.3–33.8% of pH variation across all seasons, NH_x 11.4–44.9% (dominant in autumn/winter), sulfate 8.5–10.8%; Ca²⁺ is a spring-specific driver. RH shows a non-monotonic effect with an inflection at 60–85% RH, attributed to deliquescence-driven liquid water content. Attribution rests on a thermodynamic model rather than measured pH, which is the standing caveat for this whole literature.

doi: 10.5194/acp-26-11785-2026

Wang T et al. 2026 Environ Sci Technol*Chlorine radical cycling on photoactive particulate matter amplifies photolytic NO_x release*

Bench photochemistry showing chloride on photoactive PM surfaces oxidised by photogenerated holes to Cl•, which accelerates conversion of nitrite intermediates to gaseous NO_x while chloride is regenerated. **~40-fold NO_x enhancement** under the model conditions, robust across humidity, chloride salt, substrate (including soot) and nitrogen source (ammonium, 4-nitrophenol). Rate scales with photoactive-matter loading and PM composition. Extrapolated to **0.7–6.2% of NO₂ variability** in the study region — a small number derived from a large laboratory enhancement, and the gap between the two is the result's main uncertainty.

doi: 10.1021/acs.est.6c06768

Wang et al. 2026 Atmos Chem Phys*Direct thermal versus emission-mediated pathways in heatwave-driven O₃ and SOA increases*

Explicitly decomposes the two routes by which heatwaves raise ozone and secondary organic aerosol: direct acceleration of photochemistry, versus temperature-driven biogenic VOC and soil-NO emissions. **Direct thermal and meteorological pathways account for over 80% (normalised)** of the national increases; ozone–temperature sensitivity peaks in the Yangtze River Delta (0.66 ppb °C^{−1}) and Pearl River Delta (0.95). Model-based attribution, so the decomposition inherits the chemical mechanism's biases — but the direction of the result argues against the common assumption that BVOC enhancement is the dominant heatwave pathway.

doi: 10.5194/acp-26-10789-2026

Wu et al. 2026b npj Clim Atmos Sci*Global fire activity exacerbates extreme PM_{2.5} pollution*

Metadata-only deposit. The title claim — that fire activity acts disproportionately on the *extreme tail* of the PM_{2.5} distribution rather than on the mean — is the one that matters, because attainment statistics, exposure-response functions fitted on annual means and most burden calculations are all mean-anchored and will understate a tail-loading source. It is also the regime in which low-cost networks are most useful and least characterised: sensor response at several hundred μg m^{−3} of smoke is outside the concentration range of essentially every published co-location study. Carried at tier C pending the abstract; a Nature-portfolio deposit with no abstract in Crossref is unusual and may resolve on retry.

doi: 10.1038/s41612-026-01527-0

Xu et al. 2026 Atmos Chem Phys*Multi-factor control of aerosol nitroaromatic compounds across 11 Chinese cities*

Simultaneous winter measurement of PM_{2.5}-bound nitroaromatics in 11 cities. Nitrophenols and nitrocatechols dominate, with city-varying relative abundance; totals are higher in northern cities, attributed to biomass and coal combustion. Secondary processes dominate formation everywhere, but the paper's actual target is the claim — built from chamber and single-site work — that relative humidity or aerosol liquid water *drives* that formation. At large-scale field level the relationship is conditioned by multi-factor interactions rather than being a clean RH dependence. A useful corrective to chamber-derived parameterisations.

doi: 10.5194/acp-26-10751-2026

Yadav et al. 2026 Research Square (preprint)*Airborne endotoxin on PM₁₀ in Agra: an 8-fold seasonal swing that PM mass does not track*

Full calendar year (Jan–Dec 2022) of ambient endotoxin by chromogenic LAL assay with concurrent bacterial and fungal bioaerosol, PM₁₀ and meteorology in the semi-arid Indo-Gangetic Basin. Monthly means run 0.92±0.35 (June) to 7.35±1.48 EU m^{−3} (December) on winter stagnation. Endotoxin correlates strongly with bacterial concentration ($r_s = 0.93$) and **not at all with PM₁₀ mass** ($r_s = 0.10$, $p = 0.75$) — mineral dust is not the carrier here. USEPA-model hazard quotients stay well below unity (max 0.0198, children, December). Preprint, single city, LAL assays are notoriously matrix-sensitive; the decoupling from mass is the durable point.

doi: 10.21203/rs.3.rs-10523351/v1

Yang et al. 2026 Atmos Chem Phys*Impacts of anthropogenic aerosols on a snowfall event in the Guanzhong Basin and surrounding areas*

Cloud-resolving fully coupled WRF-Chem, with emission-perturbation experiments separating aerosol–radiation from aerosol–cloud interactions. The precipitation response reverses sign between the basin and its surroundings: inside the basin, excluding ARI gives a slight precipitation increase with emissions via enhanced ice-phase processes, and ARI itself increases precipitation once the emission increase passes a threshold (higher relative humidity, larger ice water path, better survival of falling ice); outside, precipitation falls with emissions as ACI suppresses the liquid phase. Spatial distribution is largely insensitive. Relevant to anyone using precipitation as a wet-deposition assumption in a polluted basin.

doi: 10.5194/acp-26-11923-2026

Atmos Pollut Res 2026 METADATA ONLY Atmos Pollut Res*High-resolution CFD modelling of traffic-related air pollutant dispersion in an urban area of Istanbul*

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar. Carried because building-resolving CFD is the reference against which cheap dispersion proxies — Gaussian plume, land-use regression, distance-to-road — are judged, and because street-canyon gradients at the scale CFD resolves are exactly what a dense low-cost network claims to capture and what a regulatory monitor cannot. Whether the model is validated against measurement, and at what resolution, is unknown. Listed, not weighted.

doi: 10.1016/j.apr.2026.103169

You et al. 2026 Environ Pollut

Satellite-based estimation of ambient ammonia and emission inventory uncertainty in South Korea

Monthly ground-level NH_3 at 15 km resolution for 2013–2017, from Cross-track Infrared Sounder retrievals refined by a two-stage framework: a linear mixed-effects model carrying meteorology plus the Clean Air Policy Support System inventory, then a generalised additive model over the LMM residuals. **Cross-validated $R^2 = 0.73$, MAE 0.15, RMSE 0.21.** Concentrations peak in agricultural areas March–June; temperature is the dominant meteorological control (**+18.9% per +1 °C in June**), with relative humidity and wind speed strongest in March. The GAM residual surface identifies agricultural hotspots the inventory does not contain, which is the paper's actual claim — inventory gaps, not model skill. NH_3 matters here as the limiting precursor for ammonium nitrate and sulfate, so a 15 km monthly NH_3 field is a plausible predictor for secondary $\text{PM}_{2.5}$ models, but the resolution is far coarser than the sources, and validation against sparse ground NH_3 is circular with the sparsity that motivated the work.

doi: 10.1016/j.envpol.2026.128903

Yu et al. 2026 Environ Pollut

Street-canyon increment to near-road NO_2 , resolved for a whole city

Coupled WRF-CMAQ-ADMS framework with a purpose-built street-scale traffic emission inventory for Hong Kong, run in paired configurations with and without the street-canyon module to isolate the canyon-induced increment ΔC . Hotspots (segments with $\Delta C > 5 \mu\text{g}/\text{m}^3$, roughly a 1% increment in long-term NO_2 -attributable added risk) account for **12.2–18.8% of roads** in the four worst districts. Canyon effects are seasonal — mean hotspot ΔC rises **19–26% from summer to winter** under weaker ventilation — and about **184,000 residents** are exposed to substantial winter increments. **Limitations: NO_2 only, no PM increment reported**, and the canyon module is validated indirectly through the coupled system rather than against canyon-resolved measurements. The reason to read it here is siting: it identifies, at segment resolution, where a sparse regulatory network is structurally blind and where dense low-cost deployment would actually add information.

doi: 10.1016/j.envpol.2026.128871

Yu et al. 2026 Environ Sci Technol

Probabilistic assessment of $\text{PM}_{2.5}$ and ozone co-occurring extremes under wildfire smoke

Spatiotemporal Bayesian neural field (ST-BayesNF) integrating satellite smoke-coverage products, ground $\text{PM}_{2.5}$ and ozone, and meteorological reanalysis to give daily probabilities of *joint* extremes at 5 km over the contiguous US, 2004–2023. The model discriminates co-extreme from non-co-extreme days and is most sensitive to smoke density interacting with boundary-layer height and near-surface temperature; predicted co-occurrence risk rises during smoke events not only in the western US but increasingly in the Midwest and Northeast under long-range transport. No health outcome is modelled — the health premise (co-occurrence raises risk beyond either pollutant alone) is imported from prior literature and not tested here.

doi: 10.1021/acs.est.6c04403

Zaman & Mim 2026 Environ Sci Atmos

Decadal variation of atmospheric $\text{PM}_{2.5}$ over Bangladesh from MERRA-2 reanalysis

A 25-year reanalysis climatology of $\text{PM}_{2.5}$ and its speciated components across contrasting Bangladeshi microenvironments, presented by the authors as the first comprehensive treatment for a country that ranks among the most $\text{PM}_{2.5}$ -polluted globally. The Crossref abstract is truncated at source after the framing sentence, so trend magnitudes, speciation shares and the validation against ground observations are all unavailable here. The methodological caveat is generic but load-bearing for this kind of study: MERRA-2 assimilates AOD, not surface PM, and its skill over South Asia is known to be poorer than over North America or Europe — so a reanalysis-only trend for Bangladesh needs its ground-truth comparison read before the numbers are used.

doi: 10.1039/d6ea00108d

Zhang et al. 2026 Environ Sci Technol

H_2O_2 -driven sulfate formation at air–water interfaces: stepwise mechanism and accelerated kinetics

Theoretical study of the dominant global route to particulate sulfate. At the air–water interface the pre-reaction complex is favoured and the reaction proceeds stepwise — nucleophilic attack of HSO_3^- on H_2O_2 to a HOOSO_2^- intermediate, then isomerisation to sulfate — with a **rate-determining barrier of 4.1 kcal/mol**, lower than in bulk, and with the rate-determining step itself shifting. Interfacial electric field and partial solvation together account for **89% of the barrier reduction**. Limitation: entirely computational, with no experimental kinetic validation, and the idealised interface is not a deliquesced particle surface. It matters because rising H_2O_2 from wildfire activity makes this pathway more important, and because sulfate is both a large $\text{PM}_{2.5}$ mass fraction and one of its more hygroscopic — therefore optically most humidity-sensitive — components.

doi: 10.1021/acs.est.5c17902

Zhang et al. 2026 Environ Geochem Health

National-scale heavy metal contamination in urban road dust across 82 Chinese cities

13,269 samples pooled from 169 studies into an urban-agglomeration framework; geo-accumulation index for anthropogenic enrichment, potential ecological risk index, multivariate source deconvolution. Cd and Hg are the priority pollutants, jointly **>85% of total ecological risk** in every agglomeration despite low absolute concentrations relative to Mn and Zn; a Cd/Hg I_{geo} ratio is proposed as a source fingerprint. Pooling 169 studies with heterogeneous sampling depth, particle-size cut and digestion protocol is the central weakness — the fingerprint may partly index methodology. Road dust is the resuspension reservoir behind non-exhaust PM, so this pairs with Park et al. above.

doi: 10.1007/s10653-026-03422-2

Zhang et al. 2026b Environ Sci Technol

Photoinduced intraparticle chemistry drives the rapid atmospheric oxidation of biomass burning organic aerosol

Chamber isolation of biomass-burning aerosol aging pathways — UV-A irradiation, OH-driven SOA formation, heterogeneous OH oxidation, ozonolysis and evaporation — benchmarked against aircraft measurements of wildfire and prescribed-burn plumes. **UV-A irradiation drove intraparticle aging an order of magnitude faster than gas-phase oxidation**, at an estimated quantum yield of $\sim 0.4\%$ (four in a thousand absorbed photons generate particle-phase radicals). The authors estimate that **over half of oxygenated OA in biomass-burning-influenced regions may originate from photo-oxidised primary OA within the first day**. This inverts the standing assumption that gas-phase oxidation dominates BBOA aging, and it matters for exposure work because oxidation state governs both the oxidative potential of the inhaled particle and the optical properties that low-cost scattering sensors respond to — a smoke plume's sensor bias is not constant along its trajectory.

doi: 10.1021/acs.est.6c02523

Zhang et al. 2026d Atmos Chem Phys*Improved representation of isoprene-derived secondary organic aerosol in CAM6-Chem reveals regional contrasts in its long-term changes over China*

An explicit isoprene-SOA formation scheme implemented in CAM6-Chem with high- NO_x gas-phase epoxide precursors and branch-resolved heterogeneous uptake from both low- and high- NO_x pathways. Against ground observations the update **improves the SOA normalised mean bias over China from -76.7% to -51.6%** and reproduces the compositional structure of four explicit ISOA subspecies. Over 2000–2019 the national mean ISOA trend is weak because regional changes cancel: **the sharpest increase is in Southwest China, where rising biogenic isoprene emission is the leading statistical predictor at a 58.3% regression-based contribution**, with the strongest decrease in Shaanxi–Gansu–Ningxia. The honest reading of the headline number is that a -51.6% bias is still a factor-of-two underestimate — the scheme narrows the gap without closing it, and a regression-based attribution is not a causal decomposition. For exposure work the consequence is directional: any health study using a CTM-derived organic-aerosol surface over China is working with a field that is known-low by roughly half, and non-uniformly so by region.

doi: 10.5194/acp-26-11967-2026

Zheng et al. 2026 Atmos Environ*Spatially heterogeneous drivers of winter $\text{PM}_{2.5}$ in China from synchronous observation and explainable ML*

Crossref deposit of 7 August, **metadata only** — Elsevier deposits no abstract to Crossref and the record is not yet in Europe PMC, PubMed or Semantic Scholar. No summary is offered. Logged because synchronous multi-site observation paired with explainable machine learning is the design that the sensing literature most often lacks: it is the combination that could distinguish genuine spatial heterogeneity in drivers from instrument heterogeneity across a network.

doi: 10.1016/j.atmosenv.2026.122285

Zheng et al. 2026 Atmos Res*Aerosol modulation of precipitation systems migrating from mountains to plains in Northern China during early autumn*

Metadata only. Same provenance and same caveat as the entry above. The topic is adjacent to Yang et al. on this page, but nothing in the two can be compared until an abstract or full text is available.

doi: 10.1016/j.atmosres.2026.109289

Zhong et al. 2026 Environ Int*Photochemical ultrafine particles pose potential inflammatory risk despite low $\text{PM}_{2.5}$ mass*

Three years of hourly Fuzhou observations; lung-deposited surface area compared against what the local LDSA/ $\text{PM}_{2.5}$ ratio predicts, with predicted inflammation from an animal-derived surface-area dose–response. **6.6% of $\text{PM}_{2.5}$ -attainment hours ($<35\ \mu\text{g}/\text{m}^3$) carried $7.5\times$ the expected surface-area exposure, and 82% of the predicted inflammatory risk in those hours was undetected by mass.** Median $\text{PM}_{2.5}$ during them was $3.0\ \mu\text{g}/\text{m}^3$, with the dose–response projecting 27% lung neutrophil infiltration. Machine learning attributes the events to photochemical production of 50–100 nm particles on warm, high-radiation summer–autumn afternoons (O_3 and secondary organic carbon dominate importance). **The limitation that threatens the inference is the dose–response itself:** an animal-derived surface-area relation extrapolated to a human population gives a predicted, not an observed, inflammatory response, and no human endpoint is measured here. Even so, the decoupling of LDSA from mass is measured, not modelled, and it is a direct challenge to the assumption that a dense network of scattering-based mass sensors indexes health-relevant exposure.

doi: 10.1016/j.envint.2026.110469

Zhou M et al. 2026 Atmos Chem Phys*Surface $\text{PM}_{2.5}$ in 2022 India: emission updates, WRF-Chem evaluation and source attribution*

First WRF-Chem evaluation and source attribution for India for 2022, with an updated residential inventory reflecting the household clean-fuel transition, a plant-level coal-power inventory, a revised secondary-organic-aerosol scheme and improved near-surface mixing. **Annual $\text{PM}_{2.5}$ bias $0.2 \pm 16.9\ \mu\text{g}/\text{m}^3$ ($0 \pm 31\%$) across 288 South Asian monitoring sites;** simulated population-weighted annual mean $47.4\ \mu\text{g}/\text{m}^3$. Attribution: transboundary transport 27% ($12.8\ \mu\text{g}/\text{m}^3$) — larger than any single domestic sector — industry 18% ($8.6\ \mu\text{g}/\text{m}^3$), residential 15% (7.3 , still dominant in the Indo-Gangetic Plain), power 13% (6.1 , concentrated in central India). A $\pm 31\%$ annual bias is respectable for a CTM but is the same order as the sectoral shares being separated, so the ranking of industry above residential should be read as provisional. The residential demotion is nonetheless a policy-relevant result about a transition that actually happened.

doi: 10.5194/acp-26-10533-2026

da Costa Cardoso et al. 2026 Environ Monit Assess*Legacy Pb–Zn mine tailings as persistent sources of bioaccessible toxic elements in urban dust*

18 urban dust samples and reference soils from Boquira, a former Pb–Zn district in northeastern Brazil; microwave-assisted digestion and XRF for Pb, Zn, Cd, Cu, Mn, Fe, Al; geoaccumulation index, contamination and enrichment factors, ecological risk index; oral bioaccessibility by dilute-nitric-acid gastric simulation. **Pb bioaccessibility $\sim 75\%$, Cu 54.8%, Mn 28.1%, Zn 27.0%.** Oral ingestion dominated over dermal; non-carcinogenic HI exceeded 1 for children on Pb, and carcinogenic risk from Cd exceeded limits for both children and adults. Eighteen samples in one district, a single gastric extraction protocol without an intestinal phase, and no particle-size fractionation — which matters, because resuspension and inhalation-relevant dose are size-dependent and this design cannot resolve them.

doi: 10.1007/s10661-026-15743-x

de la Torre et al. 2026 J Hazard Mater*A 15-year atmospheric record of intentionally produced persistent organic pollutants in Spain*

Polyurethane-foam passive samplers at **23 sites, 2009–2023**, evaluating Stockholm Convention effectiveness. Hexachlorobenzene dominates at a median **41%** of $\sum$ POPs with a remote-enriched spatial pattern; legacy pesticides and indicator PCBs show temperature-dependent secondary volatilisation from contaminated soils by Clausius–Clapeyron analysis, and an *urban* enrichment plausibly driven by heat-island effects; PBDEs show a **35-fold** urban-to-remote gradient from late-regulated BDE-209. Early-banned compounds decline up to **-33% /yr** while PeCB, iPCBs and BDE-209 are **stalled or rising to $+20\%$** . This is semi-volatile organic chemistry, not particulate mass, and enters on method rather than substance: a passive sampler is the cheapest long-record instrument in atmospheric science and this is a clean demonstration of what a 15-year, 23-site, zero-power network buys — an altitudinal profile on Tenerife separating local secondary emission from free-troposphere clearance. The tradeoff is equally clear: no time resolution, no size information, and partitioning between gas and particle phases is unresolved throughout.

doi: 10.1016/j.jhazmat.2026.143155

NOT CARRIED BY A DAILY ISSUE — EXTRACTIVE PRECIS OR METADATA ONLY, NOT AN ASSESSED SUMMARY (1 RECORDS)

Delgado Peralta et al. 2026 Atmos Environ*Improving $\text{PM}_{2.5}$ simulations in metropolitan Sao Paulo via PMF-based emission inventory adjustment*

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1016/j.atmosenv.2026.122249

Cardiovascular & metabolic

23 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (23 OF 23 RECORDS IN THIS SUBTOPIC)

Chen et al. 2026 Research Square (preprint)*Compound cold wave-PM_{2.5} exposure and cardiovascular emergency ambulance calls*

82,714 cardiovascular emergency ambulance call records from Shijiazhuang, 1 January 2014 to 31 December 2023, with three cold-wave categories defined on temperature percentiles crossed with four PM_{2.5} pollution levels to construct compound exposure events; distributed lag non-linear model with quasi-Poisson regression for lagged and interactive effects. The compound-exposure framing is the contribution — most temperature × PM work tests a multiplicative interaction term rather than constructing joint exposure states. Preprint, not peer reviewed; ambulance calls are a severity-filtered and access-dependent outcome; and a 3×4 exposure grid over ten years leaves the extreme cells thin.

doi: 10.21203/rs.3.rs-10547139/v1

Chen et al. 2026 Ecotoxicol Environ Saf*Pollutant-specific plasma proteomic signatures and atrial-fibrillation-free life lost*

UK Biobank, ~2,900 Olink plasma proteins screened against long-term PM_{2.5}, PM₁₀, NO₂, NO_x, SO₂, benzene and O₃ under Bonferroni correction, then elastic-net regularisation to build one signature per pollutant; incident AF over a median 13.53 y by Cox regression. Signatures out-predicted the external exposure metrics and stratified risk after adjusting for ambient concentration; high-risk participants lost an estimated 1.15–1.25 AF-free years. High-throughput mediation nominates **ANGPT2, GDF15 and PLAUR**. The weakness is structural, not analytic: a signature trained on a modelled exposure inherits that model's error and its spatial confounding, so "out-performs the exposure metric" may mean "encodes residence better". No HR or interval is reported, so nothing enters today's forest plot.

doi: 10.1016/j.ecoenv.2026.120666

Chen et al. 2026 Indoor Air*Residential incense burning, indoor pollutants and cardiovascular haemodynamics*

Seventy-three adults from the YOUNG TAIwanese cohort across 60 households, with continuous 24 h indoor PM_{2.5}, PM₁₀, CO, CO₂, TVOC and formaldehyde alongside non-invasive haemodynamic measurement, analysed by mixed-effects models. Incense-burning households showed higher pollutant concentrations and worse haemodynamics — higher blood pressure and cardiac workload, lower vascular compliance. **The finding that matters is interactive: PM_{2.5} × TVOC and PM_{2.5} × CO carried the effects on myocardial contractility and vascular elasticity**, and in non-incense homes elevated CO₂ acted as an independent stressor. No effect sizes are given in the abstract. $n = 73$ with household clustering is thin for interaction terms, and residual confounding by household behaviour is not addressable here. Implication for instrumentation: a PM-only indoor node is blind to the combination that did the work.

doi: 10.1155/ina/8778576

Correa Lara et al. 2026 Thromb Haemost*Ambient PM and venous thromboembolism: updated systematic review and meta-analysis*

19 observational studies, >4.5 million participants, searched across PubMed, Embase, Web of Science and Cochrane, harmonised to 10 µg/m³ and pooled under restricted maximum likelihood with Hartung–Knapp adjustment. Overall long-term: **PM_{2.5} HR 1.42 (0.88–2.28)**, $I^2 = 97.0\%$; **PM₁₀ HR 1.18 (0.85–1.64)**, $I^2 = 96.2\%$. Restricting to studies natively reporting per 10 µg/m³: **PM_{2.5} 1.070 (1.010–1.135)** and **PM₁₀ 1.023 (1.000–1.047)**, both at $I^2 = 0\%$. Short-term exposures were null. The sensitivity analysis is the finding — near-total heterogeneity removed by excluding rescaled contrasts — but it is also a smaller, non-random subset, and a subset selected on a reporting characteristic can be homogeneous for reasons other than correctness. Inconsistent VTE case definitions across the 19 studies remain unaddressed.

doi: 10.1055/a-2945-4974

Gaggini & Vassalle 2026 Int J Mol Sci*Gene–air pollution interaction in cardiovascular disease*

Narrative PubMed review of combined genetic-susceptibility and air-pollution effects on cardiovascular risk, concentrating on detoxification and inflammation genes. Argues the area could refine cardiovascular risk prediction beyond traditional factors. Narrative, not systematic; gene–environment interaction in this literature is dominated by underpowered candidate-gene studies, which the review acknowledges as "lights and shadows" but does not quantify. Useful only as an entry point to the literature.

doi: 10.3390/ijms27156818

Kakhaia et al. 2026 Environ Int*Exposome approach to causal effects of time-varying air pollution mixtures on CVD incidence*

UK Biobank, annual PM_{2.5} and NO₂ from a Europe-wide LUR model, 10-year cumulative CVD incidence under modified treatment policies. Confounders selected by a causal structure discovery algorithm with prior knowledge; sequential doubly robust non-parametric estimation with ensemble learning and censoring adjustment. A 25% exposure reduction over 2005–2019 gave -0.62 per 100 (95% CI -0.70 to -0.55), $RR \approx 0.90$; the projected net-zero scenario reached -1.05 per 100 and $RR \approx 0.85$. **The stated sensitivity is the finding**: dropping assessment-centre region from the adjustment set more than doubled the estimate, so the headline number is contingent on a spatial-confounding choice the authors cannot adjudicate. Method-forward, estimate-fragile.

doi: 10.1016/j.envint.2026.110441

Leiser et al. 2026 JACC Adv*Ambient PM_{2.5} and preclinical change in cardiac structure and function*

1,630 participants with echocardiography at two visits (2008–11, 2015–18), median 4.3 years apart; residential PM_{2.5} and NO₂ from spatio-temporal models as annual means preceding each visit, adjusted for baseline and time-varying covariates. Per IQR of PM_{2.5}: **global longitudinal strain +0.52% (0.20–0.83)**, **LVEF $-1.05\%/year$ (-1.73 to -0.37)**, **relative wall thickness +0.12 unit (0.05–0.19)**, **left atrial volume index -1.39 (-2.25 to -0.54)**. NO₂ moves the same direction, smaller. The LAVI direction sits awkwardly with a concentric-remodelling reading and the paper does not reconcile it; treat the strain and RWT results as the load-bearing ones.

doi: 10.1016/j.jaccadv.2026.103104

Li et al. 2026 (Environ Int) Environ Int*PM_{2.5} constituents, hepatic fat and stiffness in youth, modified by PNPLA3 genotype*

113 Los Angeles Latino adolescents with obesity, MRI-measured hepatic fat fraction and liver stiffness, serum metabolomics and lipidomics, and visit-year mean exposure to **15 PM_{2.5} components** at residence; linear regression plus g-computation for mixture effects with PNPLA3 genotype as modifier, and moderated mediation. A simultaneous IQR increase across all components was associated with **10.0% higher hepatic fat fraction**, with elemental carbon, bromine, copper, potassium and lead as leading contributors — a combustion and traffic signature. **Effects concentrated in PNPLA3 GG carriers**, who showed 550 mediating metabolite features against 196, broader pathway enrichment, and the only significant total mediation. Replication in 81 mixed-ethnicity young adults held for EC and Br. Small discovery sample, cross-sectional, modelled rather than personal exposure — the genotype interaction is the hypothesis, not the estimate.

doi: 10.1016/j.envint.2026.110438

Li et al. 2026e Environ Sci Technol Lett*Sulfate-dominated PM_{2.5} constituents are associated with increased acute myocardial infarction mortality: a US nationwide difference-in-differences study*

County-level US AMI mortality against PM_{2.5} and its EC, OC, SO₄²⁻ and NO₃⁻ constituents, analysed with a causal mixture method inside a difference-in-differences design. Total mass: **RR 1.019 (95% CI 1.013–1.025) per IQR**. Constituent mixture: **RR 1.116 (1.101–1.130) per one-quartile increase**, with SO₄²⁻ carrying weight 0.762. Regionally, an eastern county sees **1.545 (0.656–2.437) excess AMI deaths per 100,000 per year** relative to the west per quartile of mixture, SO₄²⁻-driven in the east (0.707) and OC-driven in the west (0.485). Exposure is county-level and the outcome is mortality rather than incidence, so this is ecological in structure — the DiD design controls time-invariant county confounding but not time-varying composition-correlated confounders such as regional industrial decline.

doi: 10.1021/acs.estlett.6c00288

Liu et al. 2026 China CDC Wkly*Long-term ozone and cardiovascular disease: exposure windows, sex and region*

Meta-analysis of cohort studies from PubMed, Web of Science and Embase, stratified in the way most prior reviews were not. Per 10 µg/m³ ozone: CVD mortality RR 1.0105 (1.0027–1.0185), morbidity RR 1.0578 (1.0289–1.0874). The exposure-window split is the result: **warm-season ozone RR 1.0186 (1.0119–1.0253) versus an annual-average RR of 0.9971 (0.9781–1.0164)** — annual averaging destroys the signal. Female RR 1.1135 (1.0057–1.2328) versus male 1.0707 (0.9855–1.1633); Asia estimates are imprecise. Co-pollutant record; no PM estimate is reported. The averaging-window lesson transfers directly to PM network design.

doi: 10.46234/ccdcw2026.161

Liu et al. 2026b Indoor Air*Household clean cooking fuel transition and attenuated cardiopulmonary function decline in middle-aged and older adults*

CHARLS 2011–2020, with two independent subcohorts built from baseline solid-fuel users: a short-term physiological cohort (n = 6,534; 27.4% transitioned by 2013) and a long-term disease cohort (n = 6,362; 33.5% transitioned by 2015). Transition attenuated peak-expiratory-flow decline by **11.0 L/min (95% CI 4.7–17.3)** and lowered systolic blood pressure modestly ($\beta = -1.7$, $p = 0.017$). Long term: **OR 0.80 (0.7–1.0, $p = 0.044$) for chronic lung disease and 0.80 (0.7–1.0, $p = 0.028$) for heart disease**, with no association for hypertension; protective effects were larger in non-smokers and non-drinkers. The two-subcohort design is what lets the physiological intermediate and the disease endpoint be read together.

doi: 10.1155/ina/8898458

Moro et al. 2026 Environ Res*Long-term PM_{2.5}, hypertension and metabolic syndrome in rural Ghana*

Cross-sectional analysis of 1,833 AWI-Gen participants aged 40–60 in rural Ghana, with 2007–2015 mean PM_{2.5} from a satellite-based model and MetS by harmonised criteria, under sequentially adjusted multivariable logistic regression. **OR 1.06 (1.02–1.11) per 1 µg/m³ for hypertension**, a monotone quartile gradient ($p_{\text{trend}} = 0.003$) with the top quartile at 50% greater odds, and **PAF 18.6%. No association with MetS overall or with any other component** — the null is as informative as the positive, and the authors report it rather than burying it. Three limitations bound the inference: the design is cross-sectional so temporality is not established; exposure is modelled at satellite resolution over a region with almost no reference monitoring and no local validation is reported; and household air pollution from solid-fuel cooking, which dominates personal exposure in this setting, is not in the model at all.

doi: 10.1016/j.envres.2026.125484

Nong et al. 2026 PLoS One*Long-term air pollution and microvascular complications in diabetes*

UK Biobank, 9,671 participants with diabetes, 12.4-year median follow-up, 2,104 (21.8%) incident microvascular complications. Cox models adjusted for demographic, clinical and biochemical covariates give **PM₁₀ HR 1.44 (1.27–1.64)** and **NO₂ HR 1.27 (1.12–1.45)** for the highest versus lowest exposure quartile, stable across stratified and sensitivity analyses, with early divergence of the cumulative-incidence curves. Two caveats: PM_{2.5} is listed among the exposures but no PM_{2.5} estimate is reported in the abstract, and UK Biobank exposure contrasts are narrow, so the quartile framing hides how small the underlying concentration difference is. Restricted cubic splines are reported as showing risk beginning at relatively low levels — the claim that matters for guideline-adjacent argument, and the one the abstract quantifies least.

doi: 10.1371/journal.pone.0354711

Ramírez-Moreno et al. 2026 Med Sci (Basel)*Regional stroke hospitalisation and mortality against long-term PM_{2.5}*

Nationwide ecological analysis of Spain's 17 autonomous communities, 2013–2021, **306 sex-specific region-year observations**, linking INCLASNS stroke indicators to MITECO ambient pollution. Mean age-standardised hospitalisation rate 20.1 ± 5.4 per 10,000. **PM_{2.5} had the largest positive correlation with age-standardised mortality of any pollutant examined** ($r = 0.264$, $p < 0.001$), which the authors correctly describe as weak; associations with hospitalisation and with other pollutants were small and inconsistent in sign. **Regional variation in stroke burden was not explained by the available pollution indicators**. Exposure is at region-year resolution against sex-stratified outcomes, so the sex analysis is uninterpretable by construction and the authors say so. Reported here as a well-labelled null; the design cannot support individual-level inference and does not claim to.

doi: 10.3390/medsci14040475

Sewor et al. 2026 Environ Res Health*Personal household-air-pollution exposure and blood pressure in rural Rwandan children*

622 children aged 8–15 from 626 biomass-cooking households, 48 h personal PM_{2.5} and black carbon by Ultrasonic Personal Aerosol Sampler, then standardised BP measurement. Median 48 h PM_{2.5} **192.9 µg/m³** (IQR 119.7–336.0), BC 9.85 µg/m³. **No association with SBP or DBP** (PM_{2.5}-SBP 0.51 mmHg per 211.7 µg/m³ IQR, 95% CI -0.46 to 1.49) and no clear effect modification by age or sex. Yet 24% of the children were hypertensive or had elevated BP against US reference percentiles. Cross-sectional, and the exposure range sits entirely above the region of the global exposure-response curve where slope is expected — a null that constrains where measurement effort is epidemiologically productive.

doi: 10.1088/2752-5309/ae8ff9

Shi et al. 2026 Catheter Cardiovasc Interv*Ischemic heart disease in G20 countries: forecasted burden and regional disparities based on GBD 2023 estimates*

GBD 2023 descriptive burden analysis of ischemic heart disease incidence, mortality and DALYs across the G20, 1990–2023, with EAPC and joinpoint regression, population attributable fractions for modifiable risks, ARIMA and Bayesian age-period-cohort projection to 2050, and XGBoost with SHAP for DALY determinants. **8.52 M cases and 5.81 M deaths in 2023**; age-standardised incidence and mortality fell (EAPC -1.61 and -1.49) but incidence rose in women aged 15–49, with Russia highest on all three age-standardised rates and Japan projected to peak again around 2032. High systolic blood pressure dominated the risk attribution, with **particulate matter and lead named as the prominent environmental contributors**. Limitation: GBD population attributable fractions are modelled from the same exposure–response literature this watch tracks, so this is not independent evidence of a PM effect — it is that literature propagated through a burden model, and the machine-learning layer adds interpretability, not identification. Tier C: read as context, not as evidence.

doi: 10.1002/ccd.70783

Ting 2026 JACC Adv*Statistical fragility and methodological constraints in metabolomic mediation of air pollution and cardiovascular risk*

Metadata-only: correspondence deposited without an abstract. Registered because it completes an exchange this watch is already carrying — the authors' reply (10.1016/j.jaccadv.2026.103149) surfaced in the 12 August window as an Elsevier metadata-only record and remains unresolved. The subject, statistical fragility of mediation estimates in air-pollution metabolomics, is directly relevant: mediation analysis on high-dimensional metabolite panels with modest cohort sizes produces indirect effects whose confidence intervals are dominated by the product-of-coefficients variance, and the literature routinely reports them as though they were primary. No numeric claim is made from this record. Retrieve both letters together once full text is available.

doi: 10.1016/j.jaccadv.2026.103150

Wang L et al. 2026 Lipids*Joint exposure to small dense LDL-C and long-term PM_{2.5} and incident stroke*

CHARLS 2011–2020, **6,457** stroke-free adults, **710** incident strokes over a median 9 y; PM_{2.5} from the ChinaHighPM_{2.5} product; Cox models with *k*-means trajectory clustering and restricted cubic splines. Per 1 SD: **sd-LDL-C HR 1.116 (1.018–1.224)**, **PM_{2.5} HR 1.321 (1.222–1.428)**, with a significant multiplicative interaction ($p=0.037$) and **HR 1.713 (1.335–2.198)** for the dual-high stratum against dual-low. A sustained high sd-LDL-C trajectory added 36% (1.361, 1.086–1.706). sd-LDL-C outperformed conventional LDL-C for discrimination (AUC 0.578 vs. 0.534, $p < 0.001$) — though both are poor in absolute terms, which the abstract does not say. sd-LDL-C is *estimated*, not measured, and PM_{2.5} is a modelled annual mean at residence, so the interaction is between two proxies and is the result most likely to attenuate under better measurement.

doi: 10.1002/lipd.70072

Wittkopp et al. 2026 Am Heart J*Cleaner Air for Lower Cardiometabolic Risk: HEPA filtration trial protocol in prediabetes*

Protocol for a double-blind, randomised, sham-controlled trial of one month of bedroom HEPA filtration in adults with prediabetes in the New York City area (NCT05994937), with continuous indoor PM_{2.5} measured throughout. Primary outcomes are continuous glucose monitoring metrics before and after; exploratory outcomes include insulin resistance, serum biomarkers and transcriptomics. Secondary analyses will model measured PM_{2.5} as a continuous independent variable, which is the analysis that will actually be informative — the intention-to-treat contrast in filtration trials is routinely diluted by baseline indoor concentrations already being low. One month is far too short for any glycated-haemoglobin endpoint, which is presumably why continuous glucose monitoring was chosen instead. No results; logged to `trials.json`.

doi: 10.1016/j.ahj.2026.107573

Yang et al. 2026b Blood Press*Endothelial function at low-level chronic residential air pollution*

453 Flemish adults (47.7% women, mean age 52.8 y) with endothelial function by finger photoplethysmography after 5 min ischaemia, related to residential PM₁₀, PM_{2.5}, NO₂ and black carbon from high-resolution spatiotemporal interpolation. Six-year means were **15.9, 12.8, 14.3 and 1.04 µg/m³** — at or below current EU limits and near the WHO interim range. **Dmax was inversely associated with PM_{2.5}** irrespective of risk-factor adjustment; other pollutant–outcome pairs were consistently inverse but weaker ($p < 0.10$). Association sizes **weakened as pollution fell over 2010–2015** ($p \leq 0.044$), and the spatial gradient in endothelial dysfunction tracked the PM_{2.5} gradient. Both endothelial measures predicted a composite cardiovascular endpoint (HR 0.27–0.49). Modest *n*, and no confidence intervals are given for the exposure–endothelium associations themselves, which is why none appear in the forest plot. The attenuation-with-decline result is the notable one and is rarely reported.

doi: 10.1080/08037051.2026.2724736

Zhang et al. 2026 Zhonghua Yu Fang Yi Xue Za Zhi*Combined cold-spell and heavy-pollution exposure and circulatory mortality, Beijing–Tianjin–Hebei*

Daily mortality, PM_{2.5} and meteorology from 70 districts and counties, cold seasons (November–March) 2013–2018, mean daily PM_{2.5} **98.58 ± 78.19 µg/m³**. Three combined-exposure scenarios were constructed — overlapping cold spell and heavy pollution (A), a standalone pollution event within 7 d of a cold spell ending (B), and a standalone cold spell within 7 d of a pollution event ending (C) — with district-level DLNMs pooled by random-effects meta-analysis. The overlapping event gave **CRR 1.35 (1.14–1.60)** against **1.14 (1.09–1.20)** for cold spell alone and **1.08 (1.00–1.16)** for heavy pollution alone. Supra-additivity is the substantive result, and it is the case for joint rather than single-hazard alerting. Heavy pollution is defined by a concentration threshold, not by composition, so the interaction could equally be a composition effect: winter heating episodes in this region differ chemically from other high-mass days, and the design cannot separate the two.

doi: 10.3760/cma.j.cn112150-20260105-00006

Zhao et al. 2026 Medicine (Baltimore)*Ischaemic stroke burden and attributable risk factors in adults aged 20–54, 1990–2021*

GBD 2021 extraction for ischaemic stroke in the 20–54 age band, with age-period-cohort models and population-attributable fractions by risk factor. In 2021 there were **16,301,644 prevalent and 1,324,743 incident cases** (432.46 and 35.14 per 100,000), 105,859 deaths and 6.97 million DALYs. **Prevalence and incidence rose over the period while mortality and DALYs fell**, with burden concentrated in high-middle SDI regions, Eastern Europe, Central Asia and parts of North Africa and the Middle East. Air pollution enters only as one of the GBD attributable risk factors, so this contributes no exposure–response estimate and inherits every GBD modelling assumption, including sparse primary data in the highest-burden regions. Carried for the age-band signal: young- adult stroke burden is rising in exactly the regions where PM exposure is highest and monitoring is thinnest.

doi: 10.1097/MD.00000000000050104

Çulpan et al. 2026 Rev Environ Health

Long-term air pollution and blood pressure in children: updated systematic review and meta-analysis

PRISMA-guided update over prior reviews plus PubMed and Web of Science to 27 March 2025; 21 studies, 990,157 participants in the qualitative synthesis, with meta-analysis restricted to the fine and coarse fractions and NO₂ because cohort data for SO₂, O₃ and CO were too sparse. **Fine PM: +0.59 mmHg systolic per 10 µg/m³ (0.28–0.90)**, the only estimate consistent across sensitivity analyses. The coarse fraction gave nominally tighter systolic (+0.28, 0.25–0.31) and diastolic (+0.27, 0.24–0.30) associations that *did not survive* sensitivity analysis — an implausibly narrow interval for a pooled estimate, which usually indicates a single dominant study. NO₂ was null for both. Several pollutants raised the odds of hypertension as a dichotomous outcome. Two limitations are load-bearing: exposure increments and hypertension definitions were not harmonised across primary studies, and no multipollutant model could be fitted because the primary studies did not report exposure covariance, so the pollutant-specific attribution is weaker than the point estimates suggest. Note also that the PubMed deposit strips subscripts from every particle-size label, so fraction assignment here is inferred from the stated meta-analytic scope.

doi: 10.1515/reveh-2026-0032

Neuro / mental health

21 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (20 OF 21 RECORDS IN THIS SUBTOPIC)

Beydoun et al. 2026 Ecotoxicol Environ Saf

The interplay of air pollution with plasma neurodegenerative markers and metabolome for dementia, Parkinson's disease and all-cause mortality

UK Biobank, **19,645 participants aged ≥50** with complete proteomic, metabolomic and environmental data; Cox models plus multistate Weibull models for transitions from healthy to dementia, Parkinson's disease and death. Higher NfL predicted all three transitions; **GFAP was specific to dementia, HR 2.65 (95% CI 2.17–3.25)**. On the exposure side the result is largely negative: **PM × NfL interacted with mortality only at the nominal level, NO₂/NO_x × GFAP interacted negatively, and there was little evidence of PM interaction with either marker for dementia**. Metabolomic principal components (lipid, amino-acid, energy) did modify the biomarker–outcome relations and interacted with PM for PD and dementia; a branched-chain-amino-acid component attenuated the GFAP–dementia association. The interaction tests are nominal and unadjusted for multiplicity across a large grid, so the positive ones should be treated as hypothesis-generating. Cite this for the biomarker result and for the *absence* of a PM interaction, not for a PM effect.

doi: 10.1016/j.ecoenv.2026.120657

Chen et al. 2026 Environ Toxicol Pharmacol

Integrated plasma proteomics and metabolomes reveal potential biomolecules linking PM_{2.5} to depression

24,940 depression-free UK Biobank participants, exposure to PM_{2.5} and five constituents (ammonium, nitrate, sulfate, elemental carbon, organic carbon), Olink proteomics and NMR metabolomics, Cox models with mediation, functional enrichment and drug-repurposing prediction. **982 incident cases over a mean 13 years**; HRs reported “up to 1.20 per SD” with ammonium the strongest constituent. **140 proteins and 3 metabolites** were statistically significant mediators, mediation proportions up to 8.00% and 3.42%, enriched in lipid and unsaturated fatty acid metabolism, immune regulation and inflammation. Note that the abstract gives no interval for the 1.20 figure, which is why it does not appear on this issue's forest plot.

doi: 10.1016/j.etap.2026.105144

Chen et al. 2026i J Occup Environ Med

Gender and season disparities in the influence of short-term PM_{2.5} exposure on emergency room visits for mood disorders in Taiwan

Time-stratified case-crossover with distributed lag non-linear models over **2,172 emergency visits** for bipolar or depressive disorder in the Chang Gung Research Database, 2010–2020, linked to Ministry of Environment monitoring. Associations appear **exclusively in the hot season and predominantly in women**, with delayed effects at lags 3–5 and cumulative effects over lags 0–1 and 0–3 to 0–5. No effect estimates, confidence intervals or exposure contrasts appear in the abstract, which makes the result unquantifiable as reported. With 2,172 events split by sex and season the subgroup cells are small, and a hot-season-only, women-only finding from a design with no pre-registered subgroup plan is the classic shape of a multiplicity artefact. Carried at tier B on design quality, not on the strength of the reported signal.

doi: 10.1097/JOM.0000000000003848

Daoud et al. 2026 Environ Sci Pollut Res Int

Air pollution and suicide mortality in the UK

Balanced panel of **>2,200 local-authority-year observations, 2015–2023**, linking ONS suicide data to DEFRA modelled PM_{2.5} with local-authority and year fixed effects, plus nonlinear specifications. The raw cross-sectional association is *negative*; after fixed effects it reverses to $\beta = 0.221$ ($p = 0.007$), with a U-shaped form turning at $\approx 7.8 \mu\text{g}/\text{m}^3$. That sign reversal is the whole result and it is not reassuring: the estimate is driven entirely by within-area variation over a period of falling concentrations and rising suicide rates, with no time-varying socioeconomic controls (deprivation, unemployment, service provision) in the model. Ecological, modelled exposure, no individual-level confounders. The authors describe it as exploratory, which is the correct reading. Logged here because a second, independent PM–suicide record surfaced in the same window (Cai et al., below).

doi: 10.1007/s11356-026-38089-w

Delaney et al. 2026 Environ Epidemiol

Cumulative annual air pollution exposure and first hospitalisation with Parkinson's disease

Discrete-time survival with distributed lag models over 10,366,083 Medicare FFS beneficiaries 65+, contiguous US 2000–2016; 10-year exposure histories for PM_{2.5}, NO₂ and summer O₃; non-linear exposure–response. **PM_{2.5} OR 1.634 (1.489–1.792)** at 11.8 vs. 3.0 µg/m³; **NO₂ OR 1.474 (1.379–1.575)** at 31.7 vs. 3.7 ppb; O₃ limited. Associations carried by exposure ≥4 years pre-event. The endpoint is hospitalisation, not incidence — diagnosis-to-hospitalisation lag is unmeasured and plausibly differential by exposure via healthcare access, which is the main threat to the latency claim. PM_{2.5} and NO₂ are strongly correlated in Medicare-scale data and the paper does not resolve which is doing the work.

doi: 10.1097/EE9.0000000000000519

Domingo 2026 Arch Toxicol*Aging in a highly polluted world: challenges and solutions to prevent Alzheimer's disease*

BACKFILL (PubMed entry 2026-06-09; surfaced by the Amass connector). Mechanistic review drawing on human cohorts, animal models and *in vitro* neuronal/glial systems to link pollutant classes — ambient PM, heavy metals, pesticides, PFAS and micro/nanoplastics — to Alzheimer's-relevant neuropathology, framed against the 2024 *Lancet* Commission's inclusion of air pollution among 14 modifiable dementia risk factors and a projected >150 million cases by 2050. Shared pathways discussed are neuroinflammation, oxidative stress, blood–brain-barrier disruption, tau-kinase dysregulation, epigenetic reprogramming and gut–brain dysbiosis, with particular weight on reports of AD-type neuropathology in children and young adults in heavily polluted urban environments. That evidence is the review's most striking claim and also its weakest link: it rests substantially on one research programme's Mexico City autopsy series, where selection into autopsy, APOE distribution and co-exposure to metals and traffic noise are not separable from PM. Read as an argument for exposure reduction as a dementia-prevention lever, it is reasonable; read as evidence of a PM-specific causal pathway, it is not yet that.

doi: 10.1007/s00204-026-04469-x

He et al. 2026 Ecotoxical Environ Saf*Plasma proteomic signatures modifying the air pollution–epilepsy association*

Proteome-wide interaction screen: 51,428 participants, 2,254 Olink-style plasma proteins, PM_{2.5}/PM_{coarse}/NO₂/NO from land-use regression, plus a PCA-derived composite air-pollution index. Logistic models for prevalent and Cox models for incident epilepsy identify CIT-PM_{2.5}, PTPRR-NO₂ and BCL2L15-API interactions, enriched for immune, inflammatory and cell-death pathways, with hippocampal-volume and white-matter-hyperintensity correlates and neuronal/endothelial single-nucleus enrichment. The threat to inference is multiplicity: ~2,254 proteins × four exposures with no corrected significance level or interaction magnitude reported in the abstract. A candidate list, not a mechanism. Relevance here is that it is exposure misclassification in the LUR layer, not protein measurement, that would most easily generate this pattern.

doi: 10.1016/j.ecoenv.2026.120669

Hendrickx et al. 2026 Int J Environ Res Public Health*Recent air pollution exposure, children's neurocognition and endothelial function*

138 healthy children (mean age 10.3 y, 55.8% male) across seven Antwerp primary schools including one open-air school, with indoor and outdoor PM and outdoor NO₂ measured and endothelial and attentional outcomes assessed at three repeated visits. Exposure was resolved at **4 h (direct), 24 h (acute) and 1 week (recent) windows**. **Higher direct indoor PM was associated with reduced microvascular dilatation**, while **selective and sustained attention improved with direct and recent PM exposure** respectively — a sign reversal the authors report without resolving, and which most plausibly reflects the same seasonal or activity pattern driving both classroom occupancy and outdoor concentration rather than a cognitive benefit. No NO₂ associations. Excluding the open-air school shifted results, which on *n* = 138 with seven schools means the school-level clustering is doing real work. Tier B for the vascular finding; the attention result should not be cited without the full text.

doi: 10.3390/ijerph23081070

Jahanshahi et al. 2026 Int J Popul Data Sci*Dementia onset and long-term PM_{2.5} in a genuinely low-pollution setting*

Northern Ireland census data linked to administrative prescription records, 2010–2016, with dementia onset proxied by first dispensed dementia medication — an endpoint that avoids diagnostic-coding variability at the cost of missing untreated cases. Cox proportional hazards models estimated the effect of five-year moving-average PM_{2.5} and NO₂ with extensive confounder adjustment. **Clear positive associations in both unadjusted and adjusted models**, persisting across age and sex subsamples and more tentative below age 70. **Limitations: the abstract reports no numerical hazard ratios or intervals**, so the effect size cannot be entered into the forest plot; prescription-based ascertainment under-counts and lags true onset; and exposure is residence-assigned with no mobility term. The contribution is contextual — it extends the dose–response into a low-concentration range where a threshold has repeatedly been proposed and not found.

doi: 10.23889/ijpds.v11i5.3644

Lin et al. 2026 Genes Dis*A lung–brain axis hypothesis for pollution-associated epilepsy*

Hypothesis-driven review proposing that inhaled air pollutants disrupt lung microbiome homeostasis and that this dysbiosis — by analogy with the gut–brain axis — contributes to epileptogenesis. Assembles three strands: environmental associations between air pollution and epilepsy susceptibility, links between lung-intrinsic microbiota dysregulation and neurological dysfunction, and candidate molecular, immunological and neural pathways by which the lung–brain axis could modulate seizure threshold. **This is a proposal, not evidence**. No new data; the pollution–epilepsy associations it rests on are sparse and confounded, and no step of the chain (PM → lung dysbiosis → neural excitability) has been demonstrated in the same system. Carried at tier C because the mechanistic-toxicology cluster has been empty this window and because it names testable intermediate endpoints — lung microbiome composition after controlled exposure is measurable now.

doi: 10.1016/j.gendis.2026.102240

Ma et al. 2026 Ecotoxical Environ Saf*Day-on-day increases in air pollution and depression morbidity, 239 Chinese cities*

Time-stratified case-crossover on **330,610 hospital admissions and 3,726,390 outpatient visits** for depression, 239 prefecture-level-or-above cities, 2013–2017. The exposure metric is the novelty: air pollution increase between neighbouring days (APIN), the current day's absolute concentration minus the previous day's, rather than level. Per IQR increase in APIN at lag 0–3, admissions rose between 0.59% (0.06–1.13) for 8 h O₃ and 1.66% (1.09–2.22) for NO₂, with outpatient visits 0.42–0.85%; associations were robust to adjustment for absolute concentrations except for ozone. **PM_{2.5}- and PM₁₀-specific estimates are not reported in the abstract**, only the range endpoints, so no PM estimate is carried into the forest plot. The rate-of-change framing is worth borrowing regardless of the endpoint.

doi: 10.1016/j.ecoenv.2026.120707

Parkinsons Dis review 2026 Parkinsons Dis*Air pollution and Parkinson's disease pathology: clinical evidence and molecular mechanisms*

Narrative review linking long-term NO_x, PM and O₃ exposure to Parkinson's incidence and to worse prognosis, with proposed mechanisms spanning gut-microbiome alteration, oxidative stress, neuroinflammation and promotion of α-synuclein aggregation. *No pooled estimate, no systematic search strategy, no risk-of-bias assessment and no quantitative synthesis* — it is a mechanistic orientation piece, not evidence. Tier C accordingly. The gut-microbiome route is the one worth noting, because it implies an ingestion and mucosal pathway for deposited particles that inhaled-dose models do not represent, and because it predicts that the relevant exposure metric is cumulative deposited mass in a specific size range rather than ambient concentration. Carried for the register and for that single mechanistic pointer.

doi: 10.1155/padi/6914112

Parmar et al. 2026 Environ Int*Short-term PM_{2.5} and its chemical components and neuropsychiatric hospitalisations*

Ground observations fused with 5 km WRF-Chem output to build a gridded PM_{2.5} field for Delhi 2016–2019, then GAM with distributed lags against hospital neuropsychiatric admissions. Total mass gives **+0.926% (95% CI 0.097–1.762) per 10 µg/m³**; **black carbon gives +11.509% (3.272–20.402)**, ahead of organic carbon and other inorganics, and only industrial-source PM_{2.5} was significant among source contributions. Among disorder categories only anxiety reached significance. The per-unit-mass potency gap between BC and total mass is the headline, but it should be read with the exposure ranges in mind: BC concentrations are roughly an order of magnitude smaller than total mass in Delhi, so a per-10 µg/m³ contrast for BC is a large extrapolation beyond the observed spread. Single-hospital case ascertainment also limits generalisation across a city of Delhi's size.

doi: 10.1016/j.envint.2026.110464

Perera et al. 2026 Environ Health*Declining prenatal air pollution and improving cognitive scores across successive NYC cohorts*

Three successive Columbia CCCEH birth cohorts, 1998–2020, **n = 1,109**, with prenatal PM_{2.5}/NO₂ (first two cohorts) and airborne PAH (all three), and Bayley MDI at ages 1, 2 and 3. Exposures fell significantly across cohorts and MDI rose significantly; pooled associations were inverse and significant at all three ages, and cohort-specific inverse associations were clearest in the earliest, highest-exposure cohort (*n* > 700) and weaker in the 2008–2016 cohort. The design is the interesting part — using secular exposure decline as the contrast — and also the vulnerability: everything else about being a child in New York changed between 1998 and 2020, and period effects on Bayley scores are not identifiable from cohort effects here. The authors frame it as policy-relevant evidence; it is better read as consistency between two trends than as an estimated effect.

doi: 10.1186/s12940-026-01328-1

Sakhvidi et al. 2026 Ageing Res Rev*Ambient air pollution exposure and cognitive aging; mediators and possible pathways: a systematic review on the epidemiological studies*

16 studies, 72 unique exposure–mediator–outcome associations, six pollutants (PM_{2.5}, PM₁₀, NO₂, NO_x, black carbon, PM₁ components), with PM_{2.5} in 65% of analyses. Outcomes span memory, processing speed and diagnosed dementia; candidate mediators span depression/anxiety/stress, FEV₁/FVC/PEF, stroke and hypertension, CRP, type 2 diabetes, sleep and neuroanatomical change. **Cardio- and cerebrovascular disease emerged as the most consistent mediator**; mental health, sleep and lung function showed potential but unconfirmed mediation. Reporting quality was judged generally good, but most studies carried some risk of bias in outcome assessment and confounding adjustment. No pooled estimate, and 16 studies across six pollutants and this many candidate mediators is a thin base — the useful output is the map of which pathways have been tested at all, and the observation that almost none have been tested with a formal causal-mediation design.

doi: 10.1016/j.arr.2026.103330

Sullivan et al. 2026 Dev Psychopathol*Longitudinal impacts of childhood neighborhood air pollution on white matter integrity and cognition*

ABCD Study, *n* = 4,511 children (47.4% female, baseline mean age 9.9), with baseline PM_{2.5} tested against fronto-limbic fractional anisotropy and mean diffusivity across follow-up, controlling for neighbourhood factors. **The direction of association reverses with age**: at 9, higher PM_{2.5} predicted *increased* FA in anterior thalamic radiation, fornix and uncinate; by 14, *reduced* FA in all tracts. In superior longitudinal fasciculus, higher MD at 9 and reduced FA at 12 and 14. **Associations with cognition and substance use were null**. *Limitations*: baseline-only exposure ignores subsequent trajectory; neighbourhood-level assignment carries substantial misclassification; and a sign reversal is equally consistent with accelerated early myelination followed by decline, or with residual confounding that changes with age. Relevance: a design that needs longitudinal, individually resolved exposure and had only a single neighbourhood-level baseline value — the precise gap distributed personal sensing addresses.

doi: 10.1017/S095457942610159X

Wang et al. 2026e Sleep Med*Speciated fine particulate matter and self-reported sleep time in middle-aged and older Chinese adults*

19,505 CHARLS participants aged 45+, 2011–2018, with 6-month and 2-year PM_{2.5} and five constituents — BC, organic matter, SO₄²⁻, NO₃⁻, NH₄⁺ — from the Tracking Air Pollution in China product, against total sleep time (nocturnal plus napping) via linear mixed models and quantile g-computation. The secondary inorganic ions NO₃⁻, SO₄²⁻ and NH₄⁺ show the most consistent associations, and the direction is the surprise: highest versus lowest quartile of intermediate-term exposure **increases** total sleep time (PM_{2.5} +0.16 h and comparable for the constituents). Self-reported duration, no actigraphy, and an increase in sleep duration is not obviously a harm — it may index fatigue, reduced outdoor activity on high-pollution days, or reverse behavioural adaptation. Mechanism is not addressed and the finding should be treated as hypothesis-generating.

doi: 10.1016/j.sleep.2026.109229

Wu et al. 2026c Toxics*Early-life PM exposure and toddler sleep: particle size and constituents*

Birth cohort of **514 mother–child pairs** in Guangzhou followed to age two (2017–2018), with PM₁, PM_{2.5}, PM₁₀ and PM_{2.5} constituents from high-resolution ground-level grids and sleep by the Chinese Brief Infant Sleep Questionnaire. Per-IQR early-life exposure was associated with reduced nighttime sleep duration: **PM₁ β = −0.18 (95% CI −0.32, −0.04)**, **PM_{2.5} −0.23 (−0.38, −0.08)**, **PM₁₀ −0.21 (−0.35, −0.07) hours**. Quantile g-computation on the constituent mixture reproduced the decrement with **SO₄²⁻ as the dominant contributor**, and pregnancy-window findings were consistent. Outcome is parent-reported, which biases toward non-differential misclassification but not necessarily away from confounding by household environment; the three particle sizes are collinear at this resolution, so the size ordering is suggestive rather than discriminating. Sulfate as the driver is consistent with the secondary-inorganic signal this watch carried on 26 August from an unrelated endpoint.

doi: 10.3390/toxics14080653

Xu et al. 2026 Eur J Neurol*PM_{2.5} components and cerebral hypoperfusion, with mediation by inflammatory biomarkers*

967 patients with symptomatic intracranial arterial stenosis at one Suzhou hospital; hypoperfusion quantified from multimodal CT with MISTar; 3-year PM_{2.5} and component exposures from the TAP database; generalized additive models, restricted cubic splines, weighted quantile sum for the mixture, and mediation analysis. **PM_{2.5} OR 4.344 (3.323–5.734)**; **black carbon 4.396 (3.327–5.869)**, then sulfate 4.359, organic matter 4.278, nitrate 3.552, ammonium 3.313 (no intervals given). OM (44.2%) and BC (42.2%) carry 86% of the mixture effect. Mediation by neutrophils 5.37%, leukocytes 4.30%, monocytes 2.73% — i.e. inflammation accounts for roughly a tenth of the total effect, so the dominant pathway is unexplained. Single-centre clinical sample already selected on stenosis; ORs above 4 in that population are not transportable to general-population risk. Component exposures are modelled, not measured.

doi: 10.1111/ene.70716

Zhang et al. 2026 Neurotoxicology*PM_{2.5} and its components and cognitive dysfunction among rural older adults*

14,379 Henan Rural Cohort participants aged ≥60, cognitive dysfunction defined by MMSE and the Hasegawa Dementia Scale, with PM_{2.5}, black carbon, organic matter, ammonium, sulfate and nitrate from the TAP dataset; logistic and weighted quantile sum regression, then network toxicology and random forest for candidate genes. **Per-IQR increases raised CD odds by 19–43% across the six exposures**, and **BC was the leading mixture contributor under all four CD definitions** (weights 54–86%), with OM second (7–35%). Cross-sectional, so reverse causation through cognitively impaired participants' residential and behavioural patterns is unaddressed, and component fields from a single reanalysis product are strongly collinear — WQS weights under collinearity are unstable. The BC ranking is consistent with the emerging component literature; the network-toxicology arm is hypothesis-generating and labelled as such by the authors.

doi: 10.1016/j.neuro.2026.103539

NOT CARRIED BY A DAILY ISSUE — EXTRACTIVE PRECIS OR METADATA ONLY, NOT AN ASSESSED SUMMARY (1 RECORDS)

Cai et al. 2026 Atmos Environ*PM_{2.5} as the dominant air pollutant associated with suicide among people living with HIV/AIDS: province-wide case-crossover*

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1016/j.atmosenv.2026.122271

Respiratory & allergic**39 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (38 OF 39 RECORDS IN THIS SUBTOPIC)

Ali et al. 2026 Public Health Chall*Symptom burden among adult outpatients during Lahore's 2024–2025 winter smog*

400 outpatients at a tertiary hospital, October 2024 – January 2025, structured questionnaire plus city-level air quality (mean AQI 180, mean PM_{2.5} 96 µg/m³; PM_{2.5} and PM₁₀ above the WHO 24 h guideline on every monitored day). 98% reported at least one symptom — breathlessness 74%, eye irritation 56%, cough 53%. Regression on a composite vulnerability index: **smoking IRR 1.15 (1.00–1.32)**, **routine mask use IRR 0.96 (0.84–1.09)**, i.e. null. Exposure is assigned at city level to a health-seeking sample with no unexposed comparison, so no dose–response can be read from this; its value is documenting symptom prevalence in a South Asian megacity where hospital-based evidence is thin.

doi: 10.1002/puh2.70349

Aulia Anwar et al. 2026 Exp Oncol*Smoking, environmental exposures, and chronic lung disease as key risk factors for lung cancer in North Sumatra*

Hospital-based case-control across three referral hospitals in North Sumatra, August–November 2024: **240 histologically confirmed lung cancers and 240 controls** without malignancy or chronic respiratory disease, matched on sex and residential area with age adjusted in the model. Independent predictors: **unhealthy indoor environment (poor ventilation, household pollutants) OR 2.53 (95% CI 1.45–4.40)** and **occupational exposure OR 1.91 (1.13–3.25)**; tuberculosis and COPD were more prevalent among cases. **Three weaknesses threaten the inference and should be stated first.** The exposure is a self-reported composite category with no measured particulate concentration, so it is subject to recall bias in a design where cases know their diagnosis. Controls were selected to be free of chronic respiratory disease, which is a collider on the pathway from household smoke to both COPD and lung cancer and biases the household–environment OR upward. And smoking differed between groups at $p = 0.023$ yet does not appear among the adjusted independent predictors — the deposited abstract does not explain that discrepancy. Treat 2.53 as an upper bound on a poorly specified exposure, not as a household-PM effect estimate.

doi: 10.15407/exp-oncology.2026.02.152

Bogunia-Kubik et al. 2026 Allergy*Influence of environmental pollution on basophil reactivity to birch pollen allergens*

Fifty birch-allergic patients and 25 non-atopic controls followed over the 2023 and 2024 pollen seasons in Kraków, with app-recorded daily nasal, ocular and asthma symptoms against continuously monitored PM_{2.5}, PM₁₀, O₃ and birch pollen. Basophil activation tests used commercial extract, recombinant Bet v 1, and **natural pollen collected from highly and less polluted sites**. Pollen from polluted sites induced significantly stronger CD63 expression ($p < 0.0001$), and asthma symptoms correlated with both pollen and PM ($p < 0.05$, mainly 2023). *Limitations:* the central comparison is site-level, so pollution is confounded with everything else that differs between the two collection sites; $n = 75$ over two seasons with a season-specific correlation is thin; and CD63 is an ex-vivo surrogate. Relevance: if pollution modifies allergen *potency* rather than only co-occurring with it, exposure models that treat pollen and PM as independent additive terms are mis-specified.

doi: 10.1111/all.70474

Cantuaria et al. 2026 Environ Health Perspect*Source-specific PM_{2.5}, UFP, EC and NO₂ and incident COPD in two pooled Danish cohorts*

159,769 participants from the Diet, Cancer and Health cohort and the Danish National Health Survey, with complete residential address histories driving 10-year time-weighted exposures split into local-traffic and other-source contributions; Cox models adjusted for demographic, socioeconomic and lifestyle factors including smoking. Per IQR of total exposure: **PM_{2.5} HR 1.11 (1.05–1.17)**, **NO₂ 1.08 (1.04–1.13)**, **UFP 1.05 (0.99–1.11)**, **EC 1.02 (1.00–1.05)**. **The source split inverts across metrics** — non-traffic PM_{2.5} was the more harmful fraction while traffic-derived UFP and EC dominated theirs. Associations were stronger in women, never-smokers and prior asthma. Residual confounding by smoking intensity is the live threat given COPD's dominant cause. Directly relevant to siting: a traffic-gradient network resolves the wrong PM_{2.5} fraction and the right UFP one.

doi: 10.1021/EHP.6c00128

Zhongguo Fei Ai Za Zhi 2026 Zhongguo Fei Ai Za Zhi

China Wumeng Mountain lung cancer: evidence chain, clinical and molecular characteristics of regional lung cancer related to coal pollution in the Yunnan–Guizhou Plateau

Narrative synthesis assembling an exposure → component → biological-effect → clinical-molecular → prevention chain for household smoky-coal lung cancer in Xuanwei/Fuyuan, searching PubMed, Web of Science, Embase, CNKI and Wanfang to May 2026. Carcinogenic components named: **PAHs and methylated PAHs, PM_{2.5}/black carbon, inorganic minerals and metals, and carbonyls**, as a mixture. The clinical phenotype is distinctive — predominantly **never-smoking women**, high adenocarcinoma fraction, young onset, early metastasis, multifocal disease and family aggregation. So is the molecular one: total EGFR mutation is high but classical 19del and L858R are *under-represented*, while **G719X, S768I, KRAS G12C and ALK/ROS1 fusions are elevated** relative to non-Wumeng lung cancer; ARHGEF5, MYO18B and MUC16 are proposed germline susceptibility genes interacting with POT1/TERT/TERF2. Sputum microbiome shifts are reported alongside. **It is a narrative review by an unnamed author group with no formal inclusion criteria or bias assessment**, drawing on the same group's prior work, so the evidence chain is an argument rather than a synthesis. Its value is as the clearest statement in this corpus of a testable source-composition-to-mutation-spectrum hypothesis.

doi: 10.3779/j.issn.1009-3419.2026.102.16

Erickson-DiRenzo et al. 2026 J Voice

Wildfire PM_{2.5} and the human larynx: paired in-event and clean-air assessment in one adult

A single adult assessed twice — during a wildfire episode with PM_{2.5} reaching **117 µg m⁻³** and again at **<10 µg m⁻³** — with acoustic and aerodynamic voice measures, laryngeal endoscopy with stroboscopy, and *ventricular fold biopsies at both time points* analysed by qPCR for extracellular matrix and inflammatory markers. Speaking fundamental frequency fell relative to baseline, endoscopy showed visibly increased laryngeal mucus, and **MMP-1 expression roughly doubled**; other ECM and inflammatory genes moved inconsistently. *n* = 1. There is no control participant, no adjustment for phonatory load, hydration, intercurrent infection or the ordinary diurnal variation in fundamental frequency, and a two-point comparison cannot separate exposure from time. The reason to carry it is the design, not the result: paired human airway tissue across a natural exposure contrast is rare, and it establishes feasibility for a properly powered version. Treat the MMP-1 figure as a hypothesis.

doi: 10.1016/j.jvoice.2026.07.048

Feng & Chao 2026 Int J Biometeorol

Lower respiratory infection burden attributable to PM_{2.5} in China and the G20, 1990–2023

GBD 2023 deaths and DALYs from LRI attributable to ambient and household PM_{2.5}, Joinpoint regression on age-standardised DALY rates with sex subgroups and loess against SDI. Long decline reversed upward in 2021–2023 (APC China **12.76–14.23%**, G20 **14.94–16.04%** for ambient); household-attributable rates kept falling in China but rose in the G20 from 2021 (APC 8.49–11.20%). Burden concentrated in under-5s and ≥70s. The post-COVID reversal is the interesting claim and it rests on three data-years of GBD modelled estimates, which are themselves smoothed — an inflection this recent should be treated as provisional.

doi: 10.1007/s00484-026-03283-5

Govindaraju et al. 2026 J Occup Environ Med

Protective role of dietary antioxidant intake on long-term effects of extreme PM_{2.5} exposure on respiratory health

Cross-sectional analysis of 448 Hazelwood Health Study adults, with individual PM_{2.5} from the 2014 Morwell coal-mine fire reconstructed by combining modelled fire-related concentrations with contemporaneous time–location diaries — a better exposure assignment than most disaster epidemiology achieves. Vitamins A and E, magnesium and zinc attenuated the PM_{2.5}–chronic-cough association, and all but magnesium the phlegm association; omega-3 attenuated the PM_{2.5}–COPD association. *Limitations are severe*: diet was measured cross-sectionally years after a 2014 exposure, so reverse causation and healthy-diet confounding are unaddressed; **no effect sizes or intervals are reported in the abstract**; and multiple testing across several nutrients and several outcomes is not accounted for. Relevance: effect-modification analyses of this kind demand exposure precision that only time-resolved personal or time-activity-weighted measurement supplies.

doi: 10.1097/JOM.0000000000003856

Haque et al. 2026 Health Sci Rep

PM and respiratory disease in Bangladesh: what 42 studies show

Narrative review of 42 studies (2013–2025) drawn from seven databases on fine and ultrafine PM and gaseous pollutants and respiratory outcomes in Bangladesh. PM_{2.5} in Dhaka, Chattogram and Khulna is reported at **8–18× WHO guideline levels**, with persistent hotspots around brick kilns, industrial zones, transport corridors and seasonal biomass burning, and consistent associations with asthma, COPD, bronchitis, acute respiratory infection and hospital admission, concentrated in children, older adults and occupational groups. **Limitations: narrative, not systematic** — no risk-of-bias assessment, no pooling, and heavy reliance on primary studies with sparse monitoring infrastructure, so the exposure side of most included work is itself weak. Carried because it documents where the burden is largest and the measurement network is thinnest, which is the standing argument for low-cost network deployment in South Asia.

doi: 10.1002/hsr2.72925

Hatem et al. 2026 Eur J Pediatr

Child characteristics, household and home environmental factors associated with eczema

1,227 children aged 5–11 from 20 primary schools in Porto (46.5% response), analysed with mixed-effects logistic regression with a school random intercept and multiple imputation. Independently associated with higher odds: **residence within 300 m of a traffic zone, aOR 1.32 (1.11–1.48)**; **postnatal secondhand smoke, aOR 1.48 (1.26–1.73)**; family history 3.53; higher family income 1.40; all-purpose cleaner use 1.16. Lower odds: maternal education 0.68, detached housing 0.67, fur blanket use 0.85. Household dampness interacted with residence type and winter ventilation frequency. Cross-sectional and self-reported, so temporality is unavailable — but the traffic term surviving a socioeconomic adjustment set this dense is the citable part.

doi: 10.1007/s00431-026-07327-1

Hur et al. 2026 Environ Pollut

Pollen modifies the SPM–mortality association across eight Kyushu cities

Two-stage time-series over 262,288 all-cause deaths, eight cities, pollen seasons (Feb–Apr) 1989–2014, with an explicit SPM × pollen interaction term in stage one and random-effects pooling in stage two. Respiratory mortality at lag 1: **RR 1.010 (1.000–1.019)** overall, **1.021 (1.005–1.037)** on days above the 75th-percentile pollen concentration against **1.003 (0.987–1.018)** below, stable across alternative cutoffs. The threat to inference is exposure misclassification in the modifier, not the exposure: pollen is measured at a single station per city and the interaction is fitted on a binary split of a highly skewed distribution, so the contrast is a lower bound on a continuous effect. Suspended particulate matter is a Japanese regulatory metric with a 100% cut near 10 µm and is not interchangeable with PM_{2.5}. For distributed sensing the implication is concrete: an optical counter with a fluorescence or coarse-mode channel can resolve the modifier and the exposure on the same node, which no regulatory network in this study did.

doi: 10.1016/j.envpol.2026.128933

Hussein et al. 2026 J Glob Health

PM₁₀ and emergency asthma presentations in Petaling Jaya, Malaysia

Four years of ED records (Jan 2014 – Apr 2017) from a Klang Valley hospital against pollutant and meteorological data from nearby stations; generalised additive models plus distributed lag non-linear models for delayed and non-linear PM₁₀ effects. Asthma exacerbations were 2.3% of all ED visits. Relative to the Malaysian standard, daily mean PM₁₀ of 200 µg/m³ was associated with a 21.3% increase (95% CI 2.2–44.0) and 300 µg/m³ with 34.0% (4.7–71.7); the cumulative lag 0–7 exposure–response turns upward from about 150 µg/m³, and the lag structure shifts from days 2–3 at 100 µg/m³ to same-day at ≥200. Single hospital, single monitoring catchment, wide intervals — the authors correctly label the associations exploratory. The threshold shape is the transferable part: it argues for alert triggers well below current national standards.

doi: 10.7189/jogh.16.04249

Jiang et al. 2026 J Cent South Univ Med Sci

Short-term PM and respiratory-infection hospitalisation burden in kidney transplant recipients

Retrospective time-series over 874 kidney transplant recipients hospitalised for respiratory tract infection at the Third Xiangya Hospital, 2017–2023, with high-resolution daily exposure to six pollutants over the 7 days before symptom onset and DLNMs for lagged and cumulative associations. PM_{2.5} showed the strongest association with burden: at 160 µg/m³ cumulative exposure, hospitalisation cost rose by 9,544.5 yuan (95% CI 2,548.8–21,637.7); PM₁₀ showed a dose-dependent trend on length of stay (RR 1.63 at 300 µg/m³) whose interval, 0.18–15.94, is so wide it carries no information and is deliberately excluded from the forest plot. SO₂ above 75 µg/m³ also raised burden; CO, NO₂ and O₃ were null. Single centre, immunosuppressed population, ambient exposure assigned without residential history. Economic rather than clinical endpoints, which is unusual and useful.

doi: 10.11817/j.issn.1672-7347.2026.260089

Kannoth et al. 2026 J Expo Sci Environ Epidemiol

Short-term PM_{2.5} and COVID-19 pneumonia in New York City, amplified by pre-existing cardiovascular disease

15,361 NYC COVID-19 hospitalisations (March–December 2020) with short-term PM_{2.5} defined as the 7-day residential ZIP-code average from EPA CMAQ, and 15 ICD-coded circulatory conditions as the modifier. Modified Poisson regression for ARDS, pneumonia, ventilation and dialysis; Cox models for fatality and length of stay; RERI for additive interaction; TMLE for average treatment effects. PM_{2.5} alone was null (RR 1.03, 95% CI 0.95–1.12) and prior myocardial infarction alone 1.34 (1.14–1.58), but jointly 1.67 (1.41–1.99); upper-quartile PM_{2.5} raised pneumonia risk only among those with cardiovascular conditions (ATE 0.05, 0.02–0.08). The interaction is the finding and it is the sort a main-effect analysis destroys. Weaknesses are real: ZIP-code CMAQ is a coarse surrogate for personal exposure and its error is spatially structured with deprivation, the effect appears only from July onward with no mechanism offered for the seasonal split, and the cohort is hospitalised patients only, so selection on admission is unaddressed. For network design the implication is concrete — the susceptible stratum is identifiable in advance, which is an argument for siting density by cardiovascular prevalence rather than by population alone.

doi: 10.1038/s41370-026-00957-5

The Lancet Respiratory Medicine 2026 Lancet Respir Med

The far-reaching effects of wildfire smoke

Unsigned editorial. Abstract not indexed; carried as metadata only. Logged because wildfire-source attribution has been the recurring through-line of this watch since 27 July and an editorial in this journal signals where the clinical field is placing it. Full text required for any citable content.

doi: 10.1016/S2213-2600(26)00256-0

Li et al. 2026 Int J Chron Obstruct Pulmon Dis

Air pollution and COPD hospitalisation in Kashi, Xinjiang

Time-stratified case-crossover on 68,482 COPD patients in Kashi, 2012–2022, with daily pollutant concentrations matched to residential address rather than to a single city monitor. Strongest PM association at lag day 4: odds ratios 1.015 (1.002–1.032) and 1.008 (1.002–1.016) for the two PM fractions, with NO₂ and CO both at 1.034. Young patients, those with prior severe exacerbations, farmers, smokers and cool-season admissions were the susceptible strata. The honest part is the sensitivity analysis: excluding COVID-19-period hospitalisations left the PM and NO₂ associations barely significant and CO non-significant, which means a material share of the headline estimate rides on pandemic-era admission patterns. The publisher's indexing stripped every subscript from the abstract, so which PM fraction is which must be read from listing order. Kashi sits in a severe dust-storm corridor, making this one of the few large COPD series in a mineral-dust-dominated aerosol.

doi: 10.2147/copd.s617521

Makrufardi et al. 2026 World Allergy Organ J

Ambient relative humidity and particulate matter vs. asthma treatment step

288 newly diagnosed adults with asthma followed at least 6 months under GINA, with treatment step — not symptoms or spirometry — as the outcome, and both daily means and short-term differences in RH, temperature and PM_{2.5} as exposures. A 1 µg/m³ increase in the 7-day PM_{2.5} difference was associated with 1.052-fold odds of step 4 treatment and 1.056-fold odds of medium-dose ICS + LABA; a 1 µg/m³ increase in the 30-day difference gave 1.205-fold odds of triple therapy or biologics. Associations were clearer in the cold season. No confidence intervals are reported in the abstract, which is why none of these appear in the forest plot. Using variability rather than level as the exposure is the interesting choice — it maps onto what a high-time-resolution sensor measures well and a monthly average does not — but with a single centre and 288 patients this is explicitly hypothesis-generating, as the authors say.

doi: 10.1016/j.waojou.2026.101444

Morvan et al. 2026 Int J Epidemiol

Causal effect of PM_{2.5} on COVID-19 hospitalisation and death, conditioned on comorbidity

Instrumental-variable analysis over more than 3.5 million weekly municipality observations covering all of France, 2020–2021, with COVID-19 hospitalisations and deaths disaggregated by age, sex and municipality and linked to records of pre-existing conditions. Wind direction instruments local PM_{2.5} in a two-stage least squares design. Among individuals without pre-existing conditions there was no significant effect on either outcome. Conditioning on comorbidity, a 10 µg/m³ increase was associated with +1.5 COVID-19 deaths per 100,000 (95% CI 0.8–2.2), concentrated in respiratory (0.9, 0.4–1.4) and cardiovascular (0.6, 0.1–1.1) subgroups, and driven by the oldest age strata. Wind direction satisfies exclusion only if it affects COVID outcomes solely through PM_{2.5} — doubtful during a period when weather drove both mobility and indoor crowding — and the estimates are ecological rate differences, not individual risks. Still the strongest identification strategy this literature has produced.

doi: 10.1093/ije/dyag152

Ni et al. 2026 World Allergy Organ J

Socioeconomic and regional influences on eosinophilic inflammation in chronic rhinosinusitis with nasal polyps

Multicentre retrospective series, **1,047** CRSwNP patients undergoing endoscopic sinus surgery at three tertiary hospitals in Jiangsu, with paired blood and nasal-mucosa eosinophil counts at 2008, 2012 and 2016, analysed by correlation, ROC, linear regression and linear mixed-effects models. Blood eosinophils are a poor surrogate for tissue eosinophilia ($\rho = 0.28$, **AUC 0.607**, adjusted R^2 **0.053**) and vary heterogeneously by hospital and year. City-level GDP correlated with mean blood ($\rho = 0.80$, $p = 0.014$) and tissue ($\rho = 0.72$, $p = 0.037$) eosinophil counts, whereas **annual mean PM_{2.5} was associated with neither**. The null is the reason this record is carried, and it should be read as a measurement failure rather than as evidence of no effect: the exposure is a *city-level annual mean* assigned to surgical patients at three time points, which is an ecological contrast with essentially no within-city variance, over an interval (2008–2016) in which Chinese PM_{2.5} fell steeply and uniformly — so exposure and calendar time are near-collinear by construction, and any true association is absorbed by the year term. With eight city-level observations, the GDP correlations are also fragile. Tier C.

doi: 10.1016/j.waojou.2026.101440

Orr et al. 2026 (EHP) Environ Health Perspect

Wildfire-season PM_{2.5} and influenza / ILI risk across six western US states

Generalised linear distributed-lag models on state health-department influenza and ILI counts from Arizona, Colorado, Montana, Nevada, Oregon and Washington, 2010–2019, testing both preceding-wildfire-season exposure on the following flu season and short-term exposure within it. Per 10 $\mu\text{g}/\text{m}^3$: **CO 1.067 (1.056–1.078)**, **AZ 1.061 (1.026–1.100)**, **OR 1.049 (1.041–1.057)**, **MT 1.038 (1.013–1.063)** — all four states with laboratory-confirmed influenza data — against **NV 1.005 (0.920–1.097)** and **WA 0.884 (0.842–0.919)**, the two with syndromic ILI only. **The effect boundary coincides exactly with the outcome-ascertainment boundary**, so the geographic reading is almost certainly the wrong one. Ecological design, no individual exposure, and wildfire-attributable PM_{2.5} is itself a modelled quantity.

doi: 10.1021/EHP.6c00326

Rice & Miller 2026 Pediatr Pulmonol

Air pollution, microbes and respiratory infection risk in children

Narrative review joining two literatures usually kept apart: pollutant-driven epithelial barrier loss, oxidative stress and impaired innate/adaptive immunity, and the developing airway microbiome as a determinant of colonisation resistance. Argues that dysbiosis (antibiotics, air pollution, other perturbations) and pollutant exposure are partly substitutable routes to the same susceptibility phenotype, and notes that exposure to diverse environmental microbes — certain fungi included — may be protective. No quantitative synthesis and no bias assessment; useful as a framing citation for why an indoor sensing deployment in a childcare setting should log more than mass.

doi: 10.1002/ppul.71798

Shi et al. 2026 Medicine (Baltimore)

Air pollution and AECOPD hospitalisation across three Hebei cities

Ecological time series on **14,786 AECOPD hospitalisations** from 49 hospitals in Shijiazhuang, Handan and Chengde, 2018–2020 (1,096 days), with DLNMs adjusted for long-term trend, temperature, humidity, day of week, holidays and pandemic stage, then pooled by random-effects multivariate meta-analysis. Cumulative effects strengthened with lag length and peaked at lag 0–5 d for every pollutant while single-day effects were weak. **PM_{2.5} at the 99th percentile versus median gave RR 1.391 (1.169–1.658) at lag 0–5**; all five non-ozone pollutants showed non-linear cumulative exposure–response, and higher 8-h ozone was associated with *lower* admission risk, which the authors do not explain and which is most plausibly seasonal confounding. The abstract's own conclusions are internally inconsistent on susceptibility — the results state associations were more pronounced in males, the conclusion says females — so the stratified findings should not be cited without the full text. Six correlated pollutants tested across six lag structures with no multiplicity adjustment.

doi: 10.1097/MD.00000000000050080

Shukla et al. 2026 BMJ Open

Hourly NO₂ and emergency admissions for acute asthma, Birmingham and Solihull

Poisson generalised additive model with a distributed-lag non-linear term on *hourly* counts — 18,943 adult asthma admissions to four acute hospitals, June 2016–May 2022, linked through the PIONEER data hub. The mean 18 $\mu\text{g}/\text{m}^3$ NO₂ exposure over lag 0–24 h carried **RR 1.13 (1.01–1.26)** for daytime admissions, with the association concentrated at a **3–6 hour lag** and a steeper gradient in the most deprived quintile. Exposure is assigned from fixed monitors to residence, so the hourly resolution is nominal rather than personal — time-activity displacement over a 3–6 h window is precisely the scale at which residential assignment fails, which biases toward the null. The pollutant is NO₂, so this enters the watch on method rather than on substance: it is the cleanest recent demonstration that the health-relevant exposure signal lives below the 24-hour averaging period that filter-based reference methods impose.

doi: 10.1136/bmjopen-2025-110612

Silva et al. 2026 Front Sleep

Shared mechanisms linking PM_{2.5} exposure, obstructive sleep apnea and obesity

Narrative review of the three-way relationship between chronic PM_{2.5} exposure, obstructive sleep apnea and obesity, proposing systemic inflammation, oxidative stress, neuroendocrine dysregulation and gut microbiota alteration as shared pathways, and noting that intermittent hypoxia and sleep fragmentation feed back into weight gain. The mechanistic overlap is plausible and the review is explicit that it is hypothesis-organising, but **no effect estimates, no study inventory and no appraisal of confounding by adiposity** appear — and adiposity confounds both the exposure (residential location) and the outcome. Read alongside the CHARLS sleep-duration result carried on 28 August, the practical gap is identical in both: no objective sleep measurement anywhere in this literature.

doi: 10.3389/frsle.2026.1918405

Solazzo et al. 2026 Hyg Environ Health Adv

Season and pollutant interact on the nasal microbiome

Longitudinal repeated-measures design: 26 healthy Italian adults sampled weekly for three weeks in winter and three in summer, with 16S rRNA sequencing (124 samples) and whole-genome shotgun sequencing (141 samples), against weekly indoor TSP and outdoor particulate matter, black carbon, benzene and CO. The microbiota was stable within season but differed between seasons, with winter enrichment of *Moraxella* — notably among women with children. Higher pollutant levels were negatively associated with commensal abundance, particularly *Corynebacterium*, and **significant season × pollutant interactions** appeared: in summer, *Staphylococcus epidermidis* and *Cutibacterium granulosum* were negatively associated with PM. Sex explained the most variance in diversity. **Limitations:** $n = 26$ **with many taxa tested**, exposure assigned from area monitors rather than personal sampling, and season is collinear with everything from heating to indoor occupancy. Hypothesis-generating; the interaction structure is the part worth chasing with a larger cohort and personal monitors.

doi: 10.1016/j.heha.2026.100189

Tornevi et al. 2026 BMJ Open Respir Res*Short-term urban-background PM and the timing of confirmed COVID-19, three Swedish cities*

Case-crossover on **3,999** PCR-confirmed cases from a prospective adult cohort in Stockholm, Gothenburg and Uppsala, 7 Mar 2020 – 26 Jan 2022, with daily PM_{2.5}, PM₁₀ and NO₂ from urban-background stations and distributed-lag (and non-linear) models adjusted for meteorology. Mean PM_{2.5} 32.0 µg m⁻³. PM_{2.5} at lag 0: **OR 1.084 (1.011–1.162) per IQR 3.3 µg/m³**, consistent across all three cities; PM₁₀ most consistent at lag 3; NO₂ lag 0 only after dropping the final wave. The authors themselves read the same-day lag as implausible for infection and propose symptom exacerbation prompting testing — i.e. the outcome is a *test date*, not an infection date, and the estimate is at least partly a health-seeking-behaviour effect.

doi: 10.1136/bmjresp-2025-003279

Wang & Li 2026 Front Public Health*Air pollution and atopic dermatitis*

Narrative review of PubMed, Web of Science and CNKI, April 2021 to December 2025, restricted to peer-reviewed work with impact factor ≥3.0 — a screening criterion that is a proxy for nothing and silently excludes most dermatology and most regional epidemiology. Covers epidemiology, skin-barrier and microbiome mechanisms, climatic and geographical disparity, and prevention. No quantitative synthesis and no effect estimates. Carried at tier C as a citation anchor for the cutaneous endpoint, which is otherwise absent from this watch.

doi: 10.3389/fpubh.2026.1854212

Wang et al. 2026 Res Sq (preprint)*PM_{2.5} constituents and daily fever-clinic visits in Jinshan, Shanghai*

Time-series generalized additive models linking **77,637 fever-clinic visits** in 2024 to PM_{2.5} mass and speciated constituents, with single-day lags 0–7 and cumulative moving averages 01–07, adjusted for seasonality, day of week, holidays and meteorology, stratified by sex, age and season. Mean PM_{2.5} 32.0 µg m⁻³. PM_{2.5}, arsenic, cadmium, chromium, lead, manganese, nitrate and ammonium were positively associated; **cadmium showed the largest association** while nitrate and ammonium acted at delayed lags consistent with secondary inorganic aerosol. Preprint, not peer reviewed; no effect sizes or intervals in the abstract; constituents are highly collinear and single-pollutant time-series models will attribute shared variance arbitrarily among them. Fever-clinic attendance is also a behavioural endpoint sensitive to concurrent epidemic activity.

doi: 10.21203/rs.3.rs-10211219/v1

Wang et al. 2026a Front Public Health*COVID-19 social interventions amplified indoor-source pollutant–respiratory associations in Xi'an*

Six-year time series of **2,052,777** respiratory outpatient visits over 2192 days at a tertiary hospital in Xi'an (2020–2025), stratified into strict control, transition and post-pandemic phases, with quasi-Poisson GAMs and pollutant-by-phase interaction terms. Full-period associations were strongest for SO₂ (lag 4, ER **9.11%**, 95% CI 3.86–14.64) and NO₂ (lag 5, **2.72%**, 2.05–3.40). Phase-stratified, SO₂ was **18.40% (10.29–27.11)** under strict control versus **2.91% (–4.82 to 11.28)** post-pandemic; PM_{2.5} and O₃ showed no interaction ($p > 0.9$). *The authors state plainly that the interaction tests do not reach significance* ($p = 0.13$, $p = 0.15$), and single-hospital outpatient counts confound exposure effects with healthcare-seeking behaviour that the pandemic also changed. Relevance: the proposed indoor-amplification mechanism is unfalsifiable with ambient monitors alone, which is the argument for indoor and personal measurement in time-series designs.

doi: 10.3389/fpubh.2026.1896806

Wood et al. 2026 Lancet Public Health*Ultra Low Emission Zone and lung function growth in children: prospective parallel cohort*

The strongest design in the issue. **3,414 children** recruited across 84 schools (44 London, 40 Luton) between June 2018 and April 2019 — *before* ULEZ implementation — with **3,209 (94.0%) in the FEV₁ analysis**. Luton functions as a contemporaneous control city, converting the policy rollout into a difference-in-differences natural experiment rather than a before–after series. Baseline London deficit **–38 mL (95% CI –58 to –18)** against a baseline NO₂ excess of **18.95 µg/m³**; over follow-up London FEV₁ growth exceeded Luton by **10 mL/year (5–15, $p = 0.0002$)** (233 vs 223 mL/year) and mean FEV₁ reached parity at four years. Clinically impaired lung function fell 14%→9% in London and 9%→7% in Luton. Two constraints on reading it as a PM result: the exposure contrast is expressed in NO₂, the exhaust marker, so PM and NO₂ effects are not separable; and two cities differ in more than their air, which parallel-cohort adjustment mitigates but does not eliminate.

doi: 10.1016/S2468-2667(26)00163-5

Wu et al. 2026 J Thorac Dis*Air pollution and COPD-related health outcomes: bibliometric and visualisation analysis*

11,167 Web of Science articles and reviews (1964 to 31 May 2025) through CiteSpace, VOSviewer and bibliometrix: country/institution collaboration, journal co-citation, keyword clustering, burst detection. USA, China and UK lead output; Harvard most influential institution; China and climate change identified as 2021-onward frontiers. Bibliometrics measure publication behaviour, not evidence — useful only for orienting a literature search or a grant narrative.

doi: 10.21037/jtd-2026-0784

Yang et al. 2026 J Thorac Dis*COPD and LRI burden attributable to household air pollution and secondhand smoke, 1990–2021*

GBD 2021 across 204 countries with BAPC and ARIMA projection. Age-standardised COPD DALYs and mortality fell 37.00% and 37.12%; LRI fell 66.35% and 53.62%. Burden stays concentrated in LMICs, in older women for COPD and in under-15s and ≥60s for LRI; HAP-attributable COPD declined faster than SHS-attributable. Projected female COPD burden from both sources exceeds male by 2030. Standard GBD secondary analysis — the exposure is a modelled solid-fuel-use proxy, not measured PM, and the projections inherit every assumption in the parent estimates.

doi: 10.21037/jtd-2026-0284

Yang et al. 2026a J Asthma*Atmospheric humidity and air pollutants versus self-reported chronic lung disease*

16,204 CHARLS adults aged 45+, 2011/2013/2015 waves, six pollutants and six humidity indices at 1 km. Each IQR increase in pollutant concentration carried **18–45% higher odds** of self-reported CLD; PM₁₀ alone showed an approximately U-shaped exposure–response, and relative humidity an inverted-U, with a significant RH×sex interaction. Weighted quantile sum models were positive for both the pollutant and the humidity mixture. Self-reported outcome and a wide odds range quoted without per-pollutant intervals limit what can be taken from the point estimates; the reason to read it is the humidity treatment — six distinct indices rather than RH alone, which is the same distinction that separates a well-corrected from a badly-corrected low-cost nephelometer.

doi: 10.1080/02770903.2026.2721168

Yang et al. 2026a GeoHealth*Spatiotemporal risk of environmental drivers of respiratory disease, Jiangsu*

Distributed lag non-linear models coupled to geographically and temporally weighted regression on respiratory hospitalisations across Jiangsu Province, 2020–2023, a population containing over 27 million children under 15 and adults over 65. The DLNM handles the lag structure and the GTWR lets the exposure–response coefficient vary in space and time rather than assuming one province-wide slope — which is the right question for a province with this much industrial heterogeneity. The 2020–2023 window overlaps pandemic-era mobility and healthcare-seeking disruption; the abstract does not say how that is handled, and it is the main threat to the temporal component.

doi: 10.1029/2026gh001881

Zhai & Meng 2026 Int J Environ Health Res*Apparent temperature, air pollutants and pneumonia hospitalisation in rural Western China*

Distributed lag non-linear models on NRCMS surveillance data from Jiuquan, 2015–2020, with stratification and explicit pollutant $\times$ temperature interaction terms. Apparent temperature showed an inverted-U with a peak **RR 1.50 (1.41–1.60)**. Per-IQR increases gave **CO 1.11 (1.01–1.21)** and **PM₁₀ 1.05 (1.00–1.12)** as the only intervals excluding or touching unity; **PM_{2.5} 1.04 (0.97–1.12)**, SO₂, O₃ and NO₂ were null. Synergy was reported between low apparent temperature and high pollution, and among pollutants. Risks were higher in children under 7 and in autumn/winter. A rural setting is a genuine gap-filler — most Chinese DLNM work is urban — but with six correlated pollutants and an interaction screen the multiple-testing burden is not addressed, and the PM findings are the weakest of the set.

doi: 10.1080/09603123.2026.2718327

Zhang S et al. 2026 Environ Sci Technol*Respiratory and cardiovascular effects of ozone modified by PM_{2.5}*

Two complementary designs testing one hypothesis: that 30-day mean PM_{2.5} *modifies* the acute cardiorespiratory response to ozone, and that unmodelled modification is part of why the ozone literature is inconsistent. In a healthy-adult panel, short-term ozone was associated with significantly larger decrements in FEV₁, FEF_{25–75} and PEF, and larger rises in systolic and diastolic blood pressure, under **PM_{2.5} above 22 $\mu\text{g}/\text{m}^3$** . A meta-regression across published studies agreed: study areas with higher annual PM_{2.5} showed larger ozone effects on FEV₁ and blood pressure. **The abstract reports direction and significance but no interval estimates**, which is why this record contributes nothing to the forest plot. Panel size is not stated in the abstract either. Mechanistically the claim is priming rather than co-exposure confounding, and the meta-regression cannot separate the two — areas with high PM_{2.5} differ in many ways. Still the day's one health record, and the modification framing is directly relevant to designing multi-pollutant sensor nodes.

doi: 10.1021/acs.est.6c05604

Zhongguo Fei Ai Za Zhi 2026 Zhongguo Fei Ai Za Zhi*China Wumeng Mountain lung cancer: evidence chain, clinical and molecular characteristics of regional lung cancer related to coal pollution in the Yunnan–Guizhou Plateau*

Narrative synthesis assembling an exposure $\rightarrow$ component $\rightarrow$ biological-effect $\rightarrow$ clinical-molecular $\rightarrow$ prevention chain for household smoky-coal lung cancer in Xuanwei/Fuyuan, searching PubMed, Web of Science, Embase, CNKI and Wanfang to May 2026. Carcinogenic components named: **PAHs and methylated PAHs, PM_{2.5}/black carbon, inorganic minerals and metals, and carbonyls**, as a mixture. The clinical phenotype is distinctive — predominantly **never-smoking women**, high adenocarcinoma fraction, young onset, early metastasis, multifocal disease and family aggregation. So is the molecular one: total EGFR mutation is high but classical 19del and L858R are *under-represented*, while **G719X, S768I, KRAS G12C and ALK/ROS1 fusions are elevated** relative to non-Wumeng lung cancer; ARHGEF5, MYO18B and MUC16 are proposed germline susceptibility genes interacting with POT1/TERT/TERF2. Sputum microbiome shifts are reported alongside. **It is a narrative review by an unnamed author group with no formal inclusion criteria or bias assessment**, drawing on the same group's prior work, so the evidence chain is an argument rather than a synthesis. Its value is as the clearest statement in this corpus of a testable source-composition-to-mutation-spectrum hypothesis.

doi: 10.3779/j.issn.1009-3419.2026.102.16

de Souza et al. 2026 Sci Total Environ*Wildfire smoke, PM_{2.5} and respiratory morbidity: spatial mapping and time series*

1,467,172 respiratory hospital admissions in São Paulo State, 2018–2025 (DATASUS), against **20,446 INPE-recorded wildfire events** and daily CETESB PM_{2.5}, temperature and humidity, analysed by GAM on weekly aggregates with adjustment for meteorology, calendar month and a nonlinear trend. Children 0–10 and males carried the greatest burden; population-standardised rates were highest in the north and north-west, coinciding with fire hotspots, while absolute counts concentrated in the metropolitan area — a textbook illustration of why per-capita and count maps disagree. **Each 10 $\mu\text{g}/\text{m}^3$ weekly PM_{2.5} was associated with a 3.4 % increase in admission rates (RR 1.034, 95 % CI 0.889–1.204, $p = 0.054$)**, strongest at zero lag. Fire counts and PM_{2.5} correlated at $\rho = 0.732$. The GAM explains 82.8 % of deviance, but weekly aggregation of an acute exposure throws away most of the contrast and is the likeliest reason the interval spans the null.

doi: 10.1016/j.scitotenv.2026.182227

NOT CARRIED BY A DAILY ISSUE — EXTRACTIVE PRECIS OR METADATA ONLY, NOT AN ASSESSED SUMMARY (1 RECORDS)

Lu et al. 2026 Atmos Environ*Antagonistic effects of PM_{2.5} and ozone on respiratory morbidity: multi-pollutant interaction, Pearl River Delta*

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1016/j.atmosenv.2026.122270

Reproductive & developmental**19 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (19 OF 19 RECORDS IN THIS SUBTOPIC)

Atalell et al. 2026 Environ Res*Low-level PM_{2.5} and developmental vulnerability at school entry, Western Australia*

Whole-of-population linkage of the Australian Early Development Census (2009, 2012, 2015 waves) to midwives' notifications, birth registrations and hospital morbidity data; 54,458 children assigned daily PM_{2.5} at Statistical Area Level 2 across prenatal, early-childhood and cumulative windows; Bayesian multilevel logistic regression against the 2009 10th-percentile domain cut-offs. Mean exposure **7.40–7.54 $\mu\text{g}/\text{m}^3$** , below WHO guidance. Early-childhood exposure: **OR 1.04 (1.01–1.07)** for vulnerability in one or more domains and **1.06 (1.03–1.08)** in two or more, per 1 $\mu\text{g}/\text{m}^3$, with signals in physical health, social competence and language/cognitive domains. Prenatal and cumulative windows were null. The authors concede the critical limitation themselves: prenatal and early-childhood exposures are highly correlated, so the window attribution is not identified. Area-level assignment at SA2 resolution over a 1 $\mu\text{g}/\text{m}^3$ contrast is the second limitation, and the one this watch cares about.

doi: 10.1016/j.envres.2026.125493

Bello et al. 2026 Econ Hum Biol

Prenatal PM_{2.5} exposure and infant health in Quebec

Administrative birth records for Quebec 2008–2015 linked to monitoring data and tax records, in neighbourhood-by-quarter fixed-effects models — a low-pollution setting, which is the point. **Population-average effects on birth outcomes were essentially null**; the result lives entirely in the heterogeneity. Among families in the most vulnerable stratum of a cumulative socioeconomic vulnerability index, a **1 $\mu\text{g}/\text{m}^3$ increase in mean PM_{2.5} was associated with ≈ 6.6 g lower birth weight**, with further effects on prematurity and birth weight among newborns admitted to neonatal care. **Days exceeding the Canadian 24 h standard had disproportionately large effects in vulnerable families**, so peaks matter beyond the mean — directly relevant to any monitoring design optimised for annual averages. No confidence intervals are reported in the abstract, and the vulnerability index is constructed rather than pre-registered, which invites specification search across strata.

doi: 10.1016/j.ehb.2026.101631

Chen et al. 2026 Pediatr Allergy Immunol

Cord blood DNA methylation as a mediator between prenatal air pollution and infant allergic symptoms

Shanghai prospective birth cohort, 101 mother–infant pairs with spatiotemporally modelled residential exposure, caregiver-reported cumulative allergic symptoms to 6 months, and targeted bisulfite sequencing of naïve CD4⁺ T cells in a subset of 30 cord-blood samples. First-trimester PM (aOR 2.27, 1.09–4.98) and NO₂ (2.11, 1.02–4.53) were associated with allergic symptoms; third-trimester sulfate, nitrate and ammonium components were positively associated; a CpG in the PRKACB promoter showed statistical evidence of mediation for the sulfate association, supported by functional assays. The mediation arm rests on 30 samples and the outcome is caregiver-reported at 6 months, which is the weakest available allergic phenotype. The authors label the findings exploratory, correctly. Carried for the component-resolved exposure design, not the effect sizes.

doi: 10.1111/pai.70424

Chen et al. 2026g Toxics

Prenatal size-fractionated PM and fetal growth trajectories, mediated by maternal glucose

616 mother–fetus pairs in Guangzhou with repeated mid-to-late-pregnancy ultrasound (HC, AC, FL, BPD as z-scores; EFW trajectories classed average, slow or rapid) and residential PM₁, PM_{2.5}, PM₁₀ from 1 km spatiotemporal models. Each 5 $\mu\text{g}/\text{m}^3$ increase raised the odds of a rapid EFW trajectory versus average: PM₁ OR 1.56 (1.03–2.37), PM_{2.5} 1.44 (1.10–1.88), PM₁₀ 1.28 (1.05–1.56) — monotone in the inverse of particle size, as a deposition argument would predict. BPD, HC and AC z-scores moved in the same direction. **Maternal fasting glucose and insulin resistance mediated 5.0–29.4 % of the association**. The mediation range is wide enough to be uninformative at its lower end, mid-pregnancy glucose is a single measurement, and accelerated fetal growth is a less established adverse outcome than restriction, so the clinical framing is the weakest part. The size gradient is the citable result.

doi: 10.3390/toxics14080709

GBD 2023 India Stillbirth Collaborators 2026 Lancet Reg Health Southeast Asia

State-level stillbirth rates in India at ≥ 22 and ≥ 28 weeks, GBD 2023

First GBD estimation of stillbirth rate under the WHO ICD-11 ≥ 22 -week definition alongside the conventional ≥ 28 -week cutoff, disaggregated to 31 Indian geographic units. **122 data sources and 2,684 location-year observations** were modelled by spatiotemporal Gaussian process regression on the stillbirth-to-neonatal-mortality ratio, anchored on GBD 2023 fertility and all-cause neonatal mortality, with 95% uncertainty intervals and a secondary analysis against SDG3 indicators. **India carries the largest global stillbirth burden at both cutoffs**, and the ≥ 28 -week convention materially understates it. Air pollution enters only through GBD's risk-factor architecture, so no PM exposure–response is estimated here. Carried because state-level disaggregation is the resolution at which a PM–stillbirth analysis could be attempted, and because it defines the denominator such an analysis would use.

doi: 10.1016/j.lansea.2026.100819

Guo et al. 2026 Reprod Toxicol

Environmental pollutants and testicular toxicity: basic evidence and translational prospects

Narrative review of testicular-level evidence for microplastics, bisphenols and air pollution, against a reported $>50\%$ decline in global sperm concentration over five decades. The PM-relevant claims: **spermatogenesis stage I, 70–90 days before semen collection, is nominated as the PM_{2.5} vulnerability window**, with metal constituents of traffic-derived PM_{2.5} as the primary toxicity driver. Separately, microplastics are reported in human testicular tissue with a gut-dysbiosis $\rightarrow$ immune activation $\rightarrow$ blood–testis barrier pathway, and BCAT1-mediated ferroptosis is offered as a bisphenol injury mechanism. Limitation: no systematic search or quality appraisal, so the window and constituent claims inherit whatever selection bias the cited primary literature carries. Read alongside Russo et al. — convergent on timing, but not independent evidence for it.

doi: 10.1016/j.reprotox.2026.109310

Hahad et al. 2026 J Environ Health Sci Eng

Air pollution and adverse birth outcomes: a narrative review of epidemiological and mechanistic findings

BACKFILL (PubMed entry 2026-06-29; surfaced by the Amass connector). Narrative review synthesising epidemiological and mechanistic evidence linking prenatal air pollution to low birth weight, preterm birth and stillbirth, with searches across PubMed, Scopus and Web of Science. The two quoted estimates are a **–11.8 g birth-weight difference** and a **+23.1 % increase in preterm-birth risk per 10 $\mu\text{g}/\text{m}^3$ second-trimester PM_{2.5}**; mechanisms discussed are oxidative stress, inflammation, endocrine disruption, vascular dysfunction and epigenetic modification acting on placental morphology, perfusion and metabolic function. Both estimates are quoted **without confidence intervals or provenance to a specific pooled analysis**, which is why neither enters the forest plot — a narrative review is not a meta-analysis, and a point estimate lifted from one without its interval is not an effect size. The review is honest about what remains unresolved: disentangling individual pollutants, establishing causality, and identifying critical windows. Relevant to sensing because trimester-specific windows demand exposure resolution that residential-address annual means cannot deliver.

doi: 10.1007/s40201-026-00990-4

Han et al. 2026 Environ Pollut

Aerosol pH associated with preterm birth risk: unmasking the hidden role of atmospheric acidity in maternal-fetal health

Metadata only. Surfaced through the Crossref by-ISSN sweep with no abstract deposited; title, journal, authorship and DOI verified, nothing else extracted. Worth flagging by title alone: aerosol acidity as an exposure metric in a reproductive endpoint is not something this watch has seen before, and it pairs directly with Chai et al.'s pH-governed partitioning result in yesterday's issue. Queued for abstract retry.

doi: 10.1016/j.envpol.2026.129011

Hung et al. 2026 iScience*Late-pregnancy air pollution, gestational hypertension and neonatal anthropometry*

Women recruited in the second trimester and classified gestational-hypertensive or normotensive, with PM_{2.5}, PM₁₀, NO₂, SO₂, CO and O₃ over the 30 days before delivery assigned from ambient monitors, and maternal peridelivery hs-CRP, IL-1 β and TNF- α plus cord TNF- α as candidate mediators. GH women had higher hs-CRP and delivered lighter neonates with smaller chest and abdominal circumferences; PM, NO₂, SO₂ and CO were **inversely associated with birth length**. **Mediation analysis found no significant indirect effect through any measured cytokine**, which is the useful negative — the inflammatory-mediation hypothesis is the default assumption in this literature and it did not hold here. Ambient monitor assignment over a 30-day window is coarse for a length endpoint measured in centimetres, no sample size or effect sizes appear in the abstract, and a single peridelivery cytokine draw is a poor proxy for a gestation-long inflammatory exposure.

doi: 10.1016/j.isci.2026.117029

Jang et al. 2026 J Clin Endocrinol Metab*Ambient air pollution and age at natural menopause in a multi-ethnic US cohort*

SWAN, **1,457 midlife women** across six US sites 1999–2015, five racial/ethnic groups, with 1 km² annual PM_{2.5}, NO₂ and O₃ linked to residence and age at natural menopause modelled as time-to-event by accelerated failure time. **NO₂ TR 0.983 (0.971–0.995), O₃ TR 0.983 (0.971–0.996)** for the highest versus lowest exposure quartile — roughly one year earlier menopause, which the authors benchmark against the 0.8-year advance seen in current smokers. **PM_{2.5} was not significantly associated**, which is the finding this watch should record plainly rather than fold into a general air-pollution claim. Residential address as the exposure surrogate over a 16-year window in a mobile midlife population is the main measurement limitation.

doi: 10.1210/clinem/dgag340

Kessler et al. 2026 Environ Health (Wash)*Prenatal traffic-related air pollution, dietary PUFA intake and birth weight*

Atlanta Maternal–Child Cohort, **297 pregnant African American women**, first-trimester traffic-related CO, PM_{2.5} and NO_x crossed with dietary polyunsaturated fatty acid intake in a joint mixture model. At mean α -linolenic acid intake, a per-IQR PM increase was associated with a **0.08-unit decrease (0.02–0.18) in birth-weight-for-gestational-age z-score**; at twice mean ALA intake the same increase was associated with a **1.12-unit increase (0.19–2.05)**, $p_{\text{int}} = 0.04$, with an almost identical pattern for linoleic acid. The authors read this as PUFA mitigation. A cleaner reading is that a 14-fold sign-reversing interaction estimated on 297 women in the sparse upper tail of a food-frequency distribution is the signature of sparse-data bias. Worth tracking because the mechanistic story is plausible and the result will be cited widely.

doi: 10.1021/envhealth.5c00490

Li et al. 2026 Environ Pollut*Prenatal NO₂ and stillbirth in 3.98 million Iranian mother–baby pairs*

3,982,304 pairs, 2013–2018, pregnancy-average NO₂ from satellite-based gridded estimates averaged within 10 km of the delivery hospital, analysed under both the Iranian (≥ 22 weeks) and WHO (≥ 28 weeks) stillbirth definitions. Per 10 $\mu\text{g}/\text{m}^3$: **OR 1.206 (1.192–1.220)** and **1.106 (1.088–1.125)** respectively, essentially unchanged on adjustment for PM_{2.5} and ozone. The exposure–response is monotonic with *steeper slope at low concentrations*. A 10 $\mu\text{g}/\text{m}^3$ target would avert 7,686 (7,242–8,127) stillbirths, an attributable fraction of 25.8%. Exposure assigned at the delivery hospital rather than residence is a substantial misclassification channel in a country with heavy referral flows, and the effect size is large enough to warrant scepticism. Co-pollutant, not PM.

doi: 10.1016/j.envpol.2026.128908

Pai et al. 2026 Pediatr Neonatol*Household incense burning and small-for-gestational-age risk*

18,666 mother–infant pairs in the Taiwan Birth Cohort Study, incense exposure during pregnancy categorised none / occasional / daily, SGA stratified by term and preterm. SGA prevalence rose monotonically across exposure groups (8.1%, 9.4%, 10.0%). Among term infants, **aOR 1.16 (1.00–1.33) occasional and 1.25 (1.10–1.41) daily**; preterm infants showed the same direction without significance, and there was no association with preterm birth itself. Sensitivity analyses covered other indoor environmental factors, maternal medical history, gestational complications, gestational weight gain and secondhand-smoke interaction. The limitation that matters is that exposure is a self-reported frequency category with no concentration, duration or ventilation information — the dose–response is ordinal in *practice*, not in dose, and the contrast between an unventilated daily burn and a well-ventilated one sits entirely inside the top category. Pairs directly with Zhou L et al. in this issue.

doi: 10.1016/j.pedneo.2026.06.003

Russo et al. 2026 Paediatr Perinat Epidemiol*Critical windows of air-pollution exposure during spermatogenesis and IUI outcomes*

Secondary analysis of the Folic Acid and Zinc Supplementation Trial: 592 male partners in the Salt Lake City area contributing **1,223 intrauterine insemination cycles** over nine months. Residential exposure came from Community Multiscale Air Quality models, averaged over the 74-day spermatogenesis window before insemination and separately over mitosis, meiosis I–II, spermiogenesis and spermiation, with generalised linear mixed models adjusted for age, season, income and co-pollutants. Per 5 ppb across the full window, **O₃ RR 0.71 (95% CI 0.55–0.93)** and **NO₂ RR 0.73 (0.50–1.05)** for hCG+ pregnancy, concentrated in mitosis and meiosis; live-birth results were similar and no pattern emerged for pregnancy loss. **PM_{2.5} and SO₂ were less consistently associated and no numeric estimate was given**. Limitation: an infertility-treatment population is not the general population, residential modelling ignores occupational and commuting exposure, and co-pollutant models cannot cleanly separate a traffic mixture. The stage-resolved exposure definition is the methodological contribution worth borrowing.

doi: 10.1111/ppe.70185

Tanner et al. 2026 Environ Epigenet*Prenatal exposure mixture and aromatase brain-promoter methylation in cord blood*

Barwon Infant Study birth cohort, **n = 906**, testing 25 prenatal exposures across five classes — manufactured chemicals, air pollution, pharmacological, nutritional and sunlight-related — against methylation of the brain-specific CYP19A1 P1.f promoter using weighted quantile sum regression. The mixture index was associated with higher P1.f methylation, **adjusted mean difference 0.71 (95% CI 0.11–1.32)**, $p = 0.021$, interpreted as reduced brain aromatase activity. Bisphenols, reduced sunlight, household mould, phthalates, low folate and air pollution carried the largest weights. The inference-threatening limitation is structural: WQS assigns weights within a single index, so air pollution's individual contribution is neither estimated nor separable, and cord blood is a proxy for a brain-specific promoter. Tier C for this watch — PM is one component of a 25-exposure mixture.

doi: 10.1093/eep/dvag027

Wang et al. 2026 Environ Pollut

PM_{2.5} modifies the temperature–semen quality association

Prospective cohort with repeated semen analysis, linear mixed models with natural cubic splines for temperature, then piecewise regression stratified into higher and lower PM_{2.5} groups across the low- and high-temperature segments. Temperature showed **inverted-U associations with semen quality over the 0–90, 0–9 and 10–14 d windows** — 0–90 d spans a full spermatogenic cycle, 10–14 d the epididymal transit, so the window structure is biologically coherent. **Synergy between high temperature and PM_{2.5}, antagonism between low temperature and PM_{2.5}**, both acting mainly on progressive and total motility. Stratifying at a PM_{2.5} median is a crude interaction test that discards within-stratum contrast, and antagonism at low temperature is at least as easily explained by seasonal confounding (heating-season composition, behaviour, indoor time) as by a joint biological mechanism. No effect sizes or intervals are reported in the abstract, so nothing from this record enters the forest plot.

doi: 10.1016/j.envpol.2026.128910

Wang et al. 2026 Environ Health (Wash)

Ultrafine particles, hypertensive disorders of pregnancy, and homocysteine mediation

Shanghai birth cohort, **7,569 pregnancies**, 2017–2018, with individual UFP exposure from a high-spatiotemporal-resolution prediction model at residential address and serum homocysteine, folate and B₁₂ measured in early pregnancy. Poisson regression with robust error variance gave per-IQR **RR 1.15–1.23 for hypertensive disorders of pregnancy and 1.29–1.41 for pre-eclampsia** across the prepregnancy, first- and second-trimester windows. Homocysteine was associated with pre-eclampsia (**RR 1.49, 1.04–2.13**) and mediated 8.87% and 2.51% of the prepregnancy and first-trimester associations. Stratification showed the UFP–PE association only in the low-B₁₂ group. Two cautions: the exposure is modelled, not measured, so UFP–PM_{2.5} collinearity is not resolvable here; and mediated fractions this small are within the range that measurement error in the mediator can manufacture.

doi: 10.1021/envhealth.5c00760

Wu et al. 2026 Ecotoxicol Environ Saf

Prenatal PM_{2.5} constituents and early childhood obesity: organic matter dominates the mixture

339,435 mother–child pairs in Liaoning Province with prenatal PM_{2.5} and five constituents (organic matter, black carbon, ammonium, nitrate, sulfate) assigned from high-resolution atmospheric reanalysis, and overweight/obesity ascertained across the first two years. Single-pollutant models give PM_{2.5} **mass OR 1.050 (95% CI 1.036–1.064) per IQR**, with **OM 1.058 (1.043–1.072)**, **NH₄⁺ 1.048 (1.035–1.062)**, **NO₃[−] 1.048 (1.033–1.064)** and **BC 1.033 (1.020–1.047)**; sulfate is null. Quantile g-computation puts the mixture at **1.063 (1.050–1.076)** per quartile increase with **OM contributing 54.1%**, **NH₄⁺ 34.4%** and **NO₃[−] 11.5%**. The sample size makes the intervals tight, but tight intervals on reanalysis-assigned exposure are a precision claim about the model, not the exposure: constituent fields are far more uncertain than total mass and their errors are correlated with each other, which is exactly the condition under which QGC weights become unstable. Residential address at delivery stands in for the whole gestation, and no gestational windows are resolved. The direction is credible; the 54% attribution to OM is not something the exposure product can really support.

doi: 10.1016/j.ecoenv.2026.120670

Zhao et al. 2026 Chin Med J (Engl)

PM_{2.5} constituents and spontaneous abortion, modified by residential greenness

20,117 pregnancies in Nanjing with residential PM_{2.5} speciated into NO₃[−], SO₄^{2−}, NH₄⁺, BC and OM; logistic regression with propensity-score matching plus two-pollutant models, WQS and BKMR for the joint effect. Per IQR: total mass **OR 1.18 (1.11–1.26)**, ammonium **1.19 (1.11–1.26)** highest, BC and OM lowest at 1.14 — and OM and BC alone lose significance once season of conception is added. Greenness (NDVI-1000 m) interacts significantly but non-monotonically (Q1 1.30, Q2 1.02, Q3 1.29, Q4 1.11), which undercuts the attenuation reading. Depression and anxiety scores predicted SAB independently with no interaction. Constituent correlations are high, so the ordering is suggestive rather than separable.

doi: 10.1097/CM9.0000000000004259

Mechanistic toxicology

47 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (46 OF 47 RECORDS IN THIS SUBTOPIC)

Asplund et al. 2026 Toxicol Sci

Concentrated ambient PM_{2.5} reversibly suppresses erythropoiesis and proresolving lipids

Whole-body concentrated ambient particle exposure in mice with targeted splenic lipidomics. CAP suppressed erythropoietin, circulating reticulocytes and bone-marrow erythroid precursors, and reduced splenic iron and heme despite preserved splenic architecture — diminished erythrocyte turnover, not structural damage. Proresolving lipid mediators were broadly suppressed with **lipoxin A₄** most consistent, and its receptor ALX/FPR2 downregulated under prolonged exposure. **Return to filtered air normalised the lipid profile, LXA₄ and the erythroid parameters**. The recovery arm is what makes this the strongest design in the issue; the open question is whether the same reversibility holds after chronic rather than sub-chronic exposure.

doi: 10.1093/toxsci/kfag092

Ayala et al. 2026 Environ Pollut

Contrasting neurotoxic pathways triggered by PM₁₀ and organic molecular markers, Catalonia

Fifty-four PM₁₀ filter samples collected through 2022 at a rural background site (Bellver de Cerdanya), a suburban site (Manlleu) and a suburban-industrial site (Mollet del Vallès), analysed by GC-MS for 30 organic tracers and tested on SH-SY5Y neuroblastoma cells across six endpoints (viability, ROS, AChE activity, antioxidant and xenobiotic response, p53). MCR-ALS on the joint chemical-biological matrix resolved four components: winter biomass burning (levoglucosan, dehydroabietic acid, PAHs) driving cytotoxicity, oxidative stress and xenobiotic response; a year-round traffic component; a spring–summer SOA component producing **selective AChE inhibition without cytotoxicity**; and a summer POA component. PLS models were significant for five of six endpoints (R^2_{CV} 0.53–0.75, best for ROS, p53 and cell death). The strength is that exposure and effect were measured on the same extracts; the limits are one year, three sites, one immortalised cell line and organic-solvent extracts that do not represent inhaled particle dosimetry.

doi: 10.1016/j.envpol.2026.128961

Chen et al. 2026 Allergy*Gut microbiota signatures reflect PM_{2.5}-exacerbated asthma in a murine model*

Metadata-only: a research letter deposited without an abstract, so nothing below the title is verifiable from the record. Registered here because the gut–lung axis in PM-exacerbated allergic airway disease is a live mechanistic question and because the author group (Academia Sinica Air Quality Research Center with NTU clinical immunology) pairs characterised ambient aerosol with a murine model, which is unusual. No finding, effect size or exposure characterisation is claimed by this issue, and the record supports no figure or statistic in the analytics. Flagged for retrieval once the full text is indexed.

doi: 10.1111/all.70485

Chen et al. 2026a J Chin Med Assoc*PM_{2.5} suppresses IL-8-mediated epithelial migration in conjunctival cells*

Immortalised human conjunctival epithelial line exposed to PM_{2.5} with viability, apoptosis, scratch-migration and cytokine-array readouts. PM_{2.5} reduced viability time- and dose-dependently with only mild apoptosis, but markedly impaired migration and wound closure. Cytokine profiling showed *selective* remodelling rather than a general inflammatory response, with IL-8 the most strongly suppressed mediator; recombinant IL-8 rescued migration and partially rescued viability, and IL-8 neutralisation impaired migration at baseline. Single immortalised line, suspension dosing, no dose anchored to an ambient concentration — so the mechanism is credible and the dosimetry is not. The ocular surface remains an under-instrumented exposure site.

doi: 10.1097/JCMA.0000000000001418

Chen et al. 2026j Part Fibre Toxicol*Multi-omics integration reveals human placental molecular signatures associated with microplastics exposure*

Placental microplastic burden quantified by **pyrolysis-gas chromatography–mass spectrometry**, samples stratified into high- and low-burden groups, then transcriptomic, proteomic and metabolomic profiling integrated through the DIABLO framework with network analysis for regulatory hubs. High-burden placentae show multi-layer molecular perturbation centred on immune dysregulation. The design's strength is that MP burden is treated as a *graded internal dose* rather than a detection/non-detection flag, which separates it from most human-tissue microplastics work. Its limits are the usual ones for cross-sectional human tissue: burden is measured once at delivery, exposure timing is unknown, the omics contrasts are correlational, no sample size appears in the abstract, and Py-GC-MS quantifies polymer *mass* without particle size or count — so the dose metric is mass, not number or surface area.

doi: 10.1186/s12989-026-00697-2

Cho et al. 2026 J Microbiol Biotechnol*Lactacisibacillus paracasei ATG-E1 alleviates particulate-matter-induced airway inflammation in association with altered inflammatory signalling, gut microbiota and faecal metabolites*

BALB/c mice challenged intranasally with PM₁₀D (PM₁₀ plus diesel exhaust particles) on days 0, 3, 6 and 8, then treated orally with ATG-E1 or dexamethasone *after* inflammation was established — a therapeutic rather than prophylactic design, which is the methodological point in its favour. ATG-E1 reduced BALF inflammatory infiltrate and neutrophilia, lowered collagen deposition and pro-inflammatory mediators, and attenuated **IKBα and ERK phosphorylation** while leaving JNK and p38 unchanged; caspase-1 and IL-1α fell, lung RNA-seq showed broad downregulation of cytokine–cytokine-receptor signalling, and cecal 16S plus faecal metabolomics showed microbiome remodelling with increased branched-chain fatty acids and decreased branched-chain amino acids. Two limits: the microbiome and metabolite changes are correlative, with no germ-free, antibiotic or transfer arm to establish that the gut axis mediates the airway effect rather than co-occurring with it; and no effect size, n per group or variance is given in the abstract, so the magnitude is unassessable. All authors are employed by the company that owns the strain.

doi: 10.4014/jmb.2604.04029

Choi et al. 2026 Biofabrication*Degradable porous PLGA/PCL membrane enabling a lung alveoli-on-a-chip for particulate injury*

Porogen-assisted phase-separation membranes with tunable degradation (~40% surface porosity gain, ~3 μm thickness loss over culture) carrying primary human alveolar epithelium and microvascular endothelium at an air–liquid interface under breathing-mimetic cyclic strain. Apical diesel PM produced dose-dependent cytotoxicity, barrier permeability increase, elevated ROS and DNA damage *in both layers* — trans-barrier propagation, which an insert-only model cannot show. Roflumilast-N-oxide (PDE4 inhibitor) selectively attenuated oxidative and inflammatory readouts without restoring barrier integrity. Deposited dose is delivered apically in suspension and is not equivalent to inhaled dosimetry; no dose–response anchoring to ambient concentrations is attempted. Relevant here as a bench receptor for source-resolved particle samples.

doi: 10.1088/1758-5090/ae9402

Deng et al. 2026 J Biochem Mol Toxicol*Baicalin rescue of PM_{2.5}-induced testicular injury via DKK1–Wnt/β-catenin*

Oral-nasal PM_{2.5} exposure at **750 μg m⁻³, 4 h/day for 45 days**, with baicalin gavage 1 h before exposure at 50 and 100 mg/kg. PM_{2.5} lowered testicular weight and organ coefficient, sperm count and serum testosterone and downregulated StAR, CYP11A1 and CYP17A1; baicalin restored these dose-dependently alongside DKK1–Wnt/β-catenin pathway markers. The exposure concentration sits one to two orders of magnitude above ambient even in heavily polluted settings, which is the standing weakness of this whole sub-literature — the pathway may be real, the dose is not.

doi: 10.1002/jbt.71057

Forest 2026 Sci Total Environ*Molecular mechanisms of sand and dust storm toxicity*

Review of in vitro and in vivo evidence on storm-derived PM₁₀/PM_{2.5}, whose composition — quartz and clay minerals plus adsorbed bacteria, viruses and heavy metals acquired in transport — differs enough from urban PM that the author argues current regulation, which treats storm episodes as generic PM, misses hazard-specific risk. Consistent findings are cytotoxicity, oxidative stress, pro-inflammatory response and exacerbation of bacterial and viral infection. The stated gaps are the substantive content: **most in vivo work uses Asian sand dust while Saharan dust dominates global emissions**, and non-respiratory target organs are barely studied. Directly relevant to any network operating in a dust-influenced region, where the mass a sensor reports and the hazard it implies decouple.

doi: 10.1016/j.scitotenv.2026.182233

Gao et al. 2026 Zhongguo Yi Xue Ke Xue Yuan Xue Bao

SQSTM1/p62-mediated mitophagy in PM_{2.5}-aggravated asthma, and dusquetide rescue

House-dust-mite-sensitised C57BL/6 mice exposed to PM_{2.5}, assessed by whole-body plethysmography, HE/PAS histology, TOMM20-LC3B colocalisation and LC-MS/MS mitochondrial proteomics. PM_{2.5} raised airway hyperresponsiveness, inflammation and mucus and increased mitophagy signal; Mdivi-1 inhibition reduced inflammation; proteomics identified SQSTM1/p62 as the hub, and dusquetide suppressed both p62 and mitophagy while relieving hyperresponsiveness. Mdivi-1 is a poorly selective DRP1 inhibitor with known off-target effects on complex I, so the mitophagy-specificity of the rescue is not secured by that arm alone.

doi: 10.3881/j.issn.1000-503X.16948

Garcia Dominguez et al. 2026 Environ Sci Technol

Ultrafine particle deposition and clearance in an avian lung

Zebra finches exposed to 50 or 100 nm model particles at 5 or 25 °C, all within environmentally relevant ranges, with deposition quantified in lung and — in a subset — heart, liver and red blood cells. Lung deposition was highest for the smallest particles, consistent with diffusional deposition. Deposited *fraction* was unchanged by temperature but deposited *dose rate* rose in the cold, driven entirely by elevated minute ventilation: a clean demonstration that thermoregulatory demand, not respiratory physiology, sets avian dose. **44% of the initially deposited 100 nm burden remained at two weeks**, i.e. clearance is slow. Model particles, one species, small subset for extrapulmonary organs — but this is the only record in today's corpus that measures ultrafine dose in a living respiratory system, and the avian lung's cross-current architecture makes it a genuinely different test case from rodent work.

doi: 10.1021/acs.est.6c05957

Gomes et al. 2026 Cardiovasc Toxicol

Senolytics reverse PM_{2.5}-induced cell and vascular dysfunction

C57BL/6J mice inhaling concentrated ambient particles versus filtered air showed endothelial activation *in vivo* — increased leukocyte rolling on and adhesion to intact vasculature — alongside senescence of peripheral blood mononuclear cells and endothelial progenitor cells. Dasatinib + quercetin reversed the senescence readouts (β -galactosidase activity, SASP gene expression), restored EPC tube-forming capacity *in vitro*, and limited *in vivo* endothelial activation. Positions senescence upstream of early atherogenesis rather than as a late consequence. Concentrated ambient particles deliver a supra-ambient dose; the senolytic rescue demonstrates sufficiency of the senescence pathway under that dose, not its necessity at ambient concentrations. No exposure-response and no female animals reported.

doi: 10.1007/s12012-026-10171-2

Ichinose et al. 2026 J Hazard Mater

Time-dependent BALT formation and pulmonary inflammation induced by different particulate matter types

Murine time-course comparing TiO₂ nanoparticles, diesel exhaust particles and Asian sand dust, with and without ovalbumin. All three supported bronchus-associated lymphoid tissue formation, more strongly with allergen. With allergen, an early IL-1 α rise preceded type-2 responses and BALT expansion; **without allergen, BALT still formed via CXCL1 and neutrophil accumulation**, most clearly after repeated TiO₂ — two distinct routes to the same structure. 3D reconstruction showed tertiary lymphoid structures extending from large airways distally, and TiO₂ was localised in alveoli, alveolar macrophages, BALT and hilar lymph nodes. *Limitations*: instilled or high-dose murine exposure is not ambient inhalation; no dose-response to ambient concentrations; and the three particle types differ in dose, size and surface chemistry simultaneously. Relevance: particle *identity*, not mass, determines which inflammatory route is taken — the standing mechanistic case for composition-resolved monitoring.

doi: 10.1016/j.jhazmat.2026.143239

Introna et al. 2026 ACS EST Air

Novel approach to assess the toxicity of PM_{2.5} on a subway platform using a mobile air-liquid interface system and submerged exposure

A mobile ALI exposure system operated *on the platform itself*, with A549 monoculture and A549 + dTHP-1 coculture, benchmarked against conventional submerged laboratory exposure to the same metal-rich mechanical-wear particles. At the realistic in-situ ALI dose of **0.12 $\pm$ 0.07 μ g/cm² there was no change in viability or inflammatory markers at 24 h**. Submerged at higher dose, viability fell in both cell types; A549 released IL-6 and TNF α at 100 μ g/mL, and dTHP-1 released IL-8 at the lowest dose tested (10 μ g/mL) with all four cytokines elevated at 100. The authors' own reading is the right one — longer in-situ exposures are needed to reach comparable dose — and the wider implication is that the submerged literature's dose range is not the ambient one.

doi: 10.1021/acsestair.6c00018

Jerome et al. 2026 J Immunotoxicol

World Trade Center dust: neuroinflammation and myelin response at 3 and 12 months

Twelve-week-old male spontaneously hypertensive rats given intratracheal WTC dust at **2.43 $\pm$ 0.78 mg/day on two consecutive days**, then euthanised at 3 and 12 months. At 3 months, increased Black-Gold II myelin staining without change in myelin basic protein or CNPase; at 12 months, microglial activation (Iba1) and elevated brain IL-1 β . The authors read the discrepant myelin markers as a compensatory response preceding degeneration. Three limitations bound the inference: a two-day bolus instillation is not a chronic inhalation exposure and bypasses the upper airway; the SHR strain is hypertensive by design, so the neuroinflammatory phenotype is confounded with a vascular one; and males only. Useful as a mechanistic anchor for the First-Responder cognitive literature, not as a dose-response.

doi: 10.1080/1547691X.2026.2722654

Jin & Zhang 2026 Reprod Toxicol

Astragaloside IV protects against PM_{2.5}-induced spermatocyte injury via cGAS-STING/Nrf2/HMOX1

Network pharmacology and GEO reanalysis (GSE189187) to nominate targets, molecular docking, then validation in GC-2 spermatocytes. PM_{2.5} at **200 μ g/mL** raised ROS 5.35-fold and apoptosis to 42.83%; AS-IV co-treatment restored viability to 83.3%, suppressed cGAS/p-TBK1/p-IRF3 and activated Nrf2/HMOX1, reducing apoptosis to 18.27%. The docking affinities exceed the reference inhibitors G150 and H-151, which is presented as validation and is not — docking scores do not rank cellular potency. The exposure is a cell-culture suspension four orders of magnitude above any inhaled concentration, and **AUC = 1.000** for a single gene in a public transcriptomic dataset is a small-sample artefact rather than a discriminative result. Carried at tier C for the pathway hypothesis only.

doi: 10.1016/j.reprotox.2026.109328

Li & Li 2026 Zhongguo Fei Ai Za Zhi

Research advances in the pulmonary carcinogenic effects of atmospheric fine particulate matter mediated by GPER

Narrative review positioning the G protein-coupled estrogen receptor as the molecular hub linking PM_{2.5} to lung carcinogenesis. The argument: environmental estrogen-like organics and metal ions carried on PM_{2.5} activate GPER, which is overexpressed in lung tumour tissue, and the activated receptor drives **MAPK/ERK and PI3K/AKT** signalling into inflammation, oxidative stress and ferroptosis, with downstream proliferation, EMT and tumour-microenvironment remodelling. GPER inhibition is proposed as a target. **The weaknesses are structural to the format:** no search protocol, no risk-of-bias assessment, no quantitative synthesis, and a receptor-centric frame that cannot by construction weigh GPER against the AhR and oxidative-stress pathways it does not review. Read it as a hypothesis map, not as evidence of pathway dominance. The one genuinely actionable implication for exposure science is that it assigns the causal role to specific *components* — estrogenic organics and transition metals — rather than to mass, which is another argument for composition-resolved rather than gravimetric-equivalent monitoring.

doi: 10.3779/j.issn.1009-3419.2026.101.13

Li & Li 2026 Zhongguo Fei Ai Za Zhi

Research advances in the pulmonary carcinogenic effects of atmospheric fine particulate matter mediated by GPER

Narrative review positioning the G protein-coupled estrogen receptor as the molecular hub linking PM_{2.5} to lung carcinogenesis. The argument: environmental estrogen-like organics and metal ions carried on PM_{2.5} activate GPER, which is overexpressed in lung tumour tissue, and the activated receptor drives **MAPK/ERK and PI3K/AKT** signalling into inflammation, oxidative stress and ferroptosis, with downstream proliferation, EMT and tumour-microenvironment remodelling. GPER inhibition is proposed as a target. **The weaknesses are structural to the format:** no search protocol, no risk-of-bias assessment, no quantitative synthesis, and a receptor-centric frame that cannot by construction weigh GPER against the AhR and oxidative-stress pathways it does not review. Read it as a hypothesis map, not as evidence of pathway dominance. The one genuinely actionable implication for exposure science is that it assigns the causal role to specific *components* — estrogenic organics and transition metals — rather than to mass, which is another argument for composition-resolved rather than gravimetric-equivalent monitoring.

doi: 10.3779/j.issn.1009-3419.2026.101.13

Li et al. 2026 FASEB J

Size-resolved carbon-black nanoparticle effects on the maternal-fetal lung axis

Pregnant mice exposed to 30 nm or 120 nm carbon black under an explicitly high-dose mechanistic protocol, with H&E/Masson histology, dark-field hyperspectral imaging for particle localisation and chemical-isotope-labelling LC-MS metabolomics. Both sizes produced maternal inflammatory infiltration and fibrosis, **120 nm worse than 30 nm**; transplacental transfer was imaged directly in fetal lung. Fetal lungs showed no histological injury but raised oxidative and inflammatory markers, with size-specific metabolic signatures — 30 nm on histidine metabolism, 120 nm on butanoate and arachidonic acid. Two problems: the dose is not environmentally referenced, and the larger particle causing more damage inverts the surface-area expectation without a number- versus mass-dose accounting that would explain it.

doi: 10.1096/fj.202600065RR

Li et al. 2026b Environ Sci Technol

Component-resolved PM_{2.5} translocation across a human blood-brain-barrier immune platform

Microfluidic BBB-immune microenvironment platform with compartment-resolved quantification of which PM_{2.5} constituents actually cross, rather than inference from downstream effect. High doses increased barrier permeability and drove constituent translocation; translocated material produced astrocyte reactivity, microglial M1 polarisation, elevated ROS, neuroinflammation and lipid-metabolic dysregulation. Component-resolved correlation implicates **chromium** and the water-soluble organic matter fraction — specifically the condensed species C₂₈H₇O₂N and C₂₁H₁₆N₂SP₂. The dose is described only as “high”, with no concentration given in the abstract and therefore no way to judge ambient relevance, which is the standard failure mode of this literature. The compartment-resolved measurement of translocation is nonetheless a genuine methodological advance over effect-only designs.

doi: 10.1021/acs.est.6c01030

Liu et al. 2026 Environ Health (Wash)

PM_{2.5}-induced pulmonary vascular remodelling via MSC-mediated endothelial senescence

Real-ambient PM_{2.5} exposure in mice plus a pulmonary arterial endothelial cell model. Exposure raised pulmonary artery pressure and arterial wall thickness in vivo and induced senescence and mitochondrial dysfunction in mouse pulmonary arterial endothelial cells. Bioinformatic screening nominated the transcription factor MSC as the regulator; **MSC overexpression rescued mitochondrial-dysfunction-associated senescence**, and MSC was shown to act through the downstream target NDUF3, a complex I subunit — which makes the mitochondrial phenotype mechanistically coherent rather than merely correlative. **Real-ambient rather than instilled exposure** keeps the dosimetry defensible. No concentration, duration or particle characterisation appears in the abstract, so comparability to other real-ambient systems cannot be judged.

doi: 10.1021/envhealth.5c00595

Luo et al. 2026b Clin Rev Allergy Immunol

Environmental pollutants and innate immunity via group 2 innate lymphoid cells

Narrative review covering O₃, PM, diesel exhaust particles and cigarette smoke alongside nanomaterials and microplastics, organised around ILC2 biology. The proposed common route is epithelial barrier compromise followed by alarmin release — IL-33, IL-25, TSLP — with ILC2s as the sensor coupling environmental detection to type-2 inflammation, plus non-canonical regulation of ILC2 plasticity and functional adaptation. No new data, no quantitative synthesis, and PM is not separated from the other exposures, so nothing here constrains a PM-specific dose or particle-property relationship. Carried as a citation anchor for the epithelial-alarmin axis in respiratory PM work; tier C.

doi: 10.1007/s12016-026-09193-z

Mahmutovic Persson et al. 2026 Environ Res

Chronic aerosolized PM_{2.5}, plasma proteins and coronary S1P response in rats

Well-characterised urban PM_{2.5} was re-aerosolised and delivered to rats under two regimes designed to separate an acute high dose from a chronic lower dose at comparable cumulative burden, with coronary vasomotor tone measured ex vivo by wire myograph and systemic effects read from plasma sphingosine-1-phosphate and endothelin-1 plus immunohistochemistry. **Only the chronic regime moved anything:** altered S1P vasomotor response in coronary arteries and significantly decreased plasma total protein and ET-1. The design is the point — most PM inhalation toxicology uses acute high-dose exposure because it is tractable, and this says that route can miss the effect entirely. Re-aerosolised collected PM is not fresh ambient aerosol (semivolatiles are lost, morphology changes), the ET-1 *decrease* is not obviously in the direction the vascular-dysfunction hypothesis predicts and is not reconciled, and no n or dose figures appear in the abstract.

doi: 10.1016/j.envres.2026.125391

Masci et al. 2026 Environ Toxicol Pharmacol

Air pollution and undernutrition induce rat alveolar macrophage functional dysregulation and compromise their response to a viral challenge

Male Wistar rats, control vs. nutritional growth retardation (20% caloric restriction, 4 weeks), each with intranasal residual oil fly ash or vehicle; isolated alveolar macrophages assayed for viability, TNF α , oxidative stress, Nrf2 and LC3, then challenged with Poly(I:C). Undernutrition alone reproduced what ROFA did to well-nourished animals. **ROFA added little or nothing on top of undernutrition** — an attenuated rather than additive response — and after Poly(I:C), control macrophages mounted an inflammatory response that prior ROFA *amplified*, while undernourished macrophages failed to respond at all regardless of exposure. The interaction structure, not the main effect, is the finding: in a population that is already nutritionally compromised, the marginal immunological cost of PM may be small because the defence is already gone.

doi: 10.1016/j.etap.2026.105138

Mishra et al. 2026 Atmos Chem Phys

Aerosol oxidative potential and reactive species predicted with a chemical kinetics model (KM-OP)

A chemical kinetic model of oxidative potential built on literature rate coefficients and a large compilation of laboratory data for transition-metal ions, quinones and organic aerosol, predicting ROS production and consumption of ascorbic acid and DTT from PM composition. Applied to field measurements from **Grenoble, Paris and London: $R^2 > 0.75$ with $< 15\%$ model bias in 4 of 6 datasets. Organic aerosol is the strongest contributor to antioxidant-based OP assays**, with minor Cu and Fe contributions; Cu dominates H₂O₂ production but barely affects •OH, while the antioxidant assays correlate well with modelled •OH — separating the species that drives the assay from the species that drives the metal signal. Two of six datasets are not reproduced, and the model is calibrated on the same body of literature it is tested against in kind. Even so, this converts OP from a per-filter measurement into a quantity predictable from speciated composition, which is exactly what a composition-resolving sensor network would need to be worth deploying.

doi: 10.5194/acp-26-10605-2026

Mishra et al. 2026 bioRxiv (preprint)

OMA1–DELE1 as a shared node across ultrafine PM, black carbon and nanoplastic immunotoxicity

Human PBMCs exposed to ultrafine PM, black carbon and polystyrene nanoplastics under matched conditions, profiled across NRF2, HIF-1 α , PGC-1 α , TFAM, OMA1, DELE1, mitochondrial ND1, complexes I–V, NF- κ B, TNF- α and NLRP3, with PCA, hierarchical clustering and correlation networks. All three cause oxidative stress and mitochondrial dysfunction but with distinct kinetics: **UFPM gives an acute antioxidant/adaptive response, black carbon a chronic OMA1–DELE1 integrated stress response with inflammation, nanoplastics a low-grade persistent adaptation**. The comparative design under matched dosing is what makes this worth reading; it is a preprint, expression-level only, with no functional respirometry.

doi: 10.64898/2026.07.29.741662

Moonwiryakit et al. 2026 J Vis Exp

A STAT6 reporter assay for screening toxicants affecting allergic airway inflammation

A methods paper establishing a stable STAT6-responsive luciferase reporter in **16HBE14o- human bronchial epithelial cells** by lentiviral transduction, with puromycin selection, MOI optimisation and monoclonal isolation by limiting dilution. Responsiveness was validated against IL-4/IL-13, pathway specificity confirmed with selective STAT6, STAT5 and STAT3 inhibitors, and reproducibility quantified by **Z'-factor**. Applied to individual air-pollution constituents, **benzo[b]fluoranthene and particulate matter both significantly increased STAT6 reporter activity**. No dose–response, no PM characterisation (source, size, mass loading) and no Z' value are given in the abstract, so the assay's screening sensitivity cannot be assessed from it. The value is infrastructural: allergic-airway signalling is a plausible route for PM-associated asthma incidence and there has been no scalable prioritisation assay for it.

doi: 10.3791/71694

Nikaido-Landry et al. 2026 Philos Trans R Soc Lond B Biol Sci

EGFR dysregulation and climate change: pollution-promoted lung cancer

Discussion-meeting review connecting rising anthropogenic fine PM to lung cancer in never-smokers via EGFR signalling. The mechanistic core is the promotion model: **fine PM promotes malignant transformation in cells that already carry an EGFR mutation**, so the relevant population is people with oncogenic mutations in histologically normal lung tissue rather than people accumulating new mutations from exposure. That reframing matters — it predicts a susceptible subgroup rather than a population-wide dose–response, and it implies precision-prevention strategies targeted by genotype rather than universal exposure reduction. A narrative review with no systematic search and no new data; the promotion model rests principally on one line of murine and cohort work, and the “climate change” framing in the title is rhetorical rather than analytic. Tier B for the mechanistic argument.

doi: 10.1098/rstb.2024.0508

Pathak et al. 2026 medRxiv (preprint)

Airborne PM and mitochondrial DNA copy number: systematic review and meta-analysis

PRISMA, PROSPERO-registered, JBI risk-of-bias, random-effects pooling; 1,224 records screened to 24 studies and **13,092 participants**, 12 reporting percentage change and 12 absolute values. **Percentage mtDNA-CN fell (–4.90; 95% CI –7.97 to –1.82; $p = 0.002$) while absolute mtDNA-CN rose (+0.55; 0.05–1.04; $p = 0.030$)**. The two directions are not reconcilable as biology; they are an artefact of pooling across incompatible normalisation schemes, and the honest reading is that the underlying literature has not standardised its outcome. Useful as evidence that mtDNA-CN is *not yet* deployable as a PM exposure biomarker, which is the opposite of how it is usually cited.

doi: 10.64898/2026.08.20.26360559

Qi et al. 2026 J Hazard Mater

Urban PM_{2.5} worsens renal injury and fibrosis through a lung–kidney IL-6/ADCY3 axis

Repeated intratracheal urban PM_{2.5} at an environmentally relevant dose combined with renal ischaemia–reperfusion injury in mice, plus tubular epithelial cell work. PM exposure worsened renal dysfunction, tubular injury, fibrosis and lipid-droplet accumulation; the proposed route is PM → systemic inflammation with alveolar macrophages as the likely IL-6 source → suppression of ADCY3–CREB–PGC1 α → impaired fatty-acid β -oxidation, increased glycolysis, mitochondrial dysfunction and fibrosis. IL-6 blockade and ADCY3 overexpression each rescued, in vivo and in vitro. Limitation: the renal-IRI co-insult makes this a second-hit model, so the finding concerns PM aggravating existing injury rather than causing nephropathy; and intratracheal instillation again overstates deposited dose rate. The extra-pulmonary axis and the metabolic mechanism are both worth tracking.

doi: 10.1016/j.jhazmat.2026.143119

Qin et al. 2026 Biofactors*AlCl₃ aerosol exposure is associated with pulmonary lactate accumulation, neutrophil dysfunction, and fibrosis-like remodeling*

Repeated AlCl₃ aerosol exposure in rats, with and without the MCT1 inhibitor AZD3965, plus bone-marrow-derived neutrophils treated with sodium L-lactate, plus a supportive human cohort of aluminium-exposure-related ILD, non-aluminium ILD and healthy controls characterised by quantitative CT, spirometry and blood transcriptomics. Exposure produced pulmonary aluminium deposition, **lactate accumulation**, oxidative stress, DNA damage response signalling and collagen-rich fibrosis-like remodelling; in vitro, high lactate impaired neutrophil chemotaxis, oxidative burst, mitochondrial membrane potential, calcium signalling and NET responses. AZD3965 partially reversed both. Al-ILD patients showed greater CT parenchymal abnormality and lower FVC and DLCO than non-Al-ILD, with innate-immune, cytokine, ECM and NET pathways enriched in blood. **No exposure concentration, particle size distribution or duration appears in the deposited abstract**, so the animal arm cannot be placed on any dose axis, and the human arm is cross-sectional with no exposure measurement at all — it supports the mechanism narratively and cannot test it. Relevant here as occupational-aerosol toxicology and as a reminder that metal-bearing particulate has organ-level effects that mass concentration does not index.

doi: 10.1002/biof.70146

Qin et al. 2026b Environ Health (Wash)*Engineered protein coronas: ApoE enhances particulate transport across the blood-brain barrier*

Gold nanoparticles functionalised with thiolated DNA aptamers serve as a **compositionally controlled PM surrogate**, allowing reproducible assembly of coronas from three physiologically relevant proteins — human serum albumin, apolipoprotein E and complement C3b — and isolating corona identity from particle identity. In a bEnd.3 Transwell BBB model, **ApoE-enriched coronas nearly doubled apparent permeability and increased 24 h transport 1.6-fold** over bare particles, through an energy-dependent, predominantly clathrin-mediated route. ApoE precoronation also raised endothelial penetration of polystyrene particles and **increased the endothelial cytotoxicity of real industrial-source PM samples**, which is the step that connects the model to ambient aerosol. The surrogate is the limitation: gold nanoparticles share neither the surface chemistry nor the morphology of combustion PM, and a Transwell monolayer is not a neurovascular unit. Mechanistically the cleanest account yet of why particle *surface*, not particle mass, governs neural access.

doi: 10.1021/envhealth.5c00803

Seymore et al. 2026 Toxicol Appl Pharmacol*Gestational nanoparticle inhalation alters placental glucose transporters and increases glucose flux*

Pregnant Sprague Dawley rats exposed by whole-body inhalation to TiO₂ nanoparticles at $9.8 \pm 1.0 \text{ mg/m}^3$, an occupational-level concentration, from gestational day 5 to 19; GLUT1/3/4 assayed in term placenta at GD20 and ex vivo placental perfusion used as a functional test. GLUT3 mRNA and GLUT1 protein fell, but GLUT4 expression and membrane localisation of both GLUT1 and GLUT4 rose, and perfusion confirmed **increased** glucose flux (AUC 95% CI 77.4–127.5 exposed vs 39.1–73.6 control). The authors state plainly that this contradicts their hypothesis. Read as a compensatory adaptation rather than a benign finding — upregulated placental glucose delivery is not obviously protective in a fetal-growth-restriction model. The dose is roughly three orders of magnitude above ambient PM_{2.5} and the particle is a single engineered material, so this constrains occupational nanomaterial exposure and says little directly about ambient PM.

doi: 10.1016/j.taap.2026.117996

Sharma et al. 2026 J Toxicol Environ Health A*Dim light at night modulates the neuroimmune response to burn-pit emission inhalation*

32 male Swiss Webster mice housed under dark nights or dim light at night, exposed at week 4 to filtered air or burn-pit emissions (6.9 mg/m^3 mass, 4.03×10^6 particles/cc) for 4 h/day over 6 days. **No behavioural differences** in open field, Y-maze or elevated plus maze. Hippocampal *Il6*, *Tnf α* , *Il1 β* and cortical *Il6* were *decreased* in LD-exposed mice, with reduced hippocampal *Bmal1*; lung and liver were unchanged. Small *n* per cell, a single sex, and a cytokine direction opposite to the usual inflammatory prior — reported as a circadian-interaction signal, not as an inflammation result.

doi: 10.1080/15287394.2026.2708884

Shi et al. 2026 J Appl Toxicol*PM_{2.5} aggravates airway inflammation and barrier disruption in COPD mice via RAGE-dependent PERK/eIF2 α /ATF4/CHOP*

Cigarette-smoke-induced murine COPD model with subsequent PM_{2.5} exposure. PM worsened lung function, emphysema, inflammation, apoptosis and epithelial tight-junction damage, and activated the PERK/eIF2 α /ATF4/CHOP endoplasmic-reticulum stress arm. The causal claim rests on two rescue arms: the RAGE inhibitor FPS-ZM1 and the ER-stress modulator 4-PBA each attenuated the phenotype, placing RAGE upstream of ER stress rather than parallel to it. Limitation: instilled PM at a dose the abstract does not state, a smoke-primed rather than naive background, and pharmacological inhibitors of imperfect selectivity — FPS-ZM1 in particular. The RAGE→ER-stress ordering is the novel part; the downstream inflammatory readouts are well-trodden.

doi: 10.1002/jat.70358

Silva et al. 2026 Environ Toxicol Pharmacol*PM_{2.5} organic extracts in 2D monolayers versus 3D spheroids of bronchial epithelium*

16HBE cells in paired 2D and 3D culture exposed to solvent-extractable organic fractions of well-characterised ambient PM_{2.5} from three sampling periods. **Concordance between formats is endpoint-dependent, not uniform**: WST-1 metabolic activity, ROS fluorescence and extracellular metabolite exchange respond more strongly in 2D, while pro-inflammatory cytokine secretion is more evident in spheroids. Cell death is necrotic rather than apoptotic in both. The practical implication for in vitro PM toxicology is that model choice silently selects which hazard you can detect — 2D favours direct cytotoxic and oxidative readouts, 3D favours responses that need spatial organisation and exposure gradients.

doi: 10.1016/j.etap.2026.105125

Sivakumar & Kurian 2026 Environ Toxicol*Duration-dependent transition from mitochondrial adaptation to persistent cardiac toxicity*

Female Wistar rats, whole-body PM_{2.5} at $250 \mu\text{g/m}^3$, 3 h/day for 1, 7, 14 or 21 days, with cardiac mitochondrial bioenergetics, redox balance, quality-control signalling and ex vivo function assessed after each duration and after a 24 h washout. **1–7 days: adaptive and fully reversible. ≥ 14 days: suppressed respiration and ETC complex activity, mtDNA depletion, impaired biogenesis, intramitochondrial metal accumulation — none recovered within the washout.** Concentration was held constant, so duration alone moved the system across the threshold. The 24 h washout is short, so “persistent” means “not recovered in 24 h”, which the authors state plainly; a longer recovery arm would settle it. The instrumentation implication is direct: a cumulative-duration-above-threshold metric is testable with continuous logging and invisible to a 24 h filter sample.

doi: 10.1002/tox.70189

TARGET II Consortium 2026 bioRxiv (preprint)*Lifecourse sex-specific molecular response to early-life toxicant exposure*

Consortium-scale design: mice exposed from two weeks preconception through gestation and lactation to arsenic, lead, tributyltin, BPA, DEHP, TCDD or PM_{2.5}, with **1,025 omics datasets** generated from liver across three life stages. The headline is a significant sex-biased transcriptomic and epigenomic response to a common exposure. For this watch the value is comparative — PM_{2.5} is run alongside six reference toxicants under one protocol, which is the only way to judge whether a PM molecular signature is distinctive or generic. Liver-only, and mouse inhalation dosimetry does not scale simply to humans. Unreviewed.

doi: 10.64898/2026.08.20.746014

Van et al. 2026 In Vivo*MASLD exacerbates chronic low-dose PM_{2.5}-induced lung injury, inflammation and fibrosis*

C57BL/6J mice on a Western or normal diet for 28 weeks with concurrent intratracheal diesel particulate matter at **0.5 mg/kg three times weekly** (stated as environmentally relevant to ~50 µg/m³), assessed by histology, Masson's trichrome, Luminex multiplex on bronchoalveolar lavage and lung-tissue ELISA/qPCR. The Western diet induced the MASLD phenotype; PM_{2.5} did not worsen the hepatic phenotype, but **the combination markedly exacerbated pulmonary injury and fibrosis relative to PM_{2.5} alone**. The dose-equivalence claim is the weak point: bolus intratracheal instillation of diesel PM delivers a deposition profile nothing like chronic inhalation at 50 µg/m³, so the “environmentally relevant” framing should not be taken at face value. The interaction direction is the transferable result.

doi: 10.21873/invivo.14428

Wang et al. 2026 Ecotoxicol Environ Saf*Multigenerational reproductive toxicity of three vehicular PM types in C. elegans*

SRM diesel PM, WHTC diesel PM and WLTP gasoline PM, differing in morphology and elemental composition, all produced dose-dependent brood-size reduction, germ-cell apoptosis, lower sperm count and morphological abnormality in parents. Mechanistically: suppressed SPE-8/SPE-9/major sperm protein/vitellogenin, activated *ced-3/-4/-9*, disrupted *set-2*, *met-2*, *spr-5*. **Effects persist to F1–F2 and recover by F3–F4 — intergenerational carryover, not stable transgenerational inheritance**, which is the distinction most papers in this space elide. Nematode model; extrapolation to mammalian germline is speculative.

doi: 10.1016/j.ecoenv.2026.120609

Wei et al. 2026 J Hazard Mater*Seasonal source shifts drive oxidative potential of size-resolved particulate matter in coastal southeastern China*

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar. Listed, not summarised. Noted because size-resolved oxidative potential is one of the few routinely measured quantities that is both a particle property and a plausible biological effect metric, and because a source-driven seasonal shift in OP at fixed mass concentration would be direct evidence against mass as a sufficient exposure metric — which is the substantive question behind this watch's instrumentation axis. Whether the paper supports that reading is unknown until an abstract appears.

doi: 10.1016/j.jhazmat.2026.143237

Wu et al. 2026 (JHM) J Hazard Mater*TNF-α and EGF mediate endothelial activation downstream of PM_{2.5} alveolar injury*

3D alveolar–vascular barrier constructed from BEAS-2B epithelium, THP-1 macrophages and EA.hy926 endothelium, exposed to **100 µg/mL SRM2786 PM_{2.5} for 24 h**. PM_{2.5} entered epithelial cells by endocytosis, produced mitochondrial damage and apoptosis, and raised TNF-α, EGF, IL-8, TGF-α and IL-1β; extracellular TNF-α and EGF crossed the barrier and activated endothelial PI3K–Akt, with Nrf2 and CYP1A1 induction marking progressive oxidative stress. The model supplies a mechanistic route from local alveolar injury to systemic endothelial dysfunction. The single exposure concentration is far above any ambient equivalent and there is no dose–response, so the result establishes the pathway's existence, not its relevance at realistic exposures — the standard and unresolved weakness of this literature.

doi: 10.1016/j.jhazmat.2026.143204

Yan et al. 2026 Front Pharmacol*PM_{2.5} induces lung injury via mtDNA-cGAS-STING-mediated macrophage M1 polarization*

21-day subacute PM_{2.5} mouse model plus bone-marrow-derived macrophages in vitro. Exposure produced pulmonary inflammation and tissue injury with macrophage accumulation and M1 polarisation; mechanistically, mitochondrial damage released mtDNA and activated cGAS–STING. The load-bearing experiment is the rescue: **pharmacological cGAS inhibition with RU.521 attenuated STING activation, M1 polarisation and lung injury in vivo**, which converts a correlative pathway into an interventional one. Subacute rather than chronic, and a single exposure route — the dosimetry is the usual weak point.

doi: 10.3389/fphar.2026.1902928

Yibchareonporn et al. 2026 ACS Omega*Metabolic profiling of PM-induced developmental toxicity in zebrafish embryos*

Zebrafish embryos exposed 4–50 hpf to fine and coarse PM at 500 and 1000 µg/mL, with untargeted LC-MS/MS metabolomics. The fine fraction alone produced a **22 h hatching delay** and disrupted pantothenate/CoA, retinol and riboflavin metabolism; riboflavin fell significantly at the higher dose. Glutathione responses diverged by size — reduced glutathione depleted under the fine fraction but *elevated* under the coarse — which is the most interesting result here because it argues against a single shared oxidative mechanism scaled by mass. The dose is the problem: 500–1000 µg/mL of suspended PM is three to four orders of magnitude above any inhalable ambient concentration, so this establishes a mechanism, not a risk. The authors' closing suggestion of vitamin supplementation is not supported by anything in the design.

doi: 10.1021/acsomega.6c00972

Yuan et al. 2026 Innovation (Camb)*Hallmarks of lung cancer driven by inhalable particulate matter*

Critical review proposing a 12-feature “hallmarks of PM-associated lung cancer” framework spanning cell-intrinsic traits (genetic alteration, epigenetic reprogramming, metabolic remodelling, stemness) and microenvironment remodelling (inflammation, immune suppression, angiogenesis, pre-metastatic niche). Argues for integrating PM exposure into screening and risk stratification. A framework paper: the hallmark taxonomy is an organising device, not evidence, and the underlying studies are heterogeneous in particle source and dose. Useful as a citation scaffold for a mechanism section.

doi: 10.1016/j.xinn.2026.101347

Zhang et al. 2026 J Thorac Dis*Pleural anthracosis severity and the structure and function of subcarinal lymph nodes*

Thoracoscopic grading of pleural anthracosis into four grades by three thoracic surgeons with ImageJ area quantification, against paired subcarinal lymph nodes stained with HE, Masson, EVG and immunohistochemistry for Ki67, CD31, CD68 and D2-40. Black particulate accumulation in node capsule and medulla rose with anthracosis grade, but every quantitative immunostaining index **rose then fell** with advancing severity — a consistent non-monotonic pattern across proliferation, vascularity, macrophage presence and lymphatic integrity. The non-monotonicity is the finding and is also the weakness: with a graded surgical sample and no exposure history, an exhaustion interpretation and a selection interpretation are equally consistent with the data.

doi: 10.21037/jtd-2026-1274

NOT CARRIED BY A DAILY ISSUE — EXTRACTIVE PRECIS OR METADATA ONLY, NOT AN ASSESSED SUMMARY (1 RECORDS)

Saygin et al. 2026 Atmos Pollut Res

Source identification and seasonal change of trace metals, protein content and oxidative potential of fine and coarse particulates after the Kahramanmaraş earthquakes
No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1016/j.apr.2026.103164

Occupational & indoor**37 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (36 OF 37 RECORDS IN THIS SUBTOPIC)

Alqahtani et al. 2026 PLoS One*Biomarker-based assessment of tobacco smoke exposure in Saudi females via urinary cotinine*

LC-MS/MS quantification of urinary cotinine in 133 Saudi women recruited May–December 2024, validated to ICH-M10 (linearity 2–500 ng/mL, $R^2 \geq 0.9992$, accuracy 90–107%, CV <10%), with a population-specific cut-off of **7 ng/mL**. Cotinine was detected in **48 of 133** samples although **88.7% self-reported as non-smokers**, and 45.9% reported household second-hand smoke. *Limitations:* $n = 133$ is small for establishing a population cut-off; the detected range (0.02–7.693 ng/mL) sits almost entirely *below* the proposed 7 ng/mL threshold, so the cut-off classifies nearly every detected sample as non-smoking and its derivation is not defensible from the reported distribution; and cotinine's short half-life indexes recent exposure only. Relevance: a clean demonstration that self-report and biomarker diverge sharply under social desirability pressure — the same argument for objective personal measurement over questionnaires that motivates wearable PM sensing.

doi: 10.1371/journal.pone.0355166

Cuerda Barcaiztegui et al. 2026 Atmos Pollut Res*Classroom ventilation management and user activities vs. indoor air quality, Madrid primary schools*

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar at the time of this issue. A two-year classroom field study on ventilation management and occupant activity is directly on this watch's deployment axis and is the single highest-priority abstract re-check for the next run.

doi: 10.1016/j.apr.2026.103170

Gu et al. 2026b Environ Int*Nitrogen-containing organics in household solid-fuel combustion PM vs. urinary inflammatory and amino-acid profiles*

Listed on Crossref metadata only. The deposit carries no abstract, and the record is absent from Europe PMC, PubMed and Semantic Scholar as of this run, so **no finding, effect size, cohort size or location is asserted here**. The title indicates an exploratory personal exposure design pairing speciated N-containing organic compounds in solid-fuel combustion PM with targeted urinary inflammatory and amino-acid metabolites — a personal-monitoring plus targeted-metabolomics pairing that is directly on-axis for this watch. Queued for the abstract retry list; it will be summarised, not re-listed, once text is available.

doi: 10.1016/j.envint.2026.110491

Herkert et al. 2026 Environ Sci Technol*Contribution of indoor air and dust to children's PFAS exposure in the Faroe Islands*

Measurement of nonvolatile precursors, volatile neutral precursors and legacy ionic PFAS in air and dust across **40 Faroese homes**, with estimated daily intake computed for 15-year-olds. EDI ranges 0.001 / 0.012 / 0.129 ng/kg bw across low, intermediate and high scenarios, equal to roughly **0.2%, 2% and 20% of the EFSA tolerable weekly intake**, against ~60% from pilot-whale consumption. Indirect exposure to perfluoroalkyl carboxylic acids via air dominates, followed by PFOS in dust. *Limitations:* scenario-based EDIs are not measured intakes, and the 40-home sample cannot support the tail of the distribution where the 20% figure sits. Relevance: the indoor compartment supplies a fifth of a regulatory reference dose in the high scenario for a population with no local source — a direct case for indoor sampling as a routine part of exposure assessment rather than a specialist add-on.

doi: 10.1021/acs.est.6c02003

Iavazzo et al. 2026 Environ Geochem Health*Particle characterization in an urban synthetic turf: turf and vehicular wear*

XRF, SEM-EDS and dynamic light scattering on dust, turf fibre and crumb-rubber infill from an urban synthetic pitch, with stub-based sampling worn on *players*. Irregular and elongated fine, submicron and nano-sized particles were observed; Fe dominated by XRF, with Fe, Si, Al and Br-bearing particles by SEM-EDS and **Ni, Ag and Rh** recovered from the player-worn stubs — Rh being a clear three-way-catalyst tracer, so the pitch is receiving vehicular as well as turf-wear particles. Ambient PM_{2.5} averaged **10.93 µg/m³** (24-h) and O₃ reached 41.12 ppb. *Limitations:* one field, no personal inhaled dose, and the deterministic USEPA assessment — which found no significant non-carcinogenic metal risk — is mass-based and cannot speak to nanoparticle toxicity. Relevance: the contrast between an unremarkable regulatory 24-h mean and a demonstrably loaded microenvironment is the clearest argument in this issue for personal and microenvironmental sampling.

doi: 10.1007/s10653-026-03407-1

Illa et al. 2026 Environ Sci Technol*Exposure pathways when psychoactive drugs are smoked indoors*

Mechanistic indoor chemical-fate and exposure model across seven compounds spanning orders of magnitude in volatility and hydrophobicity, with scenarios varying smoker retention rate and smoking frequency, resolving first-, second- and third-hand pathways and inhalation, mouth-ingestion and dermal routes. Dominant pathway is strongly chemical-specific: volatile compounds (nicotine) go by inhalation; intermediate hydrophobicity by dermal absorption; involatile hydrophobic compounds (Δ^9 -THC) receive meaningful contributions from all three. For non-users the SHS/THS balance is set jointly by partitioning and by the ratio of active-smoking to post-smoking time. Entirely modelled — no measurements, no validation — so the useful output is the ranking of pathways, not the doses.

doi: 10.1021/acs.est.6c02953

Jin et al. 2026b Build Environ*Air change rate and air purifier type for bioaerosol control during dental ultrasonic scaling*

Metadata only. The Crossref deposit carries no abstract and the record is not yet in PubMed, Europe PMC or Semantic Scholar. Listed and counted for completeness; no finding, effect size or method detail is attributed to it, and it is on the abstract retry list.

doi: 10.1016/j.buildenv.2026.115193

Kaur et al. 2026 NanoImpact*Workplace graphene: the material you make is more toxic than the air you breathe*

Pilot study pairing occupational exposure assessment in an operating graphene production facility with in vitro toxicology on the very material collected there. Stationary and personal samplers fed elemental carbon analysis, Raman microscopy-spectroscopy (which confirmed graphene in the collected PM), and viability testing in A549 and BEAS-2B cells. **Personal EC ranged 3–484 $\mu\text{g}/\text{m}^3$** , peaking during graphite powder handling. The dissociation is the finding: **workplace-collected PM was not cytotoxic at tested concentrations, while dispersions of as-produced liquid-phase-exfoliated graphene caused dose-dependent viability loss.** Limitations: one facility, pilot scale, two cell lines, acute viability only, and collected-sample doses may simply be below any threshold rather than genuinely benign. Nonetheless the design — sample the workplace, then test that sample rather than a pristine surrogate — is the coupling this field keeps asking for.

doi: 10.1016/j.impact.2026.100642

Li & Zheng 2026 Toxics*Size-resolved PM₁₀-bound PFAS: respiratory deposition and oxidative stress in recycling workers*

Panel study at a southern-Chinese waste recycling plant: **15 size-resolved PM samples and 280 repeated first-void urines from 20 workers over 45 consecutive days**, with linear mixed models linking size-resolved PM-bound PFAS to urinary PFAS and to 8-OHdG and MDA. Short-chain PFAS predominated in both matrices; urinary ICCs were fair to excellent (>0.4). **PFAS in PM_{2.1-10} correlated more strongly with internal dose than PFAS in PM_{2.1}** — the opposite of the fine-fraction-dominates prior, and consistent with mucociliary clearance and gastrointestinal uptake of coarse-fraction deposits. **Each unit increase in ln average daily intake was associated with 10.2–12.6 % higher 8-OHdG (PM_{4.7-5.8}, PM_{9.0-10}) and 8.0–14.6 % higher MDA (PM_{5.8-9.0}).** Twenty workers, one plant, and the authors call the biomarker associations suggestive; with this many size bins the multiplicity is substantial and no correction is described. The size-resolved design is nonetheless the right one and is almost never done.

doi: 10.3390/toxics14080682

Li et al. 2026f Aerosol Air Qual Res*Non-uniform oil-mist distribution and local ventilation in a machining workshop*

Listed on Crossref metadata only — no abstract deposited, and no record in Europe PMC, PubMed or Semantic Scholar. **No quantitative claim is made here.** The title indicates spatial characterisation of oil-mist aerosol in a machining workshop with a local ventilation intervention, which is the occupational analogue of the indoor-siting problem: where a workshop's aerosol field is strongly non-uniform, a single fixed monitor is a poor estimator of personal exposure regardless of its accuracy. Queued for the abstract retry list.

doi: 10.1007/s44408-026-00161-y

Liu & Hu 2026 (preprint) Preprints.org*Multivariable determinants of indoor PM_{2.5} and infiltration dynamics in 48 residential buildings across eight cities of Northern China*

Paired indoor/outdoor PM_{2.5} at 5-minute resolution for seven consecutive days in each of four seasons across Beijing, Tianjin, Shijiazhuang, Taiyuan, Jinan, Zhengzhou, Xi'an and Anyang — **32,256 hourly observations from 48 residences**, with building characteristics, occupant behaviour and meteorology recorded alongside. A multivariable OLS model with cluster-robust standard errors ($R^2 = 0.879$) gives outdoor PM_{2.5} $\beta = 0.891$, window-open fraction $\beta = 0.572$, cooking $\beta = 0.049$ and **air-purifier operation $\beta = -0.935$** , all $p < 0.001$. Infiltration factor **0.28 ± 0.15** ; outdoor sources contribute **57–69%** of indoor PM_{2.5}, peaking in the heating season; median deposition rate 0.53 h^{-1} against a median air change rate of 0.52 h^{-1} , exceeding 2 h^{-1} with windows open. Preprint, not peer reviewed — but the sample size and the seasonal design are both well beyond what this literature usually carries.

doi: 10.20944/preprints202608.1362.v1

Makanjuola 2026 Research Square (preprint)*Personal PM exposure and spirometry among 563 Southwest Nigerian street sweepers*

Cross-sectional, non-smoking sweepers aged 18–60 from Ado-Ekiti, Ibadan and Ikeja; personal PM_{2.5} and PM₁₀ by handheld PMS5003 at breathing height; spirometry to ATS standard. Mean PM_{2.5} **$103.55 \pm 72.03 \mu\text{g}/\text{m}^3$** and PM₁₀ 105.09 ± 73.95 , roughly $20\times$ and $7\times$ the WHO 2021 annual guideline. PM_{2.5} correlates inversely with FEV₁/FVC ($r = -0.14$, $p = 0.001$) but **the association is fully attenuated in the adjusted model** ($\beta = -0.0001$, $p = 0.286$). A bare PMS5003 without RH correction at these loadings is a real measurement-error problem the paper does not address, and PM_{2.5} $\approx$ PM₁₀ is itself a red flag for the instrument. Reported for the exposure characterisation, not the health inference.

doi: 10.21203/rs.3.rs-9928829/v1

Nath et al. 2026 J Anaesthesiol Clin Pharmacol*Operating-room particulate: HEPA helps, cautery undoes it*

Portable monitoring of CO₂, TVOC, formaldehyde, PM_{2.5} and PM₁₀ across operating rooms with and without HEPA filtration. HEPA rooms were significantly cleaner (PM_{2.5} **19.78 ± 8.81 vs $29.83 \pm 4.56 \mu\text{g}/\text{m}^3$** , $p = 0.026$; PM₁₀ 29.93 ± 12.96 vs 38.61 ± 6.89 , $p = 0.04$) while other pollutants were comparable. Electrocautery raised TVOC, PM_{2.5} and PM₁₀; ORs stayed well below the control area in winter (PM_{2.5} 17.0 vs $36.66 \mu\text{g}/\text{m}^3$); average CO₂ was often high enough to matter for staff. **Limitations: small, single-centre, cross-sectional, with no personal sampling and no health endpoint**, and the HEPA and cautery comparisons appear to reuse overlapping room-groups, so the two effects are not cleanly separated. Included as a reminder that the highest-value indoor PM measurements are often minute-resolved and event-driven, which is where low-cost nodes are strongest.

doi: 10.4103/joacp.joacp_380_25

Niari et al. 2026 J Environ Health Sci Eng*Microplastic burden in beauty-salon indoor air and worker nasal lavage fluid*

Thirty-six indoor air and 18 nasal lavage samples across salons and homes in Ardabil. Salon air carried **99.4 ± 4.1 versus $36.6 \pm 3.1 \text{ MP}/\text{m}^3$** at home, with nasal lavage $12.3\text{--}23.8$ versus $9.3\text{--}20.1 \text{ MP}/\text{mL}$ and estimated daily exposure 572 ± 238 versus $212 \pm 109 \text{ MP}/\text{day}$. The nasal-lavage arm is the useful part — an internal-dose measurement rather than an inferred one — but the sample is small, no procedural blank is reported for a matrix acutely prone to airborne plastic contamination, and no polymer identification is described. Directionally credible, quantitatively soft.

doi: 10.1007/s40201-026-00999-9

O'Donnell et al. 2026 medRxiv (preprint)

Nicotine replacement therapy to reduce children's second-hand smoke exposure at home

Pilot randomised controlled trial in Scotland testing whether free NRT plus telephone-delivered support is a feasible route to a smoke-free home, with objective indoor PM_{2.5} as the exposure measure rather than self-reported smoking — which is the design decision that makes it relevant here. Feasibility, recruitment and retention are the outcomes; the trial is not powered for exposure reduction. Notable as one of very few interventions targeting the indoor source directly rather than filtering downstream of it. Preprint, not peer reviewed; no numeric claim from it is used in this issue's analytics.

doi: 10.64898/2026.08.13.26360348

Ospalek et al. 2026 Atmosphere

Air quality assessment and potential inhalation exposure across equestrian facilities

Nine measurements at 1.5 m in a stable, an indoor riding arena and an outdoor riding arena at a Slovak university equestrian centre, covering PM₁, PM_{2.5}, PM₁₀, respirable and total dust plus gaseous pollutants, with four exposure scenarios combining 18–20 h stabled with 4–6 h ridden. The **outdoor arena carried the highest concentrations of every PM fraction and the indoor arena the lowest** ($p < 0.05$) — the opposite of the expected enclosure effect, and the paper's most interesting result; ammonia was higher in the stable, as expected. Summer measurements only, one facility, nine measurements total, and the exposure scenarios are constructed rather than observed, so the inhaled-dose estimates are illustrative. Relevant here as a template for microenvironment-resolved dose reconstruction from sparse fixed-point sampling.

doi: 10.3390/atmos17090840

Park et al. 2026b Environ Pollut

Phenotypic resistance, not community turnover, tracks antimicrobial use in swine-farm bioaerosol

962 isolates across six species, six Korean swine farms stratified by defined daily dose for animals, four compartments (faecal, aerosol, pig nasal, worker nasal) and two seasons, with flow-cytometric viability and qPCR-corrected abundance feeding a Monte Carlo QMRA. *S. aureus* isolate-level resistance was higher on high-usage farms; the other four species and the community composition showed no usage association. *Escherichia* abundance fell ~four orders of magnitude on aerosolisation while *Staphylococcus* persisted and peaked in worker nasal samples. Median daily colonisation probability $1.98\text{--}8.63 \times 10^{-4}$; the model was most sensitive to aerosol *genus composition*, not total bioaerosol load. Six farms cannot resolve a farm-level effect and the authors say so. For sensing: total particle or total bioaerosol load is the wrong exposure variable here.

doi: 10.1016/j.envpol.2026.128991

Perna et al. 2026 Life (Basel)

Indoor air pollution as a determinant of airport occupational exposure

Narrative review arguing that airport occupational exposure assessment has been anchored on aircraft engine emissions and has under-weighted **indoor air quality driven by indoor-outdoor exchange and ventilation**. Ultrafine particles, black carbon, PAHs and trace metals are identified as the constituents driving oxidative stress, inflammation and xenobiotic pathway activation, with effects framed beyond the respiratory system. The operationally useful recommendation is to monitor indoor microclimatic parameters — temperature, pressure, humidity — as **event detectors for indoor air-quality deterioration**, which is a cheap addition to any fixed deployment. No systematic search, no quantitative synthesis, no exposure or effect estimates; the closing suggestion that AI-based exposure modelling is promising is asserted with the authors' own caveat that it lacks validation. Tier C: useful as a framing document for airport deployments, not as evidence.

doi: 10.3390/Life16081242

Piedra et al. 2026 Front Public Health

Informal employment versus cantonal satellite pollution as respiratory-symptom predictors

1,807 Ecuadorian workers across four sectors, ECRHS symptom instrument cross-referenced with Sentinel-5P retrievals; optimal spatial/temporal concentration percentile selected per outcome by AIC; Firth penalised logistic regression. **Informal employment dominates:** asthma aOR 12.56 (5.09–32.87), wheeze 6.78 (1.83–27.89), chronic cough 4.14 (2.26–7.74). After adjusting for cantonal altitude (itself protective, aOR 0.78, 0.66–0.93), NO₂ and SO₂ remain independent predictors of chronic phlegm and cough while ozone loses significance. Sentinel-5P carries no PM product, so the ambient arm is entirely gas-phase; cantonal assignment is coarse and the design is cross-sectional. The interesting move is using satellite retrievals as an occupational-surveillance covariate.

doi: 10.3389/fpubh.2026.1874197

Ramirez et al. 2026 Curr Allergy Asthma Rep

Airway diseases and occupational exposures in poultry industry workers: a comprehensive review

Review of occupational asthma and chronic obstructive lung disease in poultry workers, attributing risk to the barn's mixed aerosol — feather and feed dust, mites, moulds, ammonia and bacterial endotoxin — acting through airway inflammation and hypersensitivity, with a symptom spectrum from wheeze and chest tightness to chronic cough and dyspnoea. Emphasis falls on early identification through occupational history-taking and on barn control practices. **No prevalence figures, no exposure-response estimates and no search or inclusion strategy** are reported, so this is a clinical orientation piece rather than an evidence synthesis. Its relevance here is the exposure architecture: an occupational aerosol whose health-relevant fraction is defined by biological composition, which a mass-only monitor cannot resolve at all.

doi: 10.1007/s11882-026-01289-y

Ranjbar et al. 2026 Preprints.org (preprint)

1-Nitropyrene as a diesel-specific indoor exposure proxy in California homes

1-Nitropyrene and the isomeric 2-nitropyrene and 2-nitrofluoranthene measured in indoor air and house dust from **40 homes** in the eastern San Francisco Bay Area across two sampling events, alongside continuous indoor black carbon. The motivation is specificity: BC, NO_x and PM_{2.5} are all used as diesel surrogates and none is diesel-specific. Detection was **74% in air and 97% in dust**, medians **0.42 pg/m³ (IQR 0.32–0.54)** and **340 pg/g (IQR 190–620)**; air-dust correlation $r = 0.40$ ($p = 0.02$) and BC-1-NP $r = 0.38$ (air) and 0.47 (dust). Diagnostic ratios (2-NFl:1-NP < 5) indicate primary emission rather than secondary formation. Dust 1-NP associated with more spatial diesel predictors than air 1-NP did, consistent with dust integrating longer-term deposition — which is the actual argument for using it. Preprint, not peer reviewed. Forty homes, two events, no personal sampling, and a picogram-per-cubic-metre analyte whose blank and recovery performance is not summarised in the abstract.

doi: 10.20944/preprints202608.0873.v1

Sadki et al. 2026 *Cad Saude Publica*

Real-time formaldehyde exposure and acute symptoms in healthcare workers

Cross-sectional study of **180 healthcare workers across 12 hospital departments** in Morocco, with airborne formaldehyde measured over full eight-hour shifts by calibrated semiconductor dosimeter alongside a structured symptom questionnaire. **Every mean concentration was below the Moroccan occupational limit of 0.25 mg/m³** (0.048–0.196 mg/m³, 19–78 % of the limit), yet highly exposed workers reported markedly more eye irritation (72.5 %), headache (63 %) and respiratory discomfort (56.5 %). **Prolonged exposure (≥ 8 h/day, ≥ 3 days/week) independently predicted multiple symptoms (AOR 3.2, 95 % CI 1.7–5.9)**, as did inadequate ventilation (2.4) and inconsistent PPE use (1.9); years of exposure correlated with symptom count ($\rho = 0.41$). **The exposure is gaseous, not particulate**, so the AORs are excluded from this issue's forest plot. Carried because the deployment pattern — low-cost real-time dosimetry as routine occupational surveillance in a resource-limited setting — is exactly the use case for distributed sensing, and because symptom self-report against measured exposure is open to reporting bias in the obvious direction.

doi: 10.1590/0102-311XEN191325

Salcines et al. 2026 *J Environ Health Sci Eng*

Monitoring and risk assessment of airborne graphene oxide released during 3D printing with polylactic acid filaments

Tiered exposure assessment to EN 17058:2018 and OECD guidance around fused-deposition printing of graphene-oxide-loaded PLA, combining **near-field and far-field particle counters** with SEM and Raman speciation plus VOC measurement against Spanish and Finnish 2022 occupational limits, with ARIMA used to test whether the observed excursions are distinguishable from background drift. Nanoparticles and agglomerates were present at low average number concentration and **GO was detected in the respirable fraction**, below proposed limits but demonstrably released — the finding is qualitative (release occurred) rather than quantitative (how much). Weaknesses are structural: a pilot with no stated replicate count, number-concentration monitors that cannot speciate, so the GO attribution rests entirely on offline SEM/Raman of collected material, and occupational limits for GO that do not formally exist. The methodological point transfers directly — paired near-field/far-field counters with an offline speciation leg is the minimum viable design for attributing a number-concentration excursion to a source, and it is the design a distributed indoor network would need.

doi: 10.1007/s40201-026-00998-w

Seyedrezaei et al. 2026 *J Occup Environ Med*

Ecological momentary assessment plus continuous IEQ sensing across home and office

Nineteen office workers followed four months contributing **4,559 EMAs** of productivity, mood, stress and eustress–distress appraisal, against continuous CO₂, TVOC, PM_{2.5}, temperature, humidity and noise, analysed with mixed-effects and multilevel mediation. Office work associated with higher productivity and greater eustress–distress coexistence, home with higher boredom and modestly lower stress; **home workspaces had poorer air quality**, and CO₂ and TVOC indirectly explained part of the mood and stress difference. $n = 19$ is very small for multilevel mediation, and the within-person design cannot separate the environment from everything else that differs between home and office — read the IEQ mediation as hypothesis-generating. The methodological interest is the pairing of dense subjective sampling with dense environmental sampling, which is the design template for sensor-linked wellbeing work.

doi: 10.1097/JOM.00000000000003833

Shadmani et al. 2026 *J Environ Health Sci Eng*

A comprehensive health profile and exposure-risk analysis in an iron ore mining workforce

Cross-sectional study of **1,193 Iranian iron-ore mine employees**, stratified into high, moderate and low exposure *by job role*, with physical examination, clinical chemistry and haematology compared by ANOVA and χ^2 and modelled by linear and logistic regression. The workforce is young (mean **34.9 $\pm$ 7.6 y**) with low chronic-disease prevalence; the burden is localised — hearing **23.8 %**, vision **17.8 %**, dermatological **31.3 %** — while most haematological and metabolic markers did not differ by exposure group. Platelet count, LDL and blood pressure were *lower* in the high-exposure group, which the authors do not resolve and which has an obvious competing explanation the paper does not exclude: the healthy-worker survivor effect plus role-based confounding by age and seniority. The central methodological weakness for this watch is that **exposure is a job-role proxy with no measured dust concentration at all** — no gravimetric respirable sampling, no personal monitoring, no size fractionation. A three-level ordinal surrogate cannot support an exposure–response claim, and the paper's own conclusion (localised sensory and dermatological effects, not systemic metabolic ones) is a statement about a young cohort with short cumulative exposure rather than about the hazard.

doi: 10.1007/s40201-026-01000-3

Siemens et al. 2026 *Indoor Air*

A SynAir-G position statement for the improvement of indoor air quality for children

Position statement from the EU-funded SynAir-G project (18 partners, 11 countries) within the seven-project IDEAL Cluster, arguing that indoor air has been under-regulated relative to outdoor air despite most exposure occurring indoors, and that the policy-relevant unit is the **synergistic interaction between chemical and biological pollutants** rather than any single-pollutant concentration. Schools and children are the target setting. As a consensus document it presents no new measurements, no systematic evidence appraisal and no quantified synergy, so the central claim is asserted rather than demonstrated. The instrumentation implication is nonetheless concrete: a synergy framing requires co-located chemical and bioaerosol sensing at classroom scale, which no current low-cost platform delivers.

doi: 10.1155/ina/9690417

Tan et al. 2026 *Nutrients*

PM_{2.5} components, nutritional indices and leisure-time physical activity vs lower eGFR in subway workers

8,477 Wuhan subway workers; PM_{2.5} and components from the Tracking Air Pollution in China dataset; creatinine-based eGFR; GNRI and PNI nutritional indices; IPAQ-derived activity. Mixture effects by WQS, quantile g-computation and BKMR. **WQS OR 1.183 (1.106–1.266) and QGC OR 1.183 (1.083–1.291)** for lower eGFR; BKMR overall effect on eGFR of -0.045 (-0.068 , -0.022) for a 5-percentile mixture increase against medians. Subgroups by nutrition and activity differed but **no significant interaction** was found — the authors say so, and the correct reading is hypothesis generation, not effect modification. Cross-sectional, so reverse causation is unaddressed; and residential exposure is a poor proxy for the occupational exposure of people who work underground, which is the population's defining feature.

doi: 10.3390/nu18152580

Thies et al. 2026 Int J Popul Data Sci

The London Underground: very high PM_{2.5}, very little dose–response

Two retrospective cohorts from the ONS Longitudinal Study of economically active adults living and working in London (1991, 2001, 2011 Censuses). Cohort 1 ($n = 7,343$) compared Underground commuters with above-ground rail users; Cohort 2 ($n = 4,094$) tested dose–response against externally linked cumulative LU PM_{2.5} built from Transport for London data, PM_{2.5} measurements and geospatial modelling. Outcomes were all-cause mortality (1991–2017) and cancer incidence (1991–2015), analysed with inverse probability weighting and the parametric g-formula. Most LU commuters were estimated to encounter hourly PM_{2.5} above 100 $\mu\text{g}/\text{m}^3$ (8–10 $\times$ roadside), yet the survival deficit was 1.93% (95% CI 0.58–3.66), cancer differences negligible, **no dose–response** emerged, and a hypothetical 50% PM_{2.5} reduction barely moved survival. **Limitation: exposure is modelled, not measured on individuals**, and census-derived commute mode is a coarse, decadal proxy — non-differential misclassification of this size can flatten a real gradient. Either way, iron-rich mechanical-abrasion PM does not behave like ambient mass.

doi: 10.23889/ijpds.v11i5.3705

Tillery et al. 2026 Indoor Air

Indoor air nicotine, NO₂ and COVID-19 outcomes in a Great Plains tribal community

COVID-19 Wayakta He? study: 290 multigenerational Cheyenne River Sioux Tribe households, each pair of adults at least 9 years apart in age, with one week of passive air-nicotine and NO₂ badges plus saliva serology. Geometric mean air nicotine 0.052 $\mu\text{g}/\text{m}^3$ (detected in 40.1% of homes); geometric mean NO₂ 10.5 ppb, range 0.65–138, consistent with propane cooking and heating. **OR 1.21 (1.01–1.51)** for prior infection with detectable air nicotine; **OR 3.69 (1.11–16.6)** for self-reported long COVID with NO₂ exposure. The second interval spans a factor of fifteen and rests on a self-reported outcome in a cross-sectional design with one week of exposure measurement standing in for a multi-year window — it is a direction, not a magnitude. The study's real contribution is that it measured the microenvironment at all: no ambient network in the Northern Plains resolves household combustion, and community-deployed passive samplers are the only instrument that reached it.

doi: 10.1155/ina/7749725

Wood et al. 2026b Indoor Environments

Influence of ventilation rate on the potential for indoor particulate matter exposure

Metadata only. Same situation — no abstract in the Crossref deposit and no other index carries it yet. Flagged here because the title is squarely in scope for indoor PM sensing; it is on the abstract retry list and will be summarised if the abstract appears.

doi: 10.1016/j.indenv.2026.100191

Xia et al. 2026 Front Pediatr

Enhanced filtration for indoor air quality in child-serving facilities

Week-long baseline and follow-up measurements in 33 spaces across 12 facilities spanning daycare to high school, with staff surveys, as part of a community-based participatory partnership. Air change rates were low throughout (0.9–1.4 h^{-1}); baseline school-hours indoor PM_{2.5} 2.2–6.3 $\mu\text{g}/\text{m}^3$ and PM₁₀ 5.7–26.9 $\mu\text{g}/\text{m}^3$, highest in standard schools. Enhanced filtration cut PM_{2.5} **45–54%** in standard schools and the nursery and had little effect in childcare houses; a linear mixed model attributes **~51%** of the PM_{2.5} reduction to the portable air cleaners. **Single-zone mass-balance prediction held for regular classrooms and the nursery but failed for portable buildings, very large rooms and repurposed houses.** That failure boundary is the most transferable result in the paper: it maps exactly onto where a deployed sensor earns its keep. No health endpoint was measured.

doi: 10.3389/fped.2026.1875263

Yang et al. 2026b Environ Pollut

Bioaerosols at oil exploitation sites and mask filtration efficiency

Seasonal sampling in a Beijing–Tianjin–Hebei oil city carrying both heavy PM and VOC loads, characterising culturable and total bioaerosol distribution, sources and inhalation risk, plus mask filtration efficiency. **Autumn levels significantly exceeded summer while microbial activity moved the other way** — a dissociation between count and viability that matters for anyone interpreting a fluorescence channel as a proxy for biological hazard. Culture-based enumeration undercounts viable-but-non-culturable organisms by a large and site-dependent factor, which is the standing limitation of this entire measurement class. Tier C.

doi: 10.1016/j.envpol.2026.129020

Zhao et al. 2026 J Int Med Res

Occupational dust exposure and thyroid nodule screening: a predictive modelling approach for early detection in coal mining communities

Cross-sectional analysis of **1,708** coal miners examined at an occupational-disease hospital in Anhui in April 2021, combining general, laboratory and imaging data to predict nodular thyroid disease, using a proposed “Nonlinear Inverse Nearest Manifold Projection” model benchmarked against established risk models. The record is carried at tier C and should not be cited for a result: the abstract reports that the model “achieved high scores in both F1 and area under the precision–recall curve” and “significantly outperformed” comparators **without giving a single numeric value**, confidence interval, or comparator name. Beyond the missing numbers, dust exposure is never measured — occupation is the exposure proxy, with no mass concentration, no respirable fraction, no duration and no job-exposure matrix — so the paper cannot support any dose–response claim about particulate exposure and thyroid nodules. Included because occupational particulate exposure with thyroid endpoints is a thin literature and the register should show what is in it.

doi: 10.1177/03000605261473853

Zhao et al. 2026 Indoor Air

CFD-based prediction of air change rate and PM in a dynamically occupied operating room

Class 10,000 operating room in Dalian (Chinese GB 50333-2013), combining field measurement with CFD across 45 orthogonal scenarios, Sobol sensitivity, and three optimisation routes (Bayesian, multi-objective, response surface) to calibrate a non-uniformity coefficient and fit ACH–occupancy–PM. Response surface gave the lowest prediction error (**RMSE 0.032, MAE 0.024**); multi-objective optimisation was the most stable across runs (C_v 0.005). The premise is sound — fixed as-built ACH standards ignore occupancy dynamics — but the model is calibrated on one room in one climate, and the PM metric is cleanroom particle count rather than a health-relevant mass fraction.

doi: 10.1155/ina/5838379

Zhou L et al. 2026 Atmos Chem Phys*Indoor burning of Arabian incense generates abundant ultrafine particles with strong oxidative potential*

Chamber characterisation of Bakhoor under charcoal-assisted heating representative of household practice, with acellular DTT activity and a macrophage intracellular oxidative-stress assay, benchmarked against protocol-matched sidestream cigarette smoke. Normalised per gram burned: **670–1690 $\mu\text{g min}^{-1} \text{g}^{-1}$ mass and $(6\text{--}7) \times 10^{11} \text{ particles min}^{-1} \text{g}^{-1}$ number, with ultrafines 70–75% of number and UFP emission rates above sidestream cigarette smoke.** Emission magnitude followed a nonlinear power law in the hexane-soluble fraction, identifying a compositional control. **Water-soluble OP_{DTT} was $\approx 32 \text{ pmol min}^{-1} \mu\text{g}^{-1}$ — modestly below cigarette particles — yet the cellular response was stronger,** and ozone ageing raised OP for both sources at ageing equivalent to days of indoor residence. The acellular/cellular divergence is the finding with teeth: DTT alone would have ranked this source as the safer of the two. Chamber conditions and a charcoal co-source mean the absolute rates are not a household exposure estimate.

doi: 10.5194/acp-26-10587-2026

van Rheeën et al. 2026 J Hosp Infect*Microbial contamination of air-conditioning and humidifying systems in hospitals and the built environment*

Mapping review of pathogen presence, biofilm formation and dissemination in the air and water compartments of air-handling systems analogous to adiabatic humidifiers, searched across three databases: **22 of 468 studies retained, covering 184 air-handling units, none of which used adiabatic humidification.** *Pseudomonas aeruginosa* and *Burkholderia cepacia* appeared occasionally in system air; *Mycobacterium* and *Acinetobacter* on cooler surfaces and in drain pans; *Mucor* and *Aspergillus fumigatus* in water and surrounding air, *Fusarium* more airborne. **No included study examined biofilm** — the proposed mechanism is therefore unevidenced within the review's own base. The conclusion is a statement about absence of evidence for the technology actually being adopted on energy-efficiency grounds, which is a legitimate thing to publish. Bioaerosol rather than mass concentration, so it sits at the edge of this watch's scope; included because humidification is a growing indoor-air intervention with no monitoring standard.

doi: 10.1016/j.jhin.2026.07.020

NOT CARRIED BY A DAILY ISSUE — EXTRACTIVE PRECIS OR METADATA ONLY, NOT AN ASSESSED SUMMARY (1 RECORDS)

Yu et al. 2026 Atmos Pollut Res*Indoor particulate matter and associated microplastic exposure in the educational environment of Shenzhen*

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1016/j.apr.2026.103162

Burden, policy & mitigation**60 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (59 OF 60 RECORDS IN THIS SUBTOPIC)

Acat et al. 2026 Atmosphere*National trends in ambient pollutants and cause-specific mortality, Türkiye 2018–2024*

Seven annual national observations for PM_{10} , SO_2 , CO, NO_2 and O_3 , each the unweighted mean of eligible station annual means from official bulletins, against crude national mortality rates, tested by OLS with Mann–Kendall, Sen's slope and a strict common-station sensitivity analysis. **PM_{10} fell from 50.4 to 41.0 $\mu\text{g}/\text{m}^3$ and SO_2 from 13.2 to 8.5, both significant.** The all-station NO_2 indicator rose from 29.2 to 34.4 but **the trend vanished under common-station restriction** (slope -0.021 , $p = 0.950$) — a clean demonstration that a national mean over a changing monitoring network measures the network as much as the air. **$\text{PM}_{2.5}$ was excluded outright** for insufficient reporting comparability, which is itself the story for a country whose modelled $\text{PM}_{2.5}$ burden the AirQ+ record above puts at 68,440 deaths. Only neoplasm mortality declined significantly. Crude national rates against unweighted station means; no causal reading is possible and none is claimed.

doi: 10.3390/atmos17090832

Alamsyah et al. 2026 Environ Sci Pollut Res Int*Industrial intensity and traffic congestion propagate air-pollution health vulnerability across district borders in Java*

A health vulnerability index (HVI) for Java built by PCA over satellite pollutant fields, official statistics and OpenStreetMap congestion data, then regressed on industrial activity intensity and congestion in a spatial Durbin model to separate direct from spillover effects. **HVI ranges 29.2–62.3 on a 0–100 scale, highest in West Java, Jakarta and Banten and lowest in Central Java, Yogyakarta and East Java;** the exposure dimension varies most (13.18) against adaptive capacity (9.40) and sensitivity (5.17), and both industrial intensity and congestion act through **direct and indirect (spillover) channels.** The spillover result is the useful one — it says district-level mitigation will leak, which is a real argument for regional rather than municipal air management. But the index is a construct: PCA weights are data-driven rather than health-anchored, no health outcome is used to validate the HVI against observed morbidity, and OpenStreetMap congestion is an availability-biased proxy that is denser precisely in the wealthier districts. Uncertainty and sensitivity analysis is reported, which is more than most index papers do.

doi: 10.1007/s11356-026-38148-2

Anastasopoulos et al. 2026 Atmos Environ*Mortality attributed to ambient air pollution by anthropogenic emission sector in Canada*

METADATA ONLY — a comparative health-burden analysis run at multiple geographic scales with sectoral attribution. Nine authors, Environment and Climate Change Canada style. No abstract in the Crossref deposit and nothing indexed elsewhere on 20 August; the sector-resolved burden numbers, which are the point of the paper, are therefore not reportable yet. Flagged for a later issue.

doi: 10.1016/j.atmosenv.2026.122310

Anon et al. 2026d Prev Med Rep*Lifetime risk of respiratory-system cancers across 185 countries*

GLOBOCAN 2022 combined with UN life tables under an adjusted-for-multiple-primaries framework. **Global lifetime risk of developing lung cancer 4.92 % and of dying from it 4.08 %, against 0.33 % and 0.20 % for laryngeal cancer,** with a steep gradient from **0.34 % in low-HDI to 5.45 % in very-high-HDI countries.** All four pollutants were positively associated with lifetime risk in univariable linear regression, $\text{PM}_{2.5}$ strongest. Two hard limits: the HDI gradient in lung cancer incidence is dominated by historical smoking prevalence and by diagnostic capacity, and country-level pollution from **OpenWeather** is a consumer weather API rather than a validated exposure product. Univariable ecological regression across 185 countries cannot separate PM from GDP, smoking or registry completeness. Tier B for the burden estimates; the pollution association should not be cited as evidence of effect.

doi: 10.1016/j.pmedr.2026.103602

Anon et al. 2026e Res Sq (preprint)

A Harmonization Index for industrial PM emission regulations

A Regulatory Coding Framework converting legally binding documents into standardized scores across **21 indicators in three dimensions** — regulatory stringency, institutional design and enforcement capacity — aggregated into a composite Harmonization Index, applied to the EU, US, China and Uzbekistan as exemplars of technology-based, source-category-based, hybrid and dispersion-based regulatory architectures. The problem is genuine: industrial PM limits are not comparable across jurisdictions because they are not the same kind of object. The method is documentary coding of what is written, which measures statutory text and not emissions; the enforcement-capacity dimension is the one that would need external validation and is the one hardest to score from documents. Unreviewed, four jurisdictions, tier C.

doi: 10.21203/rs.3.rs-10491363/v1

Bhatta et al. 2026 Sci Total Environ

Outdoor air purification technologies: efficiency, sustainability and health impact

PRISMA-ScR review, 266 records screened to 28 studies across five technology domains. Reported bests: smog towers at **97.8% PM_{2.5} and 98.4% PM₁₀ removal under optimal conditions**; solar-chimney hybrids within 2–4% of dedicated systems while generating electricity; TiO₂-modified cementitious composites potentially carbon-negative over their lifecycle; a roadside hedge cutting leaf-surface PM_{2.5} magnetic loading 82%. The health-economics figure — 93,200 deaths averted in China at a 35 µg/m³ indoor target, positive net monetary benefit in 93% of cities — is modelled from indoor air cleaners used as an analogue for outdoor systems, which the authors flag and which should be read as a bound on plausibility, not an estimate. “Optimal conditions” removal efficiencies are chamber or near-field numbers and say nothing about neighbourhood-scale concentration. The review’s own priority list — standardised performance metrics, long-term field validation — is the honest summary.

doi: 10.1016/j.scitotenv.2026.182205

Bosso et al. 2026 J Environ Manage

AIRTREE v2.0: predicting ecosystem services provided by Italian urban parks under climate change scenarios

Multi-layer canopy model over 43 national-reforestation-programme sites under RCP4.5 and RCP8.5 to 2054, calibrated against five ICOS flux sites (Kling-Gupta efficiency up to 0.90). Under RCP8.5 net primary productivity rises 9% and evapotranspiration 8%, O₃ and NO₂ removal fall, and **PM₁₀ removal increases slightly**. Species responses diverge sharply — *Pinus halepensis* leads on PM removal, *Sorbus torminalis* on O₃ uptake, *Quercus pubescens* on carbon. Open-access tool; the species-resolved PM output is the part worth reusing.

doi: 10.1016/j.jenvman.2026.130738

Bu & Sun 2026 J Air Waste Manag Assoc

PM emissions from China’s iron and steel industry: patterns, drivers, implications

LMDI decomposition over 99 Chinese sites, separating stack from fugitive emissions into five process factors: steel output, energy intensity, waste-gas generation coefficient, stack concentration and fugitive emission factor. **National PM emissions fell 23.6% from 2011 to 2016**. Stack concentration was the dominant inhibitor (to –31,558 t in 2015–16); the fugitive factor flipped from +5,037 t (2010–11) to –23,320 t; the waste-gas coefficient pushed emissions up throughout (+19,516 t in 2015–16); energy intensity contributed under 25%. Regionally the picture inverts — South and Northwest China show annual increases, with Northwest fugitive emissions reaching 232% of the annual change. Limitation: the series ends in 2016, predating the ultra-low-emission retrofit programme entirely. The stack-versus-fugitive split is what matters for siting monitors near industry: the two have completely different spatial signatures.

doi: 10.1080/10962247.2026.2698603

Caikai et al. 2026 J Infect Dev Ctries

Risk-factor-attributable burden of respiratory infection and tuberculosis, 1990–2021

GBD 2021 across 204 countries, decomposing deaths, DALYs and age-standardised rates for respiratory infection and TB across nine risk factors by age, sex and SDI, with APC and AAPC trend estimation. Age-standardised mortality attributable to the traditional risks fell steadily — **air pollution –3.18% annually**, malnutrition –4.46%, alcohol –3.35% — while metabolic-risk-attributable deaths rose in absolute terms despite falling rates. Tobacco-attributable mortality was **3.6-fold higher in men**, and children under 5 remain the group most exposed to environmental risks. Air pollution is a GBD attributable fraction here, not a measured exposure–response, so this is a trend context record: the attributable burden is falling on rate, which the accompanying absolute counts do not support as a success narrative.

doi: 10.3855/jidc.21667

Cliff et al. 2026 GeoHealth

How do climatic factors directly influence the incidence and risk of meningococcal meningitis across the African meningitis belt?

Narrative review over the 26-country African meningitis belt, where outbreaks cluster strongly in the dry season, examining dust, wind speed, temperature, rainfall and land cover as candidate drivers. **Atmospheric dust and wind speed carry the strongest statistical association and the greatest predictive probability** of the variables examined, with mucosal damage from inhaled mineral dust the standing mechanistic hypothesis. The review’s own central caveat is the right one: these climatic variables are severely multicollinear within the dry season, so attributing incidence to dust rather than to temperature or humidity is not identified by the observational record. Carried here because it is one of the few live health endpoints where the *coarse* mineral fraction, not PM_{2.5}, is the hypothesised agent — and where satellite dust products substitute for a ground network that does not exist.

doi: 10.1029/2025gh001749

Fares et al. 2026 Environ Pollut

Can urban green infrastructure influence the urban heat island effect and air quality nexus?

Review arguing that the thermal and air-quality benefits of green infrastructure are almost never assessed together, and synthesising the interaction: shading and evapotranspiration lower leaf surface temperature, which promotes thermophoretic deposition of particles, alongside foliar uptake of gaseous pollutants and inhibition of secondary aerosol formation. Well-maintained greenspace outperforms engineered installations, whose effects are more localised. The limitations section is the honest part — BVOC emission, small effect sizes, species and climate-zone dependence, and canopy-induced night-time warming.

doi: 10.1016/j.envpol.2026.128960

Galvin et al. 2026 J Air Waste Manag Assoc

Cyclic-operation emission factors for biodiesel, renewable diesel and hydrogen-blend boilers

Emission factors developed under *representative cyclic operation* for three appliances — a residential retention-head liquid-fuel boiler, a residential modulating-condensing gas boiler and a two-stage commercial boiler — across B20, B100, renewable diesel, natural gas and a 20% hydrogen blend. PM measured by EPA Methods 5 and 202 to capture filterable and condensable fractions, supplemented with real-time instrumentation. Two results matter: **burner on–off cycling drives incomplete-combustion pollutants and is not represented in current inventory factors**, which derive from steady-state operation of larger plant; and **condensable PM can dominate total PM**, especially in advanced gas systems. An optical sensor sited near residential combustion is blind to the condensable fraction at the stack and sees it only once it forms — a real limitation for source attribution near buildings.

doi: 10.1080/10962247.2026.2657442

Gardner-Frolick et al. 2026 iScience*Double disproportionality: few emitters, and they sit next to marginalised people*

National analysis asking two questions jointly — which industrial subsectors have emissions concentrated in a few facilities, and whether those facilities are disproportionately sited near structurally marginalised groups. Emission disproportionality holds across pollutants and subsectors, and siting patterns diverge by settlement type: **oil and gas extraction disproportionately affects marginalised urban populations, mining and quarrying rural ones**. Urban Indigenous populations are strongly affected by both all-emitter and high-emitter siting, and low-income and Black populations are more likely to live near *clusters* of emitting facilities — a cumulative-hazard framing that single-facility analyses miss. **Limitations: proximity is not exposure** (no dispersion modelling, no monitoring), self-reported inventory emissions, and areal-unit demographics vulnerable to the modifiable areal unit problem. The cluster result is the one that should drive monitor siting.

doi: 10.1016/j.isci.2026.116827

Goyal et al. 2026 Nat Commun*Quantifying environmental co-benefits of nitrogen-based crop restructuring and its implications on India's interstate trade network*

Integrated nutrient, water, greenhouse-gas, economic and interstate-trade accounting for Indian agriculture, comparing nitrogen-focused against water-focused crop-restructuring pathways. The nitrogen-focused pathway cuts **nitrogen surplus by 13.4%, water use by 18.6% and greenhouse gases by 8.7%** while holding baseline calorie production, and delivers about **4.6-fold larger reciprocal co-benefits** than the water-focused alternative, worth roughly **USD 1.19 bn** in avoided nitrogen-pollution socio-environmental cost. Limitation: PM is entirely implicit — reactive nitrogen is a dominant precursor of secondary inorganic aerosol over the Indo-Gangetic Plain, but no ambient PM_{2.5} response is modelled and no health endpoint is estimated, so the PM-relevant co-benefit is asserted through the nitrogen accounting rather than computed. Carried because agricultural NH₃ is the least-constrained term in South Asian PM source apportionment.

doi: 10.1038/s41467-026-75905-w

Gu et al. 2026 J Hazard Mater*Electro-ultrasonic coupled agglomeration for removal of fine particles from biomass combustion*

Segmented continuous-flow bench platform combining bipolar charging (± 16 kV), a 300 V/400 Hz AC field and 16 kHz/142 dB ultrasound, with particle-trajectory simulation and SEM. The coupled field reached **86.4 % light transmittance**, against single-field results 33.1 and 67.2 percentage points lower, with the extinction coefficient down to 0.9 m^{-1} ; simulated cumulative particle displacement rose $>445\%$ as linear trajectories became three-dimensional helical paths, and SEM confirmed 15–20 μm agglomerates. The performance metric is the weakness: light transmittance and extinction coefficient are *optical* proxies, not gravimetric or number-resolved removal efficiency, so the headline “removal efficiency” is not directly comparable with any filtration literature that reports mass penetration. The submicron fraction claimed as the target is precisely the fraction an extinction measurement is least sensitive to.

doi: 10.1016/j.jhazmat.2026.143130

Han et al. 2026 AJPM Focus*Air quality index product use versus clinician discussion among US children with asthma*

Questionnaire administered inside the Transcript/Omics for Precision Medicine in Asthma study, youth aged 6–21 recruited 2023–2025, with bivariate testing. **75% of caregivers were aware of an AQI or alert, 64% checked it before going outdoors and 73% reported a behaviour change** — but the delivery channel is consumer, not clinical or public: a phone weather app in 75% of cases, broadcast or social alerts 44%, websites 24%, county services only 14%. Only 41% had ever discussed air pollution with a clinician and 35% received poor-air-quality recommendations. Self-report with no objective behaviour measure and a single research cohort limit generalisability. The finding that matters for network design: **the operative last mile is a commercial weather app, and public monitoring agencies reach one family in seven**.

doi: 10.1016/j.j.focus.2026.100504

He et al. 2026 Neuroepidemiology*Premature ischaemic stroke burden in adults aged 50–69 in China, India and the United States*

GBD 2023 DALYs, YLLs, YLDs and population attributable fractions for the 50–69 stratum in three countries at different stages of epidemiological transition, with ARIMA projection to 2053. Trajectories diverge: China declining, India flat at a high level, the US on a high plateau; the US has shifted to a YLD-dominant (disability) composition while India remains YLL-dominant (acute mortality); China shows a near 2:1 male:female ratio driven by male smoking. **India is characterised as facing dual pressure from persistent PM_{2.5} and rising metabolic risk**. The authors are explicit that widening intervals make the 2053 horizon scenario-based rather than forecast; a 34-year ARIMA extrapolation from modelled burden estimates is presentation, not evidence. Carried for the PAF decomposition only.

doi: 10.1159/ned/adwag005

Hou et al. 2026 J Hazard Mater*Bimodal polysulfonamide fibrous aerogel for high-temperature PM_{0.3} filtration*

Co-electrospun aerogel combining thermally stable PSA/PAN/SiO₂ coarse fibres with pure PSA fine fibres in a 3D curly network at **>98% porosity**, formed via moisture-induced phase separation and jet whipping from the mechanical mismatch between components. **PM_{0.3} filtration efficiency 98.5% at ~84 Pa pressure drop**, retained at **92.31% and 73 Pa after 12 h at 250 °C**, with acid and alkali resistance and useful thermal insulation. The gap in the report is loading behaviour: no long-duration dust-holding capacity or efficiency-versus-loading curve is given, and that is what determines service life in a real industrial stack. Relevant to source-side mitigation rather than monitoring, and to the ultrafine fraction that mass-based networks under-weight.

doi: 10.1016/j.jhazmat.2026.143218

Kim et al. 2026 J Environ Manage*Full diesel-to-BEV replacement: where the benefit lands is not where the people are*

CMAQ counterfactual in which all diesel vehicles are replaced by battery electric vehicles, evaluated with population-weighted concentration across urban core, urban periphery and rural strata — an explicit attempt to move policy appraisal off regional-average concentration change. Despite large on-road NO_x declines in the urban core and periphery, **annual daily-maximum 1-h ozone rose there through weakened NO titration** while falling in rural areas, so ozone PWC increases were concentrated where people live. **PM_{2.5} moved the opposite way**: the largest concentration reductions occurred rurally, where low population density limited the exposure benefit. The result is a genuine spatial-equity tension inside a policy usually scored as unambiguously good. **Limitations: an idealised full-replacement scenario, one domain, one meteorological year**, no fleet-turnover or grid-emissions accounting, and no health-endpoint monetisation.

doi: 10.1016/j.jenvman.2026.130597

Kokol et al. 2026 Res Sq (preprint)

Artificial intelligence in the management of climate-change-related health issues in Africa

Scoping review over Scopus to October 2025, VOSviewer clustering a global against an African literature corpus. The divergence is the finding: global work clusters on environmental modelling, air-pollution mortality tracking and exposome integration, African work on supervised classification for infectious-disease triage and food security, with a “compute deficit”, data silos and power instability named as barriers. This is bibliometrics, not evidence — a VOSviewer map measures what Scopus indexed, and Scopus under-indexes African journals, so the gap is partly an artefact of the instrument. Preprint, not peer reviewed.

doi: 10.21203/rs.3.rs-9910414/v1

Lee et al. 2026 J Korean Med Sci

Toward a unified framework for climate and health research: gaps in exposure metrics and health outcomes

Mapping review of **611** original human studies (PubMed, Cochrane, Embase; January 2000 – June 2024) on climate-related exposures and health. The distribution is sharply skewed: **526 studies on temperature and temperature-related hazards, 88 on precipitation, and only 39 on air pollution**, with extreme weather barely represented. Outcomes concentrate in all-cause mortality (249), cardiovascular disease (159) and communicable disease (148), while neurodegenerative disease (8) and endocrine disorders (4) are almost absent. Exposure-metric definitions are described as substantially heterogeneous. *Limitations*: counts are of studies, not of evidence quality, and English-only inclusion biases the geography. Relevance: the two endpoints this issue adds — thyroid function and fronto-limbic white matter — sit in the two thinnest outcome categories the review identifies, and the metric heterogeneity it documents is the same problem that makes cross-study pooling of PM exposure estimates unreliable.

doi: 10.3346/jkms.2026.41.e294

Atmos Environ 2026d METADATA ONLY Atmos Environ

Drivers of air pollutant and greenhouse gas emissions since 2010 in the Greater Bay Area, China

METADATA ONLY — no abstract retrievable at build time. Carried for the register because decomposition of co-emitted air pollutants and greenhouse gases over a 15-year window in a single megacity cluster is the analysis that determines whether observed PM improvement came from control technology, fuel switching, activity shift or economic structure — and those four have different implications for what happens next. Method, pollutant coverage and decomposition scheme are all unstated. Lowest-weight record in the issue.

doi: 10.1016/j.atmosenv.2026.122301

Liao et al. 2026 Environ Res

Beyond mortality: multi-outcome assessment of health economic benefits from air pollution control in China

Daily mortality, hospital admissions and outpatient visits across 18 Guangdong cities, 2015–2022, with generalized additive concentration–response models and valuation by value of statistical life plus direct medical cost. PM_{2.5} and NO₂ control delivered **10,305.35 and 1,017.00 million CNY** respectively; O₃ control returned *negative* net benefits on every outcome. The headline methodological number is that multi-outcome benefits of PM_{2.5} control **exceed mortality-only estimates by 31.79%** — i.e. the standard mortality-only accounting understates the case by roughly a third. Over 85% of mortality-related benefit accrues to the elderly.

doi: 10.1016/j.envres.2026.125501

Lim et al. 2026 Nat Commun

Public knowledge, attitudes and practices on air quality in the ten most populous countries

Quota sample of **10,618 respondents across 1,372 localities** in Bangladesh, Brazil, China, Indonesia, India, Mexico, Nigeria, Pakistan, Russia and the United States, linked to modelled PM_{2.5}. Exposure is *inversely* associated with knowledge about air quality but *positively* associated with protective practices, attitudes and concern — people in dirtier air act more and know less. Interactions with age, sex, education, residence, political ideology, marital, employment and smoking status are all significant. Quota sampling is not probability sampling, so national point estimates should not be read as prevalence; the cross-country contrasts are the durable part. A useful anchor for anyone arguing that a dense sensor network has a communication function as well as a measurement one.

doi: 10.1038/s41467-026-75908-7

Liu D. et al. 2026 Medicine (Baltimore)

COPD burden 1990–2050 with risk-factor attribution across age and SDI

GBD 2021 across 204 countries, analysed by joinpoint regression, age–period–cohort models and decomposition into population growth, ageing and epidemiological change, with Bayesian APC projection to 2050. Age-standardised rates fell (AAPC –0.05% incidence, –1.49% mortality, –1.48% DALYs) while absolute burden rose to **213.4 million cases and 1.69 million deaths in 2021. Ambient PM_{2.5} carries 15–25% of DALYs and household air pollution 0–55%**, against smoking at 38–55%, with an inverted-V SDI relationship peaking at SDI 0.4–0.6. The household-air-pollution range spanning zero to fifty-five percent is the whole story and is presented as a single interval; it is a statement about between-region heterogeneity, not uncertainty, and the paper does not separate the two. GBD risk-factor PAFs inherit their exposure–response functions from the integrated exposure–response curves, so this is a re-analysis of GBD outputs rather than independent evidence. Tier C accordingly.

doi: 10.1097/MD.00000000000050162

Liu M. et al. 2026 Environ Monit Assess

Divergent near-surface ozone and NO₂ trends and ozone-attributable burden, Huaihe River Basin

Twelve years (2013–2024) of high-resolution satellite-derived surface MDA8 O₃ and NO₂ across the Huaihe Basin, with pixel-wise OLS trends, the Hurst exponent as a persistence diagnostic, and BenMAP-CE for attributable premature mortality and economic loss. **O₃ rose at 2.792 μg m⁻³ yr⁻¹ (mean 104.15 μg m⁻³) while NO₂ fell at –1.385 μg m⁻³ yr⁻¹ (mean 29.86)** — the textbook VOC-limited response to NO_x-only control, here quantified at basin scale with an outward expansion of O₃-affected area and contraction of NO₂ hotspots. Cardiovascular mortality was the largest cause-specific component, concentrated in Jiangsu and Shandong. **No PM metric is estimated**, which for a basin whose PM_{2.5} and O₃ share precursors is a real gap; and the Hurst exponent applied to a 12-point annual series is a weak persistence estimator being asked to carry a forward-looking claim.

doi: 10.1007/s10661-026-15802-3

Liu et al. 2026 Environ Sci Technol

Disproportionate effects of direct air capture on the regional health cobenefits of net-zero

Grid-level (36 km²) air-pollution mortality assessment along China’s net-zero CO₂ pathway 2020–2060, contrasting scenarios with and without large-scale DACCS deployment. DACCS substitutes for fossil-fuel phase-out and therefore forgoes the co-pollutant reduction that would otherwise occur: **1.75 million additional cumulative premature deaths**, with the losses concentrated in developing provinces (**Henan –0.13 million and Hebei –0.12 million avoided deaths**) via more active emissions trading. The paradox the authors report honestly is that inter-regional *mortality-rate* inequality narrows while the distribution of forgone benefit widens. Scenario-model output over a 40-year horizon, so the intervals are structural-uncertainty bounded rather than statistical, and the result is a property of the assumed trading architecture as much as of the technology.

doi: 10.1021/acs.est.6c04728

Liu et al. 2026b Front Oncol*Lung cancer incidence and mortality trends with age-period-cohort analysis, Zibo 2014–2024*

Registry data for a heavily industrial Shandong city. Age-standardised incidence rose (AAPC +4.34%, 95% CI 2.14–6.59) while mortality fell (AAPC –2.13%, –3.53 to –0.72); male rates exceeded female throughout. Period effect increasing, cohort effect decreasing across successive birth cohorts. Rising incidence with falling mortality in a screening-active decade is the classic overdiagnosis signature and the paper does not test it; air pollution appears as narrative context with no exposure measurement at all.

doi: 10.3389/fonc.2026.1873592

Moghaddam et al. 2026 (preprint) Research Square*Reducing particulate matter emissions in landfills using novel dust suppressants*

Wind-tunnel factorial on soil from the Meyami landfill in Mashhad: three mesh sizes, four suppressants (compost, vermicompost, bentonite, nano-clay bentonite), three doses, two exposure times, two wind speeds. Mesh 20 was optimal across all suppressants (>85% reduction in wind-erosion rate); **compost at 10% dose reached 86.52%** and bentonite at 20% reached 80%, both at 4 m/s. A bench result on a single soil — no field validation and no airborne concentration measured — but the compost finding is operationally cheap enough to be worth a field trial.

doi: 10.21203/rs.3.rs-10431018/v1

Mou et al. 2026 Environ Pollut*Ship emissions and regional air quality in the post-DECA era, Yangtze River Basin*

A bottom-up 2023 emission inventory for **18,478 vessels** built from **≈230 million AIS trajectories** on the world's busiest inland shipping corridor, coupled to WRF-Chem for concentration attribution. SO₂ and PM_{2.5} ship emissions are substantially controlled post-DECA, but **NO_x reached 156.41 kt yr⁻¹, 82.0% of major-pollutant mass**, with hotspots at Yichang–Jingzhou, Wuhan and Nanjing–Shanghai. Ship contribution to NO₂ was **0.16–3.41 μg m⁻³** annual mean across 11 mainstem cities; PM_{2.5} increments were small but nitrate-dominated; O₃ fell in January and October under near-source NO titration and rose in July. The policy read is clean — fuel-sulfur control has done its work and the residual problem is NO_x — but the inventory rests on activity-based emission factors with no plume measurement to validate them, and the PM_{2.5} increment excludes secondary nitrate formed downwind of the model domain.

doi: 10.1016/j.envpol.2026.128946

Nilsson et al. 2026 Environ Epidemiol*Urban environment, health and equity: symptoms and annoyances related to air pollution, traffic noise and greenness*

46,282 adults aged 18–84 in Stockholm County responding to the 2023 Swedish National Environmental Health Survey, with geocoded residential PM_{2.5}, PM₁₀, NO₂, road/rail/aircraft noise and 300 m NDVI, dichotomised and stratified by socioeconomic status and building type. Apartment residents are more exposed to pollution and noise and have less greenness; SES-stratified patterns are exposure-specific with no consistent overall gradient. The strong associations are with *perception* — NO₂ with poor perceived outdoor air quality (**OR 3.22; 95% CI 2.75–3.79**), road noise with high annoyance (OR 4.59; 4.02–5.26). Both exposure and outcome are cross-sectional and the outcome is self-reported, so these are perception–exposure concordance estimates, not health effects, and the odds ratios should not be read as such.

doi: 10.1097/EE9.0000000000000516

Okeke et al. 2026 Environ Monit Assess*Agricultural methane emissions and non-communicable disease mortality in West Africa*

Ecological panel of 11 West African countries, 2015–2020, relating agricultural methane (Mt CO₂e, World Bank WDI) to the probability of dying from NCDs between ages 30 and 70, using pooled OLS, country fixed effects and mixed models, with The Gambia (–28.3% emissions) as a descriptive contrast. *This is a weak design and should be read as such:* 11 countries × 6 years is 66 country-years, the exposure is a national accounting aggregate with no measured concentration and no plausible direct mechanism to NCD mortality, and the outcome is a modelled WHO indicator rather than observed deaths. Ecological fallacy and confounding by development trajectory are unaddressed. Tier C, carried only because it is one of the few Sub-Saharan records in this watch and marks where the climate–health literature is expanding without measurement infrastructure — which is itself the finding.

doi: 10.1007/s10661-026-15800-5

Özmen et al. 2026 Toxics*Long-term PM_{2.5} and premature mortality across Türkiye*

Province-level cross-section of **all 81 Turkish provinces** for 2022, combining annual mean PM_{2.5} with WHO AirQ+ attributable-death estimates for adults 30+ at a 5 μg/m³ counterfactual. **Population-weighted mean PM_{2.5} was 27.0 μg/m³ — 5.4× the WHO guideline — and all 81 provinces exceeded it.** Bivariate OLS explained 41% of between-province variation in attributable mortality (**slope 3.23 additional deaths per 100,000 per 1 μg/m³, 95% CI 2.37–4.10**); a log-log model gave an elasticity of **0.80 (0.65–0.95)**. The attributable fraction of natural-cause mortality rose from **8.8% in the lowest exposure quartile to 24.6% in the highest**, totalling **68,440 premature deaths, 14.2% of natural-cause mortality**. The authors are explicit about the circularity and it is worth repeating: attributable deaths are *generated* by AirQ+ from a fixed concentration–response function, so regressing them on concentration recovers the assumed function, not independent evidence. The burden magnitude stands; the gradient is arithmetic.

doi: 10.3390/toxics14080728

Sathasivam & Veluchamy 2026 Environ Sci Pollut Res*Visual characteristics of PM from Halymenia dilatata microalgae biodiesel compared to low-sulfur diesel*

Single-cylinder Kirloskar CI engine, four microalgae biodiesel blends (B10–B25) against S10 low-sulfur diesel, PM collected on a textile element in a deceleration chamber, quantified gravimetrically and characterised by SEM for morphology, deposition and agglomeration. **PM mass reduction 10.44–69.79%** across blends. The result does not vary monotonically with blend ratio — SEM showed heavier deposition for S10, B15 and B25 but lighter for B10 and B20 — which the paper reports but does not explain, and a non-monotonic dose–response across four points with no stated replication is more likely measurement noise than combustion chemistry. Three further limits matter: a textile collection element in a deceleration chamber is not a dilution tunnel and has no characterised collection efficiency as a function of size, so the gravimetric numbers are not comparable with any regulated emissions measurement; no size distribution or number concentration was measured, so a mass reduction is fully compatible with an increase in ultrafine number; and no engine load or speed conditions are given in the abstract. Included as a tier-C source-characterisation record.

doi: 10.1007/s11356-026-38134-8

Sharma et al. 2026b Health Promot J Austr

Geospatial linkage of environmental exposures, socioeconomic profile and health outcomes among women of reproductive age, Victoria

Postcode-level linkage of 2021 Australian Census sociodemographic and health variables to PM_{2.5}, NO₂, CO, SO₂, formaldehyde, ozone, land surface temperature and greenness, with Moran's I, Wagstaff concentration indices and geospatial regression. Non-communicable disease burden in women of reproductive age is strongly socioeconomically patterned (323.0 vs. 227.3 per 1000 in most- versus least-disadvantaged postcodes; steepest for diabetes, Wagstaff -0.16), **but environmental exposures show only weak associations with outcomes** despite clear spatial clustering. Inner Melbourne has lower NCD burden despite elevated NO₂ and PM_{2.5}. This is ecological, cross-sectional and confounded by the same SES gradient it measures; the null environmental finding is uninformative rather than reassuring.

doi: 10.1002/hpja.70241

Shin et al. 2026 J Occup Environ Med

Occupational disparities in PM_{2.5}-related mortality under distributed lag models

Five years of daily all-cause mortality and PM_{2.5} for Seoul, analysed with generalized additive plus distributed lag models and stratified by age, sex and occupation. A significant positive PM_{2.5}-mortality association overall, with **blue-collar and elementary workers carrying substantially higher relative risks**. The policy argument — that concentration targets alone are insufficient and high-vulnerability strata need tailored management — is well posed, but the indexed abstract reports *no point estimates or intervals*, so the magnitude cannot be assessed and the record contributes no effect estimate. Occupation comes from death certificates, which misclassifies the late-career and retired.

doi: 10.1097/JOM.0000000000003831

Sidharaddi et al. 2026 Research Square (preprint)

Electrodeposited copper nodule films as electrostatic particulate collectors

Copper nodules electrodeposited onto steel; XRD confirms polycrystalline FCC, SEM a nodular microstructure, AFM a non-monotonic dependence of grain size (86.5→224.6 nm) and roughness (23.4→76.2 nm) on bath concentration before both fall again. The 0.075 M film (721 nm thick) gave **88.78% removal of dhoop smoke and an air-purifying factor of 54.12 over 405 s**, exceeding a stainless-steel discharge rod. Chamber test with a single combustion source, no reference-grade particle instrument, no size resolution and no ozone measurement — which for a corona-type collector is the omission that matters. Included as an engineering-control data point, tier C.

doi: 10.21203/rs.3.rs-10126181/v1

Simion et al. 2026 Lancet Reg Health West Pac

Climate-related health attribution in Southeast Asia: what is and is not being studied

Scoping review, **5,294 records screened, 77 included**. Air pollution is the subject of **70% of studies**, linked to up to 31% of all-cause mortality and an estimated **296,000 annual deaths from wildfire smoke during El Niño events**; heat is second, associated with up to 8.61% of mortality in Bangkok. Floods, droughts, extreme rainfall, storms and sea-level rise are near-absent. The paper's own framing — that the research portfolio is misaligned with the regional hazard profile — is the transferable finding; the attributable fractions themselves are heterogeneously derived across 77 studies with no standardised attribution framework, which the authors name as the central methodological gap. For a monitoring programme the operative point is that the region's dominant PM source is episodic biomass burning, which mass-only fixed networks under-sample in both space and time.

doi: 10.1016/j.lanwpc.2026.101952

Sirikul et al. 2026 J Clim Chang Health

Knowledge, attitudes and practices of caregivers regarding PM_{2.5} exposure among children in northern Thailand

Cross-sectional KAP survey of caregivers recruited at Chiang Mai Child Development Centres. Knowledge is moderate (10.44±3.80 of 22), attitudes positive (30.74±3.57 of 40), practices inconsistent (17.33±5.22 of 32). The specific gaps are the useful part: **97.0% recognise the health risk to children, but only 41.8% know surgical masks are inadequate for PM_{2.5} and only 9.7% can correctly interpret an AQI value**; air purifier ownership is 15.3% and monitoring-app use 28.2%, both with a clear education and income gradient. The abstract does not report the sample size, response rate or instrument validation, and centre-based recruitment excludes non-attending households, so absolute prevalences are not generalisable. The AQI-literacy figure is the one that should worry anyone designing a public-facing sensor dashboard.

doi: 10.1016/j.joclim.2026.100734

Slavik et al. 2026 Environ Challenges

Objective versus subjective wildfire-smoke exposure and parental protective behaviour

2,086 parents in wildfire-prone western US and Canada, surveyed and linked to 3.5 years of smoke-monitoring data (2020–2023). Perceived exposure (sight or smell) tracks measured exposure but predicts behaviour far better: $b = 1.11$ (**0.92–1.30**) **additional protective actions for perceived exposure versus $b = 0.23$ (0.06–0.40) per 10 µg/m³ of measured PM_{2.5}**. The measured-exposure relationship is non-monotonic — rise, plateau, rise. Self-reported behaviour is the obvious weakness and recall is correlated with perception by construction, which inflates the contrast. Even discounted, the direction is a design constraint on any public-facing sensor network.

doi: 10.1016/j.envc.2025.101362

Sommar et al. 2026 Int J Environ Res Public Health

Health economics of PM_{2.5} from residential wood combustion, northern Sweden

Health impact assessment for **280,742 residents of Västerbotten County** using high-resolution 2023 dispersion-modelled PM_{2.5} from small-scale residential heating, population-weighted to home-address coordinates, with the VALESOR framework and a near-source relative risk of **1.26 per 10 µg/m³** for natural-cause mortality. **Mean RWC-attributable exposure was only 0.16 µg/m³** (IQR 0.10–0.21), yielding **30 years of life lost annually** plus small morbidity counts (0.3 lung cancers, 4 childhood asthma, 3 COPD, 3 IHD, 1 stroke, 2 dementia) and a total annual cost of **~EUR 7.0 million (EUR 4.6–9.4 million)**. Two assumptions carry the result: the near-source RR of 1.26 is far above the standard ~1.08 per 10 µg/m³ and is a deliberate wood-smoke-toxicity premium that the evidence base supports only weakly, and the population-mean of 0.16 µg/m³ conceals a highly skewed near-source distribution. Fractional case counts at this precision are not meaningful.

doi: 10.3390/ijerph23081053

Song et al. 2026 Environ Sci Technol

Shifting PM_{2.5} control from mass to toxicity for health equity worldwide

Perspective synthesising toxicological, epidemiological and modelling evidence against the uniform-toxicity assumption underlying mass-based standards. The empirical hook is that age-standardised attributable death rates diverged worldwide over the past decade despite falling mass. Incomplete-combustion sources — residential solid fuel and wildfire above all — carry the highest toxicity per unit mass from primary emission through secondary ageing. Toxicity-adjusted exposure is substantially elevated in South Asia, Southeast Asia, sub-Saharan Africa, Andean Latin America and China, and the **8.2% of the world that is low-income bears 24.0% of the cumulative toxicity-adjusted exposure burden**. It is a perspective, not a quantitative assessment: the toxicity weights are inherited from a heterogeneous literature and the equity arithmetic is only as good as they are.

doi: 10.1021/acs.est.6c05282

Suryanarayanareddy et al. 2026 Environ Sci Pollut Res Int*Low-cost bio-based particle-capturing device for vehicular emissions: pilot study*

Cylindrical coconut-husk and sugarcane-pulp enclosure tapering to a conical outlet and a monitoring chamber, tested indoors under simulated junction conditions at three turbulence intensities. Single-pass removal 15–20% (PM_{2.5}) and 15–45% (PM₁₀) for halted vehicles; 20–65% and 15–65% for moving vehicles. Efficiency ordering flips with turbulence regime, which the paper reports honestly. Indoor simulation of an outdoor near-road environment, no replication count given, and no material-degradation testing on a biodegradable enclosure meant to sit at a roadside.

doi: 10.1007/s11356-026-38118-8

Tian et al. 2026 Nanomaterials (Basel)*Bioinspired honeycomb-structured nanofibrous membranes for high-efficiency PM capture*

Template-assisted electrospinning onto a mesh receiver establishes a directional electric field, producing concentrated ordered stacking plus sparse random orientation of fine-diameter fibres. **>98.51% filtration efficiency at 34 Pa pressure drop**, 85% transparency, 130.8 mm/s air permeability. Efficiency is quoted for a single challenge condition; no loading curve, no electrostatic-decay ageing, no humidity sensitivity — the three things that determine whether an electret-assisted filter holds its number in service. Bench result, not a service life.

doi: 10.3390/nano16150941

Urso et al. 2026 Lancet Reg Health Eur*Burden of dementia in Europe, 1990–2023: a systematic analysis from the Global Burden of Disease Study 2023*

GBD 2023 secondary analysis for the EU-27 and the WHO European Region, with non-fatal outcomes from DisMod-MR 2.1 and dementia-attributable mortality from an excess-mortality framework. **7.76 M (95% UI 6.69–8.72)** prevalent cases in the EU-27 and **12.28 M (10.44–13.90)** across the WHO European Region, both roughly **+90% since 1990** despite modest declines in age-standardised prevalence; dementia moved from the eighth to the **third** leading cause of death in the EU-27. **41% (24.4–57.4)** of EU-27 dementia DALYs are attributed to six modifiable risk factors, with **ambient particulate matter pollution named first**, ahead of high fasting plasma glucose and high BMI, and concentrated in socioeconomically disadvantaged populations. The limitation is the one every GBD risk-factor attribution carries and this paper does not resolve: the PM_{2.5}–dementia relative risk is imported from a comparative-risk-assessment function fitted to a thin and heterogeneous cohort literature, and the **24.4–57.4** interval is a modelling uncertainty, not a sampling one — it does not propagate uncertainty in the exposure surface, which for a 1990–2023 European back-cast is substantial. Treated here as a burden estimate, not as neurological evidence. The interval is deliberately absent from the forest plot: an attributable fraction has no position on a ratio axis.

doi: 10.1016/j.lanepe.2026.101796

Vartanyan et al. 2026 Cureus*Structural inequities and chronic disease in Fresno County: a narrative review and multi-sectoral framework*

Comparative descriptive analysis of public-health data for **2018–2021** showing Fresno County, California, exceeding state prevalence and mortality for obesity, hypertension, diabetes, asthma, coronary heart disease and stroke, attributed to poverty, low educational attainment, provider shortages, food insecurity and severe air and pesticide pollution, with disproportionate impact on racial and ethnic minority populations. Tier C, and the reason is structural rather than a criticism of execution: this is an ecological comparison of county aggregates in which air pollution is one of several correlated exposures, none measured at individual level, so no exposure-specific inference is available and none is claimed. Carried because the San Joaquin Valley is among the most PM_{2.5}-burdened airsheds in the United States and because the paper's underlying argument — that monitoring density and exposure burden are inversely related in marginalised communities — is the same one Siu et al. quantified for eastern Canada in the 5 August issue, here asserted rather than measured.

doi: 10.7759/cureus.112418

Wan 2026 Econ Hum Biol*Place-based ecological restoration and self-rated health: quasi-experimental evidence*

Six waves of the China Family Panel Studies against the staggered prefecture-level rollout of China's national ecological restoration pilot, two-way fixed-effects DiD. **+0.1697 points of self-rated health on a five-point scale**, no differential pre-trends in event study, robust to placebo tests, Goodman–Bacon decomposition, propensity-score matching, double machine learning, IV, and adjustment for overlapping environmental policies. Mediation runs through lower PM_{2.5}, greater greenness and better visibility alongside economic and psychosocial channels. The unusually thorough robustness battery is the reason to take it seriously; the mediation analysis is correlational and the PM_{2.5} share is not separately identified.

doi: 10.1016/j.ehb.2026.101636

Wang et al. 2026 Front Glob Womens Health*Trends, inequalities, and 2030 projections of women's tracheal, bronchial, and lung cancer attributable to air pollution, radon, and tobacco exposure*

GBD 2023, female tracheal/bronchial/lung cancer 1990–2023, five inhalation-related risk factors, with SDI-gradient smoothing, concentration indices for inequality, decomposition by demographic vs epidemiological change, and projection to 2030. Smoking remains the leading contributor, **ambient particulate matter second, secondhand smoke third**. Over 1990–2023 **household solid-fuel-attributable mortality fell most sharply while particulate-matter-attributable mortality rose slightly**, with China the standout for the PM-attributable share. Along the SDI gradient, PM shows a **rising burden with diminishing inequality** — the opposite pattern to radon and tobacco — and decomposition attributes the PM increase to population growth and aging rather than to rising risk. Projected to keep rising to 2030. Standard GBD caveats bind hard here: the exposure is a modelled global surface, the attributable fraction rests on assumed counterfactual and RR functions, and the 2030 projection is an extrapolation of fitted trend, not a scenario.

doi: 10.3389/fgwh.2026.1662451

Wojcik et al. 2026 ATS Sch*An air pollution curriculum for pulmonary and critical care fellows*

Kern six-step curriculum development across five domains — composition, health effects, *monitoring*, risk mitigation and environmental justice — with a 20-item knowledge scale built by modified Delphi and delivered as pre/post assessment over one academic year with spaced repetition. Knowledge rose in all five domains ($p < 0.001$). Single programme, no control arm, self-selected respondents and a knowledge rather than behaviour endpoint, so the effect size is uninterpretable; it is included because the explicit inclusion of *monitoring* as a clinical competency domain is a small but real indicator of where sensor-derived exposure data is expected to land clinically.

doi: 10.1093/atsscholar/aapag039

Wu & Cui 2026 J Glob Health

Aircraft CO, NO₂ and SO₂ emissions and public health at Chinese airports

Landing-and-take-off cycle data from the Civil Aviation Administration of China, concentrations at each airport from a Gaussian diffusion model, premature deaths from the Global Exposure Mortality Model applied to populations near airports. Attributable mortality was higher in men than women and concentrated in older age groups; CO accounted for the most respiratory deaths; the peak burden year was 2019. **No particulate metric is estimated at all**, which for an aviation source — where ultrafine particle number is the exposure of concern and is not proxied by CO — is a substantive omission rather than a scope choice. A Gaussian plume model at airport scale with no measurement validation is the second limitation. Carried at tier C for the burden framing and as a marker of where aviation-adjacent monitoring is not being done.

doi: 10.7189/jogh.16.04210

Xia et al. 2026 (preprint) arXiv

Surface PM_{2.5} mass and composition under stratospheric sulfate geoengineering

Uses the GLENS ensemble (CESM1, stratospheric SO₂ injection balancing RCP8.5 forcing) to ask what solar geoengineering does to surface PM_{2.5}. Global mean PM_{2.5} falls relative to RCP8.5, but the reduction is carried entirely by **less dust** (wetter soils, higher leaf-area index over deserts) and **less sea salt** (slower winds). Excluding those two, PM_{2.5} rises, driven by aerosol-phase secondary organic aerosol partitioning in a cooler atmosphere. Notably, the injected SO₂ does **not** raise surface sulfate. Single model, single ensemble, an RCP8.5 baseline now far from plausible, and SOA gas precursors are prescribed — so the SOA increase is partly an artefact of that setup, as the authors acknowledge. Preprint; not peer reviewed.

doi: 10.48550/arXiv.2608.14928

Xie et al. 2026 J Thorac Dis

Thirty-year change and projection of Asian tracheal, bronchus and lung cancer burden

GBD 2021 across Asian subregions with Joinpoint, age-period-cohort, decomposition, frontier, inequality and BAPC prediction. Incidence roughly doubled and deaths rose ~2.7-fold 1990–2021; ASIR +22%, ASPR +49%, ASMR +8%. Population growth was the largest contributor, then ageing. **Particulate matter pollution ranks second behind smoking among risk factors at ~10% and rising**. Six analytic frameworks on one modelled dataset multiply presentation, not information.

doi: 10.21037/jtd-2026-0740

Yoon et al. 2026 Med J Aust

Heat versus bushfire-smoke PM_{2.5} across four arms of the Australian health system

National ED presentations, inpatient admissions, MBS primary-care claims and PBS prescriptions, 2014–2020, for cardiovascular, respiratory and mental-health conditions, supporting Australia's first National Climate Risk Assessment. **Heat loads the front door** (ED RR 1.033 per °C for respiratory, 1.023 for mental health) while **PM_{2.5} loads admissions and prescribing** (RR 1.014 per 1 µg/m³ at 1-day lag for both cardiovascular and respiratory admissions; primary-care mental-health visits RR 1.049, 1.040–1.058). Mental-health services are sensitive to both exposures across every domain — the least-anticipated result and the one with the weakest mechanistic account. Ecological in exposure assignment.

doi: 10.5694/mja2.70261

Zaid et al. 2026 J Mater Sci

Fibrous aerosol filtration: theory, CFD assumptions, metrology and fabrication

Review integrating single-fibre capture mechanisms and most-penetrating-particle-size theory with electrostatic charge generation, transport and stability; 3D reconstruction of filter geometry; CFD, particle tracking, deposition and clogging simulation; and electrospinning, meltblowing, bubble electrospinning and supersonic solution blowing. The section worth reading is the critique of the homogeneity and steady-state assumptions that most published simulations inherit, alongside an explicit treatment of instrumentation limits and measurement uncertainty in filtration testing. Review, no new data; the material–process–performance framing is the organising contribution.

doi: 10.1007/s10853-026-13269-8

Zhai et al. 2026 Front Public Health

Global disparities in lifetime lung cancer risk: development, smoking and fine PM

GLOBOCAN 2022 with the adjusted-for-multiple-primaries method, giving **global lifetime risk of lung cancer incidence 4.91 % and mortality 4.07 %**, ranging from North America (7.30 % / 5.38 %) to Africa (0.99 % / 0.93 %) and higher in males on every continent. In country-level regression, smoking prevalence predicted both (LRI $\beta = 0.001$, $p < 0.001$), and **PM₁ was the strongest environmental predictor** (LRI $\beta = 0.40$, $p = 0.001$; LRM $\beta = 0.20$, $p = 0.006$), slightly ahead of PM_{2.5} (0.36 and 0.18). Higher national development correlated with higher lifetime risk. The ecological confounding is total: HDI, smoking history, registry completeness and diagnostic capacity all covary with PM at country level, and lung cancer incidence in high-HDI countries is dominated by historical smoking and by the ability to detect it. **PM₁ outranking PM_{2.5} is the one result worth carrying**, and even that is weak given how few countries have independent PM₁ products.

doi: 10.3389/fpubh.2026.1842037

Zhang et al. 2026 J Hazard Mater

Particle-number emissions from a hybrid electric vehicle across cold and warm starts

A China-6 compliant HEV tested over real-driving-emissions and RTS95 cycles with fresh and bench-aged aftertreatment. The particle-number size distribution was trimodal and dominated by an accumulation mode peaking near **90 nm**, with most emission from urban and highway segments and **engine-start events, cold and warm alike, as the dominant source**. Counter-intuitively, catalyst ageing *improved* overall PN removal: oxidation efficiency fell but gasoline particulate filter filtration rose as an ash layer established. Optimising warm-start fuel enrichment cut PN further on the aged system. Limitation: one vehicle, so this characterises a configuration, not a fleet. Directly relevant to near-road sensing — a 90 nm mode driven by discrete start events is largely invisible to mass-based instruments.

doi: 10.1016/j.jhazmat.2026.143156

Zhang et al. 2026 (preprint) medRxiv (preprint, not peer reviewed)

Health impacts of national and local air pollution control policies targeting electric generating units, mobile sources, and port activities in three US cities

Counterfactual policy evaluation for New York City, Los Angeles and Atlanta, 2005–2019. Daily PM_{2.5}, criteria gases, 1-in-3-day PM_{2.5} components and PMF-resolved PM sources at six monitoring sites were used to construct the concentrations that would have occurred without the selected EGU, mobile-source and port policies; city-site-specific multi-pollutant Poisson models then linked observed concentrations to respiratory emergency-department visits and hospitalised cardiovascular ED visits, and the fitted models were applied to both concentration series. **Median PM_{2.5} reductions of 27–62% across cities, and 7.1% (95% UI 5.4–8.8%) of cardiorespiratory visits averted**. Limitation: not peer reviewed; the counterfactual is a modelled construct whose uncertainty is not propagated into the health interval, and multi-pollutant Poisson models over highly correlated criteria pollutants cannot cleanly assign the benefit to PM. The source-resolved PMF layer is the part worth following.

doi: 10.64898/2026.08.04.26359640

Zhang et al. 2026g Nat Commun*Health and economic impacts of improved air quality from transitioning global shipping energy to hydrogen*

Integrated emission–air quality–health framework over feasible hydrogen-transition scenarios for global shipping. Full adoption averts approximately **129,500 (95% CI 68,300–202,200) premature deaths annually**, valued at **US\$306 bn (146–488)**, with India, Japan and the United States taking the largest economic benefit. The number that matters for a PM watch is the split: **75% of the averted mortality is ozone, only 25% PM_{2.5}**, with the O₃ gains concentrated in Southern Asia and the PM_{2.5} gains in Eastern Asia. The confidence interval spans a factor of three and propagates concentration–response uncertainty only, not scenario or inventory uncertainty, so the true spread is wider than stated.

doi: 10.1038/s41467-026-76134-x

Zhu et al. 2026 Atmos Pollut Res*Short-term effectiveness of urban road cleaning for fugitive dust, from TRAKER mobile monitoring*

Metadata-only deposit. TRAKER (Testing Re-entrained Aerosol Kinetic Emissions from Roads) infers road-dust emission potential from the difference between behind-tyre and background aerosol measured on a moving vehicle, which makes it one of the few mobile methods that measures an *emission* rather than a concentration. The question in the title — how long the benefit of a street-sweeping pass actually persists — is one municipalities repeatedly fund cleaning programmes without answering, and the honest prior from the existing literature is “hours to a few days, if at all”. No abstract, no site, no numbers. Carried at tier C and retained because the method class is directly relevant to mobile PM sensing.

doi: 10.1016/j.apr.2026.103190

NOT CARRIED BY A DAILY ISSUE — EXTRACTIVE PRECIS OR METADATA ONLY, NOT AN ASSESSED SUMMARY (1 RECORDS)

Faggi et al. 2026 Atmos Environ*Heatwaves and particulate matter increases: an Italian case study*

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1016/j.atmosenv.2026.122246

Other clinical endpoints

41 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (41 OF 41 RECORDS IN THIS SUBTOPIC)

Chen et al. 2026b Atmos Environ*Air pollution and sarcopenia: meta-analysis and network toxicology*

METADATA ONLY — no abstract deposited. Title indicates a meta-analysis paired with a network-toxicology arm. Pooled estimate, heterogeneity and included-study count are all unknown from the deposit; a meta-analysis carried without its numbers is a placeholder, recorded so it is not missed on a later sweep.

doi: 10.1016/j.atmosenv.2026.122312

Chen et al. 2026h Metabolites*Metabolic perturbation in elderly adults exposed to PM_{2.5}-bound ester compounds*

258 Fuzhou residents aged 60+ with 72-hour personal PM_{2.5} sampling (UPAS V2), GC-MS speciation and LC-MS plasma metabolomics, targeting non-phthalate, non-organophosphate esters that routine speciation ignores. Mean personal PM_{2.5} was **38.06 µg/m³** with esters dominating the measured organic composition; 20 high-concentration esters were prioritised. Covariate-adjusted linear regression associated ester exposure with **steroid biosynthesis, glycolysis/gluconeogenesis, glycerophospholipid and purine/pyrimidine** pathway alterations. Personal rather than ambient sampling in an elderly cohort is the strength and is uncommon; against that, the ester annotations are putative, the design is cross-sectional with a single 72-h exposure window against a single metabolome, pathway enrichment on untargeted features has a well-known false-discovery problem, and the authors state the work is hypothesis- generating. No effect sizes are reportable from the abstract.

doi: 10.3390/metabo16080553

Chin et al. 2026 Res Sq (preprint)*Impact of air quality and PM_{2.5} on hidradenitis suppurativa activity in the Sacramento region*

Retrospective analysis of **10,431** daily hidradenitis suppurativa encounters at UC Davis Health, **April 2015–March 2022**, regressing monthly encounter counts on the monthly count of days exceeding **12 µg/m³ PM_{2.5}** from the EPA Air Quality System, with lags to three months. Positive associations at **lag 0** ($\beta = 0.5658$, $P = 0.022$) and **lag 2** ($\beta = 0.6230$, $P = 0.015$); lag 1 and lag 3 null. The lag structure is the problem. A significant effect at 0 and 2 months with nothing at 1 month has no biological reading, and four lags tested without adjustment on a monthly series of ~84 points is a multiple-comparisons pattern. Three further limits: healthcare encounters measure care-seeking as much as disease activity and are strongly seasonal; a regional monitor assigned to a referral population is a substantial exposure misclassification; and no autocorrelation correction or seasonal adjustment is described, which for monthly counts over seven years will inflate significance on its own. Preprint, single centre, not peer reviewed.

doi: 10.21203/rs.3.rs-10570042/v1

Choi et al. 2026 Environ Res*Time-varying long-term PM_{2.5} and leukaemia from childhood into early adulthood*

National Health Insurance Service–National Sample Cohort, 2002–2020: **384,606** people aged 0–18 at enrolment, **272 incident leukaemias** (83 diagnosed after age 18), annual PM_{2.5} assigned to all 226 inland districts from an ML ensemble, Cox models with time-varying exposure and confounders plus two-pollutant sensitivity against NO₂ and O₃. **HR 1.40 (1.04–1.90) per 5 µg/m³**, stable when censored at 18. The exposure contrast is the problem: cohort mean **29.6 ± 3.1 µg/m³**, so a 5 µg/m³ increment exceeds 1.5 SD of the between-district spread and the estimate extrapolates beyond most of its own support, on 272 events with district-level assignment and no residential-mobility handling.

doi: 10.1016/j.envres.2026.125497

Chow et al. 2026 Nat Commun*Blood metabolomic signatures linking air pollution to lung cancer*

Prospective nested case-control across CPS-II and CPS-3, 1,357 participants, >1,100 metabolites in pre-diagnostic plasma, residential concentrations of six pollutants at blood draw. Eight metabolites associated with both exposure and subsequent lung cancer; four (γ -glutamylglutamine, phenylacetylglutamate, N-(2-furoyl)glycine, 4-vinylguaiacol glucuronide) significantly mediated PM and ozone associations at adjusted $q < 0.2$. Single-timepoint metabolomics cannot separate exposure response from early disease, and $q < 0.2$ is a permissive threshold over 1,100 features — treat the four as candidates. Smoking adjustment in a lung cancer metabolome study is the obvious residual-confounding exposure.

doi: 10.1038/s41467-026-75116-3

Curr Hematol Malig Rep 2026 Curr Hematol Malig Rep*Linking air pollution and hematologic malignancies: from epidemiology to molecular mechanisms*

Review of outdoor air pollution and haematologic malignancy across acute and chronic leukaemias, lymphomas, MDS, CHIP and myeloma, implicating PM, benzene, NO₂, SO₂, arsenic and ethylene oxide through oxidative stress, DNA damage, epigenetic dysregulation, chronic inflammation and haematopoietic stem-cell dysfunction. Evidence is graded strongest for AML with benzene; ALL associations with traffic-related pollution, NO₂ and PM are described as less consistent; evidence for chronic leukaemias and myeloma is limited. *Not systematic*: no search protocol, no pooled estimates, no bias assessment, and the authors themselves flag exposure misclassification and residual confounding. The CHIP link is the forward-looking one — a clonal intermediate would give this literature a measurable early endpoint. Relevance: benzene and PM co-vary in traffic sources, and separating them requires speciated measurement that mass monitors cannot provide.

doi: 10.1007/s11899-026-00785-2

DeAngelis et al. 2026 Soc Sci Med*Residential context and epigenetic ageing: a null for air pollution*

Health and Retirement Study participants providing venous blood in 2016 ($n = 3,187$, mean age 69), with multigroup structural equation models decomposing paths between a latent epigenetic-age-acceleration variable (GrimAge, PhenoAge, DunedinPoAm38) and five 2010 residential features: racial clustering, air pollution, greenspace, household income and perceived disorder. White respondents in areas with high White clustering showed lower EAA, about 25% of it accounted for by higher area household income; **among Black participants essentially no residential path reached significance, and air pollution was not an independent predictor in either group**. Limitations: single-timepoint methylation, exposure indexed to areal units six years before assay, and a null that is as consistent with measurement error in the area-level pollution term as with absence of effect. Carried as a negative result: coarse area-level pollution proxies are not resolving what individual-level exposure assessment might.

doi: 10.1016/j.socscimed.2026.119635

Donaldson et al. 2026 Sports Med Open*Race-day air pollution and marathon finishing time across 2.76 million finishes*

2,756,553 net finishing times from six World Marathon Majors (Berlin, Boston, Chicago, London, New York, Tokyo), 75 marathon-year events, 2010–2024, with race-day NO₂ and PM_{2.5} from CAMS global reanalysis and linear mixed models adjusting heat index, wind and temporal trend. Each 1 $\mu\text{g m}^{-3}$ NO₂ was associated with **+1.43 min (95% CI 0.66–2.21) in men and +1.56 min (0.63–2.48) in women**, with a steep gradient by ability — recreational 2.12 vs. elite 0.34 min per $\mu\text{g m}^{-3}$, persisting as a percentage of mean time (0.67% vs. 0.18%) — and by age (2.17 min in 18–29-year-olds vs. 1.32 in ≥ 60). **PM_{2.5} was null** (–0.13 min, –0.67 to 0.40 in men). The exposure is reanalysis at city scale, not course-level or personal, and NO₂ at that resolution is a traffic-proximity proxy; the NO₂/PM_{2.5} divergence is therefore at least partly a measurement-error contrast rather than a toxicological one. Still the largest exercise-performance exposure dataset assembled.

doi: 10.1186/s40798-026-01099-6

Feng et al. 2026 Joint Bone Spine*Air pollutant mixtures, plasma metabolic and proteomic signatures, and incident arthritis*

UK Biobank, **401,676** arthritis-free at baseline, **78,188** incident cases over a median 13 y. Mixture exposure quantified by weighted quantile sum regression; metabolic and proteomic signatures derived by elastic net; Cox models plus generalised propensity scores; causal mediation for the signatures. The dominant mixture contributors are subtype-specific — NO_x/NO₂ for rheumatoid arthritis, PM for osteoarthritis, NO_x and PM for gout, PM for psoriatic arthritis — and the omics signatures mediated **4.92–35.05%** of the association. The abstract reports no hazard ratios or intervals for the WQS index itself, which is why nothing from this paper appears in the forest plot. WQS indices are also not comparable across studies, and the mediation proportions rest on a no-unmeasured-confounding assumption that plasma omics cannot support. The subtype-resolved contributor pattern is the transferable result.

doi: 10.1016/j.jbspin.2026.106129

Guerrero-Morales et al. 2026 Environ Toxicol Pharmacol*Spatial distribution of long-term ultrafine particle concentrations and systemic lupus erythematosus burden in central Mexico*

Municipal-level ecological analysis: SLE mortality 1998–2023 and hospitalisations 2008–2024 from national databases against satellite-model UFP for 2010–2019, using rate ratios, Jenks natural breaks and principal component analysis. National SLE mortality 0.23–0.86 per 100,000; municipal mean UFP roughly **1,500–24,000 particles/cm³**. Higher UFP and higher hospitalisation burden co-occur in metropolitan municipalities; mortality is heterogeneous and the authors say it should be read descriptively. Ecological design, no individual exposure, explicitly hypothesis-generating — but it is one of very few UFP–autoimmunity datasets from a middle-income country.

doi: 10.1016/j.etap.2026.105143

Huang et al. 2026 Air Qual Atmos Health*PM_{2.5}-bound metals and female breast cancer incidence, New Mexico*

Census-tract analysis of New Mexico Tumor Registry breast-cancer incidence, 1997–2019, against annual PM_{2.5}-bound metal exposure intensities from a calibrated Emission Weighted Proximity Model over a **10-year pre-diagnosis window**, fitted by GEE with Gaussian estimation. All five metals were associated with higher age-adjusted incidence: **chromium RR 1.07 (1.05–1.08)**, **cobalt 1.08 (1.05–1.11)**, **copper 1.06 (1.04–1.07)**, **manganese 1.05 (1.02–1.08)**, **mercury 1.05 (1.03–1.07)**, with a monotone trend across unexposed/low/medium/high categories for every metal. The intervals are strikingly narrow, which is the tell: these are tract-level rate ratios on 23 years of registry data, so precision reflects the number of tract-years rather than the number of informative exposure contrasts, and the five metals are emitted by shared sources and are almost certainly collinear — five nominally independent estimates from one exposure surface. EWPM is a proximity model, not a measurement. Ecological throughout.

doi: 10.1007/s11869-026-02079-1

Jain et al. 2026 J Curr Glaucoma Pract*Ambient air pollution and glaucoma: systematic review and meta-analysis*

PRISMA review, PubMed and Embase to 30 December 2025; 18 observational studies, 8 pooled by generic inverse variance. **Pooled RR 1.12 (1.06–1.18)** for long-term PM_{2.5}, $I^2 = 78.7\%$. NO₂, NO_x, PM₁₀ and CO directionally positive but synthesised narratively. RNFL and GCIPL thinning and modest IOP elevation also reported. Heterogeneity at 79% with eight studies makes the pooled point estimate a weak summary; the authors say causality cannot be established and they are right. Two databases only is a real recall limitation for this literature.

doi: 10.5005/jp-journals-10078-1522

Jung et al. 2026 Ann Rheum Dis

PM_{2.5} and rheumatoid arthritis disease activity and flare

Prospective cohort at a South Korean tertiary centre, 2021–2024: 1,070 patients, 12,583 outpatient visits, monthly mean concentrations of six pollutants, generalised estimating equations adjusted for demographics, serology, medication, socioeconomic and meteorological covariates, plus a within-patient case-crossover sensitivity analysis on daily concentrations. **PM_{2.5} alone was associated with flare (adjusted OR 1.113, 95% CI 1.017–1.218)** and with higher DAS28-CRP, CDAI, and tender and swollen joint counts; the association was stronger in women and non-smokers, and the case-crossover arm reproduced it for cumulative exposure beyond two weeks. Single centre, monitor-assigned exposure; the two-design agreement is what makes it convincing.

doi: 10.1016/j.ard.2026.06.018

Kang et al. 2026 Environ Pollut

Short-term ambient air pollution and acute osteoarthritis morbidity across subgroups

Time-stratified case-crossover over **3,697,314 OA-related hospital visits** (ICD-10 M15–M19) from the Korean NHIS-NSC 2002–2019, linked to national pollutant fields, with health-screening data enabling individual-level effect modification. Per IQR: NO₂ +5.28% (5.02–5.54), CO +2.88%, SO₂ +1.36%, **PM₁₀ +0.90% (0.73–1.07)**, O₃ +0.76%, **PM_{2.5} +0.73% (0.47–1.00)** — so the particulate terms are the weakest of the six and NO₂ is the only one robust in two-pollutant models. Enormous power makes intervals tight enough to exclude the null on effects this small, which is exactly when outcome-definition error matters: the authors state plainly that claims-based visit dates index healthcare utilisation, not symptom onset. A same-day traffic-pollution association with a discretionary outpatient visit is at least as consistent with behaviour as with pathophysiology.

doi: 10.1016/j.envpol.2026.128985

Lee et al. 2026 J Expo Sci Environ Epidemiol

Short-term ambient VOC mixtures, paediatric atopic dermatitis and asthma in Seoul

METADATA ONLY — short-term effects and attributable burden of ambient volatile organic compound *mixtures* on two paediatric outcomes. Included at the edge of scope: the exposure is VOCs rather than particles, but mixture-burden methods on a shared urban monitoring backbone are directly transferable, and the same Seoul network carries PM_{2.5}. No abstract available on 20 August, so neither the mixture method nor the burden fraction is reportable.

doi: 10.1038/s41370-026-00965-5

Lees 2026 Research Square (preprint)

PM_{2.5} component mixtures and inpatient hospitalisation in chronic kidney disease

2,511 *All of Us* participants with ICD-coded CKD, 19,656 hospitalisations over 22,448 person-years (2010–2019); NASA SEDAC estimates of organic and elemental carbon, ammonium, nitrate and sulfate linked at 3-digit ZIP; weighted quantile sum regression inside a negative binomial model. Median PM_{2.5} was only 4.4 µg/m³. **IRR 1.31 (0.99–1.73, p = 0.06) per quartile of the WQS index — formally null, with 93% of the mixture weight on nitrate**, 4% each on ammonium and elemental carbon, and zero on organic carbon and sulfate. WQS weights are unstable at this sample size and 3-digit ZIP assignment is very coarse. The nitrate concentration of weight is the hypothesis worth carrying, not the point estimate.

doi: 10.21203/rs.3.rs-10438586/v1

Lei et al. 2026 Atmos Environ

Hospital admission risk and burden of acute appendicitis attributable to PM_{2.5}

METADATA ONLY — no abstract deposited. Title indicates nationwide Chinese evidence on PM_{2.5} and acute appendicitis admissions with an attributable-burden component. No effect size, lag structure or exposure model is asserted here.

doi: 10.1016/j.atmosenv.2026.122311

Li et al. 2026 Innovation (Camb)

Global and national CKD burden attributable to fine particles: updated exposure–response meta-analysis

Three-stage mixed-effects meta-analysis over 38 CKD and 16 eGFR studies, fitting a non-linear ERF and propagating it to 1990–2020 burden above a 10 µg/m³ counterfactual. **RR 1.32 (1.21–1.43) per 10 µg/m³**, predominantly log-linear; **eGFR –2.115 (0.452–3.777) mL/min/1.73 m²** with saturation at high exposure. 2020 burden: 485,000 deaths, 5.77 million incident and 243 million prevalent cases. The counterfactual floor is a choice, not a threshold, and the burden figures scale with it; the primary studies are also concentrated in high-exposure Asian settings, so the low-concentration arm of the ERF is thin.

doi: 10.1016/j.xinn.2026.101429

Liu et al. 2026 JACC Adv

Reply: statistical fragility and methodological constraints in metabolomic mediation

Correspondence from the Emory group (Liu, Chang, Jones, Quyyumi, Liang) responding to a critique of metabolomic mediation between air pollution and cardiovascular risk. **Abstract not indexed** — carried as metadata only because it sits directly against Chow et al. (same senior author, same method class) in today's harvest and the methodological dispute is the reason to read both. Full text required.

doi: 10.1016/j.jacadv.2026.103149

Mandal et al. 2026 Kidney Int Rep

Long-term ambient PM_{2.5} and kidney function in urban Indian adults

Population-representative cohort of **12,271 adults** aged 20 and over in Chennai and Delhi, with eGFR (CKD-EPI 2009) at three points between 2010 and 2016 and annual mean PM_{2.5} assigned at residential geocode from a 1 km national daily spatiotemporal model. City-specific linear mixed models with inverse-probability weighting for loss to follow-up, plus effect modification by sex, waist-hip ratio, hypertension and diabetes. Exposure contrast between cities is extreme — Chennai 33/37/30 versus Delhi 118/123/130 µg/m³ — and both cities show decline: **–0.32 (95% CI –0.49 to –0.15)** in Chennai and **–0.42 (–0.57 to –0.27) mL/min/1.73 m² per 5 µg/m³** in Delhi. That the two cities agree despite a fourfold concentration difference is reassuring for the shape of the relationship and awkward for its magnitude. CKD-EPI 2009 is poorly calibrated in South Asian populations, three measurements over six years is thin for an eGFR slope, and the by-city stratification means the between-city contrast — the largest exposure contrast in the data — is never used.

doi: 10.1016/j.ekir.2026.106693

Mao et al. 2026 Front Public Health

Component-specific PM_{2.5} and endoscopically confirmed erosive gastric lesions

Consecutive gastroscopies at Xiyang People's Hospital, 1 Jan 2024 to 31 Dec 2025: **60,789 individuals, 14,325 with erosive gastric mucosal lesions**, against gridded daily PM_{2.5} and five chemical components in generalized additive models with attributable-burden analysis. Per IQR, **total PM_{2.5} IRR 1.076 (1.024–1.132)**; after FDR correction only **ammonium 1.082 (1.026–1.141)** and **sulfate 1.070 (1.024–1.118)** survived, with **nitrate, organic matter and black carbon all at 1.012** and intervals crossing the null. Endoscopic ascertainment removes the symptom-reporting bias that weakens most digestive air-pollution studies. The limitation is real and stated: the five components are strongly collinear and share one exposure model, so their independent contributions are not identified. Still the sharpest secondary-inorganic-versus-carbonaceous contrast available.

doi: 10.3389/fpubh.2026.1891236

Negahi et al. 2026 Int J Biometeorol

Climate-sensitive exposures and cancer outcomes: systematic review and meta-analysis

Five databases searched to May 2026 for observational studies of wildfire, ambient temperature, hurricane or multi-hazard disaster, precipitation anomaly, or ambient PM_{2.5} against cancer incidence, mortality, survival or recurrence; 25 estimates from 14 studies pooled within ten exposure-outcome groups, with robust variance estimation as the primary method for groups containing dependent estimates. **Uncorrected pooled 1.07 (1.02–1.12); RVE 1.02 (0.93–1.11)** — the conventional estimate was inflated by correlated effects, with $I^2 = 97\%$. Wildfire and cancer mortality **1.11 (0.89–1.38)**; ambient PM_{2.5} associated with poorer survival; an apparently protective temperature-survival association vanished under RVE and is best read as residual confounding. GRADE certainty was low in two groups and very low in the other eight. Fourteen studies across five exposure types and four outcome classes is too thin a base for group-specific inference, and the paper says so.

doi: 10.1007/s00484-026-03293-3

Opperbeek et al. 2026 Environ Int

Exposome-wide search for epigenetic ageing — and how much is still missing

FinnTwin12: 847 participants measured at ages 12, 14, 17 and 22, with 186 unique exposures spanning lifestyle, green environments, air pollutants and demographics, analysed by exposome-wide association plus knockoff boosted trees, sNPLS and Boruta to allow non-linearity and interaction. Epigenetic age at 22 (PCGrimAge, DunedinPACE) was **~28% explained (R^2 25.7% and 30.8%)**. Smoking, alcohol, youth unemployment and green-space size predicted accelerated ageing; tree cover, vegetation index, neighbourhood age structure and **aerial black carbon predicted decelerated ageing** — a sign reversal that the authors do not resolve and that almost certainly reflects urban socioeconomic confounding rather than a protective effect of BC. Twin modelling attributes the missing variance mainly to environmental factors unshared between co-twins. **Limitations: one cohort, modest n for 186 exposures**, and BC assigned at residence. Read for the missing-exposome decomposition, not the BC coefficient.

doi: 10.1016/j.envint.2026.110430

Park et al. 2026a Gut Liver

Short-term PM_{2.5} and acute hepatic decompensation in cirrhosis

Nationwide time-stratified case-crossover on the Korean NHIS database, 2006–2021: 29,167 cirrhosis patients presenting to emergency departments with ascites, variceal bleeding, hepatic encephalopathy, jaundice or hepatorenal syndrome; control days matched on month and weekday; daily PM_{2.5} assigned on a 1 km residential grid; conditional logistic regression adjusted for temperature, humidity, holidays and weekday. **aOR 1.02 (1.00–1.04) per IQR increase**, with the top exposure quintile 5% above the bottom and *attenuation* at longer averaging windows. No subgroup interaction reached significance. The design controls time-invariant confounding well, but the lower bound sits on the null and the effect is small enough that residual exposure misclassification from grid-assigned concentration could account for it. A large, clean, honestly reported near-null.

doi: 10.5009/gnl260039

Park et al. 2026b J Endocrinol Invest

Association between annual air pollutant exposure and thyroid function: a retrospective cross-sectional study

16,193 adults (8,812 male, 7,381 female; mean age 53.8±13.1 y) screened at Seoul National University with TSH within 0.1–10.0 mIU/L, against CMAQ-estimated annual PM_{2.5}, PM₁₀, NO₂, SO₂ and CO. All five pollutants associate positively with TSH and negatively with free thyroxine ($P < 0.001$), with significant non-linearity for PM_{2.5} and NO₂ against log TSH. For the combined high-TSH/low-FT4 phenotype the per-IQR adjusted odds ratios run **PM₁₀ 1.55 (1.42–1.69)**, **SO₂ 1.38 (1.28–1.49)**, **NO₂ 1.36 (1.23–1.50)**, **PM_{2.5} 1.25 (1.17–1.34)** and **CO 1.23 (1.15–1.31)**, stronger among women. The binding limitations: exposure is a modelled annual field at residence with no personal component, the design is cross-sectional so temporality is unresolved, and the five pollutants are collinear by construction — the ordering must not be read as differential potency.

doi: 10.1007/s40618-026-02965-6

Poursafa & Jylhvä 2026 Ageing Res Rev

Air pollution and biological aging: mechanisms, biomarkers and reversibility

Review organised by biomarker class rather than by outcome, which is what makes it useful. Evidence is graded as strongest for *system-level composites* — Kleméra–Doubal biological age, PhenoAge, frailty indices — and weakest, at meta-analytic level, for *epigenetic clocks*, which the authors attribute to heterogeneity in exposure assessment, design and target-tissue cell composition rather than to absent biology. Causal mediation across independent cohorts supports biological aging as a pathway from exposure to cardiovascular, neuropsychiatric and renal disease. Pollutants of concern are PM_{2.5}, PM_{0.1}, traffic-related pollution and household combustion. Named gaps: household air pollution and low-income settings are under-represented, and longitudinal multi-omic designs are scarce. Air-quality and clean-cooking improvements are reported to partially reverse the acceleration — the strongest claim in the paper and the one most dependent on the intervention literature it summarises.

doi: 10.1016/j.arr.2026.103297

Sassano et al. 2026 Environ Health Perspect

Household solid-fuel use and kidney-disease mortality in the Golestan Cohort

50,045 adults aged 40–75 enrolled 2004–2008 in northeastern Iran and followed to April 2023, with validated fuel-use questionnaires and Cox models; 262 kidney deaths over **724,064 person-years**. Per 10 additional years of household biomass use for cooking or heating, **HR 1.20 (95% CI 1.04–1.37)** for kidney-disease mortality; kerosene 1.09 (0.95–1.24) and gas 1.00 (0.86–1.16) were null. **The biomass effect splits on ventilation** — 1.19 (1.06–1.34) without a chimney against 1.06 (0.95–1.18) with one, $P_{\text{diff}} = 0.025$ — which is the strongest evidence in the paper that the exposure is the smoke rather than a correlate of poverty, though confounding by socioeconomic position is not fully excluded. **No PM measurement of any kind:** fuel type and duration are the entire exposure model, and the ventilation interaction is precisely the variable no ambient network can observe.

doi: 10.1021/EHP.6c00520

Schlaepfer et al. 2026 Transl Androl Urol

Environmental hazards and the risk of idiopathic urethral stricture

Utah Population Database, 2,048 cases born 1996–2021 matched 1:1 on birth year and location to 1,939 controls; historical air pollution and toxics exposure assessed prenatally, in the first year of life, and in the five years before diagnosis. **Only the first-year window was associated: RR 1.21 (1.07–1.36).** Univariate signals for ethylbenzene, tetrachloroethylene and molybdenum trioxide. Three exposure windows tested with one positive is a multiplicity problem the paper does not address, and the univariate toxicant hits are unadjusted. Included as a hypothesis, not a result.

doi: 10.21037/tau-2026-0338

Traisathit et al. 2026 PLoS One

Air pollution and mortality in acute lymphoblastic leukaemia across age groups, northern Thailand

604 ALL patients diagnosed 1999–2020 in upper northern Thailand — a region with severe seasonal biomass-burning haze — stratified into paediatric (<15, n=344), adolescent and young adult (15–39, n=152) and adult (≥40, n=108). Five-year survival 64.5%, 19.6% and 6.7% respectively; adjusted HR for death 3.76 (AYA) and 8.10 (adult) versus paediatric; diagnosis before 2010 aHR 1.38. **PM_{2.5} ≥ 50 µg/m³ was associated with increased mortality in paediatric patients.** The age-group HRs are prognosis, not pollution. The pollution claim rests on comparing mean concentrations between those who died and those who did not, with a threshold analysis in one stratum; no exposure window, no treatment-era adjustment beyond the pre/post-2010 term, and ambient station concentrations as individual exposure. Signal, not estimate.

doi: 10.1371/journal.pone.0355856

Wang et al. 2026 Ecotoxicol Environ Saf

Early-life PM_{2.5} mass and components and childhood anaemia across China, India and Africa

Cross-sectional analysis of **449,894 children under five** from China, India and 26 African countries, with prenatal, postnatal and cumulative PM_{2.5} and five components (black carbon, organic carbon, sulfate, dust, sea salt) from NASA MERRA-2 reanalysis. Anaemia prevalence is **26.9% in China, 62.6% in India, 64.5% in Africa.** Per 1 µg m⁻³ of cumulative PM_{2.5}: **OR 1.019 (95% CI 1.009–1.028) in China, 1.009 (1.008–1.009) in India, 1.001 (1.000–1.002) in Africa,** with organic carbon and dust positive in all three. Population attributable fractions are **28.46%, 10.64% and 0.69%.** The regional ordering of effect size and PAF runs *opposite* to the ordering of anaemia burden, and the paper does not resolve this: the most economical explanation is that MERRA-2 at 0.5° × 0.625° has little within-country exposure contrast in Africa and correspondingly attenuated coefficients, not that African children are less susceptible. Cross-sectional design cannot order exposure and outcome, and anaemia in these settings is dominated by iron deficiency, malaria and haemoglobinopathy — confounders that co-vary strongly with dust exposure. Read the component ranking, not the PAFs.

doi: 10.1016/j.ecoenv.2026.120658

Wang et al. 2026 Atmos Environ

Long-term PM_{2.5}, aging hallmarks and age-related disease

METADATA ONLY — no abstract deposited. Title indicates a nationwide prospective cohort linking long-term PM_{2.5} to aging hallmarks and age-related disease incidence. Cross-reads directly with Poursafa & Jylhävä above, which is the reason to retrieve it in full rather than wait for indexing.

doi: 10.1016/j.atmosenv.2026.122313

Wang et al. 2026b J Hazard Mater

Ambient PM_{2.5} and thyroid health: 24,333 participants in Zhejiang, China

Community-based cross-section of **24,333 participants** (2022–2025) spanning minors and adults, with PM_{2.5} assigned from geocoded residential addresses and mixed models estimating associations with FT4, TSH, subclinical hypothyroidism and autoimmune thyroiditis. Per 10 µg/m³: FT4 **–3.57% (95% CI –4.39, –2.75), TSH +12.15% (7.92, 16.56),** subclinical hypothyroidism OR **1.27 (1.06, 1.51);** no clear association with autoimmune thyroiditis. The FT4 effect was larger in under-18s and in BMI <18.5; the TSH and hypothyroidism effects larger in those ≥50 and BMI ≥24. *Limitations:* cross-sectional, so exposure precedence is assumed not observed; residential-address assignment ignores time-activity; and iodine status is invoked as a rationale but its adjustment is not described in the abstract. The internally coherent FT4-down/TSH-up pattern argues against pure chance. Relevance: the paediatric and low-BMI susceptibility windows require exposure resolved at the individual rather than the address.

doi: 10.1016/j.jhazmat.2026.143231

Wei et al. 2026 Environ Res

Chemical composition of fine particulate matter and all-cause mortality of older lung cancer patients

528,127 SEER-Medicare adults aged ≥65 diagnosed with lung cancer 2000–2019, followed annually; two-year moving-average exposure to 15 PM_{2.5} components from high-resolution spatiotemporal models linked by residential ZIP code; generalized weighted quantile sum regression with random holdouts, adjusted for demographics, histology, stage, first-course treatment, comorbidity and neighbourhood covariates. **RR 1.011 (95% CI 1.010–1.013) per decile increase across all components.** Contribution differences were modest but silicon, nitrate, vanadium, zinc and iron led — road dust, traffic, fossil-fuel and heavy-fuel-oil combustion. Exploratory subgroups suggest a stronger association in non-small-cell, later-stage and female patients.

doi: 10.1016/j.envres.2026.125538

Wei et al. 2026 Environ Health (Wash)

Air pollution, lipid mediators and benign prostatic hyperplasia

UK Biobank, **123,037 men**, with a western-China cohort replication. **2.18 % higher BPH risk per 1 µg/m³ PM_{2.5}** (*p* = 0.0029) in UK Biobank and **2.77 %** in the replication; mixture exposure by WQS and qgcomp gave 3.53 % and 2.20 % per unit for BPH and 6.78 % and 9.17 % for lower urinary tract symptoms. Restricted cubic splines put the **BPH risk threshold at 9.83 µg/m³,** and joint high-PM/high-fat-diet exposure carried 16 % higher risk than low/low. Thirty-four lipid metabolites are nominated as mediators. Two problems: BPH ascertainment in UK Biobank is by linked diagnosis and is sensitive to healthcare-contact patterns that correlate with urbanicity and therefore with PM_{2.5}; and 34 mediators from an untargeted panel without a stated multiplicity correction is hypothesis generation, not mediation analysis.

doi: 10.1021/envhealth.5c00806

Wu et al. 2026d J Am Coll Radiol

Short-term environmental exposures and emergency department imaging utilization

Time-stratified case-crossover over **1,946,465 imaging studies** at five Toronto academic hospitals, 2013–2023, with conditional Poisson models against PM_{2.5}, NO₂, temperature, wind, rainfall and snowfall, modelling both absolute volume and per-visit rate (log daily visits as offset). Higher PM_{2.5} associates with higher imaging volume, **IRR 1.020 (95% CI 1.017–1.024) per 10 µg/m³**, alongside NO₂ 1.017 (1.014–1.020) per 10 ppb and temperature 1.040 per 10 °C; wind, rain and snow are protective, snowfall most strongly (0.959 per 10 mm). The volume effect is almost certainly mediated by visit counts rather than by per-visit imaging intensity, and the exposures are collinear with each other and with season — so this is a health-system demand signal, not evidence of a PM-specific clinical mechanism. Its value is operational: PM_{2.5} forecasts carry usable information for ED resource planning.

doi: 10.1016/j.jacr.2026.08.014

Yang et al. 2026 J Pain

Long-term exposure to multiple air pollutants and chronic pain in middle-aged and older Chinese adults

China Health and Retirement Longitudinal Study, 2011–2020: **10,378 participants free of chronic pain at baseline, 8,047 incident cases over a mean 7.39 years** — an incidence approaching 78%, itself the study's most important number and a strong hint that the outcome definition is broad. Time-dependent Cox and accelerated failure time models were combined with weighted quantile sum regression over the five-pollutant mixture. PM_{2.5}, PM₁₀, SO₂ and CO were each associated with higher risk and earlier onset, and **SO₂ was the predominant mixture contributor**, not PM. Limitation: no interval estimates appear in the abstract, self-reported pain invites differential misclassification, and in a highly correlated mixture WQS weights identify the best-measured proxy as readily as the causal agent — read “SO₂ drives it” as a hypothesis.

doi: 10.1016/j.jpain.2026.106400

Zewdie et al. 2026 Cancer Epidemiol

Wildfire smoke PM_{2.5} and cancer history prevalence at census-tract level

Daily wildfire-specific smoke PM_{2.5} aggregated to cumulative (2006–2020) and recent (2018–2020) means, linked to 2022 tract-level cancer history prevalence for all contiguous-US tracts; quasi-Poisson GAMs with random effects and spatial splines, adjusted for sociodemographics, behaviours, health status, density, climate and region. **No association in fully adjusted models for either exposure window**. Regional effect modification produced small positive estimates in the Midwest (PR 1.020, 1.013–1.026) and Northeast (PR 1.040, 1.032–1.049) — the two regions with the least smoke exposure, a spatial pattern more consistent with residual confounding than with dose. Cross-sectional and ecological, with prevalence of *ever* having cancer set against a 15-year exposure window that cannot precede diagnosis for most cases; latency alone makes this design close to uninformative for incidence. The value here is a well-powered, honestly reported null in a literature prone to amplification.

doi: 10.1016/j.canep.2026.103199

Zhang & Yu 2026 (preprint) Res Sq (preprint)

Air pollution is detectable in ED demand and useless for forecasting it

Single-centre daily time-series at Tongji Hospital Guanggu Branch, Wuhan, 1 Jan 2024 to 26 Jun 2026: **906 dates, 391,344 visits**, mean 431.9/day (SD 59.2), falling to 319.3/day over Spring Festival. Quasi-Poisson models give, per IQR of lag 0–2 pollutant-specific AQI, RR **1.028 (1.015–1.042)** for PM_{2.5}, 1.030 (1.017–1.043) for PM₁₀, 1.049 (1.034–1.064) for NO₂ and an inverse 0.980 (0.965–0.996) for O₃; each 1 °C is +0.85% visits. The contribution is the temporally-ordered validation: a history-plus-calendar model achieves MAE 33.6 visits/day, meteorology reduces it to 33.2, and **adding air-pollution AQI does not improve it at all**. Single centre, city-level AQI rather than catchment exposure, unrefereed. The inverse O₃ term is almost certainly seasonal confounding. Read as a warning that a statistically robust exposure–response can carry no incremental predictive information.

doi: 10.21203/rs.3.rs-10496137/v1

Zhang et al. 2026 Environ Sci Technol

PM_{2.5} exposure and the prognosis of severe fever with thrombocytopenia syndrome

A cohort of **3,583 hospitalised SFTS patients** with long-term pre-admission ambient PM_{2.5}: **HR 1.16 (95% CI 1.04–1.30) per 10 µg/m³** for SFTS-attributable death, with exploratory mediation implicating viral load. A counterfactual burden assessment estimates **137 avoidable SFTS deaths in Xinyang, 2010–2022** had annual PM_{2.5} met the Chinese national standard. The experimental arm shows PM_{2.5} pre-exposure increasing SFTSV replication and tissue injury; transcriptomics implicate mitochondrial redox and ROS pathways, and ROS/MDA measurement plus N-acetylcysteine intervention support oxidative priming as the upstream step. Emerging-infection prognosis is a genuinely new endpoint class for this watch.

doi: 10.1021/acs.est.6c03214

Zhang et al. 2026f Front Public Health

Plasma metabolomic signatures of air pollution and rheumatoid arthritis risk

403,332 RA-free UK Biobank participants followed a median **14.01 years**, with residential annual PM_{2.5}, PM₁₀, NO₂, NO_x and a composite air pollution score, plasma metabolomics, machine-learning-derived pollutant-specific metabolic signatures, Cox models, generalised propensity score balancing and mediation analysis. Per SD of signature: **PM_{2.5}-related HR 1.208 (1.180–1.236)**, NO₂ 1.141, NO_x 1.174, APS 1.184, stable under propensity balancing. **Signatures accounted for 15.7–21.1 %** of the pollution–RA associations. The critical reading: the hazard ratios are for the *signature*, a metabolite score trained to predict exposure, not for exposure itself — a signature can outperform its own target because it absorbs correlates of exposure (diet, adiposity, smoking) that the pollutant variable does not. Single-timepoint metabolomics, UK Biobank's healthy-volunteer selection, and a mediation framework that assumes no exposure–mediator confounding.

doi: 10.3389/fpubh.2026.1901951

Zinan'dala et al. 2026 Int J Mol Sci

Urinary metabolomic alterations across contrasting ambient pollution regions, Thailand

Untargeted ¹H-NMR urinary metabolomics comparing **51 adults in Chiang Mai (high exposure) with 52 in Songkhla (low exposure)** during a regional pollution episode, using long-term residence as the exposure proxy. PLS-DA separated the groups ($R^2 = 0.873$ but $Q^2 = 0.447$ — good fit, modest prediction, and the gap is the standard warning sign of overfitting in this method). Eight metabolites differed at $p < 0.05$, six surviving Benjamini–Hochberg: **L-tryptophan, hippuric acid, xanthine and 5-HIAA remained significant after covariate adjustment**, with L-arginine and L-cystathionine lower in the high-exposure group. The design's fatal ambiguity is acknowledged by the authors: **region is the exposure variable**, and Chiang Mai and Songkhla differ in diet, water, altitude and season as much as in PM_{2.5}. Hippuric acid in particular is a dietary polyphenol marker before it is anything else. Tier B for method, not for attribution.

doi: 10.3390/ijms27167193

Pooled corpus register — August 2026

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Beydoun et al.	The interplay of air pollution with plasma neurodegenerative markers and metabolome for dementia, Parkinson's disease and all-cause mortality risks and transitions: the UK Biobank study	<i>Ecotoxical Environ Saf</i>	Cohort	Ambient PM _{2.5} / PM ₁₀ and NO ₂ /NO _x , modelled residential	United Kingdom
Cai et al.	PM _{2.5} as the dominant air pollutant associated with suicide among people living with HIV/AIDS: province-wide case-crossover	<i>Atmos Environ</i>	Metadata only (no abstract)	PM _{2.5} / PM ₁₀	China
Chen et al.	Integrated plasma proteomics and metabolomes reveal potential biomolecules linking PM _{2.5} to depression	<i>Environ Toxicol Pharmacol</i>	Cohort	PM _{2.5} + 5 constituents (NH ₄ , NO ₃ , SO ₄ , EC, OC)	UK
Chen et al.	Gender and season disparities in the influence of short-term PM _{2.5} exposure on emergency room visits for mood disorders in Taiwan	<i>J Occup Environ Med</i>	Time-stratified case-crossover + DLNM	Daily PM _{2.5} , lags 0-5	Taiwan
Daoud et al.	Air pollution and suicide mortality in the UK: implications for environmental and mental health policy	<i>Environ Sci Pollut Res Int</i>	Ecological panel, fixed effects	PM _{2.5} (modelled, DEFRA)	UK
Delaney et al.	Cumulative annual air pollution exposure and first hospitalization with Parkinson's disease	<i>Environ Epidemiol</i>	Registry cohort + geospatial	PM _{2.5} , NO ₂ , summer O ₃	United States
Domingo	Aging in a highly polluted world: challenges and solutions to prevent Alzheimer's disease	<i>Arch Toxicol</i>	Mechanistic review	Ambient PM + metals, PFAS, micro/nanoplastics	Multi-region
He et al.	Large-scale plasma proteomics identifies protein signatures linking air pollution to epilepsy	<i>Ecotoxical Environ Saf</i>	Prospective cohort	PM _{2.5} , PM _{coarse} , NO ₂ , NO from land-use regression	Not stated
Hendrickx et al.	Effects of recent air pollution exposure on children's neurocognitive functions and the possible link with endothelial function	<i>Int J Environ Res Public Health</i>	Observational - cohort	Indoor and outdoor PM over 4 h, 24 h and 1 week windows; monthly NO ₂	Belgium
Jahanshahi et al.	Long-term PM _{2.5} and NO ₂ and dementia onset in a low-pollution setting	<i>Int J Population Data Science</i>	Registry cohort + geospatial	PM _{2.5} , NO ₂ (5-yr moving average)	Northern Ireland
Lin et al.	Lung-brain axis and lung dysbiosis as a route from inhaled pollutants to epileptogenesis	<i>Genes & Diseases</i>	Critical review	Inhaled air pollutants (unspecified)	n/a
Ma et al.	Associations between short-term absolute increases in ambient air pollution and hospital admissions and outpatient visits for depression: A nationwide case-crossover study	<i>Ecotoxical Environ Saf</i>	Time-stratified case-crossover	Day-on-day absolute increase (APIN) in PM _{2.5} and PM ₁₀	China
Parkinsons Dis review	Air pollution and Parkinson's disease pathology: clinical evidence and molecular mechanisms	<i>Parkinsons Dis</i>	Mechanistic review	PM, NO _x , O ₃	Global
Parmar et al.	Short-term PM _{2.5} and its chemical components and neuropsychiatric hospitalisations	<i>Environ Int</i>	Time-series	Ground-WRF-Chem fused PM _{2.5} at 5 km, speciated into BC, OC, OIN	India
Perera et al.	Declining prenatal air pollution and improving cognitive scores across successive NYC cohorts	<i>Environ Health</i>	Prospective cohort	Prenatal personal/modelled PM _{2.5} , NO ₂ and airborne PAH	USA
Sakhvidi et al.	Ambient air pollution exposure and cognitive aging; mediators and possible pathways: a systematic review on the epidemiological studies	<i>Ageing Res Rev</i>	Systematic review	PM _{2.5} , PM ₁₀ , PM ₁ components, black carbon, NO ₂ , NO _x	Global
Sullivan et al.	Longitudinal impacts of childhood neighborhood air pollution exposure on white matter integrity and cognition	<i>Dev Psychopathol</i>	Prospective cohort	PM _{2.5}	United States
Wang et al.	Speciated fine particulate matter and self-reported sleep time in middle-aged and older Chinese adults	<i>Sleep Med</i>	Prospective cohort (CHARLS)	6-month and 2-year PM _{2.5} plus BC, OM, SO ₄ , NO ₃ , NH ₄ (TAP dataset)	China

First author	Focus	Journal	Design	Metric	Region
Wu et al.	Early-life exposure to particulate matter and sleep in Chinese toddlers: the roles of particle size and particulate constituents	<i>Toxics</i>	Observational - cohort	Early-life PM ₁ , PM _{2.5} , PM ₁₀ and PM _{2.5} constituents from high-resolution grids	China
Xu et al.	Association between PM _{2.5} components and cerebral hypoperfusion: a mediation role of inflammatory biomarkers	<i>Eur J Neurol</i>	Retrospective clinical series	PM _{2.5} , black carbon, sulfate, nitrate, ammonium, organic matter	China
Zhang et al.	Association of PM _{2.5} and its components with cognitive dysfunction among rural older adults	<i>Neurotoxicology</i>	Cross-sectional (cohort baseline)	PM _{2.5} , BC, OM, NH ₄ , SO ₄ , NO ₃ (TAP)	China
Cardiovascular & metabolic					
Chen et al.	Compound cold wave-PM _{2.5} exposure and cardiovascular emergency ambulance calls	<i>Research Square (preprint)</i>	Ecological panel	PM _{2.5}	China
Chen et al.	Pollutant-specific plasma proteomic signatures, mediators and atrial-fibrillation-free life lost	<i>Ecotoxical Environ Saf</i>	Prospective cohort	Long-term modelled PM _{2.5} and PM ₁₀ (plus NO ₂ , NO _x , SO ₂ , benzene, O ₃)	UK
Chen et al.	Residential Incense Burning, Indoor Air Pollution, and Cardiovascular Hemodynamic Alterations	<i>Indoor Air</i>	Repeated-measures panel, 24 h indoor monitoring	24 h indoor PM _{2.5} and PM ₁₀ with CO, CO ₂ , TVOC, HCHO	Taiwan
Correa Lara et al.	Ambient particulate matter exposure and risk of venous thromboembolism: an updated systematic review and meta-analysis	<i>Thromb Haemost</i>	Review / synthesis	Long- and short-term PM _{2.5} and PM ₁₀ , harmonised to 10 µg/m ³	Global / multi-region
Gaggini & Vassalle	Gene-air pollution interaction in cardiovascular disease	<i>Int J Mol Sci</i>	Narrative review	PM _{2.5} , PM ₁₀ , traffic-related pollutants	Global
Kakhaia et al.	Causal effects of time-varying air pollution mixtures on cardiovascular disease incidence	<i>Environ Int</i>	Prospective cohort (mixtures)	PM _{2.5} , NO ₂	United Kingdom
Leiser et al.	Ambient PM _{2.5} and preclinical re-modelling of cardiac structure and function	<i>JACC Adv</i>	Observational - cohort	PM _{2.5} and NO ₂ (spatio-temporal model, annual mean at residence)	North America
Li et al. 2026 (Environ Int)	PM _{2.5} constituents, hepatic fat and stiffness in youth, and effect modification by PNPLA3 genotype	<i>Environ Int</i>	Cross-sectional imaging	15 PM _{2.5} components, visit-year mean at residence	North America
Li et al.	Sulfate-dominated PM _{2.5} constituents are associated with increased acute myocardial infarction mortality: a US nationwide difference-in-differences study	<i>Environ Sci Technol Lett</i>	Quasi-experimental (difference-in-differences)	PM _{2.5} mass and EC / OC / SO ₄ / NO ₃ constituent mixture, per IQR	United States
Liu et al. 2026 (CCDC)	Long-term ozone and cardiovascular disease: warm-season vs annual-average exposure windows	<i>China CDC Wkly</i>	Review / synthesis	Ozone (co-pollutant; no PM estimate reported)	Global / multi-region
Liu et al.	Household clean cooking fuel transition and attenuated cardiopulmonary function decline: a national longitudinal study of middle-aged and older adults in China	<i>Indoor Air</i>	Cohort	Household solid-fuel to clean-fuel cooking transition	China
Moro et al.	Long-term ambient PM _{2.5} exposure, hypertension and metabolic syndrome among Ghanaian adults	<i>Environ Res</i>	Cross-sectional (cohort baseline)	PM _{2.5} (satellite-based model, 2007-2015 mean)	Ghana
Nong et al.	Air pollution and diabetic microvascular complications: a prospective cohort study	<i>PLoS One</i>	Prospective cohort	PM ₁₀ , PM _{2.5} , NO ₂ (annual, quartiles)	UK
Ramirez-Moreno et al.	Regional disparities in stroke hospitalization and mortality associated with long-term fine particulate matter exposure: a nationwide ecological study in Spain	<i>Med Sci (Basel)</i>	Ecological	Regional annual mean PM _{2.5} (MITECO), region-year resolution	Spain
Sewor et al.	Null: personal HAP exposure and blood pressure in rural Rwandan children at 193 µg/m ³ median PM _{2.5}	<i>Environ Res Health</i>	Observational - cross-sectional	PM _{2.5} and black carbon (48 h personal, Ultrasonic Personal Aerosol Sampler)	Sub-Saharan Africa
Shi et al.	Ischemic heart disease in G20 countries: forecasted burden and regional disparities based on GBD 2023 estimates	<i>Catheter Cardiovasc Interv</i>	Modelling / inventory	Unspeciated PM	Multi-country

First author	Focus	Journal	Design	Metric	Region
Ting	Statistical fragility and methodological constraints in metabolomic mediation of air pollution and cardiovascular risk	<i>JACC Adv</i>	Method review	Ambient PM (methodological commentary)	Not stated
Wang L et al.	Joint exposure to small dense LDL-C and long-term PM _{2.5} and risk of incident stroke	<i>Lipids</i>	Prospective cohort + trajectory clustering	Annual mean PM _{2.5} from the ChinaHighPM _{2.5} dataset, 2011-2020	China
Wittkopp et al.	Cleaner Air for Lower Cardiometabolic Risk: protocol for a double-blind, randomized, sham-controlled trial of HEPA filtration in adults with prediabetes	<i>Am Heart J</i>	Trial / intervention	PM _{2.5} (indoor, continuous)	USA
Yang et al.	Endothelial function in relation to low-level chronic residential air pollution in a general population: a cohort study	<i>Blood Press</i>	Observational - cohort	6-year residential PM ₁₀ 15.9, PM _{2.5} 12.8, NO ₂ 14.3, BC 1.04 µg/m ³ (spatiotemporal interpolation)	Belgium
Zhang et al.	Acute impact of combined exposure to cold spells and heavy pollution on circulatory disease mortality in the Beijing-Tianjin-Hebei region	<i>Zhonghua Yu Fang Yi Xue Za Zhi</i>	Distributed-lag time-series	PM _{2.5} , heavy-pollution thresholds >75 / >115 / >150 µg/m ³	China
Zhao et al.	Ischaemic stroke burden and attributable risk factors in adults aged 20-54, 1990-2021	<i>Medicine (Baltimore)</i>	Burden estimation (GBD)	Air pollution as GBD risk factor (PAF)	Global / multi-region
Çulpan et al.	Long-term air pollution and blood pressure in children: updated systematic review and meta-analysis	<i>Rev Environ Health</i>	Systematic review (PRISMA)	PM _{2.5} , PM ₁₀ , NO ₂ (long-term, per 10 µg/m ³)	Global
Reproductive & developmental					
Atalell et al.	Exposure to low-level fine particulate matter air pollution and developmental vulnerability at school entry: a population-based cohort study in Western Australia	<i>Environ Res</i>	Registry cohort + geospatial	PM _{2.5}	Australia
Bello et al.	Prenatal exposure to PM _{2.5} and infant health: evidence from Quebec	<i>Econ Hum Biol</i>	Registry cohort, FE models	PM _{2.5} (monitor-assigned)	Canada
Chen et al.	Cord blood DNA methylation as a potential mediator between prenatal ambient air pollution and early infant allergic symptoms in a Shanghai birth cohort	<i>Pediatr Allergy Immunol</i>	Prospective birth cohort	PM _{2.5} , PM _{2.5} components (SO ₄ , NO ₃ , NH ₄), NO ₂	China
Chen et al.	Prenatal exposure to size-fractionated particulate matter and fetal growth trajectories: mediation by maternal glucose metabolism in a birth cohort	<i>Toxics</i>	Observational - cohort	Residential PM ₁ , PM _{2.5} , PM ₁₀ from 1 km spatiotemporal models	China
GBD 2023 India Still-birth Collab.	State-level stillbirth rates in India at >=22 and >=28 weeks, GBD 2023	<i>Lancet Reg Health South-east Asia</i>	Burden estimation (GBD)	Risk-factor attribution incl. ambient and household PM	South Asia
Guo et al.	Environmental pollutants and testicular toxicity in male reproductive health: basic evidence and translational prospects	<i>Reprod Toxicol</i>	Review / synthesis	PM _{2.5} (traffic-derived metal constituents)	Global / multi-region
Hahad et al.	Air pollution and adverse birth outcomes: a narrative review of epidemiological and mechanistic findings	<i>J Environ Health Sci Eng</i>	Narrative review	PM _{2.5} , trimester-specific	Multi-region
Han et al.	Aerosol pH associated with preterm birth risk: unmasking the hidden role of atmospheric acidity in maternal-fetal health	<i>Environ Pollut</i>	Metadata only (no abstract)	Aerosol acidity (pH) (not extractable)	Not stated
Hung et al.	Late-pregnancy air pollution exposure and neonatal anthropometry with and without gestational hypertension	<i>iScience</i>	Prospective birth cohort	PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , CO, O ₃ (30 d pre-delivery, ambient monitors)	Taiwan
Jang et al.	Ambient air pollution and age at natural menopause in a multi-ethnic US cohort	<i>J Clin Endocrinol Metab</i>	Prospective cohort	1 km ² modelled annual PM _{2.5} , NO ₂ and O ₃ at residence	USA
Kessler et al.	Joint Mixture Associations of Prenatal Air Pollution and Polyunsaturated Fatty Acid Intake on Birth Weight Outcomes in an African American Cohort	<i>Environ Health (Wash)</i>	Prospective birth cohort (mixtures)	First-trimester traffic-related PM _{2.5} , CO, NO _x	United States
Li et al. 2026 (Environ Pollut)	Prenatal NO ₂ and stillbirth in 3.98 million Iranian mother-baby pairs	<i>Environ Pollut</i>	Observational - cohort	NO ₂ (satellite-based gridded, 10 km buffer); PM _{2.5} and O ₃ as co-pollutants	Middle East & N. Africa

First author	Focus	Journal	Design	Metric	Region
Pai et al.	Household incense burning and small for gestational age risk in newborn infants	<i>Pediatr Neonatol</i>	Prospective birth cohort	Incense PM (indoor, unspecified)	Taiwan
Russo et al.	Critical windows of air pollution exposure during spermatogenesis and intrauterine insemination outcomes	<i>Paediatr Perinat Epidemiol</i>	Cohort (secondary trial analysis)	PM _{2.5} , NO ₂ , O ₃ , SO ₂ (CMAQ, 74-day spermatogenesis window)	USA
Tanner et al.	Prenatal environmental determinants of aromatase brain-promoter methylation in cord blood: chemical, airborne, pharmacological, and nutritional factors	<i>Environ Epigenet</i>	Birth cohort + weighted quantile sum regression	Air pollution as 1 of 25 prenatal exposures in a WQS mixture index	Australia
Wang et al.	Synergistic and antagonistic interactions of PM _{2.5} with high and low temperatures on sperm motility	<i>Environ Pollut</i>	Observational - cohort	PM _{2.5} with temperature, 0-90 / 0-9 / 10-14 d exposure windows	China
Wang et al.	Ambient Ultrafine Particles and Hypertensive Disorders of Pregnancy: A Mediation Analysis of Homocysteine in a Birth Cohort	<i>Environ Health (Wash)</i>	Prospective birth cohort	Modelled ultrafine particle number, high spatiotemporal resolution	China
Wu et al.	Prenatal PM _{2.5} and its chemical constituents and early childhood overweight/obesity: organic matter dominates the mixture effect	<i>Ecotoxicol Environ Saf</i>	Prospective cohort (mixtures)	Gestational PM _{2.5} mass + OM, BC, NH ₄ ⁺ , NO ₃ ⁻ , SO ₄ ²⁻ (reanalysis)	China
Zhao et al.	PM _{2.5} constituents and spontaneous abortion, modified by residential greenness	<i>Chin Med J (Engl)</i>	Prospective cohort (mixtures)	Residential annual PM _{2.5} with NO ₃ ⁻ , SO ₄ ²⁻ , NH ₄ ⁺ , BC and OM fractions	China
Respiratory & allergic					
Ali et al.	Symptom burden among adult outpatients during the 2024-2025 winter smog season in Lahore	<i>Public Health Chall</i>	Cross-sectional (multistage)	City-level PM _{2.5} (mean 96 µg/m ³), PM ₁₀ , CO, AQI	Pakistan
Aulia Anwar et al.	Smoking, environmental exposures, and chronic lung disease as key risk factors for lung cancer in North Sumatra	<i>Exp Oncol</i>	Case-control	Self-reported unhealthy indoor environment (ventilation, household pollutants)	Indonesia
Bogunia-Kubik et al.	Influence of environmental pollution on basophil reactivity to birch pollen allergens in atopic and non-atopic subjects	<i>Allergy</i>	Panel study	PM _{2.5} , PM ₁₀ (and O ₃)	Poland
Cantuaria et al.	Source-specific PM _{2.5} , NO ₂ , UFP and elemental carbon and incident COPD in two pooled Danish cohorts	<i>Environ Health Perspect</i>	Prospective cohort	PM _{2.5} , UFP, EC, NO ₂ (10-y time-weighted, address history)	Denmark
China Lung Oncology Group	China Wumeng Mountain Lung Cancer: Evidence Chain, Clinical and Molecular Characteristics of Regional Lung Cancer Related to Coal Pollution in the Yunnan-Guizhou Plateau	<i>Zhongguo Fei Ai Za Zhi</i>	Multi-omics regional evidence synthesis	Household coal-combustion PM, source-resolved mineral composition	China
Erickson-DiRenzo et al.	Wildfire PM _{2.5} and the human larynx: paired in-event and clean-air voice, endoscopy and ventricular-fold biopsy in one adult	<i>J Voice</i>	Repeated-measures biomarker	Wildfire episode PM _{2.5} up to 117 µg/m ³ vs <10 µg/m ³ baseline	USA
Feng & Chao	Lower respiratory infection burden attributable to PM _{2.5} in China and the G20, 1990-2023	<i>Int J Biometeorol</i>	GBD secondary analysis	Ambient PM _{2.5} , household PM _{2.5}	China
Govindaraju et al.	Protective role of dietary antioxidant intake on long-term effects of extreme PM _{2.5} exposure on respiratory health	<i>J Occup Environ Med</i>	Cross-sectional (cohort baseline)	Fire-sourced PM _{2.5}	Australia
Haque et al.	Airborne PM and respiratory health in Bangladesh: narrative review	<i>Health Science Reports</i>	Narrative review	PM _{2.5} , PM ₁₀ , UFP, gaseous co-pollutants	Bangladesh
Hatem et al.	Child characteristics, household, and home environmental factors associated with eczema: a cross-sectional study	<i>Eur J Pediatr</i>	Cross-sectional survey	Traffic-zone proximity (<300 m) and postnatal secondhand smoke	Porto, Portugal
Hur et al.	Pollen modifies the association between suspended particulate matter and mortality across eight Kyushu cities	<i>Environ Pollut</i>	Time-series / case-crossover	SPM (daily, Japanese suspended PM)	Kyushu, Japan
Hussein et al.	PM ₁₀ and emergency visits for asthma exacerbation in Petaling Jaya, Malaysia	<i>J Glob Health</i>	Time-series / case-crossover	PM ₁₀ , daily mean, DLNM lag 0-7	Southeast Asia

First author	Focus	Journal	Design	Metric	Region
Jiang et al.	Short-term PM _{2.5} , PM ₁₀ and SO ₂ and respiratory-infection hospitalisation burden in kidney transplant recipients	<i>J Cent South Univ Med Sci</i>	Time-series / case-crossover	PM _{2.5} , PM ₁₀ , SO ₂ , NO ₂ , CO, O ₃ ; 7-day pre-symptom cumulative	China
Kannoth et al.	Short-term PM _{2.5} and COVID-19 pneumonia in New York City: risk amplified by pre-existing cardiovascular disease	<i>J Expo Sci Environ Epidemiol</i>	Retrospective cohort	7-day residential ZIP-code CMAQ PM _{2.5} prior to admission	USA
Lancet Respir Med	The far-reaching effects of wildfire smoke	<i>Lancet Respir Med</i>	Metadata only (no abstract)	Wildfire PM _{2.5}	Global
Li et al.	Associations Between Air Pollution and Hospitalization of Patients with Chronic Obstructive Pulmonary Disease in Kashi, China: A Time-Stratified Case-Crossover Study	<i>Int J Chron Obstruct Pulmon Dis</i>	Time-stratified case-crossover	Address-matched daily PM _{2.5} and PM ₁₀	China
Lu et al.	Antagonistic effects of PM _{2.5} and ozone on respiratory morbidity: multi-pollutant interaction, Pearl River Delta	<i>Atmos Environ</i>	Metadata only (no abstract)	PM _{2.5} / PM ₁₀	China
Makrufardi et al.	Effect of ambient relative humidity and particulate matter on the levels of asthma therapy	<i>World Allergy Organ J</i>	Prospective cohort	PM _{2.5}	Taiwan
Morvan et al.	When air pollution meets comorbidities: evidence on the causal effect of PM _{2.5} on COVID-19 hospitalizations and deaths	<i>Int J Epidemiol</i>	Spatial econometric model	Municipal weekly PM _{2.5} , instrumented by wind direction (2SLS)	France
Ni et al.	Socioeconomic and regional influences on eosinophilic inflammation in chronic rhinosinusitis with nasal polyps	<i>World Allergy Organ J</i>	Retrospective clinical series	Annual mean PM _{2.5} (city-level)	China
Orr et al. 2026 (EHP)	Wildfire-season PM _{2.5} and influenza / influenza-like-illness risk across six western US states	<i>Environ Health Perspect</i>	Distributed-lag time-series	Wildfire-specific PM _{2.5} (per 10 µg/m ³)	North America
Rice & Miller	Air pollution, microbes and respiratory infection risk in children	<i>Pediatr Pulmonol</i>	Narrative review	PM _{2.5} and indoor combustion PM (qualitative)	Global
Shi et al.	Air pollution and acute exacerbation of COPD in Hebei Province: a multi-city time-series study	<i>Medicine (Baltimore)</i>	Distributed-lag time-series	PM _{2.5} , PM ₁₀ , SO ₂ , NO ₂ , CO, 8-h O ₃	China
Shukla et al.	Hourly ambient NO ₂ and emergency admissions for acute asthma, Birmingham and Solihull	<i>BMJ Open</i>	Distributed-lag time-series	NO ₂ (hourly, co-pollutant proxy)	Birmingham, UK
Silva et al.	Shared mechanisms linking PM _{2.5} exposure, obstructive sleep apnea and obesity	<i>Front Sleep</i>	Mechanistic review	Chronic ambient PM _{2.5}	Global / multi-region
Solazzo et al.	Season-by-pollutant interaction on the nasal microbiota of healthy adults	<i>Hygiene & Environmental Health Advances</i>	Repeated-measures biomarker	Indoor TSP; outdoor PM, BC, benzene, CO	Italy
Tornevi et al.	Short-term urban-background PM _{2.5} and the timing of confirmed COVID-19, three Swedish cities	<i>BMJ Open Respir Res</i>	Case-crossover	Daily urban-background PM _{2.5} and PM ₁₀ from fixed reference monitors	Sweden
Wang & Li	Advances in research on the association between global air pollution and atopic dermatitis	<i>Front Public Health</i>	Narrative review	Ambient PM _{2.5} / PM ₁₀ and gaseous co-pollutants	Multi-region
Wang et al.	PM _{2.5} constituents and daily fever-clinic visits: a time-series study in Jinshan, Shanghai	<i>Res Sq (preprint)</i>	Time-series	PM _{2.5} mass + As, Cd, Cr, Pb, Mn, nitrate, ammonium	Shanghai, China
Wang et al.	COVID-19 social interventions were associated with amplified, not attenuated, indoor-source pollutant-respiratory associations in Xi'an	<i>Front Public Health</i>	Retrospective time series	PM _{2.5} , SO ₂ , NO ₂ , O ₃	China
Wood et al.	Ultra Low Emission Zone and lung function growth in children: parallel cohort study	<i>Lancet Public Health</i>	Prospective cohort	Modelled residential annual NO ₂ and PM (LAEI dispersion model)	UK
Wu et al.	Air pollution and COPD-related health outcomes: biometric and visualisation analysis	<i>J Thorac Dis</i>	Mapping review	PM _{2.5} , PM ₁₀ , household PM	Global
Yang et al.	COPD and LRI burden attributable to household air pollution and second-hand smoke, 1990-2021	<i>J Thorac Dis</i>	GBD secondary analysis	Household PM from solid fuel	Global
Yang et al.	Interactive associations of atmospheric humidity and air pollutants with self-reported chronic lung disease	<i>J Asthma</i>	Prospective cohort (mixtures)	PM _{2.5} , PM ₁₀ + 4 gases at 1 km, with 6 humidity indices	China

First author	Focus	Journal	Design	Metric	Region
Yang et al.	Analysis of Spatiotemporal Risks of Environmental Driving Factors of Respiratory Diseases	<i>GeoHealth</i>	DLNM + geographically-temporally weighted regression	Ambient PM and co-pollutants, prefecture daily	China
Zhai & Meng	Impacts of apparent temperature and air pollutants on daily pneumonia hospitalizations and their interaction in rural areas of Western China	<i>Int J Environ Health Res</i>	Distributed-lag time-series	PM _{2.5} , PM ₁₀	China
Zhang S et al.	Respiratory and cardiovascular effects of ozone modified by PM _{2.5}	<i>Environ Sci Technol</i>	Panel study	30-day mean PM _{2.5} as effect modifier of short-term O ₃	Guangzhou, China + literature
Zhongguo Fei Ai Za Zhi	China Wumeng Mountain lung cancer: evidence chain, clinical and molecular characteristics of regional lung cancer related to coal pollution in the Yunnan-Guizhou Plateau	<i>Zhongguo Fei Ai Za Zhi</i>	Narrative review	Household smoky-coal PM _{2.5} / black carbon, PAHs and methylated PAHs	China
de Souza et al.	Wildfire smoke, PM _{2.5} exposure and respiratory morbidity in a densely populated urban setting: spatial mapping and time-series evidence	<i>Sci Total Environ</i>	Time-series	Weekly mean CETESB PM _{2.5} ; INPE active-fire counts	Brazil
Mechanistic toxicology					
Asplund et al.	Concentrated ambient PM _{2.5} suppresses erythropoiesis and splenic proresolving lipid mediators, reversibly	<i>Toxicol Sci</i>	Murine inhalation exposure	Concentrated ambient particles (whole-body)	United States (Louisville CAP facility)
Ayala et al.	Contrasting neurotoxic pathways triggered by PM ₁₀ in relation with organic molecular markers in suburban and rural sites in Catalonia	<i>Environ Pollut</i>	Exposure assessment + in vitro	PM ₁₀	Spain
Chen et al.	Gut microbiota signatures reflect particulate matter 2.5-exacerbated asthma in a murine model	<i>Allergy</i>	Animal model	PM _{2.5} (murine asthma model)	Taiwan
Chen et al.	PM _{2.5} suppresses IL-8-mediated epithelial migration of human conjunctival epithelial cells	<i>J Chin Med Assoc</i>	In vitro (cell lines)	PM _{2.5} (suspension, time- and dose-response)	Laboratory
Chen et al.	Multi-omics integration reveals human placental molecular signatures associated with microplastics exposure	<i>Part Fibre Toxicol</i>	Human tissue multi-omics (transcriptome/proteome/metabolome)	Placental microplastic mass burden by Py-GC-MS, high vs low strata	China
Cho et al.	Lactacaseibacillus paracasei ATG-E1 alleviates particulate matter-induced airway inflammation in association with altered inflammatory signalling, gut microbiota and faecal metabolites	<i>J Microbiol Biotechnol</i>	Animal model	PM ₁₀ D (PM ₁₀ + diesel exhaust particles)	East Asia
Choi et al.	Degradable PLGA/PCL membrane alveoli-on-a-chip for particulate-induced alveolar injury	<i>Biofabrication</i>	In vitro (cell lines)	Diesel particulate matter	In vitro
Deng et al.	Baicalin rescue of PM _{2.5} -induced testicular injury via DKK1-Wnt/beta-catenin signalling	<i>J Biochem Mol Toxicol</i>	Murine inhalation exposure	PM _{2.5} , 750 µg/m ³ , 4 h/day, 45 days (oral-nasal)	China
Forest	Toxicological impact of sand and dust storms on human health: A review on the molecular mechanisms involved	<i>Sci Total Environ</i>	Mechanistic review	Storm-derived PM ₁₀ and PM _{2.5} (mineral, biological, metal fractions)	Multi-region
Gao et al.	SQSTM1/p62-mediated mitophagy in PM _{2.5} -aggravated HDM asthma, and dusquetide rescue	<i>Zhongguo Yi Xue Ke Xue Yuan Xue Bao</i>	Murine inhalation exposure	PM _{2.5} (HDM-sensitised C57BL/6)	China
Garcia Dominguez et al.	Ultrafine airborne particle deposition and clearance in an avian lung under ecologically relevant conditions	<i>Environ Sci Technol</i>	Animal inhalation exposure + dosimetry	50 and 100 nm model ultrafine particles at 5 or 25 degrees C	Sweden
Gomes et al.	Senolytics reverse PM _{2.5} -induced cell and vascular dysfunction	<i>Cardiovasc Toxicol</i>	Animal + in vitro	Concentrated ambient PM _{2.5}	United States
Ichinose et al.	Time-dependent bronchus-associated lymphoid tissue formation and pulmonary inflammation induced by different particulate matter types	<i>J Hazard Mater</i>	Animal model	TiO ₂ nanoparticles, diesel exhaust particles, Asian sand dust	Japan
Introna et al.	Novel approach to assess the toxicity of PM _{2.5} on a subway platform using a mobile air-liquid interface system and submerged exposure	<i>ACS EST Air</i>	In vitro exposure	Subway-platform PM _{2.5} , metal-rich; ALI dose 0.12 +/- 0.07 µg/cm ²	Sweden (Stockholm Univ. affiliation; platform city not stated)
Jerome et al.	Prolonged brain cytokine elevation and microgliosis following exposure to World Trade Center dust particulate matter in male spontaneously hypertensive rats	<i>J Immunotoxicol</i>	Animal exposure study	WTC dust, intratracheal, 2.43 (+/-0.78) mg/day x 2 d	United States

First author	Focus	Journal	Design	Metric	Region
Jin & Zhang	Astragaloside IV protects against PM _{2.5} -induced spermatocyte injury via cGAS-STING/Nrf2/HMOX1	<i>Reprod Toxicol</i>	In vitro exposure	PM _{2.5} suspension at 200 ug/mL in GC-2 spermatocytes	China
Li & Li	Research advances in the pulmonary carcinogenic effects of atmospheric fine particulate matter mediated by GPER	<i>Zhongguo Fei Ai Za Zhi</i>	Narrative review	PM _{2.5} estrogen-like organic components and metal ions	China
Li & Li	Research Advances in the Pulmonary Carcinogenic Effects of Atmospheric Fine Particulate Matter Mediated by GPER	<i>Zhongguo Fei Ai Za Zhi</i>	Mechanistic review	PM _{2.5} estrogen-like components and metal ions	China
Li et al.	Size-resolved carbon-black nanoparticle effects on the maternal-fetal lung axis	<i>FASEB J</i>	Animal model	Ultrafine carbon-black nanoparticles, 30 nm and 120 nm, high-dose mechanistic exposure	Laboratory (murine)
Li et al.	Component-resolved PM _{2.5} translocation across a human blood-brain barrier immune platform	<i>Environ Sci Technol</i>	In vitro organ-on-chip	Ambient PM _{2.5} extract, compartment-resolved constituent quantification	China
Liu et al.	Ambient PM _{2.5} Induced Pulmonary Vascular Remodeling via MSC-Mediated Cellular Senescence in Pulmonary Arterial Endothelial Cells	<i>Environ Health (Wash)</i>	Real-ambient murine + cell model	Real-ambient PM _{2.5} exposure	China
Luo et al.	Impacts of Environmental Pollutants on Innate Immunity: The Role of Group 2 Innate Lymphoid Cells	<i>Clin Rev Allergy Immunol</i>	Narrative review	PM, diesel exhaust particles, O ₃ , cigarette smoke, nano-materials, microplastics	Multi-country
Mahmutovic Persson et al.	Chronic exposure to aerosolized PM _{2.5} in rats alters plasma protein levels and ex vivo coronary artery response to sphingosine-1-phosphate	<i>Environ Res</i>	Animal model	Re-aerosolized urban PM _{2.5} (acute high-dose vs chronic low-dose)	Europe
Masci et al.	Air pollution and undernutrition induce rat alveolar macrophage functional dysregulation and compromise their response to a viral challenge	<i>Environ Toxicol Pharmacol</i>	Animal experiment	Residual oil fly ash (ROFA), intranasal, sub-chronic	Argentina
Mishra et al.	Aerosol oxidative potential and reactive species predicted with a chemical kinetics model	<i>Atmos Chem Phys</i>	Kinetic model + field validation	PM composition, OP-DTT / OP-AA	France / UK
Mishra et al.	OMA1-DELE1 as the shared node across ultrafine PM, black carbon and nanoplastic immunotoxicity	<i>bioRxiv (preprint)</i>	Experimental / toxicology	Ultrafine PM, black carbon, polystyrene nanoplastics (matched conditions)	South Asia
Moonwiriyaakit et al.	Establishment of a STAT6 reporter assay for screening environmental toxicants affecting allergic airway inflammation	<i>J Vis Exp</i>	Toxicological (in vitro)	PM and B[b]F as test constituents; no ambient exposure metric	Not stated
Nikaido-Landry et al.	Epidermal growth factor receptor dysregulation and climate change: insights into pollution-promoted lung cancer	<i>Philos Trans R Soc Lond B Biol Sci</i>	Review / synthesis	Ambient fine PM as a promoter in EGFR-mutant histologically normal lung tissue	Global / multi-region
Pathak et al.	Effect of airborne particulate matter on mtDNA copy number: A systematic review and meta-analysis	<i>medRxiv (preprint)</i>	Systematic review & meta-analysis	Ambient PM _{2.5} / PM ₁₀ , study-specific contrasts	Multi-region
Qi et al.	Exposure to urban PM _{2.5} exacerbates renal injury and fibrosis via the lung-kidney axis through IL-6/ADCY3 signalling	<i>J Hazard Mater</i>	Animal + in vitro	Urban PM _{2.5} (repeated intra-tracheal)	China
Qin et al.	AlCl ₃ aerosol exposure is associated with pulmonary lactate accumulation, neutrophil dysfunction, and fibrosis-like remodeling	<i>Biofactors</i>	Animal + in vitro + human cohort	Repeated AlCl ₃ aerosol inhalation (dose not deposited)	China
Qin et al.	Engineered protein coronas reveal that apolipoprotein E enhances transport of particulate pollutants across the blood-brain barrier	<i>Environ Health (Wash)</i>	Toxicological (in vitro)	Aptamer-functionalised gold nanoparticles as PM surrogate; polystyrene and industrial-source PM	Not stated

First author	Focus	Journal	Design	Metric	Region
Saygin et al.	Source identification and seasonal change of trace metals, protein content and oxidative potential of fine and coarse particulates after the Kahraman-maras earthquakes	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	PM _{2.5} / PM ₁₀	Türkiye
Seymore et al.	Gestational inhalation of nanoparticles alters placental glucose transporter expression and increases glucose flux in rats	<i>Toxicol Appl Pharmacol</i>	Animal model	TiO ₂ nanoparticles, 9.8 +/- 1.0 mg/m ³ whole-body inhalation	USA
Sharma et al.	Dim light at night reshapes the neuroimmune response to burn-pit emission inhalation	<i>J Toxicol Environ Health A</i>	Experimental / toxicology	Burn-pit emissions, 6.9 mg/m ³ mass and 4.03e6 particles/cc	North America
Shi et al.	PM _{2.5} exacerbates airway inflammation and epithelial barrier disruption in COPD mice through the RAGE-dependent PERK/eIF2a/ATF4/CHOP pathway	<i>J Appl Toxicol</i>	Animal experiment	PM _{2.5} (instilled)	China
Silva et al.	2D monolayers and 3D spheroids disagree by endpoint, not by magnitude, for PM _{2.5} organic extracts	<i>Environ Toxicol Pharmacol</i>	Experimental / toxicology	Solvent-extractable organic fraction of characterised ambient PM _{2.5}	Latin America
Sivakumar & Kurian	Repeated PM _{2.5} inhalation drives a duration-dependent transition from mitochondrial adaptation to persistent cardiac toxicity	<i>Environ Toxicol</i>	Animal model	PM _{2.5} 250 µg/m ³ , 3 h/day, 1-21 days (whole-body inhalation)	Laboratory
TaRGET II Consortium	Lifecourse sex-specific molecular response to early-life exposures of toxic substances	<i>bioRxiv (preprint)</i>	Multi-omics animal consortium	PM _{2.5} among seven toxicants, preconception through lactation	United States
Van et al.	MASLD exacerbates chronic low-dose PM _{2.5} -induced lung injury, inflammation and fibrosis	<i>In Vivo</i>	Animal model (chronic instillation)	Diesel PM 0.5 mg/kg intratracheal 3x/week (50 µg/m ³ equivalent)	Taiwan
Wang et al.	Vehicular PM reproductive toxicity carries to F2 but not F3 in <i>C. elegans</i>	<i>Ecotoxicol Environ Saf</i>	Experimental / toxicology	SRM diesel PM, WHTC diesel PM, WLTP gasoline PM	China
Wei et al.	Seasonal source shifts drive oxidative potential of size-resolved particulate matter in coastal southeastern China	<i>J Hazard Mater</i>	Metadata only (no abstract)	Size-resolved PM oxidative potential	China
Wu et al. 2026 (JHM)	TNF-alpha and EGF mediate endothelial activation downstream of PM _{2.5} alveolar injury in a 3D barrier model	<i>J Hazard Mater</i>	Toxicological (in vitro)	SRM2786 PM _{2.5} , 100 ug/mL, 24 h	Computational / laboratory
Yan et al.	PM _{2.5} induces lung injury via mtDNA-cGAS-STING-mediated macrophage M1 polarization	<i>Front Pharmacol</i>	Animal + in vitro	PM _{2.5} , 21-day subacute exposure	Laboratory (mouse + BMDM)
Yibcharoenporn et al.	Metabolic profiling of PM _{2.5} - and PM ₁₀ -induced developmental toxicity in zebrafish embryos	<i>ACS Omega</i>	Toxicological (in vitro)	PM _{2.5} + PM ₁₀ jointly (500, 1000 ug/mL suspensions)	Thailand (laboratory)
Yuan et al.	Hallmarks of lung cancer driven by inhalable particulate matter	<i>Innovation (Camb)</i>	Critical review	PM _{2.5} , PM ₁₀	Global
Zhang et al.	Impact of pleural anthracosis severity on structure and function of subcarinal lymph nodes	<i>J Thorac Dis</i>	Retrospective clinical series	Deposited carbonaceous particles	China
Occupational & indoor					
Alqahtani et al.	Biomarker-based assessment of tobacco smoke exposure in Saudi females via urinary cotinine	<i>PLoS One</i>	Repeated-measures biomarker	Tobacco smoke (urinary cotinine)	Saudi Arabia
Cuerda Barcaiztegui et al.	Classroom ventilation management and user activities vs. indoor air quality in Madrid primary schools	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	PM (unspecified) + CO ₂ , indoor	Madrid, Spain
Gu et al.	Nitrogen-containing organic compounds in household solid-fuel combustion PM and residential urinary inflammatory and targeted amino-acid profiles: an exploratory personal exposure study	<i>Environ Int</i>	Metadata only	Personal PM from household solid-fuel combustion; N-containing organic speciation	Not stated (metadata only)
Herkert et al.	Contribution of indoor air and dust to children's PFAS exposure in the Faroe Islands	<i>Environ Sci Technol</i>	Field measurement	Indoor air and settled dust, PFAS	Faroe Islands
Iavazzo et al.	Particle characterization in an urban synthetic turf: insights into turf and vehicular wear	<i>Environ Geochem Health</i>	Field measurement	Fine, submicron and nano-sized particles; Fe, Si, Al, Br, Ni, Ag, Rh	Italy

First author	Focus	Journal	Design	Metric	Region
Illa et al.	How are we exposed to chemicals when drugs are smoked indoors?	<i>Environ Sci Technol</i>	Mechanistic indoor fate and exposure model	Indoor smoke aerosol; 7 compounds spanning volatility and hydrophobicity; first-, second- and third-hand pathways	Global (computational)
Jin et al.	Synergistic optimization of air change rate and air purifier type for bioaerosol control during dental ultrasonic scaling	<i>Build Environ</i>	Metadata only	Bioaerosol transport and deposition during dental ultrasonic scaling	China
Kaur et al.	Workplace graphene aerosol: personal EC exposure paired with in vitro lung toxicity	<i>NanoImpact</i>	Exposure assessment + in vitro	Airborne graphene / elemental carbon	Graphene production facility (Europe)
Li & Zheng	Size-resolved PM ₁₀ -bound PFASs: respiratory deposition and association with oxidative stress biomarkers among industrial waste recycling workers	<i>Toxics</i>	Observational - cohort	Size-resolved PM-bound PFAS (PM _{2.1} , PM _{2.1-10} and cut-point bins); ln-ADI	China
Li et al.	Study on the Non-uniform Distribution Characteristics of Oil Mist and Local Ventilation Method in Machining Workshop	<i>Aerosol Air Qual Res</i>	Metadata only	Oil-mist aerosol, spatial distribution	China
Liu & Hu 2026	Multivariable determinants of indoor PM _{2.5} and infiltration dynamics in 48 residential buildings across eight cities of Northern China	<i>Preprints.org</i>	Measurement campaign	Paired indoor/outdoor PM _{2.5} ; infiltration factor F _{inf} 0.28 +/- 0.15	Northern China (8 cities)
Makanjuola	Street sweepers at 20x the WHO PM _{2.5} guideline, with spirometry that does not follow	<i>Research Square (preprint)</i>	Observational - cross-sectional	Personal PM _{2.5} and PM ₁₀ (PMS5003 handheld, breathing height)	Sub-Saharan Africa
Nath et al.	Operating-room PM _{2.5} /PM ₁₀ and the effect of HEPA filtration and cautery	<i>J Anaesthesiol Clin Pharmacol</i>	Environmental surveillance campaign	PM _{2.5} , PM ₁₀ , CO ₂ , TVOC, HCHO	India
Niari et al.	Microplastic burden in beauty-salon indoor air and worker nasal lavage fluid	<i>J Environ Health Sci Eng</i>	Observational exposure study	Airborne microplastics (MPs/m ³); nasal lavage MPs/mL	Ardabil, Iran
O'Donnell et al.	Use of nicotine replacement therapy to reduce children's exposure to second-hand smoke in the home: a pilot randomised controlled trial in Scotland	<i>medRxiv (preprint)</i>	RCT process evaluation	PM _{2.5} (indoor)	Scotland
Ospalek et al.	Air quality assessment and potential inhalation exposure across equestrian facilities: a case study	<i>Atmosphere</i>	Cross-sectional environmental survey	PM ₁ , PM _{2.5} , PM ₁₀ , respirable and total dust at 1.5 m; NH ₃	Slovakia
Park et al.	Within-species phenotypic resistance rather than community turnover tracks antimicrobial usage in swine-farm bioaerosol	<i>Environ Pollut</i>	Environmental surveillance campaign	Viable bioaerosol abundance (flow cytometry + qPCR-corrected)	South Korea
Perna et al.	Impact of indoor air pollution on occupational exposure: rethinking the major health threat for airport workers	<i>Life (Basel)</i>	Review / synthesis	Ultrafine PM, black carbon, PAHs and trace metals in airport indoor microenvironments	Global / multi-region
Piedra et al.	Informal employment dwarfs cantonal satellite pollution as a respiratory-symptom predictor	<i>Front Public Health</i>	Observational - cross-sectional	Sentinel-5P NO ₂ , SO ₂ , O ₃ (cantonal percentiles; no PM product)	Latin America
Ramirez et al.	Airway diseases and occupational exposures in poultry industry workers: a comprehensive review	<i>Curr Allergy Asthma Rep</i>	Comprehensive review	Poultry-barn organic dust: feather, mite, feed dust, mould, endotoxin, NH ₃	Global / multi-region
Ranjbar et al.	Evaluating diesel exhaust exposure in California homes using 1-nitropyrene in indoor air and dust	<i>Preprints.org (preprint)</i>	Measurement campaign	1-Nitropyrene, 2-nitropyrene, 2-nitrofluoranthene; continuous indoor black carbon	USA
Sadki et al.	Real-time monitoring of formaldehyde exposure and acute health symptoms in healthcare workers in Morocco: a cross-sectional study	<i>Cad Saude Publica</i>	Cross-sectional survey	Shift-long airborne formaldehyde by calibrated semiconductor dosimeter (gas, not PM)	Morocco

First author	Focus	Journal	Design	Metric	Region
Salcines et al.	Monitoring and risk assessment of airborne graphene oxide released during 3D printing with polylactic acid filaments	<i>J Environ Health Sci Eng</i>	Field measurement	Nanoparticle number conc. + respirable fraction	Spain
Seyedrezaei et al.	Ecological momentary assessment and continuous IEQ sensing across home and office workspaces	<i>J Occup Environ Med</i>	Repeated-measures biomarker	PM _{2.5} , CO ₂ , TVOC, T, RH, noise	United States
Shadmani et al.	A comprehensive health profile and exposure-risk analysis in an iron ore mining workforce	<i>J Environ Health Sci Eng</i>	Cross-sectional survey	Job-role dust exposure fertile (proxy)	Iran
Siemens et al.	A SynAir-G position statement for the improvement of indoor air quality for children	<i>Indoor Air</i>	Consensus position statement	Indoor chemical and biological pollutants, bioaerosols, synergistic mixtures	Europe
Tan et al.	PM _{2.5} components, nutritional indices and physical activity vs lower eGFR in subway workers	<i>Nutrients</i>	Cross-sectional (cohort baseline)	PM _{2.5} component mixture	China
Thies et al.	London Underground commuting, cumulative PM _{2.5} and long-term mortality	<i>Int J Population Data Science</i>	Retrospective cohort	PM _{2.5} (metro, iron-rich; modelled cumulative)	London, UK
Tillery et al.	Indoor air nicotine, NO ₂ and COVID-19 outcomes in a Great Plains tribal community	<i>Indoor Air</i>	Cross-sectional survey	Indoor air nicotine + NO ₂ (1-week passive badges)	South Dakota, USA
Wood et al.	Influence of ventilation rate on the potential for indoor particulate matter exposure	<i>Indoor Environments</i>	Metadata only	Indoor PM exposure potential as a function of ventilation rate	United Kingdom
Xia et al.	Effectiveness of enhanced filtration interventions for indoor air quality in child-serving facilities	<i>Front Pediatr</i>	Trial / intervention	Indoor PM _{2.5} and PM ₁₀ , week-long pre/post continuous	USA
Yang et al.	Bioaerosols in the atmospheric environment of oil exploitation sites: spatial and temporal distribution, sources, influencing factors, health risks, and reduction efficiency of masks	<i>Environ Pollut</i>	Seasonal field sampling + filtration test	Culturable and total bioaerosol, particle-bound microorganisms	China
Yu et al.	Indoor particulate matter and associated microplastic exposure in the educational environment of Shenzhen	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	PM _{2.5} / PM ₁₀	China
Zhao et al.	Occupational dust exposure and thyroid nodule screening: a predictive modelling approach for early detection in coal mining communities	<i>J Int Med Res</i>	Cross-sectional + prediction model	Occupational coal dust (unmeasured)	China
Zhao et al.	CFD-based prediction of air change rate and particulate matter in a dynamically occupied operating room	<i>Indoor Air</i>	Physical exposure model	Indoor PM (Class 10,000 cleanroom)	Dalian, China
Zhou L et al.	Indoor burning of Arabian incense generates abundant ultrafine particles with strong oxidative potential	<i>Atmos Chem Phys</i>	Chamber emission characterisation	UFP number + PM mass, DTT/cellular OP	Chamber (Gulf practice)
van Rheenen et al.	Microbial contamination of air conditioning and humidifying systems in hospitals and the built environment	<i>J Hosp Infect</i>	Mapping review	Bioaerosol (bacteria, fungi, protozoa)	Multi-country
Sensing, forecasting & instrumentation					
Acton-Bond et al.	Rolling positive matrix factorisation for aerosol optical apportionment	<i>Atmos Environ</i>	Source-apportionment method	Absorption coefficient / optical apportionment	Europe
Adams et al.	Sampling sub-pixel scale variability in total column NO ₂ within a coastal urban environment	<i>Environ Sci Atmos</i>	Field campaign / remote sensing	Total column NO ₂	United States
Adeniji et al.	Twelve years of hourly criteria-pollutant and meteorological coupling in Abeokuta, Nigeria	<i>PLOS Glob Public Health</i>	Monitoring-record analysis	PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , CO, O ₃ , hourly 2012-2023	Sub-Saharan Africa
Akaezuwu et al.	SFFO-RL intelligent environmental monitoring: smart sensing networks with artificial intelligence	<i>Environ Monit Assess</i>	Algorithm development	Unspecified pollutant concentration	Not stated
Albashir 2026	A physics-informed digital twin framework for health vigilance and indoor air quality decision support	<i>Res Sq (preprint)</i>	Simulation (system-level)	Indoor PM and IAQ composite	Not stated
Alexa et al.	Simultaneous determination of gaseous nitric acid and particulate nitrate in ambient air using continuous samplers and chemiluminescent online analysers	<i>Talanta</i>	Multi-instrument field evaluation	Gaseous HNO ₃ and particulate NO ₃ ⁻ , 1 s chemiluminescence	Czech Republic

First author	Focus	Journal	Design	Metric	Region
Alharbi et al.	Multi-Technique Characterization of Atmospheric Aerosol Particles from the Coastal Area of Jeddah, Saudi Arabia: Morphology, Surface Chemistry, and Mineralogy	<i>Atmosphere</i>	Field campaign	PM _{2.5} morphology, surface oxidation state, crystalline phase (SEM/EDS/XPS/XRD)	Saudi Arabia
Anon et al.	Assessment Campaign of the Gravimetric Method (PM ₁₀ and PM _{2.5}): Analyzing Environmental Artifacts and Metrological Consistency for the 2030 Regulatory Challenge	<i>Atmosphere</i>	Interlaboratory comparison	Gravimetric PM ₁₀ and PM _{2.5} (EN 12341)	Italy
Aparicio et al.	Sensor system design and deployment for environmental and visitor monitoring in cultural heritage sites	<i>Sensors (Basel)</i>	Network deployment + field evaluation	Low-cost pollution monitors + T/RH, VWC, wind; static and robot/drone-borne	Europe
Astaneh et al.	Source-tagged PM tracers in random-forest bias correction of ozone design-value predictions	<i>Atmos Environ</i>	Metadata only (no abstract)	PM (unspecified), source-tagged tracers	USA
Ayele et al.	Fine-scale variations in black carbon and TSP chemical composition at a heavy-traffic intersection, Addis Ababa	<i>Environ Sci Atmos</i>	Metadata only (no abstract)	Equivalent black carbon + total suspended particulate organics	Ethiopia
Babu et al.	Capturing the invisible: hyperlocal variability of size-resolved particulate matter in a smart city	<i>Environ Monit Assess</i>	Mobile monitoring campaign	PM ₁ , PM _{2.5} , PM ₁₀ + size distribution	India
Bajaj & Settle	Rapid assessment of air pollution data quality using a composite index	<i>Environ Monit Assess</i>	Monitoring-record analysis	PM _{2.5}	USA
Balasubramanian et al.	Calibration model for PurpleAir-monitored PM _{2.5} across the United States	<i>Environ Pollut</i>	Calibration model comparison	PM ₁ , PM _{2.5} , PM ₁₀	United States
Band & Mohammadi 2026	Shift-aware dual-encoder transfer learning for short-horizon PM _{2.5} forecasting in data-limited target domains	<i>arXiv (preprint)</i>	Deep-learning model	Hourly regulatory PM _{2.5} , 10 US source sites -> 77 Taiwan target stations	Taiwan
Barkjohn et al.	Air Sensor Network Analysis Tool: an R-Shiny application for sensor QA/QC and reference comparison	<i>Atmosphere</i>	Software / tool description	PM _{2.5} , O ₃ (low-cost sensor vs regulatory reference)	United States
Benjamin et al.	Limitations of environmental extrapolation in low-cost carbon monoxide and PM _{2.5} sensors	<i>Sensors (Basel)</i>	Calibration model comparison	PM _{2.5} , CO	United States
Blaga et al.	City-scale calibration of a low-cost PM _{2.5} network for regulatory-compliant air-quality assessment	<i>PLOS Clim</i>	Sensor co-location + seasonal MLR calibration	PM _{2.5} , Clarity Node-S vs NEPA reference stations	Romania
Bobbia et al.	Multi-source dataset of NO ₂ and PM from mobile and fixed sensors in Rouen	<i>Data in Brief</i>	Network deployment / data descriptor	PM ₁₀ , PM _{2.5} , NO ₂ + RH/T/P	France
Buettner et al.	PIA v3.0.0: automated quality assurance and analysis for PINE ice-nucleating-particle data	<i>Atmos Meas Tech</i>	Software / tool description	INP concentration; optical particle size discrimination	Germany (multi-site PINE)
Burke et al.	Polar-grid nonparametric trajectory analysis for near-source pollutant attribution	<i>Atmospheric Environment</i>	Methods / near-road field data	PM _{2.5} , BC, NO _x	St. Louis, MO, USA
Cao et al.	On-road NO ₂ and particulate matter across urban road environments from city-scale mobile monitoring	<i>Atmos Environ</i>	Metadata only (no abstract)	PM _{2.5} , NO ₂	Not stated
Chen et al.	Study on the pollution characteristics of aerosols and VOCs along the lower reaches of the Yangtze River based on shipborne observations	<i>Atmos Environ</i>	Metadata only (no abstract)	Shipborne aerosol and VOC concentrations (not extractable)	China
Chen et al.	Lake Breeze Effects on the Urban Environment in Chicago	<i>Environ Sci Technol</i>	Sensor-network event analysis (lake-breeze compositing)	PM _{2.5} and O ₃ from the Microsoft Eclipse low-cost sensor network plus meteorological stations, 1 year	United States
Cho et al.	Greenspace, land use and PM _{2.5} across pandemic phases: a community scientist-led mobile sensing campaign	<i>Sci Total Environ</i>	Mobile monitoring campaign	PM _{2.5} (low-cost mobile sensors, street-level)	USA
Chung et al.	PurpleAir deployment trends, longevity and uncertainty across the United States, 2016-2025	<i>Sensors</i>	Network deployment + field evaluation	PM _{2.5} (PurpleAir PA-II)	United States (national)
Chyzykhov et al.	Particle-bound water and its bias on gravimetric PM mass	<i>J Hazardous Materials</i>	Laboratory metrology	PM constituents, artificial PM, ambient PM ₁ filters	Laboratory

First author	Focus	Journal	Design	Metric	Region
Clark & Bashar	How IoT enables the Protective Action Decision Model during a wildfire event	<i>Risk Anal</i>	Cross-sectional survey	Hyper-local real-time PM _{2.5} (IoT)	Oregon, USA
Colarusso et al.	Mobile urban sensing: spatio-temporal observability analysis and optimization	<i>Internet of Things</i>	Method / algorithm	Reconstructed pollutant concentration maps	Italy
Dal Porto et al.	Humidified CAPS-PM _{2.5} SSA with a revised truncation-correction methodology	<i>Aerosol Sci Technol</i>	Metadata only (no abstract)	Aerosol optical extinction and scattering, single-scattering albedo	Chamber / laboratory
DeMarsh et al. 2026 (backfill)	Community PurpleAir network versus regulatory monitors for PM _{2.5} characterisation in Fresno County	<i>Atmosphere</i>	Sensor co-location	PurpleAir PA-II PM _{2.5} vs regulatory FEM	North America
Diana et al.	Validated multi-technique workflow for inorganic characterisation of brake-wear PM ₁₀ and source-marker identification	<i>Talanta</i>	Bench / device experiment	WLTP brake-dynamometer PM ₁₀ , low-steel and NAO pads on cast-iron/coated discs	Italy
Eger et al.	Applicability of Compact Low- and Mid-Cost Sensors to Monitor Air Pollutant Emissions from Inland Ships	<i>Aerosol Air Qual Res</i>	Year-long co-location field evaluation	BC, ultrafine particle number, NO/NO ₂ , CO ₂ in ship plumes	Germany
Fahim & Karim	Cost-Aware Evaluation of Time-Series Foundation Models for Urban Air-Quality and Temperature Forecasting	<i>Res Sq (preprint)</i>	Benchmark / forecasting model comparison	Hourly PM _{2.5} concentration	Multi-country (29 cities)
Fawaz et al.	Surface cleaning of SAW-based microparticle sensors in a cascade impactor via SAW-induced droplet displacement	<i>Sensors (Basel)</i>	Instrument development	Size-segregated microparticles	France
Fei et al.	SHAP-interpretable machine learning integrating exposures and multi-omics reveals immune alterations and biomarkers in COPD-lung cancer comorbidity	<i>Front Immunol</i>	Machine-learning model	Particulate-matter pollution as a GBD exposure risk factor	Global
Filioglou et al.	Particle linear depolarisation ratio at 1565 nm from Halo Doppler lidar: smoke vs volcanic ash	<i>Atmos Chem Phys</i>	Multi-instrument field evaluation	Lidar depolarisation ratio and aerosol layer optical properties at 1565 nm	Finland
Gao et al.	Feasibility of an air sensor array for real-time detection and characterisation of VOCs	<i>Atmos Meas Tech</i>	Instrument development	VOC sensor response (non-PM)	Laboratory
Giannetti et al.	Ants as bioindicators of urban traffic pollution	<i>Environ Pollut</i>	Field study	Magnetic susceptibility of traffic-derived PM on ant exoskeleton	Parma, Italy
Gilardoni et al.	First continuous aerosol measurements at Testa Grigia at 3480 m a.s.l.: aerosol populations and dust transport dynamics in the southern European Alps	<i>Atmos Chem Phys</i>	Measurement campaign	Number size distribution + absorption coefficient, 1 h	Europe
Ginsburg et al.	Transferability of US PurpleAir correction equations worldwide; regional recalibration	<i>Aerosol Science & Technology</i>	Multi-site co-location evaluation	PM _{2.5} (PurpleAir vs reference)	Global, 6 continents
Gomez et al.	Machine- and deep-learning calibration of low-cost PM ₁₀ /PM _{2.5} nodes against a T640X reference	<i>Sensors</i>	Calibration model comparison	PM ₁₀ , PM _{2.5} (LCS vs T640X)	Bogota, Colombia
Guo et al.	Mie-theory optical calibration plus aerosol liquid water separation for RH correction	<i>ACS ES&T Air</i>	Mechanistic correction framework	PM _{2.5} (nephelometer + OPC vs EPA reference)	United States (EPA CSN sites)
Gäbel & Hertig	Recalibration of low-cost O ₃ and PM _{2.5} sensors: linking practices to recent air sensor test protocols	<i>Atmos Meas Tech</i>	Field campaign / remote sensing	PM _{2.5}	Europe
Habib et al.	Stable carbon isotopic composition of PM ₁ emissions from biomass and coal combustion under monitored combustion conditions	<i>Atmos Environ</i>	Metadata only (no abstract)	PM ₁ , stable carbon isotopes	Not stated
Hsieh et al.	Identifying winter PM _{2.5} sources in an urban environment using receptor modeling and nonparametric trajectory analysis	<i>Aerosol Air Qual Res</i>	Metadata only (no abstract)	PM _{2.5} , source-resolved	Not stated
Huang et al.	Metal ions and pH as controls on the optical properties of atmospheric brown carbon	<i>Atmos Pollut Res</i>	Chamber / laboratory	Brown carbon absorption spectra	Computational / laboratory
Jafari et al.	Joint optimization of bus-based mobile sensing and reference sensor deployment for urban air quality monitoring	<i>IEEE Open J Intell Transp Syst</i>	Modelling / inventory	Unspeciated PM	Europe

First author	Focus	Journal	Design	Metric	Region
Jin et al.	Emissions, chemistry and health impact of aged wildfire smoke in Missoula, Montana	<i>Atmos Chem Phys</i>	Measurement campaign	Hourly PM _{2.5} , CO, NO _x , O ₃ and 75 speciated VOCs; GEOS-Chem	North America
Kafi et al.	Evaluating the suitability of Sentinel-5P satellite data for air quality monitoring in Tehran	<i>Sci Total Environ</i>	Satellite retrieval evaluation	TROPOMI CO, NO ₂ , SO ₂ , O ₃ and UV Aerosol Index (no direct PM product)	Iran
Kalkavouras et al.	Long-term (10-year) trends in ultrafine particle number concentrations and size distributions in Athens	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Ultrafine particle number concentration + size distribution	Europe
Kang et al.	Emission characteristics and human health impacts of vehicle originated condensable particulate matter	<i>J Hazard Mater</i>	Multi-instrument field evaluation	Real-time condensable PM mass and number (PNU-CPM system); CPM emission factors	South Korea
Kirchhoff et al.	Evaluation of DMSO as working fluid in condensation particle counters	<i>Atmos Meas Tech</i>	Laboratory metrology	Ultrafine particle number	Laboratory
Kumpika et al.	Correction of humidity and sensor aging effects in low-cost PM _{2.5} light-scattering sensors	<i>Environ Pollut Bioavailab</i>	Field co-location + chamber	PM _{2.5} (PMS7003 vs BAM)	Thailand
Kumpika et al.	Joint humidity and sensor-aging correction for PMS7003 light-scattering sensors	<i>Environ Pollutants & Bioavailability</i>	Field co-location + chamber	PM _{2.5} (PMS7003 vs BAM)	Chiang Mai, Thailand
Kumpika et al.	Joint humidity and sensor-aging correction for low-cost light-scattering PM _{2.5} sensors	<i>Environ Pollut Bioavailab</i>	Mechanistic correction framework	PM _{2.5} (PMS7003 vs BAM reference)	Chiang Mai, Thailand
Laffolley et al.	A 3D-printed two-stage virtual impactor (uSPLIT) for radioactive aerosol size classification and direct analysis	<i>ACS Omega</i>	Instrument development	Aerodynamic diameter, 3 classes (>10, 1-10, <1 µm)	Japan
Le et al.	Operational performance of the Vaisala CL61 ceilometer for atmospheric profiling	<i>Atmos Meas Tech</i>	Multi-instrument field evaluation	Attenuated backscatter, depolarisation ratio	Finland / Germany
Lefevre et al.	Urban microclimate dataset from low-cost sensor networks in a hot and humid tropical climate: Reunion Island	<i>Data Brief</i>	Network deployment + field evaluation	None; T, RH, wind, GHI, grey-globe temperature at 10-min cadence	Reunion Island (France)
Legaria-Santiago et al.	Leveraging remote traffic data for local air pollutant estimation: a scenario-based machine learning study across London monitoring sites	<i>Atmosphere</i>	Machine-learning model	NO ₂ , PM ₁₀ , PM _{2.5} , O ₃ from remote traffic counts	London, UK
Lei et al.	Eight-year speciated PM ₁ record in Dublin: heating emissions vs trans-boundary transport	<i>Atmos Chem Phys</i>	Measurement campaign	PM ₁ (chemically speciated, ACSM + BC)	Europe
Li et al.	Summertime ice-nucleating particle concentrations over the Eurasian-Arctic Seas	<i>Atmos Chem Phys</i>	Field campaign / shipborne	INP number concentration (immersion mode)	Arctic Ocean
Li et al.	Spatiotemporally synchronized dual-wavelength photoacoustic-scattering measurement of aerosol optical properties	<i>Photoacoustics</i>	Laboratory metrology	Absorption & scattering coefficients, 450/532 nm	China
Li et al.	Crowdsourced, community-engaged VOC monitoring and source apportionment near a landfill	<i>Environ Sci Technol</i>	Field measurement	Gas-phase BTEX by EPA TO-17, not PM	USA
Li et al. 2026 (Front Pediatr)	Interpretable ML for 90-day paediatric allergic-rhinitis exacerbation from clinical + ambient exposure	<i>Front Pediatr</i>	Modelling / inventory	PM _{2.5} , PM ₁₀ , O ₃ , NO ₂ (same-day and 0-7 d lag)	North America
Li et al.	Early warning of cucumber angular leaf spot by estimating airborne pathogen aerosols with particulate matter sensors	<i>Plants (Basel)</i>	Network deployment + field evaluation	Low-cost PM sensor concentration as a physical proxy for pathogen aerosol load	China
Liu T et al.	Aerosol solute strength drives transition-metal-ion-catalyzed sulfate aerosol formation	<i>Environ Sci Technol</i>	Aerosol flow reactor + model implementation	Aqueous aerosol sulfate from SO ₂ + O ₂ with Fe(III)/Mn(II) catalysis; ionic strength threshold 16 mol/kg	Global / multi-region

First author	Focus	Journal	Design	Metric	Region
Liu et al. 2026 (ACP)	From kappa to chi: PartMC evaluation of hygroscopicity-based mixing-state retrieval	<i>Atmos Chem Phys</i>	Modelling / inventory	HTDMA kappa distributions; mixing-state index chi (50-250 nm)	Global / multi-region
Liu et al.	Polarization calibration of spaceborne lidar using dense cirrus-scattered solar background	<i>Atmos Meas Tech</i>	Method / algorithm	Aerosol depolarization ratio	Global
Liu et al.	LRM-YOLO: lightweight YOLOv10n model for forest fire smoke detection in UAV images	<i>Sensors (Basel)</i>	Method / algorithm	Smoke plume (image proxy)	China
Magdaong et al.	Developing criteria and an algorithm for a low-cost IoT-based air quality sensor network for near-road air quality monitoring	<i>ISPRS Ann Photogramm Remote Sens Spat Inf Sci</i>	Method / algorithm	PM (low-cost sensor network siting)	Philippines
Marcon et al.	Citizen science in air pollution monitoring: methodological approaches, effectiveness, and integration with health research	<i>Curr Environ Health Rep</i>	Critical review	Low-cost sensor networks, wearables, AI-assisted calibration	Global / multi-region
Masoom et al.	Long range transport of Canadian wildfire smoke to Europe in 2023: aerosol properties and spectral features of smoke particles	<i>Atmos Chem Phys</i>	Field campaign / remote sensing	AOD; Angstrom exponent; single scattering albedo; volume size distribution	Multi-country
Mohapatra et al.	IoT-machine learning proof-of-concept framework for real-time urban particulate matter prediction	<i>IEEE Access</i>	Proof-of-concept deployment	PM ₁ / PM _{2.5} / PM ₁₀ (PMS7003)	India
Moteki et al.	Constraining the refractive index of fractal-like aggregates from complex forward-scattering amplitudes of mass-classified aerosol particles: application to black carbon samples	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Complex refractive index of mass-classified black carbon aggregates	Not stated
Neuberger et al.	Importance of hydrated aerosol particles for aerosol-fog relationships in the Italian Po Valley	<i>Atmos Chem Phys</i>	Field campaign + LES modelling	Hygroscopicity parameter kappa; hydrated vs activated particle diameter	Italy
Ng et al.	Impacts of droughts on biomass burning emissions, air quality, and public health in the Amazon	<i>Atmos Chem Phys</i>	Satellite + chemical transport model	Biomass-burning PM _{2.5} and CO (GFEDv4s / GEOS-Chem HP)	Amazon, Brazil
Nimmala et al.	Drone and ground sampling of transgenic switchgrass pollen with Lagrangian dispersal modelling	<i>Environ Monit Assess</i>	Release-recapture field experiment	Bioaerosol (OFP-tagged pollen), high/low-volume samplers	Oliver Springs, TN, USA
Nurseitov et al.	Design, deployment and field evaluation of a low-cost IoT monitoring network for urban particulate matter	<i>Information</i>	Network deployment + field evaluation	PM _{2.5} / PM ₁₀ (ZPHS01B)	Kazakhstan
Orr et al. 2026 (JMIR)	Feasibility and acceptability of a digital in-home air quality monitor in families of children with asthma	<i>JMIR Pediatr Parent</i>	Sensor deployment (pilot)	Indoor PM (consumer digital monitor)	North America
Oybek et al.	Chamber quantification of T/RH sensitivity across three low-cost sensor chemistries	<i>Am J Applied Science & Technology</i>	Controlled chamber factorial	PM _{2.5} (OPC), NO ₂ (MOx), CO ₂ (electro-chemical)	Laboratory
Park et al.	Diagnosing explainability of machine learning: transformer-based PM _{2.5} estimation in Seoul	<i>Atmos Environ</i>	Metadata only (no abstract)	PM _{2.5} / PM ₁₀	Republic of Korea
Perez-Pastor et al.	Robust quantification of anhydrosugars and reference material evaluation for biomass burning tracers	<i>Environ Sci Pollut Res Int</i>	Laboratory metrology	PM biomass burning tracers	Spain
Perisic et al.	Explainable AI resolves ten recurrent environmental settings behind urban PM ₁₀ variability at a single regulatory station	<i>Sci Total Environ</i>	Algorithm development	Regulatory PM ₁₀ /PM _{2.5} + NO ₂ , SO ₂ , O ₃ , meteorology, 2020-2023	Europe
Qian et al.	AutoML multi-stage calibration for indoor low-cost PM _{2.5} sensors	<i>Atmospheric Measurement Techniques</i>	Chamber co-location, AutoML calibration	PM _{2.5} (indoor low-cost sensors)	Indoor chamber
Qian et al.	Enhancing accuracy of indoor air quality sensors via automated machine learning calibration	<i>Atmos Meas Tech</i>	Chamber co-location, AutoML calibration	PM _{2.5} (two LCS models vs reference)	Chamber
Rapp et al.	Curve fitting algorithm for multimodal particle size distributions: a theoretical basis	<i>Atmos Meas Tech</i>	Measurement campaign	Mobility particle size distributions, number-based	USA
Rehbein et al.	Source identification and triangulation from a low-cost sensor network in Ile-Ife	<i>Clean Air J</i>	Source apportionment	PM _{2.5} , PM ₁₀ , particle number concentration	Ile-Ife, Nigeria

First author	Focus	Journal	Design	Metric	Region
Reis	Low-power embedded sensor node for environmental monitoring with on-board machine-learning inference	<i>Sensors</i>	Simulation (system-level)	PM + gas, multi-parameter node	Simulation
Reis	Low-power sensor node with on-board quantised neural inference and adaptive LoRaWAN	<i>Sensors</i>	Simulation (system-level)	PM ₁ , gas, T/RH (embedded node)	Simulation
Roska et al.	Beyond targeted calibrations: comprehensive quantification of oxygenated organic compounds in urban atmospheres	<i>Environ Sci Technol</i>	Instrument development	OVOC precursors of SOA; NH ₄ ⁺ -adduct CIMS	United States + Germany
Rubio et al.	LoRa-connected micromobility pollution monitor for the Internet of Vehicles	<i>IEEE Consum Electron Mag</i>	Proof-of-concept deployment	PM _x , CO ₂ , meteorology	Cartagena, Spain
Saad et al.	Short-term monitoring of air pollutants at the port-urban exposure zone, Alexandria Port	<i>Environ Monit Assess</i>	Portable sensor monitoring campaign	PM _{2.5} , PM ₁₀ , CO ₂ , CO, NO ₂ , SO ₂ , NH ₃ and VOC by portable multi-gas monitor and particulate sensors	Egypt
Samburova et al.	Effects of western US wildfire smoke on urban air quality in northern Nevada	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	PM _{2.5} only (wildfire-influenced)	Nevada, USA
Sanders et al.	Nonionic surfactant composition and hygroscopic growth of supermicron NaCl particles	<i>Aerosol Sci Technol</i>	Chamber / laboratory	Hygroscopic growth factor, supermicron NaCl	Computational / laboratory
Saravagna Sai Lakshmi et al.	CORTA-Net: hybrid deep learning with attention-based interpretability for PM _{2.5} forecasting	<i>J Vis Exp</i>	Machine-learning model	Hourly regulatory-station PM _{2.5} plus MODIS fire counts	India
Senarathna et al.	Enhancing low-cost PM _{2.5} air quality monitoring sensors through sensor calibration using regression and echo-state network models	<i>Environ Monit Assess</i>	Sensor co-location + regression / echo-state network calibration	PM _{2.5} , TSI BlueSky vs reference, wet and dry season	Sri Lanka
Sengupta et al.	A temporal deep learning framework for calibration of low-cost air quality sensors	<i>arXiv (preprint)</i>	Sensor co-location + LSTM sequence calibration	PM _{2.5} , PM ₁₀ and NO ₂ , LCS vs co-located reference (OxAria)	United Kingdom
Shirkhani et al.	Design and development of a high-throughput multipattern aerosol deposition device (MARTHA) for respiratory toxicity and health-risk assessment	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Aerosol deposition (unspecified)	USA
Shlipak et al.	Spatial and temporal trends of ambient PM _{2.5} and source-apportioned black carbon in Addis Ababa	<i>ACS EST Air</i>	Network deployment + field evaluation	PM _{2.5} + black carbon (BC)	Addis Ababa, Ethiopia
Silva et al.	Low-cost sensors for indoor air quality monitoring: systematic review of accuracy, applications and limitations	<i>J Air Waste Manag Assoc</i>	Systematic review (PRISMA)	PM (OPC) and CO ₂ (NDIR)	Multi-country
Silva et al.	Accuracy, applications and limitations of low-cost sensors for indoor air quality	<i>J Air Waste Manag Assoc</i>	Systematic review (PRISMA)	PM _{2.5} , PM ₁₀ , CO ₂ (OPC and NDIR)	Global
Siu et al.	Disparities in fine particulate matter (PM _{2.5}) monitoring capacity in eastern Canada	<i>Environ Res Health</i>	Spatial analysis / GIS	PM _{2.5}	North America
Somnugpong	When seasonal refinement helps (and does not help) Stineman imputation of gappy air-quality series	<i>Environ Monit Assess</i>	Modelling / inventory	PM _{2.5} (UCI Air Quality; Beijing PM _{2.5})	China
Suryawanshi et al.	Spatiotemporal variation and hotspot identification of PM _{2.5} from low-cost sensors	<i>Int J Environ Res</i>	Metadata only (no abstract)	PM _{2.5} (low-cost sensors)	India (urban)
Tian et al.	When lidar-based CCN profiles fail: a heterogeneity index for vertically layered aerosol	<i>Atmos Meas Tech</i>	Measurement campaign	CCN number concentration; lidar ratio and depolarisation ratio	North America
Tsai et al.	Vision-based vertical PM _{2.5} retrieval from a 5G-connected UAV with generative-AI interpretation	<i>IEEE Sens J</i>	Proof-of-concept deployment	PM _{2.5} (dark-channel-prior transmittance proxy)	Taiwan
Vivanco et al.	Assessing the responsiveness of an air quality model to real-world emission changes for air quality planning	<i>Environ Pollut</i>	CTM evaluation against observation-based counterfactuals (COVID-19 emission shift)	Modelled NO ₂ and O ₃ response to a real-world emission reduction; PM not reported in abstract	Europe
Wang et al.	Traffic-induced environmental health assessment at intersections using roadside LiDAR	<i>Environ Monit Assess</i>	Modelling / inventory	Unspecified PM	China

First author	Focus	Journal	Design	Metric	Region
Wang et al.	Measuring heavy metals in atmospheric particulate matter: operational definitions, method comparability and environmental interpretation	<i>Atmos Environ</i>	Method review	Heavy-metal content of PM	Not stated
Wang et al.	Remote calibration of a nationwide dense low-cost sensor network for hyperlocal air surveillance	<i>Environ Sci Technol</i>	Machine-learning model	Low-cost sensor PM _{2.5} vs regulatory reference monitors	USA
Wang et al.	Perchloric acid (HClO ₄) drives atmospheric new particle formation enhanced by dimethylamine, ammonia and sulfuric acid	<i>Atmos Chem Phys</i>	Kinetic model + data reanalysis	Cluster formation rate for PA-DMA / PA-NH ₃ / PA-SA nucleation	Arctic / coastal marine
Wei et al.	Stacked Transformer-XGBoost model with SHAP interpretability for air quality index prediction	<i>iScience</i>	Machine-learning model	AQI driven by PM _{2.5} , PM ₁₀ , NO ₂	Linfen, China
Werderman et al.	Laboratory evaluation of Sensirion SEN66, Piera IPS-7100, PurpleAir PIXEL and OPC-N3 low-cost aerosol sensors across aerosol types and concentration ranges	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Chamber PM across aerosol types (not extractable)	Not stated
Wongsasuluk et al.	PM _{2.5} exposure and respiratory health risk for vulnerable populations in an urban underpass: a predictive MLR approach, Chiang Mai	<i>Environ Sci Pollut Res Int</i>	Network deployment + statistical model	PM _{2.5} and PM ₁₀ from five DustBoy low-cost stations, 10-min cadence	Thailand
Wu & Qu	Interpretable short-term PM _{2.5} forecasting using meteorology-pollution coupling across Beijing stations	<i>Environ Monit Assess</i>	Method / algorithm	PM _{2.5}	Beijing
Xiong et al.	In situ real-time determination of SO ₂ photochemical oxidation in nanoscale sea salt aerosols based on dark-field microscopy	<i>Atmos Chem Phys</i>	Laboratory instrument development	Single-nanoparticle hygroscopic growth factor; first-order rate constant	China
Yan	Review of machine-learning calibration for low-cost air quality sensor networks	<i>Sensor Review</i>	Narrative review	PM, O ₃ , NO ₂ (optical, electrochemical, MOx, NDIR)	Global
Yang et al.	Diurnal sea breeze worsens coastal air quality and complicates monitoring of remote North Atlantic air	<i>Atmos Chem Phys</i>	Field campaign / remote sensing	PM _{2.5} ; PM ₁₀ ; total aerosol number	Europe
Yang et al.	Dust transport and local anthropogenic emissions differently shape ice-nucleating particles in an industrial urban atmosphere	<i>Atmos Chem Phys</i>	Measurement campaign	Immersion-mode INP concentration, size distribution, PM _{2.5} PMF factors	China
Yaqoob et al.	Multi-city bi-pollutant ML calibration of PurpleAir and Alphasense OPC-N3	<i>IEEE Access</i>	Calibration model comparison	PM _{2.5} , PM ₁₀ (PurpleAir, OPC-N3 vs TEOM/FIDAS)	Antwerp, Oslo, Zagreb
Yaqoob et al.	Semiparametric hybrid air quality forecasting using Prophet ensemble and feature selection	<i>Sci Rep</i>	Machine-learning model	PM _{2.5}	South Korea
Yee et al.	UCB-GLOBES: open-access GC/EI-MS spectral library for identified and unidentified organic aerosol	<i>Atmos Meas Tech</i>	Measurement campaign	Semi-volatile organic aerosol (TMS derivatives, GC/EI-MS)	Global / multi-region
Yohanna & Lim	Regional and seasonal variation of PM _{2.5} and ozone across Malaysia using XGBoost and SHAP	<i>Chemosphere</i>	Gradient-boosting model + SHAP interpretability	Daily PM _{2.5} and O ₃ from the DOE monitoring network plus ERA5 reanalysis, 2018-2023	Malaysia
Yu et al.	Machine learning-based prediction of occupant symptom risk using indoor and outdoor environmental quality data in university dormitories	<i>Indoor Air</i>	Continuous IEQ monitoring + supervised classification	Five indoor and five outdoor environmental parameters incl. PM; distributional shape features	Italy
Yukhymchuk et al.	Low-cost particulate matter sensor performance under polluted conditions in Northeast China	<i>Aerosol Air Qual Res</i>	Sensor co-location	Low-cost PM sensor vs reference	China
Zeng et al.	Aerosol vertical distributions determine the seasonal relationships between column AOD and surface PM _{2.5} over Europe	<i>Atmos Environ</i>	Metadata only (no abstract)	Column AOD vs surface PM _{2.5} (not extractable)	Europe
Zhang et al.	High frequency of urban new particle formation on the Tibetan Plateau: quantifying formation rates, growth rates and CCN production	<i>Atmos Chem Phys</i>	Supersite observation	Particle number size distribution, CCN	China

First author	Focus	Journal	Design	Metric	Region
Zhang et al.	Wearable device for direct assessment of particle deposition in the human respiratory system	<i>Environ Sci Technol</i>	Field measurement	Paired inhaled and exhaled size-resolved PM, PM ₁ -PM ₁₀	China
Zhang et al.	Impact of aerosol-type assumption on Landsat 8 atmospheric correction over land	<i>Atmos Meas Tech</i>	Model intercomparison	Aerosol optical depth	Global
Zhang et al.	Air pollution transport from absorbing aerosol index across multiple payloads	<i>Atmos Environ</i>	Metadata only (no abstract)	Absorbing aerosol index	Global
Zhang et al.	Buffering of atmospheric nanoparticle growth by temperature-dependent shifts in molecular composition, volatility and diffusivity	<i>Atmos Chem Phys</i>	Kinetic model + data reanalysis	Nanoparticle growth rate (nm/h) vs condensable organic vapour	Global / multi-region
Zhang et al.	Contrasting nighttime heterogeneous and daytime photochemical aging drive the optical evolution of black carbon	<i>Atmos Chem Phys</i>	Field campaign, single-particle instrumentation	rBC mass, core MMD, coating-to-core mass ratio, MAC at 532 nm (SP2 + SPAMS)	China
Zhao et al.	WHA air-LSTM: composite morbidity-driven pollution index for next-day respiratory outpatient forecasting	<i>Biomed Environ Sci</i>	Modelling / inventory	Composite air pollution index (WHA air), city-specific ER curves	China
Zhao et al.	Monte-Carlo inversion of aerosol complex refractive index from multi-wavelength lidar plus size distribution	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Complex refractive index, multi-wavelength lidar backscatter, particle size distribution	Not stated
Zheng et al.	Augmented transformer framework for PM _{2.5} chemical composition via one-step LDI single-particle aerosol mass spectrometry	<i>Atmos Environ</i>	Metadata only (no abstract)	PM _{2.5} / PM ₁₀	China
Zhivkov et al.	Distance-dependent calibration-transfer uncertainty in a hybrid reference/low-cost network	<i>Atmosphere</i>	Spatial cross-validation, operational network	PM _{2.5} , PM ₁₀ , NO ₂ , O ₃ (AirThings vs EEA reference)	Sofia, Bulgaria
Zhou et al.	Reliability-constrained identification of primary and elevated aerosol structures from a single tilted Mie lidar under dust-influenced conditions	<i>Atmos Environ</i>	Metadata only (no abstract)	Backscatter / aerosol layer structure	Not stated
Zundel et al. 2026	Wearable personal PM _{2.5} monitoring during wildfire smoke episodes in Metro Detroit	<i>medRxiv</i>	Proxy validation vs personal exposure	Personal PM _{2.5} (AirBeam3) vs PurpleAir	Detroit, USA
d'Hondt et al.	Neural downscaling of air-quality simulations requires structural correction before spatial refinement	<i>Research Square (preprint)</i>	Deep-learning model	PM _{2.5} , PM ₁₀ , NO ₂ , O ₃	Europe
Exposure assessment & modelling					
Abadi et al.	Future projections of surface dust concentrations in major dust source regions of the Mesopotamian plains	<i>Atmos Res</i>	Metadata only (no abstract)	Projected surface mineral dust (not extractable)	Iraq
Adeniji et al.	Long-term air quality and meteorological coupling dynamics in Abeokuta, Nigeria	<i>PLOS Glob Public Health</i>	Monitoring-record analysis	PM _{2.5} , PM ₁₀ + 4 gases, hourly	Nigeria
Agpaoa et al.	Occurrence, temporal variation and meteorological drivers of airborne fungal spores in Laoag City, northern Philippines	<i>Environ Monit Assess</i>	Bioaerosol monitoring campaign	Volumetric spore counts (spores/m ³), six spore types	Philippines
Alaimo et al.	Elemental composition and Pb isotopic signatures in pine needles (<i>Pinus pinea</i> L.) from the industrialized Milazzo area	<i>Atmosphere</i>	Deposition sampling	Atmospheric trace-element deposition; Pb isotope source apportionment	Milazzo, Italy
Albugami	Dust event characteristics, transport pathways and source regions over Riyadh	<i>Atmosphere</i>	Field campaign / remote sensing	Coarse-mode dust (AOD, Angstrom exponent)	Riyadh, Saudi Arabia
Alshehri & El-Hamid 2026	Sentinel-5P and MODIS land-surface coupling of pollutant fields over Riyadh, 2018-2025	<i>Research Square (preprint)</i>	Spatial analysis / GIS	TROPOMI NO ₂ /SO ₂ /CO/O ₃ + Absorbing Aerosol Index; MODIS LST/NDVI	Middle East & N. Africa
Anon et al.	Severe winter PM _{2.5} pollution in remote Canadian mountain communities: Insights into the role of condensable particulate matter under subzero conditions	<i>Environ Int</i>	Field campaign	PM _{2.5} including condensable fraction, subzero conditions	Canada

First author	Focus	Journal	Design	Metric	Region
Anon et al.	Metal Processing and Combustion Dominate Source-Specific PM _{2.5} Oxidative Potential in a National Shipbuilding Complex	<i>Atmos Pollut Res</i>	Field campaign	Source-apportioned PM _{2.5} oxidative potential	South Korea
Ashta et al.	Wet and dry atmospheric deposition of microplastics at urban, suburban, rural and mountainous sites	<i>Atmos Chem Phys</i>	Deposition sampling	Microplastics 20-215 um	Switzerland
Awan et al.	BRIQ-PM _{2.5} : quantile-mapping attribution of industrial excess PM _{2.5} across Pakistan	<i>Aerosol Air Qual Res</i>	Spatial analysis / GIS	Satellite-derived surface PM _{2.5}	Pakistan
Baldo et al.	Chemical and mineralogical composition of Southern African mineral dust	<i>Atmos Chem Phys</i>	Laboratory metrology	Mineral dust (Si/Al, (Ca+Mg)/Al, K/Al, clay)	Southern Africa
Bhardwaj & Subramanian	Identifiability-aware source apportionment for sparse city-scale sensor networks	<i>arXiv (preprint)</i>	Source apportionment	PM (wind-conditioned lagged inverse problem)	Method (city-scale, synthetic + real)
Blanco et al.	Influence of on-road mobile monitoring design on ultrafine particle exposure models and cognitive health inferences	<i>J Expo Sci Environ Epidemiol</i>	Design resampling + cohort analysis	UFP particle number concentration	United States
Blanco-Alegre et al.	Emissions from prescribed fires in two Mediterranean shrublands: chemical and morphological analysis	<i>Atmos Chem Phys</i>	Field measurement	Gravimetric PM _{2.5} emission factors + OC/EC, ions, SEM morphology	Spain
Blanco-Villafuerte et al. 2026 (backfill)	Hybrid low-cost-sensor, satellite and CTM modelling of ambient PM _{2.5} in Peru	<i>Curr Environ Health Rep</i>	Machine-learning model	LCS + satellite AOD + CTM hybrid, 5 km daily	Latin America
Boogaard et al.	Advancing long-term outdoor air pollution exposure assessment for health studies	<i>Curr Environ Health Rep</i>	Programme-level synthesis of exposure-model comparisons	Long-term outdoor PM _{2.5} , NO ₂ and BC; mobility- and infiltration-adjusted exposure models	Global / multi-region
Buhendwa et al.	Modeling the coupled PM _{2.5} -ozone system in China: chemical mechanisms, persistent model biases, and future perspectives	<i>Environ Pollut</i>	Review / synthesis	PM _{2.5} and O ₃ (chemical transport models)	China
Cai et al.	Molecular characteristics and formation pathways of particulate organosulfur compounds in Shanghai	<i>Atmos Chem Phys</i>	Measurement campaign	Particulate organosulfates, FT-ICR MS	China
Cai et al.	Effect of ammonia on reactive oxygen species generation in secondary organic aerosols	<i>Atmos Environ</i>	Metadata only (no abstract)	SOA reactive oxygen species yield (not extractable)	Not stated
Chai et al.	Role of ammonia, aerosol pH and liquid water content in secondary inorganic aerosol formation in suburban Nanjing	<i>Environ Pollut</i>	Measurement campaign	PM _{2.5} secondary inorganic aerosol (mean 34.4 µg/m ³)	Nanjing, China
Chebaicheb et al.	A phenomenology of submicron organic aerosol components in urban France from long-term online observations and modelling	<i>Environ Int</i>	Source apportionment	PM ₁ organic aerosol, ACSM, rolling PMF (HOA/BBOA/COA/LO-OOA/MO-OOA)	France
Chen et al.	Inventory uncertainty and transboundary transport of biomass-burning black carbon in southern China from radiocarbon plus modelling	<i>Environ Sci Technol</i>	CTM simulation + source attribution	Black carbon, 14C source-resolved	China
Chen et al.	Source-specific oxidative potential of PM _{2.5} in Xi'an: roles of water-soluble metals revealed by DTT assay and interpretable machine learning	<i>Toxics</i>	Source apportionment	Volume-normalised DTT oxidative potential (DTTv) of ambient PM _{2.5}	China
Chi et al.	Heterogeneous nitrosation of amines driven by dinitrogen tetroxide: a missing source of particulate nitrosamines	<i>Atmos Chem Phys</i>	Born-Oppenheimer molecular dynamics + metadynamics	Particulate nitrosamines from methylamine / dimethylamine + N ₂ O ₄ at the air-water interface	Global (computational)
Chillrud et al.	pcpr: principal component pursuit for exposure pattern recognition and outlier detection	<i>Environ Epidemiol</i>	Software / tool description	26 PM _{2.5} constituents	United States
Chuang et al.	Simulating PM _{2.5} in southern Taiwan under high-pressure return flow during the Kao-Ping Experiment	<i>Atmos Res</i>	Metadata only (no abstract)	PM _{2.5} (not extractable)	Kao-Ping, Taiwan

First author	Focus	Journal	Design	Metric	Region
Cui et al.	Daily 1 km satellite-based PM _{2.5} exposure estimates in Bangladesh, 2020-2025	<i>Environ Pollut</i>	Satellite-surface fusion, machine learning	Gap-filled MA-IAC AOD -> ground PM _{2.5} (LightGBM + TabNet)	Bangladesh
Defrance 2026	Beyond the fire perimeter: the smoke footprint that constrains outdoor activity dwarfs the burned area of the July 2026 Gironde-Landes wildfire episode	<i>Research Square</i>	Modelling / inventory	CAMS analysis PM _{2.5} + 40 background monitors	Gironde-Landes, France
Delgado Peralta et al.	Improving PM _{2.5} simulations in metropolitan Sao Paulo via PMF-based emission inventory adjustment	<i>Atmos Environ</i>	Metadata only (no abstract)	PM _{2.5} / PM ₁₀	Brazil
Deng et al.	Impact of black carbon on daytime valley and slope winds in idealised WRF-Chem simulations	<i>Atmos Chem Phys</i>	Theory / simulation	Black carbon, radiative absorption	Idealised terrain
Dias et al.	Micro-scale spatial patterns of PM _{2.5} across urban green and non-green areas in an arid city	<i>Bull Environ Contam Toxicol</i>	Measurement campaign	Mobile-monitored PM _{2.5} along a fixed transect	Argentina
Ding et al.	Urban-rural balanced PM _{2.5} estimation corrects station-density bias	<i>Geo-spatial Information Science</i>	Machine-learning model	PM _{2.5} from AOD + reanalysis + ground	China, 2015-2021
Do et al.	Field-constrained diurnal viscosity variability of urban PM _{2.5} and implications for N2O5 uptake	<i>Environ Sci Technol</i>	Laboratory metrology	PM _{2.5} phase state / viscosity	Ansan, South Korea + 14 megacities
Dovrou et al.	Wintertime wildfires reshape urban atmospheric chemistry: evidence from the January 2025 Los Angeles fires	<i>Environ Sci Atmos</i>	Metadata only (no abstract)	Wildfire-influenced urban aerosol (not extractable)	USA
Duan et al.	Regime-dependent aerosol pH variability in ammonia-rich urban Beijing from explainable machine learning	<i>Atmos Chem Phys</i>	Machine-learning model	PM1 mass and composition + thermodynamic pH	China
Environ Sci Atmos	Multi-scale variability and lagged meteorological controls of PM _{2.5} in Delhi from wavelet coherence and causality analysis	<i>Environ Sci Atmos</i>	Monitoring-record analysis	PM _{2.5}	India
Fan et al.	Santa Ana winds control wildfire smoke transport and related health impacts in California during January 2025	<i>Atmos Environ</i>	Chemical transport modelling	Wildfire-sourced PM _{2.5}	North America
Fan et al.	Multiple amides synergistically enhance sulfuric acid-ammonia nucleation by an order of magnitude over single-base systems	<i>iScience</i>	Chemical transport modelling	Modelled SA-NH3-MPA / MAA cluster formation (new particle formation)	China
Florencio et al.	Comprehensive analysis of PM ₁₀ in the Metropolitan Area of Sao Paulo, Part I	<i>Atmos Environ</i>	Metadata only (no abstract)	PM ₁₀	Brazil
Goncalves et al.	Extreme Saharan dust outbreak: air quality crisis in NW Spain	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Saharan mineral dust (not extractable)	NW Spain
Gong et al.	Regional analysis of urban physical factors influencing air pollution: evidence from Seoul's living areas	<i>Res Sq (preprint)</i>	Spatial regression (GWR)	Composite AQ index (incl. PM _{2.5} , PM ₁₀)	East Asia
Gould & Jagarnath	Satellite monitoring for air quality and public health research in Africa: a scoping review	<i>Research Square (preprint)</i>	Scoping review	Satellite-derived PM _{2.5} and AOD	West Africa
Guo et al.	Meteorology-driven long-range transport to an ultra-low emission region during the monsoon	<i>Atmos Environ</i>	Metadata only (no abstract)	Not extractable from registered metadata	China
Guo et al.	Resolving systematic errors in sulfate source apportionment: a field-validated kinetic isotope fractionation framework	<i>Atmos Chem Phys</i>	Kinetic model + field validation	Seasonal PM _{2.5} sulfate; stable-isotope Bayesian mixing plus Rayleigh fractionation	China
Hao et al.	Source-specific black carbon responses to short-term interventions and cross-year emission changes in coastal Qingdao	<i>Atmos Environ</i>	Metadata only (no abstract)	Black carbon, source-resolved	China
Helbich et al.	Socioeconomic patterning of the urban external exposome depends on how you define it	<i>Environ Int</i>	Observational - cross-sectional	PM and NO2 within an 8-exposure external exposome	Europe
Hernandez Pardo et al.	Downward transport of tropical upper-tropospheric aerosols: multi-year idealized simulations	<i>Atmos Chem Phys</i>	Global chemistry-climate tracer simulation	19 idealised aerosol tracers, 20-100 nm, released in the tropical upper troposphere	Global / multi-region

First author	Focus	Journal	Design	Metric	Region
Hernandez et al.	Urbanization and humidity impact atmospheric deposition of anthropogenic microparticles (including microplastics) in a humid subtropical city	<i>Atmos Environ</i>	Deposition sampling	Anthropogenic microparticle deposition flux	North America
Hopke et al.	A PM _{2.5} magnitude-rank index for summarizing annual concentration-frequency structure	<i>Environ Monit Assess</i>	Metric development + retrospective application	PMRI: largest k with $\geq k$ days at $\geq k \mu\text{g}/\text{m}^3$	United States
Hu et al.	Hourly 1-km seamless PM _{2.5} from ground, satellite AOD and WRF-Chem fusion	<i>Geo-spatial Information Science</i>	Machine-learning model	PM _{2.5} , AOD, WRF-Chem	China
Ianiri et al.	Street- vs roof-level deposition fluxes of POPs and metals, normalised to sedimentable PM	<i>Environ Pollut</i>	Deposition sampling	Settleable particulate matter, monthly deposimeter collection	Italy
Ibrahim	Air pollution status and temporal patterns in Bukit Rambai and Muar, 2013-2017	<i>Research Square (preprint)</i>	Ecological panel	PM ₁₀ , O ₃ , CO, SO ₂ , NO ₂	Malaysia
Isworo & Oetari	A risk-integrated framework for industrial air quality assessment linking dispersion modeling and health-based environmental pressure indicators	<i>Environ Sci Pollut Res Int</i>	Modelling / inventory	Gaussian-plume PM _{2.5} and PM ₁₀ ; hazard-quotient-weighted AEPI	Indonesia
Jin et al.	Hourly 2 km Himawari-8 aerosol fine-mode fraction retrieval improves PM _{2.5} mapping	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Satellite AOD fine-mode fraction to surface PM _{2.5}	China
Joshi et al. 2026	Environmental drivers of foliar particulate accumulation, leaf structural and biochemical responses in urban trees in Delhi	<i>Research Square</i>	Deposition sampling	Leaf-deposited coarse (10-100 μm) and fine (2.5-10 μm) PM	Delhi, India
Kalumbi et al.	Age-specific time-spent patterns among residents living near gold mine tailings storage facilities in the West Rand, South Africa	<i>J Expo Sci Environ Epidemiol</i>	Cross-sectional time-activity survey	None measured; indoor/outdoor time fractions as exposure-weighting inputs	South Africa
Kan et al.	Short-term observations of airborne microplastics in Phnom Penh: concentrations, aerodynamic size distribution, and polymer composition	<i>Atmosphere</i>	Measurement campaign	Airborne microplastics in TSP and PM _{2.5} (uFTIR-ATR)	Phnom Penh, Cambodia
Kang et al.	Characteristics and dry deposition fluxes of particulate parent, alkyl, and nitro-polycyclic aromatic hydrocarbons in the Bohai Sea Strait, China	<i>Atmos Environ</i>	Metadata only (no abstract)	Particle-bound parent / alkyl / nitro-PAH (not extractable)	China
Kim et al.	Wildfire emissions suppress downwind anthropogenic sulfate via OH scavenging	<i>Environmental Pollution</i>	CTM simulation + source attribution	PM _{2.5} sulfate	Northeast Asia (Siberian fire, Jul 2014)
Kim et al.	Application of machine learning-based wind corrections to coupled WRF-CMAQ simulations: a comparison of dynamic assimilation and static replacement	<i>Atmos Environ</i>	Metadata only (no abstract)	Not extractable	Not stated
Kommula et al.	Comparable organic aerosol mass enhancement during photochemical and dark oxidation of biomass burning emissions	<i>Environ Sci Technol</i>	Chamber emission characterisation	PM1 organic aerosol / BB-SOA	Chamber (Finland)
Lazo et al.	Biomonitoring trace metal deposition with <i>Ligustrum lucidum</i> : Albanian fire incidents	<i>J Environ Qual</i>	Deposition sampling	Fire-derived trace-metal deposition on leaves (ICP-MS)	Elbasan, Albania
Leibensperger et al.	Sea surface warming suppresses primary sea spray aerosol number production	<i>Atmos Chem Phys</i>	Controlled wave-channel experiment	Bubble and SSA number concentration and emission flux, 0.008-20 μm	Global / multi-region
Li Q et al.	Health risk of fine particulate matter across China with a pollutant-specific index, 2013-2023	<i>Environ Sci Technol</i>	Machine-learning exposure model + meta-analysis + risk index	Daily 1 km PM _{2.5} , 2013-2023, machine-learning estimate; PM _{2.5} -specific AQHI	China
Li et al.	Multi-media multi-pathway PAH exposure and carcinogenic risk under industrial, urban and rural scenarios	<i>Environ Monit Assess</i>	Physical exposure model	PAHs in air, water, soil, dust, food (BaP-eq)	China
Li et al.	Upgrading diesel vehicle after-treatment technology significantly changed the composition of reactive nitrogen in exhaust	<i>Environ Sci Technol</i>	Vehicle emission experiment	Reactive nitrogen speciation (NO, NO ₂ , NH ₃ , N ₂ O)	China

First author	Focus	Journal	Design	Metric	Region
Li et al.	Study on the transport mechanism of surface particulate matter based on the closed earth-air model	<i>Environ Pollut</i>	Measurement campaign	PM ₁ , PM _{2.5} and PM ₁₀ in a semi-enclosed basement; earth-air flow, CO ₂ , 222Rn	China
Lin et al.	Elevated aerosol layers within the daytime mixed layer over ecologically sensitive areas of northwest China: diurnal variation, formation mechanisms and regional effects	<i>Atmos Chem Phys</i>	Measurement campaign	Aerosol vertical profile, 0.6-2 km layer	China
Lin et al.	Three-millennia alpine-lake records of divergent atmospheric Cd, Pb, Sb and Hg pollution in southwestern China	<i>Environ Sci Technol</i>	Environmental surveillance campaign	Atmospheric metal deposition: Cd, Pb, Sb, Hg	China
Liu et al.	Identification of dominant weather patterns and circulation regulation mechanisms of winter PM _{2.5} pollution in Iran	<i>Atmos Res</i>	Metadata only (no abstract)	Winter PM _{2.5} under synoptic weather types (not extractable)	Iran
Lu et al.	Moving-vehicle wakes and noise-barrier interaction in traffic emission dispersion	<i>Atmospheric Environment</i>	Metadata only (no abstract)	Traffic emissions / near-road dispersion	Not stated in deposit
Madalipay et al.	Establishing PM _{2.5} conversion factors for South Korea at provincial level	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	PM _{2.5} + PM ₁₀ jointly	South Korea
Mahia et al.	Real-time effects of street geometry on elderly well-being under combined air-heat exposure	<i>J Environ Manage</i>	Observational exposure study	Personal portable PM ₁₀ (to 18.77 µg/m ³) and NO ₂ , wearable physiology	Italy
Mainka & Mainka	Urban cycling vs. PM _{2.5} exposure: does the benefit outweigh the risk?	<i>Preprints.org</i>	Proxy validation vs personal exposure	Trip-resolved personal PM _{2.5} from low-cost sensors; inhaled dose via ventilation rate	Poland
Makacha et al.	Unequal exposure: spatio-temporal and structural determinants of personal PM _{2.5} exposure in three sub-Saharan African geographies	<i>Front Glob Womens Health</i>	Observational exposure study	PM _{2.5}	The Gambia / Kenya / Mozambique
Mandalapu et al.	Joint Heat and PM _{2.5} Exposure Across US Metropolitan Areas: Multi-Stressor Disparities, Historical Redlining, and a Multi-Metric Assessment Framework	<i>medRxiv (preprint)</i>	Ecological cross-sectional, copula tail dependence	Summer mean PM _{2.5} , tract-level, jointly with surface heat	United States
McCerery et al.	Arctic transport of North American and Canadian wildfire aerosols	<i>Atmos Environ</i>	Metadata only (no abstract)	PM (unspecified) wildfire aerosol	Arctic / North America
Meena et al.	Transplanted lichen trace-metal accumulation and pigment response along a Jaipur urban gradient	<i>Environ Pollut</i>	Biomonitoring transplant	Zn, Pb, Cd, Cr, Cu, Ni by ICP-MS; chlorophyll / carotenoid	South Asia
Mohammed	Spatiotemporal dynamics of atmospheric vulnerability in hot arid cities: introducing the Urban Microclimate Air Stress Index	<i>Build Environ</i>	Metadata only (no abstract)	Composite microclimate/air-stress index (not extractable)	Hot arid cities (not stated)
Moryani et al.	PM _{2.5} -bound Cu, Cd, Ni, Pb and Zn in 20 schools across two contrasting South China cities	<i>J Appl Toxicol</i>	Source apportionment	Heavy-metal content of PM _{2.5} , summer and winter	China
Moryani et al.	Inhalation bioaccessibility, pollution assessment, and speciation of heavy metals in PM _{2.5} road dust from urban and semi-urban areas of Pakistan	<i>Toxics</i>	Measurement campaign	PM _{2.5} -fractionated road dust; ALF-bioaccessible Cr, As, Pb	Pakistan
Mueangsai et al.	Factors Associated with Health Risk Quotient among Communities Exposed to PM _{2.5} from Stone Crushing Plants in Chonburi Province Thailand	<i>Res Sq (preprint)</i>	Community cross-sectional + USEPA HQ	Fixed-site PM _{2.5} , two stations, community average	Thailand
Mushore et al.	Daily outdoor PM _{2.5} exposure and the road-edge effect across three sub-Saharan campaigns	<i>Atmos Environ X</i>	Measurement campaign	PM _{2.5} (personal, interpolated to surfaces)	Sub-Saharan Africa
Myat et al.	Ground, satellite and model assessment of PM _{2.5} dynamics and attributable mortality in Myanmar	<i>Environ Pollut</i>	Physical exposure model	PM _{2.5} speciated, TROPOMI HCHO/NO ₂ , GAM	Southeast Asia
Naghizade et al.	Multi-class organic marker profiling of urban PM ₁₀ in Strasbourg	<i>Atmos Environ</i>	Metadata only (no abstract)	PM ₁₀ organic markers (not extractable)	Strasbourg, France
Nguyen et al.	Systematic review of satellite-plus-ML PM _{2.5} prediction and explainable AI	<i>Engineering Research Express</i>	Systematic review (PRISMA)	AOD, TOA reflectance, Angstrom exponent -> PM _{2.5}	Global
Parajuli et al.	Soil NO/HONO parameterisation choice and its effect on modelled O ₃ and PM	<i>Atmospheric Environment</i>	Model intercomparison	PM _{2.5} , O ₃ , NO ₂ (CMAQ)	Continental United States

First author	Focus	Journal	Design	Metric	Region
Park et al.	City-scale non-exhaust PM ₁₀ from emissions to exposure with a coupled chemistry-CFD framework	<i>Environ Pollut</i>	CTM simulation + source attribution	Non-exhaust PM ₁₀	Seoul
Pavlovic et al.	Aligning satellite Earth-observation capability, use case and capacity for air-quality health decisions	<i>J Air Waste Manag Assoc</i>	Mapping review	Satellite AOD-derived PM _{2.5} , trace gases	Global (GEO Health CoP)
Phalen et al.	Inhaled aerosol dosimetry update	<i>Inhal Toxicol</i>	Review / synthesis	Unspeciated PM; ultrafine	Multi-country
Pinakana et al.	Influence of transboundary wildfire smoke on fine particulate matter concentrations at South Texas border schools	<i>NPJ Clean Air</i>	Field campaign / remote sensing	PM _{2.5}	North America
Qing et al.	Seasonal and urban-suburban characteristics, emission and spatial sources of PM _{2.5} -bound PAHs and n-alkanes in Wuhu City	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	PM _{2.5} -bound PAHs and n-alkanes (not extractable)	Wuhu, China
Rahmani et al.	Airborne microplastics: a comprehensive analysis of indoor and outdoor pollution patterns	<i>Rev Environ Health</i>	Systematic review + random-effects meta-analysis	Airborne microplastic number concentration (MP/m ³), indoor vs outdoor	Multi-country
Rai et al.	Land surface temperature, land cover and aerosol optical depth over Ahmedabad, 1993-2023	<i>Environ Monit Assess</i>	Satellite remote sensing	MODIS AOD (2002-2023), Landsat LST	South Asia
Raptaki et al.	Atmospheric BTEX in Heraklion, Crete: levels, sources and contribution to secondary aerosol formation	<i>Atmos Environ</i>	Metadata only (no abstract)	Secondary organic aerosol precursors (BTEX)	Heraklion, Greece
Rashid et al.	Ultrafine particles and mortality: a scoping review of epidemiologic evidence and exposure assessment limitations	<i>Toxics</i>	Review / synthesis	Ambient ultrafine particle number concentration (<100 nm)	Global / multi-region
Ray et al.	Evaluating CCN activity through size-resolved hygroscopicity: synergistic observations and simulations in a tropical montane region	<i>Atmos Environ</i>	Metadata only (no abstract)	Size-resolved hygroscopicity (kappa), CCN	India
Ren et al.	Contrasting chemical characteristics and secondary formation of PM _{2.5} from mountain and urban sites in central China	<i>iScience</i>	Source apportionment	PM _{2.5} mass and SNA / OC-EC speciation, annual	China
Reyna-Bello et al.	Seasonal variability of PM _{2.5} -bound metals and meteorology in an urban-industrial area of northern Mexico: a PCA assessment	<i>Environ Monit Assess</i>	Filter sampling + composition	MiniVol low-volume PM _{2.5} mass; 10 metals by FAAS	Mexico
Roky et al.	Distance-resolved Pb dispersion and crop uptake from open battery-plate burning	<i>Environmental Pollution</i>	Dispersion model + soil sampling	PM _{2.5} , PM ₁₀ , Pb deposition	Kushtia, Bangladesh
Saarikoski et al.	Role of hydrocarbons and oxygenated organic species in traffic-derived aerosol	<i>Atmos Chem Phys</i>	Measurement campaign	Submicron organic aerosol (AMS)	Helsinki, Finland
Saeedi et al.	Feature selection and cross-validation strategy in land-use regression for mobile-monitored ultrafine particles	<i>Environ Sci Technol</i>	LUR model comparison	Ultrafine particle number (mobile)	Canada
Stevens et al.	Biomass burning marker fingerprints for Australian hardwoods, softwoods and grasses	<i>Atmos Environ</i>	Metadata only (no abstract)	Biomass burning PM tracers	Australia
Sui et al.	Contrasting annual variation of organic and elemental carbon in the Pearl River Delta across the COVID-19 lockdown	<i>J Environ Manage</i>	Source apportionment	PM _{2.5} organic and elemental carbon, PMF	China
Temu et al.	A six-domain framework for lichen biomonitoring including symbiont and microbiome response	<i>Environ Sci Pollut Res</i>	Literature review	Not applicable (framework)	Global / multi-region
Walker et al.	Divergent social patterning of directly measured environmental exposures across Rhode Island communities	<i>medRxiv (preprint)</i>	Direct-measurement field campaign	Short-term PM _{2.5} alongside LAeq/LCeq noise and night-time illuminance	United States
Wandji Nyamsi et al.	Broadband AOD reconstructed from sunshine-duration records since 1900	<i>Atmos Chem Phys</i>	Physical exposure model	Broadband aerosol optical depth	Global (2,700 sites)
Wang Q et al.	Quantitative analysis of aerosol acidity and its driving factors in the Guanzhong area	<i>Atmos Chem Phys</i>	Measurement campaign	PM _{2.5} water-soluble inorganic ions, pH	Guanzhong Plain, China
Wang T et al.	Chlorine radical cycling on photoactive particulate matter amplifies photolytic NOx release	<i>Environ Sci Technol</i>	Controlled chamber factorial	Photoactive PM surfaces, soot	Laboratory
Wang et al.	Direct thermal vs emission-mediated pathways in heatwave-driven O3 and SOA increases	<i>Atmos Chem Phys</i>	Chemical transport model	SOA, O3 (BVOC and soil-NO pathways)	China

First author	Focus	Journal	Design	Metric	Region
Wu et al.	Global fire activity exacerbates extreme PM _{2.5} pollution	<i>npj Clim Atmos Sci</i>	Metadata only (no abstract)	Extreme-percentile PM _{2.5} attributable to fire activity (not extractable)	Global / multi-region
Xu et al.	Multi-factor control of aerosol nitroaromatic compounds across 11 Chinese cities in winter	<i>Atmos Chem Phys</i>	Supersite observation	PM _{2.5} -bound nitrophenols and nitrocatechols	11 cities, China
Yadav et al.	Airborne endotoxin on PM ₁₀ in Agra: an 8-fold seasonal swing that PM mass does not track	<i>Research Square (preprint)</i>	Measurement campaign	PM ₁₀ mass; endotoxin (LAL); bacterial and fungal bioaerosol	South Asia
Yang et al.	Impacts of anthropogenic aerosols on a snowfall event in the Guanzhong Basin and surrounding areas	<i>Atmos Chem Phys</i>	Modelling / inventory	Anthropogenic aerosol (ARI and ACI pathways)	Guanzhong Basin, China
Yavuz et al.	High-resolution CFD modelling of traffic-related air pollutant dispersion in an urban area of Istanbul	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Traffic-related PM and NOx	Turkiye
You et al.	Satellite-based estimation of ambient ammonia reveals meteorological influences and emission inventory uncertainties in South Korea	<i>Environ Pollut</i>	Satellite remote sensing	Ground-level NH ₃ (secondary PM _{2.5} precursor), CrIS, 15 km monthly	South Korea
Yu et al.	Street-canyon increment to near-road NO ₂ quantified citywide with WRF-CMAQ-ADMS	<i>Environmental Pollution</i>	Chemical transport model	NO ₂ (traffic-related; canyon increment)	Hong Kong
Yu et al.	Probabilistic assessment of PM _{2.5} and ozone co-occurring extremes under wildfire smoke	<i>Environ Sci Technol</i>	Bayesian + geospatial regression	PM _{2.5} + O ₃ joint extremes	Contiguous USA
Zaman & Mim	Decadal variation of atmospheric PM _{2.5} over Bangladesh from MERRA-2 reanalysis	<i>Environ Sci Atmos</i>	Machine-learning model	MERRA-2 reanalysis PM _{2.5} and speciated components	Bangladesh
Zhang et al.	H ₂ O ₂ -driven sulfate formation at air-water interfaces: stepwise mechanism and accelerated kinetics	<i>Environ Sci Technol</i>	Theory / simulation	Particulate sulfate formation rate	Computational
Zhang et al.	National-scale heavy metal contamination in urban road dust across 82 Chinese cities	<i>Environ Geochem Health</i>	Review / synthesis	Road dust heavy metals	China
Zhang et al.	Photoinduced intraparticle chemistry drives the rapid atmospheric oxidation of biomass burning organic aerosol	<i>Environ Sci Technol</i>	Chamber + aircraft	Biomass-burning organic aerosol oxidation state	United States
Zhang et al.	Improved representation of isoprene-derived secondary organic aerosol in CAM6-Chem reveals regional contrasts in its long-term changes over China	<i>Atmos Chem Phys</i>	Chemical transport modelling	Isoprene-derived SOA mass and subspecies composition	China
Zheng et al.	Spatially heterogeneous drivers of winter PM _{2.5} in China from synchronous observations and explainable machine learning	<i>Atmos Environ</i>	Machine-learning model	Winter PM _{2.5} , synchronous multi-site observation	China
Zheng et al.	Aerosol modulation of precipitation systems migrating from mountains to plains in Northern China during early autumn	<i>Atmos Res</i>	Metadata only (no abstract)	Aerosol-precipitation interaction (not extractable)	Northern China
Zhong et al.	Photochemical ultrafine particles pose potential inflammatory risk despite low PM _{2.5} mass	<i>Environ Int</i>	Observational + ML	Lung-deposited surface area vs PM _{2.5} mass	Fuzhou, China
Zhou M et al.	Surface PM _{2.5} air pollution in 2022 India: emission updates, WRF-Chem evaluation, and source attribution	<i>Atmos Chem Phys</i>	CTM simulation + source attribution	PM _{2.5} (WRF-Chem, speciated)	India
da Costa Cardoso et al.	Legacy Pb-Zn mine tailings as persistent sources of bioaccessible toxic elements in urban dust	<i>Environ Monit Assess</i>	Field measurement	Resuspended urban dust, Pb, Zn, Cd, Cu, Mn	Brazil
de la Torre et al.	A 15-year atmospheric record of intentionally produced persistent organic pollutants in Spain	<i>J Hazard Mater</i>	Environmental surveillance campaign	Semi-volatile POPs (passive PUF samplers)	Spain
Burden, policy & mitigation					
Acat et al.	National trends in ambient air pollutants and cause-specific mortality in Türkiye, 2018-2024: an ecological time-trend study	<i>Atmosphere</i>	Monitoring-record analysis	Unweighted national mean station PM ₁₀ (50.4 → 41.0 µg/m ³); PM _{2.5} excluded	Turkiye
Alamsyah et al.	Industrial intensity and traffic congestion propagate air-pollution health vulnerability across district borders in Java	<i>Environ Sci Pollut Res Int</i>	Ecological panel	Satellite pollutant fields + OpenStreetMap congestion, district level	Indonesia

First author	Focus	Journal	Design	Metric	Region
Anastasopoulos et al.	Comparative health burden analysis at multiple geographic scales: mortality attributed to ambient air pollution by sector	<i>Atmos Environ</i>	Metadata only (no abstract)	Sectoral anthropogenic PM _{2.5} (not extractable)	Canada
Anon et al.	Lifetime risk of respiratory system cancers across 185 countries: Global variation, socioeconomic inequalities, and ecological associations with air pollution	<i>Prev Med Rep</i>	Ecological (GLOBOCAN secondary analysis)	Country-level annual PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂	Multi-region
Anon et al.	A Harmonization Index for Comparative Assessment of Industrial Particulate Matter Emission Regulations	<i>Res Sq (preprint)</i>	Regulatory coding framework	Industrial PM emission limits across regulatory architectures	Multi-region
Bhatta et al.	Outdoor air purification technologies for urban PM and gaseous pollutant mitigation: a comprehensive review	<i>Sci Total Environ</i>	Systematic review (PRISMA)	PM _{2.5} and PM ₁₀ removal efficiency across five technology domains	Global
Bosso et al.	AIRTREE v2.0: predicting ecosystem services provided by Italian urban parks under climate change scenarios	<i>J Environ Manage</i>	Modelling / inventory	PM ₁₀ and PM _{2.5} removal by urban canopy	Italy (43 sites)
Bu & Sun	Particulate matter emissions from the iron and steel industry of China: patterns, drivers, and implications	<i>J Air Waste Manag Assoc</i>	Emission inventory / decomposition	PM (stack and fugitive)	China
Caikai et al.	Risk-factor-attributable burden of respiratory infection and tuberculosis, 1990-2021	<i>J Infect Dev Ctries</i>	Burden estimation (GBD)	Air pollution as GBD risk factor (ASMR, ASDR)	Global / multi-region
Cliff et al.	How do climatic factors directly influence the incidence and risk of meningococcal meningitis across the African meningitis belt? A narrative literature review	<i>GeoHealth</i>	Narrative review	Atmospheric mineral dust load and wind speed	Sub-Saharan Africa
Faggi et al.	Heatwaves and particulate matter increases: an Italian case study	<i>Atmos Environ</i>	Metadata only (no abstract)	PM _{2.5} / PM ₁₀	Italy
Fares et al.	Can urban green infrastructure influence the urban heat island effect and air quality nexus?	<i>Environ Pollut</i>	Review / synthesis	PM deposition and thermophoresis onto green infrastructure	Global
Galvin et al.	Cyclic-operation emission factors for biodiesel, renewable diesel and hydrogen-blend boilers	<i>J Air Waste Manag Assoc</i>	Chamber emission characterisation	Filterable + condensable PM (EPA M5/M202) + real-time PM	United States
Gardner-Frolick et al.	Double disproportionality: concentrated PM _{2.5} emitters and marginalised siting in Canada	<i>iScience</i>	Spatial econometric model	PM _{2.5} and PM _{2.5} precursor emissions (NPRI)	Canada
Goyal et al.	Quantifying environmental co-benefits of nitrogen-based crop restructuring and its implications on India's inter-state trade network	<i>Nat Commun</i>	Modelling / inventory	Precursor emissions (N)	India
Gu et al.	Electro-ultrasonic coupled agglomeration for removal of toxic fine particles from biomass combustion	<i>J Hazard Mater</i>	Chamber / laboratory	PM (unspecified) from biomass combustion	China
Han et al.	Use of air quality index products versus clinician discussion among US children with asthma	<i>AJPM Focus</i>	Cross-sectional survey	Self-reported AQI product use	North America
He et al.	Premature ischaemic stroke burden in adults aged 50-69 in China, India and the United States	<i>Neuroepidemiology</i>	GBD secondary analysis	Ambient PM _{2.5} (population attributable fraction)	Global
Hou et al.	Bimodal polysulfonamide fibrous aerogel for high-temperature PM _{0.3} filtration	<i>J Hazard Mater</i>	Materials / filtration	PM _{0.3} filtration efficiency, pressure drop	Computational / laboratory
Kim et al.	Spatial equity and cost-effectiveness of full diesel-to-BEV replacement	<i>J Environmental Management</i>	Chemical transport model	PM _{2.5} , O ₃ (population-weighted concentration)	Republic of Korea
Kokol et al.	Artificial intelligence in the management of climate-change-related health issues in Africa: scoping review	<i>Res Sq (preprint)</i>	Scoping review	PM (unspecified), bibliometric corpus	Sub-Saharan Africa
Lee et al.	Toward a unified framework for climate and health research: gaps in exposure metrics and health outcomes	<i>J Korean Med Sci</i>	Mapping review	Climate exposure metrics incl. air pollution	Global
Li et al.	Drivers of air pollutant and greenhouse gas emissions since 2010 in the Greater Bay Area, China	<i>Atmos Environ</i>	Metadata only (no abstract)	PM and co-emitted pollutants, emissions	China
Liao et al.	Beyond mortality: a multi-outcome assessment of health economic benefits from air pollution control in China	<i>Environ Res</i>	Burden / economic model	PM _{2.5} , NO ₂ and O ₃ concentration-response	Guangdong, China (18 cities)

First author	Focus	Journal	Design	Metric	Region
Lim et al.	Public responses about air quality in the world's ten most populous countries	<i>Nat Commun</i>	Multi-country quota survey + exposure linkage	Modelled ambient PM _{2.5} linked to respondent locality	Global / multi-region
Liu D. et al.	COPD burden 1990-2050: joinpoint, age-period-cohort and Bayesian APC modelling of diverging risk-factor attribution	<i>Medicine (Baltimore)</i>	Burden estimation (GBD)	Ambient PM _{2.5} and household air pollution as GBD risk factors (PAF)	Global
Liu M. et al.	Divergent trends in near-surface ozone and nitrogen dioxide and ozone-attributable health burden in the Huaihe River Basin, 2013-2024	<i>Environ Monit Assess</i>	Satellite remote sensing	Satellite-derived surface MDA8 O ₃ and NO ₂ (no PM metric)	China
Liu et al.	Disproportionate effects of direct air capture on regional health cobenefits of net-zero	<i>Environ Sci Technol</i>	Health impact model	Grid-level (36 km ²) modelled PM _{2.5} and O ₃ attributable mortality	China
Liu et al.	Lung cancer incidence and mortality trends with age-period-cohort analysis, Zibo 2014-2024	<i>Front Oncol</i>	Ecological panel	Ambient air pollution (contextual)	China
Moghaddam et al. 2026	Reducing particulate matter emissions in landfills using novel dust suppressants	<i>Research Square</i>	Chamber / laboratory	Wind-eroded fugitive dust (wind-tunnel)	Mashhad, Iran
Mou et al.	Impacts of ship emissions on regional air quality in the post-DECA era: the Yangtze River Basin	<i>Environ Pollut</i>	CTM simulation + source attribution	Ship-source PM _{2.5} , SO ₂ , NO _x , nitrate; WRF-Chem increments	China
Nilsson et al.	Urban environment, health and equity: symptoms and annoyances related to air pollution, traffic noise and greenness	<i>Environ Epidemiol</i>	Cross-sectional population survey	Dichotomised residential PM _{2.5} , PM ₁₀ , NO ₂ at geocoded address	Sweden
Okeke et al.	Agricultural methane emissions and non-communicable disease mortality in West Africa, 2015-2020	<i>Environ Monit Assess</i>	Ecological panel, fixed effects	Agricultural CH ₄ (Mt CO ₂ e)	West Africa
Ozmen et al.	Long-term ambient PM _{2.5} exposure and premature mortality across Türkiye	<i>Toxics</i>	Burden estimation (GBD)	Province-level annual mean PM _{2.5} , counterfactual 5 µg/m ³ (WHO AirQ+)	Türkiye
Sathasivam & Veluchamy	Visual characteristics of PM from <i>Halymenia dilatata</i> microalgae biodiesel compared to low-sulfur diesel fuel	<i>Environ Sci Pollut Res</i>	Chamber emission characterisation	Gravimetric PM mass + SEM morphology	India
Sharma et al.	Geospatial linkage of environmental exposures, socioeconomic profile and health outcomes among women of reproductive age, Victoria	<i>Health Promot J Austr</i>	Ecological geospatial linkage	Postcode-level PM _{2.5} , NO ₂ , CO, SO ₂ , HCHO, O ₃ , LST, greenness	Australia
Shin et al.	Occupational disparities in PM _{2.5} -related mortality under distributed lag models	<i>J Occup Environ Med</i>	Time-series	PM _{2.5} (daily, city-wide)	Seoul, South Korea
Sidharaddi et al.	Electrodeposited copper nodule films as electrostatic PM collectors	<i>Research Square (preprint)</i>	Experimental / toxicology	Dhoop-smoke particulate (chamber, unspiced)	South Asia
Simion et al.	Attributing climate-related health burdens in Southeast Asia: a scoping review	<i>Lancet Reg Health West Pac</i>	Scoping review	Air pollution, heat and extreme-weather attribution studies	Southeast Asia
Sirikul et al.	Knowledge, attitudes and practices of caregivers regarding PM _{2.5} exposure among children in northern Thailand	<i>J Clim Chang Health</i>	Cross-sectional survey	None measured; AQI literacy and protective behaviour	Thailand
Slavik et al.	Perceived smoke beats measured smoke as a predictor of parental protective action	<i>Environ Chall</i>	Observational - cross-sectional	PM _{2.5} on smoke days (monitoring data) vs self-reported sight/smell	North America
Sommar et al.	Health economic assessment of PM _{2.5} originating from residential wood combustion - a case study in northern Sweden	<i>Int J Environ Res Public Health</i>	Burden estimation (GBD)	Population-weighted RWC-attributable PM _{2.5} , 0.16 µg/m ³ (IQR 0.10-0.21)	Sweden
Song et al.	Shifting PM _{2.5} control from mass to toxicity for health equity worldwide	<i>Environ Sci Technol</i>	Perspective / evidence synthesis	Source-specific PM _{2.5} toxicity weighting applied to global exposure surfaces	Global / multi-region

First author	Focus	Journal	Design	Metric	Region
Suryanarayanareddy et al.	Low-cost bio-based particle-capturing device for vehicular emissions: pilot study	<i>Environ Sci Pollut Res Int</i>	Bench / device experiment	PM _{2.5} , PM ₁₀ , CO, CO ₂	India
Tian et al.	Bioinspired honeycomb-structured nanofibrous membranes for high-efficiency PM capture	<i>Nanomaterials (Basel)</i>	Bench / device experiment	PM _{0.3} , PM _{2.5}	Laboratory
Urso et al.	Burden of dementia in Europe, 1990-2023: a systematic analysis from the Global Burden of Disease Study 2023	<i>Lancet Reg Health Eur</i>	Burden estimation (GBD)	Ambient PM (GBD risk-factor exposure)	EU-27
Vartanyan et al.	Structural inequities and chronic disease in Fresno County: a narrative review and multi-sectoral framework	<i>Cureus</i>	Narrative review	Air + pesticide pollution (qualitative)	North America
Wan	Ecological restoration as a population health intervention, with PM _{2.5} as one mediator	<i>Econ Hum Biol</i>	Ecological	PM _{2.5} concentration (mediator alongside greenness and visibility)	China
Wang et al.	Trends, inequalities, and 2030 projections of women's tracheal, bronchial, and lung cancer attributable to air pollution, radon, and tobacco exposure	<i>Front Glob Womens Health</i>	Burden estimation	Ambient particulate matter and household solid fuel (GBD risk factors)	Global
Wojcik et al.	An air pollution curriculum for pulmonary and critical care fellows	<i>ATS Sch</i>	Educational intervention	PM (unspecified), curriculum content	USA
Wu & Cui	Impact of aircraft CO, NO ₂ and SO ₂ emissions on public health in Chinese airports	<i>J Glob Health</i>	Burden estimation (GBD)	CO, NO ₂ , SO ₂ (no PM metric)	China
Xia et al. 2026	Surface PM _{2.5} mass and composition under stratospheric sulfate geoengineering	<i>arXiv (preprint)</i>	Chemical transport modelling	Modelled surface PM _{2.5} mass and speciation (dust, sea salt, OC, sulfate, SOA)	Global
Xie et al.	Thirty-year change and projection of Asian tracheal, bronchus and lung cancer burden	<i>J Thorac Dis</i>	GBD secondary analysis	Ambient particulate matter pollution	Asia
Yoon et al.	Heat vs bushfire-smoke PM _{2.5} across four arms of the Australian health system	<i>Med J Aust</i>	Ecological	PM _{2.5} (bushfire smoke) and temperature, short-term lags	Oceania
Zaid et al.	Fibrous aerosol filtration: where CFD assumptions and measurement uncertainty diverge from practice	<i>J Mater Sci</i>	Review / synthesis	Ultrafine particles; most penetrating particle size; pressure drop	Global / multi-region
Zhai et al.	Global disparities in lifetime risk of lung cancer: links to socioeconomic development, smoking, and fine particulate matter from GLOBOCAN 2022	<i>Front Public Health</i>	Ecological	Country-level PM ₁ and PM _{2.5} from global environmental databases	Global / multi-region
Zhang et al.	Particulate emission risks of HEVs during cold-starts and warm-restarts: particulate number under fresh and aged aftertreatment systems	<i>J Hazard Mater</i>	Vehicle emission experiment	Particle number; particle number size distribution; sub-23 nm PN	China
Zhang et al. 2026	Health impacts of national and local air pollution control policies targeting electric generating units, mobile sources, and port activities in three US cities	<i>medRxiv (preprint)</i>	Time-series / case-crossover	PM _{2.5} ; PM sources (PMF)	North America
Zhang et al.	Health and economic impacts of improved air quality from transitioning global shipping energy to hydrogen	<i>Nat Commun</i>	Integrated emission-air quality-health modelling	Modelled PM _{2.5} and O ₃ concentration change under hydrogen scenarios	Global / multi-region
Zhu et al.	Short-term effectiveness of urban road-cleaning operations for fugitive road dust control: evidence from TRAKER-based mobile monitoring	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	TRAKER-derived road-dust emission potential (not extractable)	Not stated
Other clinical endpoints					
Chen et al.	Air pollution and sarcopenia: a meta-analysis and network toxicology analysis	<i>Atmos Environ</i>	Metadata only (no abstract)	Air pollution mixture (not extractable)	Not stated
Chen et al.	Untargeted metabolomics reveals metabolic perturbations in community-dwelling elderly exposed to PM _{2.5} -bound ester compounds	<i>Metabolites</i>	Observational - cross-sectional	72-h personal PM _{2.5} (UPAS V2), mean 38.06 µg/m ³ ; 20 GC-MS-prioritised esters	China
Chin et al.	Impact of air quality and PM _{2.5} on hidradenitis suppurativa activity in the Sacramento region	<i>Res Sq (preprint)</i>	Retrospective time series	Days above 12 µg/m ³ PM _{2.5} (monthly count)	North America

First author	Focus	Journal	Design	Metric	Region
Choi et al.	Time-varying long-term PM _{2.5} and leukaemia incidence from childhood into early adulthood	<i>Environ Res</i>	Nationwide cohort	Annual district PM _{2.5} from a machine-learning ensemble, 226 inland districts	South Korea
Chow et al.	Blood metabolomic signatures linking air pollution to lung cancer	<i>Nat Commun</i>	Prospective cohort	PM _{2.5} , O ₃ , plus four co-pollutants	United States
Curr Hematol Malig Rep	Linking air pollution and hematologic malignancies: from epidemiology to molecular mechanisms	<i>Curr Hematol Malig Rep</i>	Mechanistic review	PM, benzene, NO ₂ , SO ₂ , arsenic, ethylene oxide	Global
DeAngelis et al.	Residential context and epigenetic age acceleration: a null for air pollution	<i>Social Science & Medicine</i>	Cross-sectional (multistage)	Neighbourhood air pollution (1 of 5 contextual features)	United States
Donaldson et al.	Air Pollution Impairs Marathon Performance: A Cross-Sectional Analysis of 2.7 Million Finishers Across Six World Marathon Majors	<i>Sports Med Open</i>	Observational - cross-sectional	Race-day NO ₂ and PM _{2.5} from CAMS global reanalysis, per 1 µg/m ³	Multi-country
Feng et al.	Plasma metabolic and proteomic signatures of mixed ambient air pollutant exposure and incident arthritis subtypes	<i>Joint Bone Spine</i>	Prospective cohort (mixtures) + omics mediation	Annual PM _{2.5} , PM ₁₀ , NO _x , NO ₂ combined in a weighted quantile sum index	United Kingdom
Guerrero-Morales et al.	Spatial distribution of long-term ultrafine particle concentrations and systemic lupus erythematosus burden in central Mexico	<i>Environ Toxicol Pharmacol</i>	Ecological / spatial	Satellite-modelled UFP, 1,500-24,000 particles/cm ³	Central Mexico
Huang et al.	Long-term exposure to PM _{2.5} -bound metals and female breast cancer incidence in New Mexico, USA	<i>Air Qual Atmos Health</i>	Ecological	Census-tract PM _{2.5} -bound Cr, Co, Cu, Mn, Hg over a 10-year pre-diagnosis window (EWPM)	United States
Jain et al.	Ambient air pollution and glaucoma: systematic review and meta-analysis	<i>J Curr Glaucoma Pract</i>	Systematic review (PRISMA)	PM _{2.5} , PM ₁₀ , NO ₂ , NO _x , CO	Global
Jung et al.	PM _{2.5} and rheumatoid arthritis flare: 12,583 visits with a case-crossover sensitivity arm	<i>Ann Rheum Dis</i>	Observational - cohort	PM _{2.5} , PM ₁₀ and four gases (monthly means; daily for case-crossover)	East Asia (ex-China)
Kang et al.	Short-term ambient air pollution and acute osteoarthritis morbidity across subgroups	<i>Environ Pollut</i>	Case-crossover	District-level PM ₁₀ , PM _{2.5} , NO ₂ , SO ₂ , O ₃ , CO	South Korea
Lee et al.	Short-term effects and attributable burden of ambient VOC mixtures on paediatric atopic dermatitis and asthma	<i>J Expo Sci Environ Epidemiol</i>	Metadata only (no abstract)	Ambient VOC mixture (not extractable)	Seoul, South Korea
Lees	Null with a nitrate signal: PM _{2.5} component mixtures and hospitalisation in CKD	<i>Research Square (preprint)</i>	Observational - cohort	PM _{2.5} components OC, EC, NH ₄ , NO ₃ , SO ₄ (NASA SEDAC, 3-digit ZIP)	North America
Lei et al.	Hospital admission risk and burden of acute appendicitis due to fine particulate matter pollution	<i>Atmos Environ</i>	Metadata only (no abstract)	PM _{2.5} (not extractable)	China
Li et al.	Global and national CKD burden attributable to fine particles: updated exposure-response meta-analysis	<i>Innovation (Camb)</i>	Umbrella review / meta-analysis	PM _{2.5}	Global
Liu et al.	Reply: statistical fragility and methodological constraints in metabolomic mediation	<i>JACC Adv</i>	Metadata only (no abstract)	PM _{2.5}	United States
Mandal et al.	Long-term exposure to ambient particulate matter and kidney function in urban Indian adults	<i>Kidney Int Rep</i>	Observational - cohort	Annual mean PM _{2.5} , 1 km x 1 km national spatiotemporal model at residential geocode	India
Mao et al.	Component-specific effects of PM _{2.5} on endoscopically identified erosive gastric mucosal lesions: a time-series study in Yixing, China	<i>Front Public Health</i>	Time-series (generalized additive model)	Gridded daily PM _{2.5} and five chemical components	China
Negahi et al.	Climate-sensitive exposures and cancer outcomes: a systematic review and meta-analysis	<i>Int J Biometeorol</i>	Systematic review (PRISMA)	PM _{2.5} , wildfire smoke	Global

First author	Focus	Journal	Design	Metric	Region
Opperbeek et al.	Exposome-wide search for drivers of epigenetic ageing, and the size of the missing exposome	<i>Environment International</i>	Exposome-wide association	Aerial black carbon among 186 exposures	Finland (FinnTwin12)
Park et al.	Short-term fine particulate matter and hepatic decompensation in cirrhosis: a nationwide case-crossover study	<i>Gut Liver</i>	Case-crossover	Daily PM _{2.5} on a 1 km residential grid	South Korea
Park et al.	Association between annual air pollutant exposure and thyroid function: a retrospective cross-sectional study	<i>J Endocrinol Invest</i>	Retrospective cross-sectional (health screening)	Annual mean PM _{2.5} and PM ₁₀ from CMAQ, per IQR	South Korea
Poursafa & Jylhava	Aging in the air we breathe: mechanisms and consequences of air pollution exposure for biological aging	<i>Ageing Res Rev</i>	Mechanistic review	PM _{2.5} , PM _{0.1} , TRAP, household combustion (qualitative)	Global
Sassano et al.	Household solid-fuel use and kidney disease mortality in the Golestan Cohort	<i>Environ Health Perspect</i>	Prospective cohort	Household fuel type and duration (biomass / kerosene / gas), questionnaire	Middle East & N. Africa
Schlaepfer et al.	Environmental hazards and risk of idiopathic urethral stricture	<i>Transl Androl Urol</i>	Retrospective cohort	Ambient PM, VOCs, metal oxides	Salt Lake City
Traisathit et al.	Air pollution and mortality in acute lymphoblastic leukaemia across age groups, northern Thailand	<i>PLoS One</i>	Retrospective cohort	PM _{2.5} , PM ₁₀ , O ₃ , CO	Thailand
Wang et al.	Early-life PM _{2.5} mass and components and childhood anaemia across China, India and 26 African countries	<i>Ecotoxical Environ Saf</i>	Cross-sectional (multistage)	MERRA-2 pre-natal/postnatal/cumulative PM _{2.5} + BC, OC, sulfate, dust, sea salt	Multi-country
Wang et al.	Long-term exposure to PM _{2.5} , aging hallmarks and age-related diseases: a nationwide prospective cohort study	<i>Atmos Environ</i>	Metadata only (no abstract)	Long-term PM _{2.5} (not extractable)	Not stated
Wang et al.	Ambient PM _{2.5} and thyroid health: a cross-sectional study of 24,333 participants in Zhejiang, China	<i>J Hazard Mater</i>	Cross-sectional (multistage)	PM _{2.5}	China
Wei et al.	Chemical composition of fine particulate matter and all-cause mortality of older lung cancer patients	<i>Environ Res</i>	Cohort	15 PM _{2.5} chemical components, 2-year moving average	USA
Wei et al.	Prospective Metabolomic Cohort of 123,037 Men Reveals Lipid Mediators Linking Air Pollution to Benign Prostatic Hyperplasia	<i>Environ Health (Wash)</i>	Prospective cohort (mixtures)	Annual PM _{2.5} , PM ₁₀ , NO ₂ , NO _x ; WQS and qgcomp mixtures	United Kingdom
Wu et al.	Short-term environmental exposures and emergency department imaging utilization	<i>J Am Coll Radiol</i>	Time-stratified case-crossover	Daily PM _{2.5} , per 10 µg/m ³	Canada
Yang et al.	Associations of long-term exposure to multiple air pollutants with chronic pain among middle-aged and older adults in China	<i>J Pain</i>	Cohort	PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , CO (5-pollutant mixture, WQS)	China
Zewdie et al.	Wildfire smoke PM _{2.5} and cancer history prevalence at census-tract level	<i>Cancer Epidemiol</i>	Ecological panel	Daily modelled wildfire-specific smoke PM _{2.5} , 2006-2020	USA
Zhang & Yu 2026	Environmental and calendar correlates of daily emergency department demand in Wuhan, China: a single-centre time-series study with temporal validation	<i>Res Sq (preprint)</i>	Time-series / case-crossover	Pollutant-specific AQI (PM _{2.5} , PM ₁₀ , NO ₂ , O ₃), lag 0-2, per IQR	China
Zhang et al.	PM _{2.5} exposure and the prognosis of severe fever with thrombocytopenia syndrome: an integrated epidemiological and experimental study	<i>Environ Sci Technol</i>	Cohort	Long-term pre-admission ambient PM _{2.5}	Xinyang, China
Zhang et al.	Plasma metabolomic signatures of air pollution and rheumatoid arthritis risk	<i>Front Public Health</i>	Observational - cohort	Residential annual PM _{2.5} , PM ₁₀ , NO ₂ , NO _x and a composite air pollution score	United Kingdom
Zinan'dala et al.	Urinary metabolomic alterations associated with contrasting ambient air pollution exposure in Thai adults: an untargeted 1H-NMR study	<i>Int J Mol Sci</i>	Observational - cross-sectional	Regional residence as proxy for PM _{2.5} contrast (Chiang Mai vs Songkhla)	Thailand

Provenance and method — read this before citing any count above

This issue keeps the standing rolup contract. Unlike the July monthly, no API was re-queried to build it: every one of the 556 records was published in one of the **31 daily issues covering 1–31 August 2026**, each of which harvested its own single-day entry window, and this rollup is pure aggregation over `state/corpus/2026-08-*.json` and `state/metrics.csv`. There is no backfill batch and no curated sample: subject to the source limitations below, every proportion here is a proportion of what the daily windows actually recovered.

Sources across the month. PubMed E-utilities on `[Date - Entry]`, two axes (health, instrumentation), queried daily. **Europe PMC** on `CREATION_DATE`; productive early in the month, returning zero on several late-August days. **Crossref by-ISSN** across 25 tracked non-PubMed journals, windowed on `created` — this channel was built during August in response to July's stated gap and is the reason *Atmos Chem Phys* and *Atmos Meas Tech* appear at all. **arXiv** relevance query with a local recency screen. **OpenAlex** returned HTTP 403 on every call all month and contributed nothing; its coverage was routed through **Consensus**, which has no entry-date filter and therefore contributes explicitly labelled dated backfill rather than windowed harvest. **Google Scholar** was never a dependency.

Known limits on every count. (a) **Eight within-August duplicate records** across seven DOIs are included in all totals; see defect (i) in the gaps box. Counts are over-stated by roughly 1.4%. (b) **Sixty-five records (12%) are classification-only**, with no abstract available at issue time. (c) Deposit timing at a few high-volume publishers drives most day-to-day variation, so daily counts are not a publication-rate signal — read week totals. (d) Subtopic, design, particle-metric and geography labels are single-label judgements from title, abstract, keywords and MeSH terms, not a controlled vocabulary. (e) Effect estimates were transcribed verbatim from abstracts and only where an interval was reported; 148 qualified, from 63 of 556 records. (f) Every DOI in every daily issue passed `check_dois.py` — PMID-authority and resolution checks — before that issue shipped.

Trial watch aggregates `state/trials.json` over seven sweeps; the ClinicalTrials.gov connector has no reliable last-update filter on some axes, so parts of the watch are relevance-ranked rather than windowed and the list is a floor. Content derived from PubMed and Crossref metadata; abstracts remain the copyright of their respective publishers.